Contradiction Fields: CRT Rigidity, Prime Matrix Geometry, and Zeta-Zero Escape

Project manuscript

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Abstract

This monograph draft unifies two contradiction-field research programs. The first studies prime existence in CRT prime matrices through row, column, diagonal, anchor, and rough-composite rigidity. The second studies an off-critical zero of the Riemann zeta function through a global contradiction-field ledger for prime anomalies. The common method is to convert a hypothetical counterexample into a structured covering field, decompose it into local rigidity layers, and exclude all free escape channels by capacity, periodicity, and no-cycle bookkeeping.

This is a synthesis and dependency manuscript. It does not assert that the Riemann Hypothesis or the prime-matrix row/column theorem has already been accepted as a final unconditional theorem. Each claim is marked by status in the accompanying status table: proved in manuscript, reduced to a structured input, external, computationally certified, or requiring independent referee verification.

Contents

Chapter 1

Guiding Principle: The Contradiction Field

A contradiction field is a structured ledger attached to a hypothetical counterexample. Instead of estimating a single exceptional set, it records every possible explanation for the counterexample as a named channel. A proof closes when every channel is either absorbed, forced to descend in a finite potential, transferred with source deletion, or contradicts a global capacity lower/upper bound.

The common rigidities used throughout this manuscript are:

1. CRT skeletons and nonzero-residue balance;
2. local short-window non-reuse of large prime factors;
3. small-prime shell migration and diagonal locking;
4. rough-composite and double-anchor decompositions;
5. capacity ledgers after baseline subtraction;
6. no-cycle potentials for internal escape mechanisms.

Chapter 2

Critical-Density Feedback Bridge

This chapter records the critical-density principle used to organize the current frontier. It is a bridge from heuristic insight to proof obligations, not a replacement for the remaining row/column, two-point, or RH exit proofs.

Definition 2.1 (Self-sieving density model)**.** Let a self-sieving set have local density

$$\rho(x) \sim \frac{C}{(\log x)^{\alpha}}.$$

Its previous elements impose a first-order one-residue sieve intensity

$$S(x) = \sum_{a \le x,\ a \in A} \frac{1}{a} \sim \int_2^x \frac{\rho(t)}{t}\, dt.$$

The associated model survivor density is $\exp(-S(x))$.

Lemma 2.2 (Critical fixed point)**.** *Inside the self-sieving density model, the only logarithmic fixed-point exponent is*

$$\rho(x) \asymp \frac{1}{\log x}.$$

More precisely, if $\rho(x) \sim C/(\log x)^{\alpha}$ and the model survivor density is required to have the same logarithmic order as $\rho(x)$, then $\alpha = 1$; at that layer, exponent matching gives $C = 1$.

Proof. If $\alpha > 1$, then

$$\int_2^x \frac{C\, dt}{t(\log t)^{\alpha}}$$

converges, so the model survivor density stays bounded away from zero while $C/(\log x)^{\alpha}$ tends to zero. This is not self-consistent.

If $\alpha < 1$, then the same integral is

$$\frac{C}{1-\alpha}(\log x)^{1-\alpha} + O(1),$$

so the survivor density is

$$\exp\{-c(\log x)^{1-\alpha} + O(1)\},$$

which is smaller than any fixed negative power of $\log x$. This is again not of order $C/(\log x)^{\alpha}$.

Thus $\alpha = 1$. Then $S(x) \sim C \log\log x$, and the survivor density is $(\log x)^{-C+o(1)}$. Matching the logarithmic exponent with $C/\log x$ forces $C = 1$ at the model level. □

Remark 2.3 (Proof boundary)**.** The lemma is a fixed-point calculation for the self-sieving model. It does not by itself prove a short-interval prime, a prime pair, or RH. To become a theorem, the fixed-point pressure must be upgraded into local lower bounds, new-modulus anti-concentration estimates, or controlled-exit inequalities.

Reduction-closed Statement 2.4 (Critical feedback bridge for the three programs)**.** *The critical-density principle routes the three current frontiers as follows.*

1. *For the prime-matrix row/column program, a fully covered row or column is a local collapse below the* $1/\log x$ *fixed point. Such a collapse must be explained by fixed CRT concentration, by moving high-factor slots that pay Rankin/SAE mass, or by a failure of the local survivor lower bound. The active narrow gate is*

   $$\mathtt{AffineTwinFillResidueArrivalBoundOrFillCatchUpMassPDECExclusion}.$$

2. *For the two-point secondary sieve, the corresponding fixed point is the two-residue product*

   $$\prod_{q\le y}\left(1-\frac{\nu_q(w)}{q}\right)\asymp\frac{\mathfrak{S}(w)}{(\log y)^2}.$$

   The remaining theorem-level input is not a one-point density estimate but the true-residual new-modulus correlation bound in the I3-Core/TRC package.

3. *For the RH contradiction field, an off-critical zero creates a smooth prime anomaly relative to the same* $1/\log x$ *fixed point. The anomaly must pass through the recorded sparse, dense, tail, internal, and global exits; the manuscript can be promoted only after those controlled exits are independently verified under one load convention.*

Status and reference. This is a routing statement. The detailed version is archived as

`docs/monograph/three-claims-critical-density-feedback-frontier.md`.

It identifies the next prime-matrix input as a critical fill-arrival anti-overfilling theorem: persistent AffineTwin threshold crossing must either reuse a residue and become a reset/ColumnCRT/PDEC certificate, or introduce new fill residues whose Rankin/SAE mass exceeds the critical-density carrying capacity. That final carrying-capacity inequality is the new proof obligation; it is not asserted here as already proved. □

Reduction-closed Statement 2.5 (Critical fill-arrival pressure packet frontier)**.** *In the AffineTwin branch, write* $g=q-2$*,* $f=q$*, and let* A_g, A_f *be the generator-side and fill-side residue counts. The safe integer gate is*

$$A_gA_f\le T_q:=\lfloor\sqrt{gf}\rfloor.$$

For a threshold-crossing route with final counts A'_g, A'_f*, crossing is equivalent to*

$$A'_gA'_f>T_q,\qquad \textit{hence}\qquad \frac{(A'_g)^2}{g}\frac{(A'_f)^2}{f}>1.$$

Thus fill catch-up is not merely a raw residue-arrival event. It is a threshold-relevant pressure packet: for fixed A'_g *the minimum fill target is*

$$A'_{f,\min}(A'_g)=\left\lfloor\frac{T_q}{A'_g}\right\rfloor+1.$$

Consequently a persistent row/column counterexample in this branch must enter one of four named exits: repeated-residue reset/ColumnCRT PDEC, fixed pressure packet recurrence PDEC, moving pressure packet SAE/Rankin, or non-sparse AffineTwin pressure demand.

Status and reference. The product gate and the pressure form are algebraic equivalents. The minimum fill target follows by solving $A'_g A'_f > T_q$ in integers. The repeated-residue exit is the previous fill-catchup dichotomy; the fixed-packet exit is the AffineTwin slot phase lock, where the combined modulus $q(q-2)$ exceeds the common phase support; and the moving-packet exit is the existing moving-family SAE/ColumnCRT split. The remaining unproved input is the global pressure-packet carrying ceiling, recorded in

`docs/monograph/prime-matrix-critical-fill-arrival-anti-overfilling-router.md`.

□

Reduction-closed Statement 2.6 (Pressure-packet carrying ceiling)**.** *Let $M_q = A_g A_f$ be the pressure-packet multiplicity for an AffineTwin modulus q, so that q and $q-2$ are prime. The packet SAE mass is*

$$\mu_q = \frac{M_q}{q(q-2)}.$$

If the square-root packet gate

$$M_q^2 \le q(q-2)$$

holds, then

$$\mu_q \le \frac{1}{\sqrt{q(q-2)}} \le \frac{1}{q-2}.$$

Consequently, any standard Brun–Selberg upper-bound sieve of the form

$$\Pi_2(X) := \#\{q \le X : q,\ q-2 \text{ prime}\} \le C\frac{X}{(\log X)^2}$$

implies by partial summation that the moving pressure-packet tail is summable. If the square-root packet gate fails, then

$$M_q^2 > q(q-2),$$

which is exactly the named SuperSqrt/PressureProduct PDEC exit.

Status and reference. The first inequalities are immediate from the square-root gate. The Brun–Selberg bound gives

$$\sum_{\substack{q \ge Q \\ q,\ q-2 \text{ prime}}} \frac{1}{q} \le \frac{\Pi_2(Q)}{Q} + \int_Q^\infty \frac{\Pi_2(t)}{t^2}\,dt \ll \frac{1}{(\log Q)^2} + \frac{1}{\log Q},$$

and $1/(q-2) \ll 1/q$ for $q \ge 5$. Thus all packets satisfying the square-root gate are absorbed by the reciprocal ceiling. The complementary case is the previous pressure-product inequality with product > 1, hence the registered PDEC/ColumnCRT exit. The detailed route is recorded in the pressure-packet carrying-ceiling router under `docs/monograph/`. This still leaves two explicit tasks before final row/column promotion. First, internalize or cite the Brun–Selberg reciprocal ceiling under the monograph's convention. Second, exclude the SuperSqrt/PressureProduct PDEC family. □

Reduction-closed Statement 2.7 (Critical-error frontier). *After the critical-density fixed point has been fixed, the remaining frontier is best measured by a normalized critical error*

$$\mathcal{E} = \frac{\textit{actual load}}{\textit{critical capacity}} - 1.$$

A positive critical error is not allowed to remain an anonymous remainder in the contradiction field. In the prime-matrix AffineTwin branch it is the pressure product

$$\frac{A_g^2}{g}\frac{A_f^2}{f} - 1;$$

in the two-point sieve it is the excess of the large-incidence mean above the Buchstab semiprime transition constant $K(\alpha)$, together with the terminal gap to the threshold 1; in the RH branch it is the off-critical zero load after zero-frequency CRT subtraction, normalized by the capacity of each controlled exit. Thus every positive critical error must become a fixed CRT/PDEC defect, a moving SAE/Rankin or analytic-average packet, a denominator-floor defect, or a named controlled exit.

Status and reference. This is a frontier reduction and audit rule, not a final theorem. The detailed ledger is recorded in

`docs/monograph/three-claims-critical-error-frontier-audit.md`.

It sharpens the current next tasks to three interfaces: a non-PDEC square-root phase-support bound for the prime-matrix pressure packets, a critical denominator-floor check for the two-point BMD-to-TLI transfer, and a common baseline-subtracted load convention for every RH controlled exit. □

Reduction-closed Statement 2.8 (Actual-packet square-root phase support). *In the prime-matrix AffineTwin branch, distinguish the formal product*

$$M_q^{\mathrm{form}} = A_g A_f$$

from the actual non-PDEC packet count N_q. The latter counts only generator–fill packets whose two slot conditions are compatible, whose common phase support is nonempty, and which have not already become a reset or ColumnCRT/PDEC event. For a primitive AffineTwin slot pair with $q \geq 13$, the recorded depth identities give common support width

$$W_q = \frac{q+9}{2}.$$

Since

$$W_q \leq \sqrt{q(q-2)}$$

for all $q \geq 13$, and since a repeated actual packet projection inside this support is by definition a PDEC/ColumnCRT event, every non-PDEC primitive packet family satisfies

$$N_q \leq W_q \leq \sqrt{q(q-2)}.$$

Thus the square-root gate is closed for actual primitive packets. A remaining formal SuperSqrt failure must be either product-accounting looseness (M_q^{form} overcounts N_q), a projection-collision PDEC, or a primitive support escape.

Status and reference. The inequality follows by squaring:

$$4q(q-2)-(q+9)^2 = 3q^2 - 26q - 81 \geq 0$$

for $q \geq 13$. The injection of actual packets into the primitive support is the non-PDEC projection rule: two actual packets with the same primitive projection would be a repeated fixed CRT phase and therefore a reset/ColumnCRT/PDEC event. The detailed audit is archived in

`docs/monograph/prime-matrix-nonpdec-sqrt-phase-support-reduction.md`.

This does not yet prove the row/column theorem. It reduces the remaining Prime-Matrix obligation to product-accounting tightening and to excluding or routing primitive-support escape packets. □

Reduction-closed Statement 2.9 (Formal envelope versus actual load)**.** *Across the three programs, a critical contradiction can only be triggered by actual load, not by a formal envelope. If F is a formal upper envelope, A is the actual load, and C is the critical capacity, then*

$$0 \leq A \leq F.$$

The case $F \leq C$ is absorbed. The case $F > C$ but $A \leq C$ is only an accounting issue and requires tightening the envelope to the actual load. Only the case $A > C$ can force a structural exit. In the prime-matrix AffineTwin branch this is the distinction between $(A_g A_f)^2$ and N_q^2; in the two-point sieve it is the distinction between model numerator/denominator envelopes and the actual ratio

$$|U_Y(I)|^{-1} \sum_{x \in U_Y(I)} D_Y^P(x);$$

in the RH branch it is the distinction between routed atom envelopes and source-deleted final loads in the GEE ledger.

Status and reference. This is a status-discipline and reduction rule. It prevents any of the three frontiers from closing by a coarse upper envelope alone. The detailed cross-program audit is recorded in

`docs/monograph/three-claims-formal-to-actual-critical-load-frontier.md`.

The resulting active interfaces are actual packet enumeration for the prime-matrix branch, actual denominator-floor control for the two-point BMD-to-TLI transfer, and actual final-load normalization for the RH controlled exits. □

Reduction-closed Statement 2.10 (Actual-load closure contracts)**.** *The current frontier is organized into three actual-load contracts. In the prime-matrix branch, PM–ALC requires that the actual AffineTwin packet count N_q replace the formal product $A_g A_f$ in the critical load; the current machine contract records candidate $q = 31, 43, 103$, total formal product upper bound 40, but only one actual packet in the current sweep. In the two-point branch, TP–ALC requires both a BMD/KLS numerator bound and a denominator floor for $|U_Y(I)|$ in the same Buchstab convention, with*

$$K(\alpha) + \varepsilon(\alpha) < 1 - \delta(\alpha).$$

In the RH branch, RH–ALC requires every controlled exit to be written with baseline-subtracted source-deleted final load. Failure of any contract is not an anonymous error; it is a named PDEC/SAE/parity/contro exit obligation.

Status and reference. The PM current-sweep contract is generated by

`experiments/prime_matrix_affine_twin_actual_packet_contract.py`.

The unified contract ledger is archived in

`docs/monograph/three-claims-actual-load-closure-contracts.md`.

This is a reduction and evidence-organizing step. It does not close the row/column theorem, the two-point theorem, or RH. □

Reduction-closed Statement 2.11 (AffineTwin formal-pair pruning). *In the current prime-matrix AffineTwin sweep, the formal product universe has total size* 40*, while the actual packet count is* 1*. The difference is now fully explained at the current level:* 11 *formal residue pairs are removed by an exact CRT-window emptiness check, and* 28 *formal residue pairs have no matching gap-fill source materialization in the required orientation. Thus the current formal-to-actual gap is not an unaccounted critical load.*

Status and reference. The pruning audit is generated by

`experiments/prime_matrix_affine_twin_formal_pair_pruning_audit.py`.

For $q = 31$, the audit enumerates the $3 \cdot 4$ formal residue pairs and finds only the pair $(19, 8)$ with CRT representative 2687 in the support interval $[2669, 2688]$. The remaining current candidates $q = 43, 103$ have no matching gap-fill source in the required direction. The global obligation remains: future formal pairs must be routed to CRT-window emptiness, source materialization failure as PDEC/SAE, or primitive-support escape. This does not close the row/column theorem. □

Reduction-closed Statement 2.12 (AffineTwin source materialization gate). *The current source-materialization failures in the AffineTwin audit are not unclassified missing mass. An actual AffineTwin packet must have a matching gap-fill source with gap q, generator modulus $q - 2$, fill modulus q, the prescribed orientation, and affine delay*

$$p_{\text{fill}} - p_{\text{gen}} = \frac{11q - 21}{4}.$$

In the current sweep, $q = 31$ is the only candidate passing this gate. The candidate $q = 43$ has a same-gap source, but it uses generator 47*, orientation plus-to-minus, and delay* 74 *instead of generator* 41*, orientation minus-to-plus, and delay* 113*. The candidate $q = 103$ has no gap-fill source. Thus the* 28 *source-unmaterialized formal pairs from the pruning audit are fully classified at the current level.*

Status and reference. The source gate audit is generated by

`experiments/prime_matrix_affine_twin_source_materialization_gate_audit.py`.

It records 16 blocked formal pairs for the wrong-source $q = 43$ row and 12 blocked formal pairs for the no-source $q = 103$ row. The remaining global obligation is to prove that every future source-gate failure routes to SameGapWrongSource-PDEC/SAE or NoGapSource-PDEC/SAE, rather than contributing to actual load. This does not close the row/column theorem. □

Reduction-closed Statement 2.13 (AffineTwin CRT window gap)**.** *After the source gate, the remaining current AffineTwin formal pairs are controlled by a CRT window test. For $q = 31$, the exact source has generator modulus 29, fill modulus 31, combined modulus 899, and common phase support $[2669, 2688]$ of width 20. Among the 12 formal residue pairs, exactly one pair, $(19, 8)$, has a CRT representative in the support, namely 2687. The other 11 pairs have nearest CRT representatives at positive distance from the support; the minimum such distance is 40.*

Status and reference. The CRT-window distance audit is generated by

`experiments/prime_matrix_affine_twin_crt_window_gap_audit.py`.

It upgrades current CRTWindowEmpty rows from a Boolean non-hit to positive phase-gap data. The global obligation remains to prove that future source-materialized non-hits have a uniform CRT window gap, or otherwise route the failure to the named PDEC/SAE exits recorded in the audit. This does not close the row/column theorem. □

Reduction-closed Statement 2.14 (Window edge collision)**.** *The current CRT-window non-hits can be expressed as residue-grid displacement requirements. For $q = 31$, the phase window $[2669, 2688]$ induces 20 target residue pairs. Each of the 11 non-hit formal pairs has a nearest target pair; the least L^1 displacement is 1, realized by*

$$(19, 9) \longrightarrow (19, 8).$$

Thus the narrowest current edge-collision atom is a one-step fill-residue collision with the already realized actual pair $(19, 8)$.

Status and reference. The edge-collision displacement audit is generated by

`experiments/prime_matrix_affine_twin_window_edge_collision_audit.py`.

It records two non-hit pairs whose nearest target is the existing actual pair, and nine non-hit pairs whose nearest target requires unused target-residue arrival. The global obligation remains to prove that such residue movements cannot be supplied persistently by a counterexample chain, or else to route the failure to residue-collision, target-arrival, or support-motion exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.15 (Existing actual CRT jump)**.** *In the current AffineTwin edge-collision audit, the existing-actual branch is not a small boundary slip. The two non-hit formal pairs whose nearest target is the already realized pair $(19, 8)$ have CRT phase jumps 120 and 435, whereas the common phase support has width 20. In particular the narrowest residue atom*

$$(19, 9) \longrightarrow (19, 8)$$

has L^1 residue displacement 1, but the fill-residue unit step in the combined CRT modulo $29 \cdot 31$ is 435.

Status and reference. The CRT jump audit is generated by

`experiments/prime_matrix_affine_twin_existing_actual_collision_jump_audit.py`.

It records generator and fill CRT unit steps 465 and 435, respectively, and verifies the exact signed jumps from the empty-pair nearest representatives to the existing actual point. This closes the current-sweep existing-actual collision jump ledger. The global obligation remains to prove that such jumps cannot be reset persistently by moving support, or to route the failure to repeated-residue, ColumnCRT, PDEC, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.16 (Unused target arrival). *In the current AffineTwin edge-collision audit, the unused-target branch also has no hidden actual load. The* 9 *non-hit formal pairs whose nearest targets are unused collapse to* 5 *target residue pairs:*

$$(10, 30),\ (16, 5),\ (17, 6),\ (18, 7),\ (20, 9).$$

None of these pairs belongs to the current formal product. To make them available, one must introduce new generator residues

$$10, 16, 17, 18, 20$$

and new fill residues

$$5, 6, 7, 30.$$

Moreover every corresponding CRT jump has absolute size at least 59*, while the common phase support has width* 20*.*

Status and reference. The unused-target arrival audit is generated by

`experiments/prime_matrix_affine_twin_unused_target_arrival_audit.py`.

It records that all unused targets are outside the current formal product, all need a new side residue, all need a new generator residue, and all CRT jump identities exceed the support width in the current sweep. The global obligation remains to control new generator/fill residue arrival or route the failure to support-motion, PDEC, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.17 (Support motion depth). *In the current AffineTwin CRT-window audit, support motion is not a free escape. The common support is the intersection of*

$$[2669, 2693] \quad \textit{and} \quad [2659, 2688],$$

namely $[2669, 2688]$*. Therefore an empty CRT representative can enter by support motion only if the corresponding generator endpoint and shifted-fill endpoint are both released. In the current sweep all* 11 *empty pairs require such a two-endpoint release, and the common side depth required ranges from* 58 *to* 435*.*

Status and reference. The support-motion depth audit is generated by

`experiments/prime_matrix_affine_twin_support_motion_depth_audit.py`.

For the narrowest row $(19, 12)$, the nearest CRT representative is 2629. To include it, the generator left depth must increase by 40 and the fill left depth by 30, so the total endpoint release is 70. The audit records the analogous depth-inflation vector for every empty row. This closes the current-sweep support-motion depth ledger. The global obligation remains to prove non-persistence of such synchronized depth inflation, or to route it to moving-support, endpoint-coupling, PDEC, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.18 (Support motion primitive defect). *The current support-motion branch cannot be absorbed by the fixed* $q = 31$ *AffineTwin primitive key. The slot-lock identities give*

$$d_g^- = 18, \quad d_g^+ = 6, \quad d_f^- = 28, \quad d_f^+ = 1.$$

Every support-motion candidate requires a common side depth different from the corresponding generator and fill depths. The smallest defect is the row $(19, 12)$*: it requires left common depth* 58*, producing defects*

$$58 - 18 = 40, \qquad 58 - 28 = 30.$$

Thus even the narrowest support-motion atom breaks both fixed primitive depth identities.

Status and reference. The primitive-depth defect audit is generated by

`experiments/prime_matrix_affine_twin_support_motion_primitive_defect_audit.py`.

It records 11 support-motion candidates, all of which break both the generator and fill affine-depth identities for the fixed $q = 31$ primitive key. This closes the current-sweep fixed-key absorption route. The global obligation remains to exclude moving primitive keys, or to route them to moving-key, depth-inflation, endpoint-coupling, PDEC, or SAE exits. This does not close the row/column theorem. $\square$

Reduction-closed Statement 2.19 (Moving-key depth formula)**.** *Even if the AffineTwin key is allowed to move, same-orientation depth absorption is blocked in the current support-motion rows. On the lower side, a common depth* D *would have to satisfy*

$$(q + 5)/2 = D, \qquad q - 3 = D,$$

so $q = 2D - 5$ *and* $q = D + 3$ *must agree. This occurs only at* $D = 8$*. The narrowest lower support-motion row has* $D = 58$*, giving* $q = 111$ *from the generator formula and* $q = 61$ *from the fill formula. On the upper side, the fill depth is fixed at* 1*, while the current required depths are at least* 342*.*

Status and reference. The moving-key depth formula audit is generated by

`experiments/prime_matrix_affine_twin_moving_key_depth_formula_audit.py`.

It records 11 support-motion candidates. None admits a common moving q with the same orientation and both depth formulas. This closes the current-sweep same-orientation moving-key absorption route. The global obligation remains to exclude orientation changes, source rematerialization, or other primitive-key migration, or to route them to PDEC/SAE/endpoint-coupling exits. This does not close the row/column theorem. $\square$

Reduction-closed Statement 2.20 (Moving-key source rematerialization)**.** *The one-sided* q *values produced by the moving-key depth formulas also fail to become AffineTwin sources with the same orientation in the current sweep. Across the* 11 *support-motion rows the formulas produce* 19 *distinct one-sided candidate keys. Thirteen are composite. The six prime candidates*

$$61, \ 181, \ 293, \ 761, \ 1499, \ 1747$$

all fail the AffineTwin prime/source gate: none satisfies the full same-orientation requirement that q *and* $q - 2$ *are prime,* $q \equiv 3 \pmod 4$*, the delay* $(11q - 21)/4$ *is integral, and a matching gap-fill source with generator* $q - 2$*, fill* q*, and sides minus-to-plus exists.*

Status and reference. The moving-key source-rematerialization audit is generated by

`experiments/prime_matrix_affine_twin_moving_key_source_rematerialization_audit.py`.

It records no exact rematerialized candidate. The narrowest prior depth atom 19:12 gives $q = 111$ and $q = 61$; 111 is composite, while 61 has $61 \equiv 1 \pmod 4$ and hence a nonintegral AffineTwin delay. The nearest available gap source has gap 59, with generator 61, fill 59, and the wrong orientation for a $q = 61$ AffineTwin source. Thus the current-sweep same-orientation moving-key source-rematerialization route is closed. The global obligation remains to exclude orientation changes or persistent source rematerialization, or to route them to the named PDEC/SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.21 (Endpoint-release critical error). *In the current AffineTwin sweep, support motion has no anonymous critical-error channel left. If an empty CRT representative is pulled into the primitive support by endpoint motion, the actual load is the total generator and shifted fill endpoint release. The critical capacity is the current common support width, which is* 20. *All* 11 *support-motion candidates are supercritical: the total endpoint-release load is* 4929 *against total capacity* 220*, and the narrowest atom* 19:12 *already has load* 70*, hence critical error*

$$\frac{70}{20} - 1 = \frac{50}{20} = 2.5.$$

Status and reference. The endpoint-release critical-error audit is generated by

`experiments/prime_matrix_affine_twin_endpoint_release_critical_error_audit.py`.

It records that every support-motion row requires both endpoints to release, that every endpoint-release load exceeds the support width, and that the fixed primitive-depth defect total matches the endpoint-release load row by row. Together with the moving-key source-rematerialization audit, this eliminates same-orientation source rematerialization as a hidden carrier for the positive local error. Thus, in the current sweep, the positive endpoint-release critical error is routed to `EndpointReleaseCoupling-PDEC`. The global obligation remains to exclude persistent endpoint-release coupling, orientation changes, and source rematerialization, or to route them to named exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.22 (Endpoint-release feedback loss). *Even after the current support-motion rows are given full geometric credit for the larger endpoint movement, a supercritical feedback loss remains. For each row, write*

$$L = \textit{generator release} + \textit{fill release}, \qquad G = \max(\textit{generator release}, \textit{fill release}).$$

The credited support gain is G, so the unrecovered feedback loss is

$$L - G = \min(\textit{generator release}, \textit{fill release}).$$

In the current audit, the total release load is 4929*, the total geometric credit is* 2512*, and the remaining feedback loss is* 2417 *against support-width budget* 220*. The narrowest atom* 19:12 *has load* 70*, credit* 40*, and feedback loss* 30*, still larger than the support width* 20*.*

Status and reference. The endpoint-release feedback-loss audit is generated by

`experiments/prime_matrix_affine_twin_endpoint_release_feedback_loss_audit.py`.

It records that all 11 feedback losses are positive, equal to the smaller synchronized endpoint release, and still exceed the original support width. Thus the current sweep has no self-feedback absorption channel after geometric support gain is fully credited. The remaining positive residual is routed to `EndpointReleaseFeedbackLoss-PDEC`. The global obligation is to exclude that persistent feedback-loss family, or route it to the named moving-key, orientation-change, source-rematerialization, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.23 (Endpoint-release feedback horizon)**.** *The endpoint feedback loss has an equivalent phase-horizon form in the current sweep. A support-motion representative can be absorbed by endpoint self-feedback only if its distance from the current support window is at most*

$$\textit{support width} + |\textit{generator side depth} - \textit{fill side depth}|.$$

For all 11 *support-motion rows the representative lies outside this horizon. The narrowest row is* 19:12*: its representative is at distance* 40*, while the support width is* 20 *and the side-depth skew is* 10*, so the feedback horizon is* 30 *and the phase-horizon surplus is* 10*.*

Status and reference. The endpoint-release feedback-horizon audit is generated by

`experiments/prime_matrix_affine_twin_endpoint_release_feedback_horizon_audit.py`.

It proves, for the current sweep, the identities

$$\text{feedback loss} = \text{window distance} - \text{side-depth skew},$$

and

$$\text{post-credit overcritical units} = \text{window distance} - (\text{support width} + \text{side-depth skew}).$$

Thus the residual critical error is exactly the distance by which the CRT representative crosses the feedback horizon. All rows have positive horizon surplus, so the current self-feedback absorption route is closed and the residual is routed to `EndpointReleaseFeedbackHorizon-PDEC`. The global obligation remains to exclude persistent horizon surplus, or to route it to orientation-change, source-rematerialization, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.24 (Endpoint-release feedback-horizon slack)**.** *In the same current sweep, the feedback-horizon surplus is exactly the missing side-depth skew needed for absorption. If a representative has distance d from the current support window and the support width is W, then absorption requires total side-depth skew at least*

$$\max(0, d - W).$$

Hence the required extra skew is

$$\max(0, d - W) - |\textit{generator side depth} - \textit{fill side depth}|,$$

which equals the phase-horizon surplus row by row. The total required skew is 2292*, while the current skew supplies only* 95*; the missing extra skew is* 2197*. The narrowest row is again* 19:12*, where distance* 40 *and support width* 20 *require total skew* 20*, but the current skew is* 10*.*

Status and reference. The endpoint-release feedback-horizon slack audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
feedback_horizon_slack_audit.py
```

It verifies that all 11 rows have positive required extra skew, that this extra skew equals the phase-horizon surplus, and that a pure orientation flip does not change the absolute skew. It also inherits from the moving-key source-rematerialization ledger that same-orientation exact rematerialization is absent. Thus the current anonymous slack absorption route is closed and the remaining obstruction is routed to `EndpointSkewGrowth-PDEC/OrientationChangingPrimitiveKey-SAE`. The global row/column theorem remains open until those named exits are excluded or absorbed. □

Reduction-closed Statement 2.25 (Endpoint-release bidirectional skew hull)**.** *The current slack rows also have a simultaneous CRT-hull form. Each formal generator/fill residue pair gives a local alignment condition in the variable P,*

$$P \equiv r_g \pmod{q-2}, \qquad P \equiv r_f - \Delta \pmod{q},$$

where Δ is the affine delay. For the current $q = 31$ atom, the combined modulus is 899 and the primitive support interval is $[2669, 2688]$, of width 20. The nearest representatives of all 12 formal pairs lie in the hull $[2304, 3122]$, of width 819. Thus a single support hull would need left and right extensions 365 and 434; after the feedback horizons are credited, positive two-sided deficits 335 and 409 remain. Moreover

$$899 - 819 = 80,$$

which equals the affine delay.

Status and reference. The bidirectional skew-hull audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
bidirectional_skew_hull_audit.py
```

It imports the inverse-alignment equivalence only as a local CRT-alignment discipline: the global statement that the minimal zero-row alignment x exceeds P is not used as a theorem. The audit proves that the current formal-pair system cannot be absorbed by one-sided skew growth, because below and above representatives both remain outside the credited feedback horizons. The remaining obstruction is routed to `BidirectionalSkewHull-PDEC/ColumnCRT` or to the moving-family multiplicity exits. This is a current-sweep closure, not a final unconditional row/column theorem. □

Reduction-closed Statement 2.26 (Endpoint-release circular aperture)**.** *The bidirectional hull also has to be read on the circle modulo $q(q-2)$, not only in the current linear cut. For the current $q = 31$ atom, the previous interval $[2304, 3122]$ has width 819 in one chosen cut of the modulus 899. Removing the largest circular open gap instead gives the minimal circular alignment arc*

$$[3029, 3586],$$

of width 558. The largest open gap runs from the actual pair 19:8 to the pair 13:9 and has open width 341. The affine delay gap of width 80 is present, but it is only the third largest circular gap, so the previous equality $899 - 819 = 80$ is a linear-cut rigidity, not the complement of the circular minimal arc.

Status and reference. The circular-aperture audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
circular_aperture_audit.py
```

It verifies that even after the optimal circular cut, the minimal arc has width 558, whereas the primitive support has width 20. After optimally shifting the support interval to $[3568, 3587]$, the total extension still required is 539; after crediting the most favorable one-sided feedback horizon, a conservative deficit 509 remains. Thus the current single-sided circular-aperture absorption route is closed for this sweep. The remaining obstruction is routed to `CircularAperture-PDEC/ColumnCRT`, `SAE`, or the already named moving-family multiplicity exits. This refinement corrects the linear-hull wording but does not close the global row/column theorem. □

Reduction-closed Statement 2.27 (Endpoint-release anchored circular depth)**.** *The circular aperture can be sharpened by keeping the unique actual packet as an anchor. In the current sweep the minimal circular arc is* $[3029, 3586]$*, and its right endpoint* 3586 *is the actual pair* $19{:}8$ *after one period of the modulus* 899*. Hence an actual-anchored absorption of the whole arc requires the generator phase and the shifted-fill phase to both reach the left endpoint* 3029*. The required common left depth is*

$$D = 3586 - 3029 = 557.$$

The current generator and fill left depths are 18 *and* 28*, so the required endpoint releases are* 539 *and* 529*, for a total of* 1068*. This is* 53.4 *times the primitive support width* 20*.*

Status and reference. The anchored circular-depth audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
anchored_circular_depth_audit.py
```

It shows that the previous one-sided circular support extension 539 hides a second endpoint release 529: the intersection support moves only when both underlying phase intervals move. The same-orientation moving-key explanation also fails at this anchored depth. The lower-side AffineTwin depth formulas would require

$$q = 2D - 5 = 1109, \qquad q = D + 3 = 560,$$

so there is no common odd-prime key. Thus the current same-orientation anchored circular-depth absorption route is closed for this sweep. The remaining obstruction is routed to anchored circular-depth PDEC, ColumnCRT, SAE, orientation-changing keys, or the moving-family multiplicity exits; the global row/column theorem remains open. □

Reduction-closed Statement 2.28 (Anchored parity no-go)**.** *For a same-orientation lower-side AffineTwin key to absorb an anchored common left depth* D*, the generator and fill depth formulas must give the same prime parameter:*

$$q_g = 2D - 5, \qquad q_f = D + 3.$$

Thus a common key forces $2D - 5 = D + 3$*, hence* $D = 8$ *and* $q = 11$*. The actual-anchored circular arc in the current sweep instead forces* $D = 557$*. It gives*

$$q_g = 1109, \qquad q_f = 560.$$

The fill-side candidate is an even integer larger than 2*, so it cannot be an odd-prime AffineTwin key.*

Status and reference. The anchored parity no-go audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
anchored_parity_nogo_audit.py
```

It records both obstructions: the forced depth is 549 away from the only common-depth solution $D = 8$, and its odd parity makes the fill-side candidate $D + 3$ even. Therefore the current same-orientation anchored parity absorption route is closed for this sweep. The remaining obstruction is a global family version of this no-go, or one of the already named exits: orientation-changing keys, ColumnCRT/PDEC, SAE, or moving-family multiplicity. The global row/column theorem remains open. □

Reduction-closed Statement 2.29 (Endpoint-release cut-anchor sweep). *The anchored parity obstruction can be checked against every circular cut, not only the optimal one. In the current $q = 31$ atom there are 12 adjacent CRT cuts on the circle modulo 899. The unique actual pair 19:8 is retained in every lifted arc: it is the left endpoint for one cut, the right endpoint for one cut, and an interior point for the remaining ten cuts. For each cut we anchor the actual representative in the lifted arc and charge both the generator interval and the shifted-fill interval for the left and right depths needed to cover that arc.*

The smallest arc is still the previous cut $19{:}8 \to 13{:}9$, with width 558 and actual pair at the right endpoint. Its total endpoint release is $1068 = 53.4 \cdot 20$, where 20 is the primitive support width. Across all cuts, the endpoint release ranges from 1068 to 1737, so no cut fits inside the support budget.

Status and reference. The cut-anchor sweep audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
cut_anchor_sweep_audit.py
```

It also checks the same-orientation moving-key formulas for every cut. On the left lower side a common key can occur only at $D = 8$; the closest positive left depth in the sweep is $D = 58$. On the right above side the shifted-fill right depth is fixed at $D = 1$; the closest positive right depth in the sweep is $D = 342$. Hence changing the circular cut or moving the actual endpoint does not reopen the same-orientation absorption channel. Endpoint cuts fall into the left anchored parity no-go or the right fixed-fill no-go; interior cuts require both side releases simultaneously. The current sweep is closed, but the global theorem still requires a family version of this cut-anchor no-go, or one of the named exits: orientation-changing keys, ColumnCRT/PDEC, SAE, or moving-family multiplicity. □

Reduction-closed Statement 2.30 (Cut-anchor ColumnCRT compression). *The circular cut is a gauge choice, not a new actual residue. In the current cut-anchor sweep all 12 cuts retain the same actual pair 19:8. The actual representative is either 2687 or $3586 = 2687 + 899$, and in both cases*

$$P \equiv 889 \pmod{899}.$$

Thus the fixed $q = 31$ actual slot contributes one ColumnCRT atom of mass $1/899$, not twelve independent atoms of total mass $12/899$. The apparent extra mass

$$\frac{11}{899}$$

is only cut multiplicity.

Status and reference. The cut-anchor ColumnCRT compression audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
cut_anchor_columncrt_compression_audit.py
```

It registers the persistent fixed-residue escape as a `CutAnchorColumnCRT-PDEC` object with phase $P \equiv 889 \pmod{899}$, equivalently $P \equiv 19 \pmod{29}$ and $P \equiv 21 \pmod{31}$ after the fill shift. Therefore, after same-orientation cut-anchor absorption is closed, changing the cut cannot supply additional capacity. If q or the residues move, the case leaves this fixed ledger and returns to the existing AffineTwin moving-family SAE/ColumnCRT ledger. The current sweep closes cut multiplicity as an independent capacity source; it does not yet exclude the global ColumnCRT/PDEC or orientation-changing exits. □

Reduction-closed Statement 2.31 (Actual-anchor replacement)**.** *After cut-anchor compression, keeping the actual pair* 19:8 *gives the single fixed residue*

$$P \equiv 889 \pmod{899}.$$

If the counterexample chain tries to abandon this anchor, then some other formal pair must become actual, or else an unused target residue must arrive. Both alternatives require a CRT phase jump larger than the primitive support width 20*. Among existing formal pairs, the closest replacement is* 19:12*, at CRT distance* 58*, and it requires total endpoint release* 70*. Among unused targets, the closest replacement is* 20:9*, at CRT jump* 59*, and it already needs a new side residue.*

Status and reference. The actual-anchor replacement audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
actual_anchor_replacement_audit.py
```

It imports the support-motion and unused-target ledgers only as audited local interfaces. The first interface covers the 11 unsupported formal pairs: every replacement CRT jump exceeds the support width, every replacement requires both endpoint releases, and the minimum release is $70 = 3.5 \cdot 20$. The second interface covers the unused target residues: every CRT jump again exceeds the support width, and every target requires a new side residue. Thus actual-anchor replacement is closed for the current sweep. The global theorem still requires a family version of this no-go, or exclusion of the named support-motion, unused-target, ColumnCRT/PDEC, and moving-family exits. □

Reduction-closed Statement 2.32 (Near-miss 61/59 phase fracture)**.** *The closest apparent bridge between actual-anchor replacement and moving-key rematerialization is not a bridge in the current sweep. The closest existing formal replacement is still* 19:12*: its CRT jump is* 58*, its required common side depth is* 58*, and its endpoint release is* 70*. The corresponding one-sided moving-key formulas produce* $q = 61$ *and* $q = 111$*. Meanwhile the closest unused-target replacement is* $19{:}12 \to 20{:}9$*, with CRT jump* 59*, and the nearest available gap source to the candidate* $q = 61$ *also has gap* 59*.*

Status and reference. The near-miss phase-fracture audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
near_miss_61_59_phase_fracture_audit.py
```

It records that $q = 61$ is prime and $q-2 = 59$ is prime, but $61 \equiv 1 \pmod 4$, so the AffineTwin delay $(11q-21)/4$ is not integral, and there is no matching gap-61 source. The nearest actual source has signature

$$\text{gap} = 59, \quad \text{generator} = 61, \quad \text{fill} = 59, \quad \text{sides} = \text{minus} \to \text{minus}, \quad p \text{ delay} = 70.$$

This differs from the $q = 61$ AffineTwin source signature

$$\text{gap} = 61, \quad \text{generator} = 59, \quad \text{fill} = 61, \quad \text{sides} = \text{minus} \to \text{plus}.$$

The other moving-key candidate, $q = 111$, is composite. Thus the integer 59 appears in two nearby roles but not in the same phase role: as the nearest available gap source to $q = 61$, and as the unused-target CRT jump. The latter still exceeds the support width 20 and introduces a new side residue. Hence the current-sweep 61/59 near miss cannot carry hidden actual load. The global theorem still requires a family version of this no-go, or routing to source-rematerialization, unused-target, ColumnCRT/PDEC, or moving-family exits. □

Reduction-closed Statement 2.33 (Phase-scale bridge exhaustion). *In the current sweep there is no other exact phase-scale bridge among the moving-key candidates, the available gap sources, and the unused-target CRT jumps. Across all one-sided moving-key candidates, no candidate q is equal to an available gap-source scale, and no candidate q is equal to an unused-target jump scale. The only exact offset bridge of the form $q-2=$ gap source $=$ unused-target jump is the already fractured case $q=61$, $q-2=59$.*

Status and reference. The phase-scale bridge exhaustion audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
phase_scale_bridge_exhaustion_audit.py
```

It compares the 19 moving-key candidates

$$61, 65, 96, 111, 119, 123, 154, 181, 235, 293, 297, 355, 386, 575, 699, 761, 1375, 1499, 1747$$

with the available source scales

$$31,\ 43,\ 59$$

and the unused-target jump scales

$$59,\ 60,\ 90,\ 150,\ 281,\ 343,\ 345,\ 374,\ 375.$$

The audit records

$$\{q : q = \text{gap source}\} = \varnothing, \qquad \{q : q = \text{unused jump}\} = \varnothing,$$

and

$$\{q : q-2 = \text{gap source} = \text{unused jump}\} = \{61\}.$$

The previous near-miss audit already closes this unique offset bridge: the gap-59 source has the wrong role and orientation for a $q=61$ AffineTwin source, and the unused-target jump 59 still introduces a new side residue. Therefore the current sweep has no anonymous exact or offset scale bridge. The global theorem still requires a family version of this exhaustion, or a route to source-rematerialization, unused-target, ColumnCRT/PDEC, or moving-family exits. □

Reduction-closed Statement 2.34 (Support-width near-scale fracture). *Even after exact phase-scale bridges are exhausted, the current sweep has no support-width near-scale bridge. With support width* 20*, the only moving-key candidates that are simultaneously within distance* 20 *of an available source scale and an unused-target jump scale are $q=61$ and $q=65$. The first is the already fractured prime-but-not-AffineTwin case; the second is composite.*

Status and reference. The support-width near-scale audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
support_width_nearscale_fracture_audit.py
```

It enumerates all events satisfying

$$|s-q| \le 20, \quad |s-(q-2)| \le 20, \quad |j-q| \le 20, \quad |j-(q-2)| \le 20,$$

where s is a current gap-source scale and j is an unused-target CRT jump. It records

$$\{q : \text{near source}\} = \{61, 65\},$$

$$\{q : \text{near unused jump}\} = \{61, 65, 96, 111, 154, 293, 297, 355, 386\},$$

and

$$\{q : \text{near both source and unused jump}\} = \{61, 65\}.$$

There is no viable near-scale bridge. For $q = 61$, the AffineTwin source gate fails by the same delay, congruence, and missing-source invariants as in the near-miss audit. For $q = 65$, q is composite. The remaining near-jump candidates have no near source and cannot form a three-way phase bridge. Thus support-width proximity cannot replace exact CRT/source signatures in the current sweep. A global proof must still show the corresponding family near-scale no-go, or route failures to source-rematerialization, unused-target, ColumnCRT/PDEC, or moving-family exits. □

Reduction-closed Statement 2.35 (Orphan near-jump source deficit). *The remaining support-width near-scale candidates that are near an unused-target jump but not near a source scale are source-deficient in the current sweep. The orphan near-jump candidates are*

$$96,\ 111,\ 154,\ 293,\ 297,\ 355,\ 386.$$

For all of them, the nearest source scale is farther than the support width 20*. The smallest source distance is* 35*, attained at* $q = 96$*, leaving a positive source deficit* $35 - 20 = 15$*.*

Status and reference. The orphan near-jump source-deficit audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
orphan_nearjump_source_deficit_audit.py
```

It records that the seven orphan near-jump candidates split as six composite candidates and one prime-but-not-AffineTwin candidate. Every corresponding unused-target near-jump event introduces a new side residue, and no moving source is materialized. Thus target-side proximity does not create a three-way bridge: the source side remains outside the support-width tolerance. Any attempt to repair the orphan near-jump would have to move or create a source family, which is precisely the source-rematerialization or moving-family exit, not an anonymous primitive-support absorption. A global proof still requires the corresponding family no-go or routing to named PDEC/SAE exits. □

Reduction-closed Statement 2.36 (Near-jump carrier exhaustion). *All near-jump carriers are exhausted in the current sweep. The carrier set is*

$$61,\ 65,\ 96,\ 111,\ 154,\ 293,\ 297,\ 355,\ 386.$$

It splits into two disjoint parts:

$$\{61, 65\}$$

near both a source scale and a target jump, and

$$\{96, 111, 154, 293, 297, 355, 386\}$$

near only a target jump. The first part fails the source gate; the second part has source distance greater than the support width. In every case the target-side near-jump event requires a new side residue.

Status and reference. The near-jump carrier exhaustion audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
nearjump_carrier_exhaustion_audit.py
```

It records the source-status histogram

$$\{\text{near-source gate fracture} : 2, \quad \text{orphan source deficit} : 7\}$$

and the moving-route histogram

$$\{\text{CompositeQ} : 7, \quad \text{PrimeButNotTwinAffine} : 2\}.$$

The nearest orphan source distance is 35, still larger than the support width 20, and all carrier jump events need new side residues. Hence a near-target phase cannot by itself carry actual load. Repairing it forces one of the named exits: source rematerialization, unused-target residue arrival, ColumnCRT/PDEC, or moving-family migration. This closes the current-sweep near-jump carrier but not the global row/column theorem. □

Reduction-closed Statement 2.37 (Carrier-arrival routing)**.** *Every target-side near-jump event from the exhausted carrier set routes to an already registered unused-target arrival atom. There are* 27 *such events, using all* 9 *arrival atoms and* 5 *target pairs. The missing-match count is* 0*. The event-level new-side-residue demand is* 50*, compressed to generator residues*

$$10,\ 16,\ 17,\ 18,\ 20$$

and fill residues

$$5,\ 6,\ 7,\ 30.$$

The only exact zero-phase event is $q = 61$, $q - 2 = 59$, jump 59*; it still requires the new generator residue* 20*.*

Status and reference. The carrier-arrival routing audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
carrier_arrival_routing_audit.py
```

It matches each carrier event against the unused-target arrival ledger by source pair, target pair, and CRT jump. All matches are present, and every matched target is outside both the current formal product and the current actual support. Therefore near-target proximity is only a repeated projection of the unused-target arrival packet, not a new actual-load carrier. A persistent version must prove a global new-residue arrival bound or route to new-generator, new-fill, ColumnCRT/PDEC, or moving-family exits. This closes the current-sweep anonymous target-side carrier but not the global row/column theorem. □

Reduction-closed Statement 2.38 (Arrival pressure product)**.** *If the carrier-arrival packet is materialized as side residues, then the current sweep crosses the square-root pressure gate. The base side counts are*

$$(A_g, A_f) = (3, 4), \qquad A_g A_f = 12 < \sqrt{29 \cdot 31}.$$

The carrier packet adds generator residues

$$10,\ 16,\ 17,\ 18,\ 20$$

and fill residues

$$5,\ 6,\ 7,\ 30.$$

Thus full materialization gives side counts $(8, 8)$ *and*

$$(8 \cdot 8)^2 - 29 \cdot 31 = 3197 > 0.$$

Moreover the minimal crossing is exact to one unit: any two two-sided target atoms give counts $(5,6)$ *and*

$$(5\cdot 6)^2 - 29\cdot 31 = 1.$$

Status and reference. The carrier-arrival pressure-product audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
carrier_arrival_pressure_product_audit.py
```

It starts from the already routed carrier-arrival packet and updates the formal side-residue counts exactly. Since the updated product is above the square-root gate, the packet cannot remain an anonymous unused-target arrival. It is a named `SuperSqrt/PressureProduct-PDEC` atom. Therefore, after excluding anonymous target-side carriers, the remaining global obligation is to exclude this PDEC family or keep it as the explicit named exit; the row/column theorem is not yet unconditionally closed. □

Reduction-closed Statement 2.39 (Arrival projection deficit)**.** *The preceding square-root crossing is not an actual overload in the current sweep. Project the formal side product back to the common support window. For each minimal crossing row the formal product is* 30*, while the actual projection consists of exactly three points:*

$$\{\textit{actual anchor } 19{:}8\} \cup \{\textit{the two selected target atoms}\}.$$

Thus the projection deficit is 27*, and the actual square-root slack is* $29-3=26$*. For the full carrier packet the formal product is* 64*, but the actual projection has only* 6 *hits, with deficit* 58 *and slack* 23*.*

Status and reference. The carrier-arrival projection-deficit audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
carrier_arrival_projection_deficit_audit.py
```

It enumerates the side-residue Cartesian product for each minimal crossing and intersects it with the 20 residue pairs actually present in the common support window. The excess $30 > 29$ is therefore a formal-product excess, not an actual-load excess. Any global continuation must either tighten product accounting to actual projection, or produce a named projection-collision or support-escape PDEC. The current sweep has no anonymous actual overload. □

Reduction-closed Statement 2.40 (Support-graph cap)**.** *In the current fixed AffineTwin slot the common support is a graph, not a two-dimensional product set. The target window consists of the* 20 *pairs*

$$(1,21), (2,22), \ldots, (10,30), (11,0), \ldots, (20,9),$$

so each generator residue has at most one fill partner and conversely. Hence any actual projection from side sets is bounded by the graph size 20*. Since*

$$20 \le \lfloor\sqrt{29\cdot 31}\rfloor = 29,$$

the fixed slot cannot produce an actual square-root overload.

Status and reference. The support-graph cap audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
support_graph_cap_audit.py
```

It verifies that the common window is functional in both directions and has constant fill offset modulo 31. The general fixed-slot inequality is also recorded: for an AffineTwin modulus $q \geq 13$, the support width $W = (q+9)/2$ satisfies $W^2 \leq q(q-2)$, equivalently $3q^2 - 26q - 81 \geq 0$. Therefore the current ProductAccounting problem is closed for fixed slots. Any remaining escape must move or change the support graph, which is the named primitive-slot support-escape exit, not an anonymous load surplus. □

Reduction-closed Statement 2.41 (Moving-slot graph-cap route). *In the current endpoint-release sweep there is no anonymous moving-slot actual overload. A fixed or same-orientation AffineTwin slot remains bounded by the graph cap; the candidate fixed-family moduli are* $q = 31, 43, 103$*, and each satisfies* $W^2 \leq q(q-2)$*. If the support is truly moved while keeping the current primitive identity, the narrowest atom already requires endpoint release* 70 *against support width* 20.

Status and reference. The moving-slot route audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
moving_slot_graph_cap_route_audit.py
```

It records that all eleven support-motion candidates require both endpoint releases and that the fixed primitive depth identities break on both sides. Allowing a same-orientation moving key does not repair this: at the narrowest atom the depth formulas give $q_g = 111$ and $q_f = 61$. Taking all one-sided candidate q values through the source-rematerialization gate gives no exact rematerialized source. Thus the present sweep routes every moving-slot escape to a named exit: orientation-changing primitive key, source rematerialization, ColumnCRT/PDEC, SAE, or unused-target arrival control. This is a reduction of the remaining interface, not a final unconditional closure of the row/column theorem. □

Reduction-closed Statement 2.42 (Moving-family persistence pressure). *The current moving-family escape is not an anonymous capacity reservoir. Among the candidate moduli* $q = 31, 43, 103$*, only* $q = 31$ *is source-materialized. The candidate* $q = 43$ *has a same-gap source but with the wrong orientation and delay, and* $q = 103$ *has no gap source. The non-realized candidates have one-sided generator pressure, but their paired pressure products remain below the square-root gate.*

Status and reference. The moving-family persistence pressure audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
moving_family_persistence_pressure_audit.py
```

It combines the source-materialization gate, epoch-pair sparse ledger, paired pressure ledger, threshold route classifier, and fill catchup budget. The source gate blocks 28 formal pairs: 16 from $q = 43$ and 12 from $q = 103$. All minimal threshold-crossing routes require fill-side residue arrival; if all three candidates crossed, the minimum fill-arrival demand would be 11 new fill residues, with Rankin mass 23339/137299. Repeated fill arrival is routed to reset or ColumnCRT-PDEC. Hence the present sweep routes moving-family persistence to named exits: fill-arrival control, high-density epoch-pair PDEC/ColumnCRT, pressure-product PDEC, or source rematerialization. This is still a reduction boundary, not a final global proof. □

Reduction-closed Statement 2.43 (Fill-arrival projection gate). *In the current sweep, fill-side residue arrival alone cannot create an actual overload. The only source-materialized candidate is* $q = 31$*, and its minimal threshold crossing is not fill-only: it requires at least two new generator residues. When such co-arrivals are counted formally, the minimal side product is* 30*, but its actual projection has only* 3 *hits and remains below* $\lfloor\sqrt{29 \cdot 31}\rfloor$.

Status and reference. The fill-arrival projection gate audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
fill_arrival_projection_gate_audit.py
```

For $q = 31$, every fill-arrival threshold crossing requires generator co-arrival and then returns to the support-graph projection ledger. The unused-target ledger shows that the needed target residues also require new generator residues and have CRT jumps beyond the current support width. For $q = 43$ and $q = 103$, formal fill-only routes exist, but they are blocked before projection: $q = 43$ has the wrong source signature and $q = 103$ has no gap source. Thus the present sweep routes fill-arrival persistence to generator co-arrival, ProductAccounting tightening, source rematerialization, or ColumnCRT/PDEC. This remains a reduction boundary rather than a global exclusion theorem. □

Reduction-closed Statement 2.44 (Generator-coarrival projection accounting)**.** *In the current sweep, forcing fill arrival to co-arrive with generator residues still does not create an actual overload. For the realized candidate $q = 31$, all 31 nonempty subsets of the five unused-target atoms have been enumerated. There are 22 formally super-square-root subsets, but none projects above $\lfloor\sqrt{29 \cdot 31}\rfloor$.*

Status and reference. The generator-coarrival projection-accounting audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
generator_coarrival_projection_accounting_audit.py
```

The base packet has formal side product $3 \cdot 4 = 12$ and one actual projection hit, the anchor $19 : 8$. The minimal coarrival crossing has formal product 30, but its actual projection has only 3 hits and slack 26 below $\lfloor\sqrt{29 \cdot 31}\rfloor = 29$. The full carrier packet has formal product 64, but only 6 actual hits and slack 23. Moreover every projection hit is exactly the anchor plus the selected target atoms; no hidden projection collision is being counted as free capacity. Thus the present sweep routes generator coarrival to ProductAccounting tightening, source rematerialization, ColumnCRT/PDEC, or moving-family persistence. This remains a local closed audit and not a global unconditional row/column proof. □

Reduction-closed Statement 2.45 (Coarrival schema)**.** *For every fixed AffineTwin support graph with $q \geq 13$, generator coarrival cannot by itself create an actual overload. The graph width is $W = (q + 9)/2$, and*

$$W^2 \leq q(q-2) \quad \Longleftrightarrow \quad 3q^2 - 26q - 81 \geq 0.$$

The right hand side is already positive at $q = 13$ and is increasing for $q \geq 13$.

Status and reference. The family-schema audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
generator_coarrival_family_schema_audit.py
```

The current candidate moduli $q = 31, 43, 103$ all pass this fixed-graph inequality. Hence the only way for coarrival to become an actual overload is to leave the fixed functional support graph. The present sweep routes those escapes to named ledgers: support motion requires endpoint release 70 against support width 20, source materialization blocks 28 formal pairs, the fixed actual anchor compresses to the single ColumnCRT atom $P \equiv 889 \pmod{899}$, and moving-family persistence remains a named pressure/source/fill exit. This closes the current-sweep schema, but the global row/column theorem still needs the schema promoted to all persistent AffineTwin families and the named exits excluded or absorbed. □

Reduction-closed Statement 2.46 (Persistent-family promotion). *The current sweep promotes the coarrival schema through the active persistent AffineTwin family ledger. Fixed support graphs are bounded by the graph-width inequality; fixed q and fixed residue recurrence is routed to ColumnCRT/PDEC; moving q or moving residues are routed to epoch-pair SAE unless they become high-density.*

Status and reference. The promotion audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
persistent_family_promotion_audit.py
```

In the current ledger the candidate moduli are $q = 31, 43, 103$, with realized modulus 31. Their epoch-pair occupancy upper sum is 0.023577117628562343, below the threshold $\eta = 0.025$, and there is no high-density epoch pair. The fixed-slot recurrence count is 0; the 12 fixed-residue recurrences all drift in slot and are therefore the next ColumnCRT/PDEC input. Source materialization blocks 28 formal pairs, paired pressure does not cross, and fill arrival is routed to new-fill or reset PDEC. Thus the current sweep has no anonymous persistent-family capacity source. The remaining global obligations are an epoch-pair multiplicity bound and exclusion of fixed-residue slot drift by ColumnCRT/PDEC or SAE/Rankin. □

Reduction-closed Statement 2.47 (Transport-frontier integration). *In the current AffineTwin endpoint-release sweep, fixed-residue slot drift is not an anonymous ColumnCRT capacity source. The* 12 *slot-drift pairs match* 12 *distinct transport cells, with no transport-cell recurrence and no exact phase translate. Chained recurrence has finite forward lifetime; non-chained recurrence must repeat the full transport-cell key, and the current reset-PDEC atom count is* 0.

Status and reference. The transport-frontier integration audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
transport_frontier_integration_audit.py
```

It imports the persistent-family promotion ledger and the fixed-residue transport-cell, iteration-drift, reset-PDEC, and persistence-frontier partition ledgers. The aggregate values are: 12 fixed-residue slot-drift pairs, 12 transport cells, 12 unique transport cells, 0 transport-cell recurrences, 0 exact phase translates, maximum forward transition count 8, and reset-PDEC atom count 0. The frontier partition has 324 physical records, split as 24 transport-cell records and 300 singleton-residue records, with no unclassified record. Thus the present sweep routes the fixed-residue drift branch to TransportResetPDECExclusion or SingletonResidue SAE/Rankin, while the global row/column theorem still requires those exits and the epoch-pair multiplicity bound to be proved globally. □

Reduction-closed Statement 2.48 (Remaining-frontier bridge). *After transport-frontier integration, the computable current-sweep remainder bridges to three narrower ledgers. Transport reset has no current atom; singleton residue SAE is localized to the active prime-band endpoint problem; and AffineTwin epoch-pair multiplicity is currently below the $\eta = 1/40$ sparse gate.*

Status and reference. The bridge audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
remaining_frontier_bridge_audit.py
```

It reads seven ledgers: transport integration, singleton profile, Rankin budget, one slot capacity, active source, active band, and epoch-pair multiplicity. The current singleton profile has 300 singleton packets and Rankin mass about 4.328474446, with one slot mass about 4.291749691 and two slot share about 0.008484457. The one slot route has 40 active epochs, maximum occupancy ratio about 0.309859155, and minimum spare ratio about 0.690140845.

The active ℓ source is exactly the prime band $23, \ldots, 109$, with core $29, \ldots, 107$ and minus-only endpoints $23, 109$. The AffineTwin candidate epoch-pair product-mass upper sum is $0.023577118 < 0.025$, with no high-density row.

Thus the current remainder is sharpened to endpoint growth or reset PDEC, transport reset PDEC exclusion, global epoch-pair multiplicity, and moving residue SAE/Rankin. These remain global obligations, not a completed unconditional proof. □

Reduction-closed Statement 2.49 (Endpoint motion stencil). *In the current H-lower singleton route, endpoint motion of the active prime band has only named exits. The outward neighbor primes* 19 *and* 113 *have no current record; the endpoints* 23 *and* 109 *are minus-side singleton atoms; and the inward core edges* 29 *and* 107 *already have both-side support.*

Status and reference. The endpoint-motion stencil audit is generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_endpoint_motion_stencil_audit.py
```

The six-point stencil is

$$19 \to 23 \to 29, \qquad 107 \to 109 \to 113.$$

The outer entries $19, 113$ are empty in the current sweep. The endpoint entries $23, 109$ each have one minus-side record, one residue, and one slot. The inward entries $29, 107$ have both-side core support, with 4 and 6 records respectively. Hence current endpoint motion cannot be counted as anonymous capacity: outward motion requires a new endpoint arrival, while inward motion is absorbed into the core. The global obligations are endpoint outward arrival control, endpoint atom PDEC/SAE exclusion, and core-edge multiplicity control. □

Reduction-closed Statement 2.50 (Endpoint motion gap). *In the current H-lower singleton route, endpoint expansion of the active prime band creates only three temporary internal prime gaps. The missing prime moduli are* $31, 43, 59$, *and all are filled within the current sweep; the largest observed fill delay is* 80 *in the* P *parameter.*

Status and reference. The endpoint-motion gap router is generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_endpoint_motion_gap_router.py
```

It orders one-slot ℓ values by their first activation. The three non-interval snapshots are: activation of 47 at $P = 2063$ leaves 43 missing until delay 74; activation of 29 at $P = 2687$ leaves 31 missing until delay 80; activation of 61 at $P = 3187$ leaves 59 missing until delay 70. Every 1000-level prefix snapshot from 3000 to 10000 is already a complete prime interval, and the final active band is $23, \ldots, 109$. Thus the current sweep routes endpoint motion to a gap-fill bound or to a Gap-PDEC/SAE if such a gap persists globally. □

Reduction-closed Statement 2.51 (Gap-fill pair and corridor). *In the current H-lower singleton route, every endpoint-motion gap is repaired by the next first activation in* ℓ*-activation order, and every repair lies inside the short corridor*

$$P_{\text{fill}} - P_{\text{gen}} \le 3\ell_{\text{gap}}.$$

The three gap-fill keys are distinct in the current sweep.

Status and reference. The gap-fill pair and repair-corridor routers are generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_gap_fill_pair_router.py
```

and

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_gap_repair_corridor_router.py.
```

The three repairs are $47 \to 43$ with delay 74, $29 \to 31$ with delay 80, and $61 \to 59$ with delay 70. Each has activation-rank delay 1, and each restores the exact prime interval after the fill. The corresponding full GapFillPair keys are pairwise distinct. The corridor ratios are bounded by 3: the tight row is $\ell_{\text{gap}} = 31$, where $80 \le 3 \cdot 31$ with margin 13; a $2\ell_{\text{gap}}$ corridor is already false there. Thus the global remaining obligation is a short gap-repair corridor bound, or exclusion of a Corridor-PDEC/SAE when that bound fails. □

Reduction-closed Statement 2.52 (Corridor phase budget)**.** *In the current H-lower singleton route, the short repair corridor decomposes into the exact three-term identity*

$$P_{\text{fill}} - P_{\text{gen}} = d_{\text{gen,right}} + g_{\text{bridge}} + d_{\text{fill,left}}.$$

All three current gap repairs satisfy this identity and keep positive $3\ell_{\text{gap}}$ slack. The unique component excess is the phase bridge for $\ell_{\text{gap}} = 31$, but that excess is absorbed by the neighboring depth spares.

Status and reference. The corridor phase-budget router is generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_corridor_phase_budget_router.py.
```

The three identities are

ℓ_{gap}	$d_{\text{gen,right}}$	g_{bridge}	$d_{\text{fill,left}}$	$P_{\text{fill}} - P_{\text{gen}}$
43	12	40	22	74
31	6	46	28	80
59	31	31	8	70.

The minimum remaining $3\ell_{\text{gap}}$ slack is 13. The only component excess is $46 - 31 = 15$ in the $\ell_{\text{gap}} = 31$ phase bridge; the other two components provide total spare 28, leaving the same positive corridor slack 13. Hence the current short-corridor obstruction has been sharpened to a phase-budget obstruction: a global proof must bound the bridge component by neighboring depth spare, or route any persistent failure to a PhaseBridge-PDEC or SAE exit. □

Reduction-closed Statement 2.53 (Phase-bridge excess return to AffineTwin)**.** *The unique current phase-bridge excess is not a new anonymous capacity source. It collapses to the already registered AffineTwin fixed slot atom with*

$$q = 31, \qquad q - 2 = 29.$$

The excess is 15, the absorbing spare is 28, and the post-absorption slack is 13. The same atom satisfies the primitive affine identities and the coprime CRT lock

$$P \equiv 19 \pmod{29}, \qquad P \equiv 21 \pmod{31}.$$

Hence the combined modulus $29 \cdot 31 = 899$ exceeds the common support width 20 and isolates the fixed representative.

Status and reference. This return chain is generated by the phase-bridge excess atom/source routers, the margin-slot normal-form and primitive-identity routers, the primitive affine-collapse router, and the AffineTwin slot-phase-lock router:

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_phase_bridge_excess_atom_router.py

experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_phase_bridge_excess_source_router.py

experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_margin_slot_primitive_identity_router.py
```

and the downstream AffineTwin phase-lock and moving-family routers. They record

$$\text{excess} = 15 = \text{generator margin} = 3\rho_{\text{jump}}, \qquad \text{slack}_{\text{after}} = |\Delta b| = 13,$$

and the affine primitive identities

$$\ell_{\text{fill}} = \ell_{\text{gap}} = 31, \quad \ell_{\text{gen}} = \ell_{\text{gap}} - 2 = 29,$$
$$m_{\text{gen}} = (\ell_{\text{gap}} - 1)/2 = 15, \quad |\Delta b| = (\ell_{\text{gap}} - 5)/2 = 13, \quad |\Delta u| = (\ell_{\text{gap}} - 3)/4 = 7.$$

Thus the bridge-excess atom is exactly the $q = 31$ AffineTwin atom. Since 29 and 31 are coprime, the two slot congruences have modulus 899; the pair-support width is only 20, so the fixed slot contributes a single representative, already recorded as the fixed actual-anchor ColumnCRT atom. If the same q and residues recur, recurrence is a fixed-modulus ColumnCRT/PDEC object. If q or the residues move, the packet returns to the AffineTwin epoch-pair SAE/Rankin and moving-family exits. This closes the new current-sweep anonymous phase-bridge outlet, but the global row/column theorem still requires the moving-family multiplicity, endpoint-growth/reset, transport-reset, and moving-residue exits to be bounded or excluded. □

Reduction-closed Statement 2.54 (AffineTwin sparse SAE envelope)**.** *After the phase-bridge outlet returns to the AffineTwin moving-family ledger, the current candidate epoch-pairs pass the* $\eta = 1/40$ *sparse gate. The candidate moduli are*

$$q = 31, 43, 103,$$

the current product-mass upper sum is approximately

$$0.023577117629 < 0.025,$$

and there is no high-density epoch-pair row in the current sweep. Moreover, the single fixed-slot SAE mass has the telescoping envelope

$$\frac{1}{q(q-2)} = \frac{1}{2}\left(\frac{1}{q-2} - \frac{1}{q}\right),$$

so the odd tail from $q \geq Q$ *is at most* $1/(2(Q-2))$.

Status and reference. The epoch-pair multiplicity and sparse-SAE envelope routers are generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_affine_twin_epoch_pair_
multiplicity_router.py
```

and

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_affine_twin_sparse_sae_
global_envelope_router.py.
```

The current realized atom is the $q = 31$ fixed slot with mass $1/899$, while the single-atom tail from $q \geq 31$ is bounded by

$$\frac{1}{2(31-2)} = \frac{1}{58}.$$

This is a genuine global summability statement for one fixed AffineTwin atom per modulus. It does not, however, close the moving-family problem by itself: the η sparse gate alone would still permit $O(\eta q(q-2))$ atoms at a fixed q, and each such atom has mass $1/(q(q-2))$. A saturated sparse row could therefore contribute about η for each q, which is not summable over infinitely many moduli. Hence the latest narrow obstruction is not the single-atom SAE mass; it is the per-q multiplicity bound. A global proof must show $O(1)$, decay, or another summable bound for realized AffineTwin atoms per q, or route excess multiplicity to a HighDensityEpochPair PDEC/ColumnCRT exit. □

Reduction-closed Statement 2.55 (AffineTwin square-root product gate)**.** *The per-q multiplicity target can be weakened to a square-root product gate. Let M_q be the candidate two-side residue product upper bound for one AffineTwin modulus q. If*

$$M_q^2 \leq q(q-2),$$

then the corresponding SAE contribution is bounded by

$$\frac{M_q}{q(q-2)} \leq \frac{1}{\sqrt{q(q-2)}} \leq \frac{1}{q-2}.$$

In the current sweep the candidate moduli $31, 43, 103$ *all pass this gate. The largest product-to-capacity ratio is about* 0.400222408.

Status and reference. The square-root product and Brun-conditional router is generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_affine_twin_sqrt_product_
brun_router.py.
```

It records

q	M_q	$q(q-2)$	$M_q^2 \leq q(q-2)$
31	12	899	yes
43	16	1763	yes
103	12	10403	yes.

The minimum squared slack is 755. Thus the current finite sweep has no SuperSqrtEpochPair row. If the square-root product gate is proved globally, then accepting the classical Brun convergence of reciprocal twin-prime sums would make the AffineTwin SAE tail summable, since AffineTwin moduli satisfy that both q and $q-2$ are prime. This is recorded only as an external-input lane: the author-side self-contained route has not accepted Brun as an inline proof and has not proved the global square-root product bound. Therefore the latest self-contained obstruction is sharper: prove $M_q^2 \leq q(q-2)$ for every persistent AffineTwin epoch-pair, or route every violation to a SuperSqrtEpochPair PDEC/ColumnCRT exit. □

Reduction-closed Statement 2.56 (Return to actual-packet support)**.** *The square-root product hard point is an actual-packet question, not merely a formal product question. Let $M_q^{\text{form}} = A_g A_f$ be the side-residue Cartesian product upper bound, and let $N_q = |\Pi_q|$ be the number of actual non-PDEC packets passing the two slots, CRT compatibility, common phase support, and non-reuse gates. If every such packet lies in the primitive AffineTwin two-slot support, then*

$$N_q \le \frac{q+9}{2} \le \sqrt{q(q-2)} \qquad (q \ge 13).$$

Status and reference. The reduction is recorded in

`docs/monograph/prime-matrix-nonpdec-sqrt-phase-support-reduction.md`

and is compatible with the pressure-packet carrying ceiling file. The primitive support width is $W_q = (q+9)/2$. For $q \ge 13$,

$$W_q \le \sqrt{q(q-2)} \quad \Longleftrightarrow \quad 3q^2 - 26q - 81 \ge 0,$$

and the right side is positive at $q = 13$ and then increasing. A non-PDEC packet has a primitive phase projection into the common support set Ω_q. If two distinct actual packets had the same primitive projection, the recurrence would be a repeated-residue/reset event or a fixed projection ColumnCRT/PDEC event. After those named exits are removed, the projection is injective:

$$\Pi_q \hookrightarrow \Omega_q, \qquad N_q \le |\Omega_q| \le W_q.$$

Thus a formal SuperSqrt row has only three interpretations: product accounting used M_q^{form} where N_q is required; an actual projection collision gives a PDEC/ColumnCRT certificate; or an actual packet escaped the primitive two-slot support. The latest narrow task is therefore PrimitiveTwinSlotSupportExhaustion, together with replacing remaining SAE and Rankin ledgers by the actual-packet count N_q where they still use the formal product upper bound. □

Reduction-closed Statement 2.57 (Current actual-packet contract)**.** *In the current H-lower AffineTwin sweep, the formal product upper count is*

$$\sum_q M_q^{\text{form}} = 40,$$

but the actual non-PDEC packet count is only

$$\sum_q N_q = 1.$$

The single actual packet occurs at $q = 31$; the candidate rows $q = 43$ and $q = 103$ have no current actual packet.

Status and reference. The actual-packet contract is generated by

`experiments/prime_matrix_affine_twin_actual_packet_contract.py`

and archived at

`docs/monograph/prime_matrix_affine_twin_actual_packet_contract.md.`

The current table is

q	M_q^{form}	N_q	W_q	$q(q-2)$
31	12	1	20	899
43	16	0	26	1763
103	12	0	56	10403.

Every current actual packet is in primitive support and passes the actual square-root gate $N_q \leq W_q \leq \sqrt{q(q-2)}$. The formal-to-actual gap is 39, and there is no current projection-collision PDEC row. Therefore the current SuperSqrt pressure is product-accounting looseness, not actual load. The global obligations remain: prove the product-accounting tightening for all persistent AffineTwin families, prove primitive twin-slot support exhaustion, or route support escapes to their named PDEC/SAE exits. □

Reduction-closed Statement 2.58 (Current formal-pair pruning)**.** *In the current sweep, the whole formal-to-actual gap*

$$40 - 1 = 39$$

is accounted for by named gates:

$$39 = 11 + 28,$$

where 11 *formal pairs are CRT-window empty and* 28 *formal pairs fail source materialization. There is no unresolved current formal pair.*

Status and reference. The pruning certificate is generated by

```
experiments/prime_matrix_affine_twin_formal_pair_pruning_audit.py
```

and archived at

```
docs/monograph/prime_matrix_affine_twin_formal_pair_pruning_audit.md.
```

The current rows are

q	M_q^{form}	N_q	CRT empty	source unmaterialized
31	12	1	11	0
43	16	0	0	16
103	12	0	0	12.

For $q = 31$, the only supported pair is the residue pair $(19, 8)$; after the fill shift it gives the CRT class $889 \bmod 899$, represented in the support window $[2669, 2688]$ by 2687. The other eleven formal pairs have no support-window representative. For $q = 43$, the same gap exists only with the wrong generator/orientation/delay data; for $q = 103$, there is no current gap-fill source. Hence the present product-accounting gap is not hidden actual load. To globalize this step one must prove that every persistent AffineTwin formal pair either materializes as an actual primitive-supported packet, is CRT-window empty, fails source materialization as a named PDEC/SAE, or is explicitly routed as a primitive support escape. □

Reduction-closed Statement 2.59 (Current formal-to-actual cutset)**.** *In the current sweep, the formal-to-actual pipeline has no remaining anonymous capacity source. The formal product universe has* 40 *pairs, the actual packet count is* 1*, and the gap*

$$39 = 11 + 28$$

is exactly the sum of CRT-window-empty pairs and source-unmaterialized pairs. The active CRT modulus is 899*, the support width is* 20*, and the minimum empty-window distance is* 40*.*

Status and reference. The cutset router is generated by

`experiments/prime_matrix_formal_to_actual_global_cutset_router.py`

and archived at

`docs/monograph/prime_matrix_formal_to_actual_global_cutset_router.md.`

It imports the actual-packet, formal-pruning, source-gate, CRT-window, moving-slot, remaining-frontier, and endpoint-atom ledgers. The current support escape branch is already routed: the fixed support width is 20, the square-root floor is 29, exact rematerialized moving q-values are empty, transport-reset atoms are zero, the active singleton band is $23, \ldots, 109$, and the endpoint band has two minus-only atoms. Thus a current counterexample cannot keep the formal product as hidden actual load. The global obligation is to promote this cutset to all persistent families and to exclude or summably absorb the named exits: source failure, CRT-window failure, primitive-support escape, endpoint/reset, transport reset, epoch-pair multiplicity, and moving-residue SAE/Rankin. This is still a reduction boundary, not a global unconditional row/column proof. □

Reduction-closed Statement 2.60 (Cutset completeness)**.** *The formal-to-actual cutset has a deterministic completeness law. For a fixed AffineTwin modulus $q \geq 13$, let*

$$F_q = G_q \times H_q$$

be the formal generator/fill residue product. Then F_q is the disjoint union of source failures, CRT-window failures, primitive actual packets, and named primitive-support or projection escapes:

$$F_q = S_q \sqcup C_q \sqcup P_q \sqcup E_q.$$

Consequently

$$M_q^{\mathrm{form}} = |S_q| + |C_q| + |P_q| + |E_q|, \qquad N_q = |P_q|.$$

Status and reference. The deterministic lemma is archived at

`docs/monograph/prime_matrix_formal_to_actual_cutset_`
`completeness_lemma.md.`

For a formal pair $a \in F_q$, first test whether the matching gap-fill source exists with the required gap, generator, fill, orientation, delay, and slot-lock key. If not, $a \in S_q$. If the source exists, combine the generator and shifted-fill congruences modulo $q(q-2)$. If no representative lies in the primitive pair support, $a \in C_q$. If the representative exists and the primitive depth identities, non-reuse, and non-ColumnCRT conditions all hold, then $a \in P_q$. The remaining cases are precisely source-and-CRT hits that fail the primitive actual conditions; they form E_q, routed to ProjectionCollision/ColumnCRT/PDEC or PrimitiveTwinSlotSupportEscape-PDEC/SAE. The four alternatives are selected by the first failed gate, hence are disjoint and exhaustive. Thus formal overcount is never hidden actual load. What remains open is not classification, but global exclusion or summable absorption of the named exits. □

Reduction-closed Statement 2.61 (After-cutset named-exit frontier)**.** *After the deterministic cutset split, the current sweep has no anonymous post-cutset capacity or phase escape. The* 40 *formal pairs split as*

$$40 = 1 + 28 + 11,$$

where 1 *pair is an actual primitive packet,* 28 *pairs are source failures, and* 11 *pairs are CRT-window empty. The latter* 11 *pairs are routed to the named frontier consisting of window-edge or unused-target arrival, support motion and endpoint release, primitive identity shift, moving-slot families, transport reset, epoch-pair multiplicity, and moving-residue or singleton SAE.*

Status and reference. The frontier router is generated by

`experiments/prime_matrix_after_cutset_named_exit_frontier_router.py`

and archived at

`docs/monograph/prime_matrix_after_cutset_named_exit_frontier_router.md.`

The current numerical gates are: edge-collision candidates 11, support motion candidates 11, minimum endpoint release $70 = 3.5 \cdot 20$, minimum affine depth defect 70, no fixed highfactor slot isolation failure at the replay base, no transport reset atom, epoch-pair mass $0.023577117628562343 < 1/40$, 37 moving residue shapes, and 300 singleton residue packets. Hence the current after-cutset frontier is closed as a classification and routing statement. The next global obligation is still to exclude or summably absorb these named exits: SourceMaterializationFailure, CRTWindowEmpty, WindowEdge/UnusedTargetArrival, SupportMotion, PrimitiveIdentityShift, MovingSlotFamily, TransportReset, EpochPairMultiplicity, MovingResidueShape, and SingletonResidue. This is a sharper reduction boundary, not an unconditional row/column theorem. □

Reduction-closed Statement 2.62 (Terminal named-exit choke set)**.** *The ten after-cutset named exits in the current frontier compress to five terminal chokes: Source/CRT materialization, support-motion primitive identity, transport–singleton active-ℓ, epoch-pair paired pressure, and moving-residue SAE. All five are closed in the current sweep as routing and finite-capacity statements, but none is yet globally excluded.*

Status and reference. The terminal choke router is generated by

`experiments/prime_matrix_global_named_exit_terminal_choke_router.py`

and archived at

`docs/monograph/prime_matrix_global_named_exit_terminal_choke_router.md.`

Its current explicit margins are:

quantity	current value
unresolved formal pairs	0
CRT phase margin	879
minimum extra endpoint release over support width	50
transport reset atoms	0
one-slot total spare ratio	0.9038893044128646
one-slot minimum epoch spare ratio	0.6901408450704225
paired-pressure slack	0.8398220244716351.

Thus the latest obstruction is not a local capacity deficit. A persistent counterexample family must repeatedly realize one of the five terminal chokes; the task is to exclude those recurrences or summably absorb them through the registered PDEC/SAE/ColumnCRT ledgers. This is still a global reduction frontier, not an unconditional row/column proof. □

Reduction-closed Statement 2.63 (Terminal choke amplification barrier)**.** *In the current terminal choke set, the narrowest free phase jump is the CRT-window jump of size $40 = 2 \cdot 20$, but that jump is not a terminal escape: it is routed to unused-target arrival or window-edge PDEC/SAE. Among the terminal capacity barriers, the narrowest one-slot transport epoch is the minus $\ell = 71$ epoch, which uses 22 of 71 residues and therefore needs 50 new residues to overflow. The support-motion endpoint release barrier has the same extra size $70 - 20 = 50$.*

Status and reference. The amplification-barrier router is generated by

`experiments/prime_matrix_terminal_choke_amplification_barrier_router.py`

and archived at

`docs/monograph/prime_matrix_terminal_choke_amplification_barrier_router.md.`

The current ordered barriers are

barrier	factor	additive units
CRTWindowPhaseJump	2	40
OneSlotTransportOverflow	71/22	50
SupportEndpointRelease	70/20	50
EpochPairPairedPressure	899/144	1 to 8 fill units
FixedMovingSlotCRT	731/104	703.

Thus the current narrow terminal obstruction is a fifty-unit cross-lock: persistent recurrence must pay a common release/residue package of size 50, or else fall into reset, PDEC, SAE, or ColumnCRT. This identifies the next global interface, but does not yet exclude that interface for all persistent families. □

Reduction-closed Statement 2.64 (Fifty-unit cross-lock carrier separation)**.** *The fifty-unit cross-lock is not an immediate same-carrier contradiction in the current sweep. The support side is the* $q = 31$ *atom with source pair* $(19, 12)$*, generator prime* $P = 2687$*, and endpoint release* $70-20 = 50$*. The one-slot side is the minus* $\ell = 71$ *epoch, which uses* 22 *of* 71 *residues and needs* 50 *new residues to overflow. These two carriers have distinct moduli, distinct phase taxonomy, and disjoint current* P*-bands; therefore any persistent recurrence must pass through the cross-carrier modulus* $31\cdot 71 = 2201$*.*

Status and reference. The carrier-separation router is generated by

`experiments/prime_matrix_fifty_unit_cross_lock_carrier_separation_router.py`

and archived at

`docs/monograph/prime_matrix_fifty_unit_cross_lock_carrier_separation_router.md.`

It records

`same_q_or_ell=false`, `same_side_taxonomy=false`, `same_p_band=false`,
`direct_same_carrier_contradiction=false`, `combined_carrier_modulus=2201`.

The fixed delay of the support atom is 80. Modulo 71 this is 9, and $\gcd(80, 71) = 1$, so the first 50 transported residues are distinct. The first such block fitting the current minus-71 epoch runs from $P = 4207$ to $P = 8127$, inside the recorded epoch range $4177 \le P \le 9257$. Thus a short repeated-residue PDEC is not automatic. If, however, these 50 transported arrivals are all new residues in the same minus-71 epoch, then $22 + 50 > 71$, so one-slot capacity fails. If they are not new, or if they cannot stay synchronized, the failure is routed to TransportReset-PDEC, SingletonResidue-SAE, support-motion/unused-target exits, or moving-carrier ColumnCRT. The next global interface is therefore

`CrossCarrierFiftyUnitSynchronizationPDECOrSAE`.

This closes the same-carrier shortcut and sharpens the remaining obstruction; it is not yet an unconditional row/column proof. □

Reduction-closed Statement 2.65 (Cross-carrier fifty-unit residue saturation)**.** *After reconstructing the complete singleton physical records, the current cross-carrier synchronization does not overflow the minus-*71 *epoch inside the recorded* P*-band. The already-used minus-*71 *residue set has* 22 *elements. The support lattice points admitted by the current epoch range contribute* 64 *distinct residues;* 17 *of them are already used, so the combined set has size* 69*, leaving exactly two missing residues,* 0 *and* 62*.*

Status and reference. The saturation router is generated by

`experiments/prime_matrix_cross_carrier_fifty_unit_residue_saturation_router.py`

and archived at

`docs/monograph/prime_matrix_cross_carrier_fifty_unit_residue_saturation_router.md.`

It records

`used_residue_count_reconstructed=22`, `exact_used_matches_capacity_ledger=true`,
`admitted_step_range=[19,82]`, `admitted_distinct_residue_count=64`,
`admitted_intersection_existing_count=17`, `admitted_new_residue_count=47`,
`union_size_after_admitted_band=69`, `spare_after_admitted_band=2`.

Every contiguous 50-step block inside the current admitted band remains below capacity: the largest block union has size 62. Thus the branch "fifty arrivals are automatically fifty new residues" is false on the actual frontier. The precise remaining phase data are:

$$P = 9647 \mapsto 62, \qquad P = 9727 \mapsto 0, \qquad P = 9807 \mapsto 9.$$

The first two arrivals fill the two missing residues; the third is an old residue and hence must be routed to transport reset-PDEC if the same epoch persists. Therefore the next interface is

`TwoResidueSpareEndpointExtensionOrTransportResetPDEC`.

This is a sharper current-frontier obstruction, not a proof of the global row/column theorem. □

Reduction-closed Statement 2.66 (Two-residue spare prime-anchor filter)**.** *The two-residue near-fill route is not a valid actual-chain closure after the prime-anchor condition on* P *is imposed. In the same admitted lattice* $19 \le t \le 82$*, only* 17 *of the* 64 *values* $P = 2687 + 80t$ *are prime anchors. They contribute only* 13 *new residues to the minus-*71 *epoch; together with the* 22 *existing residues this covers* 35 *of the* 70 *nonzero residue classes. The two near endpoint arrivals from the previous router are composite or structurally impossible as prime anchors.*

Status and reference. The prime-anchor filter router is generated by

`experiments/prime_matrix_two_residue_spare_prime_anchor_filter_router.py`

and archived at

`docs/monograph/prime_matrix_two_residue_spare_prime_anchor_filter_router.md.`

It records

`admitted_lattice_step_count=64`, `admitted_prime_anchor_count=17`,
`admitted_prime_anchor_new_residue_count=13`, `prime_filtered_union_size=35`,
`prime_filtered_nonzero_spare=35`, `residue_zero_prime_anchor_impossible=true`.

The previous near endpoints satisfy

$$9647 \mapsto 62, \qquad 9727 \mapsto 0, \qquad 9807 \mapsto 9,$$

but all three P-values fail the prime-anchor test. Moreover the residue 0 arithmetic progression has $P \equiv 4047 \pmod{5680}$ and $\gcd(4047, 5680) = 71$, so for $P > 71$ it cannot contain a prime anchor. The first prime anchor for residue 62 is delayed to $P = 26687$. Full nonzero-residue coverage in the scanned progression is delayed until $P = 98047$, and the first prime-anchor repeat after that is $P = 98207$. Thus the current interface is

`PrimeAnchorFilteredNonzeroResidueCoverageOrTransportResetPDEC`.

This closes the two-residue near-fill shortcut. It still leaves a global obligation: exclude persistent long arithmetic-progression prime-anchor coverage, or route the first repeat after coverage to transport reset-PDEC. □

Reduction-closed Statement 2.67 (Prime-anchor post-band immediate repeat)**.** *The long prime-anchor nonzero-residue coverage branch is preempted in the current primitive epoch. After the admitted band $19 \le t \le 82$, the first prime anchor in the progression $P = 2687 + 80t$ is*

$$t = 90, \qquad P = 9887, \qquad \text{residue}_{71}(P) = 18.$$

The residue 18 is already in the original minus-71 used-residue set. Thus no new missing nonzero residue is added before the first repeat.

Status and reference. The immediate-repeat router is generated by

`experiments/prime_matrix_prime_anchor_postband_immediate_repeat_router.py`

and archived at

`docs/monograph/prime_matrix_prime_anchor_postband_immediate_repeat_router.md.`

It records

`first_postband_prime_anchor=(90,9887,18)`, `first_postband_prime_is_repeat=true`,
`first_postband_prime_is_original_used_repeat=true`,
`new_prime_anchor_count_before_first_repeat=0`,
`missing_nonzero_remaining_at_first_repeat=35`,
`coverage_before_reset_possible_in_same_epoch=false`.

The intervening candidates $t = 83, \ldots, 89$, namely

$$9327, 9407, 9487, 9567, 9647, 9727, 9807,$$

are composite. Therefore, if the same reset-free epoch persists past the admitted band, it hits a used residue before it hits any new missing nonzero class and must enter transport reset-PDEC. If the epoch does not persist to that anchor, the branch has already moved to endpoint motion/SAE. The latest interface is

`ImmediatePrimeAnchorRepeatTransportResetPDECOrEndpointMotionSAE`.

This closes the current long-coverage shortcut. It does not globally exclude transport reset or endpoint motion. □

Reduction-closed Statement 2.68 (Prime-anchor repeat reset atom). *The first post-band repeat is an exact one-slot reset atom. The original minus-*71 *singleton record with residue* 18 *occurs at* $P = 7757$*. The first post-band prime anchor has* $P = 9887$ *and the same residue. Their difference is*

$$9887 - 7757 = 2130 = 30 \cdot 71.$$

Thus the first prime anchor after the admitted band is not a new coverage packet, but a lift by 30 *full* ℓ*-periods of an existing residue packet.*

Status and reference. The repeat-reset atom router is generated by

```
experiments/prime_matrix_prime_anchor_repeat_reset_atom_router.py
```

and archived at

```
docs/monograph/prime_matrix_prime_anchor_repeat_reset_atom_router.md.
```

It records

`original_record_p=7757`, `repeat_prime_anchor_p=9887`, `repeat_residue=18`,
`p_delta=2130`, `p_delta_over_ell=30`, `exact_same_residue_ell_translate=true`,
`reset_atom_instantiated=true`, `missing_nonzero_remaining_at_reset_atom=35`.

The associated original slot key is `644:128:71`, and the formal unit is

```
one-slot-repeat-reset|side=minus|ell=71|residue=18|
lift_delta=30|p=7757->9887 .
```

Consequently, if the same reset-free epoch extends to $P = 9887$, the reset-PDEC atom is already present. To avoid it, the endpoint must cut before step 90, leaving only the composite buffer $t = 83, \ldots, 89$. The next interface is

`OneSlotPrimeAnchorRepeatResetPDECExclusionOrEndpointMotionSAE`.

This instantiates the local reset atom. It does not yet globally exclude all one-slot reset atoms or endpoint-motion escapes. □

Reduction-closed Statement 2.69 (Endpoint cut zero-gain). *If the endpoint is cut before the first post-band prime anchor in order to avoid the instantiated reset atom, the cut has no prime-anchor gain in the current primitive epoch. The only buffer before the reset is*

$$t = 83, \ldots, 89, \qquad P = 9327, 9407, 9487, 9567, 9647, 9727, 9807,$$

and none of these P*-values is prime.*

Status and reference. The zero-gain endpoint router is generated by

```
experiments/prime_matrix_endpoint_cut_zero_gain_router.py
```

and archived at

```
docs/monograph/prime_matrix_endpoint_cut_zero_gain_router.md.
```

It records

$$\texttt{cut\_buffer\_step\_count=7},\quad \texttt{formal\_gap\_composite\_row\_count=2},$$
$$\texttt{formal\_gap\_composite\_residues=[62,0]},$$
$$\texttt{actual\_prime\_anchor\_count\_before\_reset=0},$$
$$\texttt{actual\_new\_prime\_anchor\_count\_before\_reset=0},$$
$$\texttt{endpoint\_cut\_actual\_gain\_zero=true},$$
$$\texttt{reset\_atom\_instantiated\_at\_first\_prime\_anchor=true}.$$

The apparent gap residues 62 and 0 occur at $P = 9647$ and $P = 9727$, respectively, but 9647 has factor 11 and 9727 is divisible by 71. Thus they are formal gaps only, not actual prime-anchor arrivals. If the endpoint is not cut, the next value $P = 9887$ is the one-slot repeat-reset atom. Hence the local dichotomy is

$$\texttt{reset-PDEC} \quad \text{or} \quad \texttt{zero-gain endpoint cut SAE}.$$

The latest interface is

$$\texttt{ZeroGainEndpointCutSAEOrOneSlotResetPDECExclusion}.$$

This closes the current endpoint-gain explanation, not the global SAE or reset-PDEC exclusion. □

Reduction-closed Statement 2.70 (Endpoint cut no-payload SAE)**.** *In the current primitive epoch, the endpoint-cut branch has empty actual payload. The cut buffer* $t = 83, \ldots, 89$ *contains no prime-anchor row, no new prime-anchor row, and no filtered repeat prime-anchor row. The apparent gap residues* 62 *and* 0 *do not survive prime-anchor filtering.*

Status and reference. The no-payload endpoint router is generated by

`experiments/prime_matrix_endpoint_cut_no_payload_sae_router.py`

and archived at

`docs/monograph/prime_matrix_endpoint_cut_no_payload_sae_router.md.`

It records

$$\texttt{actual\_prime\_anchor\_count\_before\_reset=0},$$
$$\texttt{actual\_new\_prime\_anchor\_count\_before\_reset=0},$$
$$\texttt{filtered\_repeat\_prime\_anchor\_count\_before\_reset=0},$$
$$\texttt{actual\_payload\_empty=true},\quad \texttt{actual\_payload\_mass=0},$$
$$\texttt{missing\_nonzero\_before\_cut=35},\quad \texttt{missing\_nonzero\_after\_cut=35},$$
$$\texttt{missing\_nonzero\_set\_preserved\_by\_cut=true}.$$

Thus the endpoint cut cannot supply the capacity required by the counterexample chain; it is a no-payload SAE branch. If that branch is not taken, the first prime anchor is already the one-slot reset-PDEC atom. The current local dichotomy is

$$\texttt{one-slot reset-PDEC} \quad \text{or} \quad \texttt{no-payload endpoint SAE}.$$

The latest interface is

$$\texttt{OneSlotResetPDECExclusionOrNoPayloadEndpointSAESummability}.$$

This closes the current endpoint-cut actual-load exit, not the global reset or SAE summability problem. □

Reduction-closed Statement 2.71 (One-slot reset prefix no-relief). *If the endpoint branch is not cut, the line first accepts the one-slot reset at $t = 90$, $P = 9887$, residue 18. After that accepted reset, there is still no new missing nonzero residue until $t = 115$, $P = 11887$, residue 30.*

Status and reference. The reset-prefix no-relief router is generated by

`experiments/prime_matrix_one_slot_reset_prefix_no_relief_router.py`

and archived at

`docs/monograph/prime_matrix_one_slot_reset_prefix_no_relief_router.md.`

It records

`reset_step=90`, `reset_p=9887`, `reset_residue=18`,

`prefix_prime_anchor_count_before_first_relief=7`,

`prefix_new_missing_nonzero_count_before_first_relief=0`,

`prefix_repeat_prime_anchor_count_before_first_relief=7`,

`first_relief_step_gap_after_reset=25`, `first_relief_p_gap_after_reset=2000`.

Thus a new actual-capacity path must first cross the accepted reset-PDEC atom; before the first relief all actual prime anchors remain old residues. If the reset is rejected, the endpoint branch has already been reduced to a no-payload SAE. The latest interface is

`AcceptedResetPDECExclusionOrDelayedReliefSupportMotionSAE`.

This closes the explanation that reset immediately supplies fresh capacity, not the global exclusion of accepted reset-PDEC atoms. □

Reduction-closed Statement 2.72 (Accepted reset full relief horizon). *After accepting the one-slot reset at $P = 9887$, complete relief of the remaining 35 missing nonzero residues is not local. It first completes at $t = 1192$, $P = 98047$, after 1102 synchronized steps.*

Status and reference. The accepted-reset full-relief horizon router is generated by

`experiments/prime_matrix_accepted_reset_full_relief_horizon_router.py`

and archived at

`docs/monograph/prime_matrix_accepted_reset_full_relief_horizon_router.md.`

It records

`missing_nonzero_count_at_reset=35`,

`first_relief=(step=115,P=11887,residue=30)`,

`full_relief=(step=1192,P=98047,residue=67)`,

`full_relief_step_gap_after_reset=1102`,

`full_relief_p_gap_after_reset=88160`,

`new_relief_prime_anchor_count_until_full_relief=35`,

`repeat_prime_anchor_count_until_full_relief=225`.

In addition, 101 formal missing-residue hits before full relief are composite and cannot be counted as actual load. Hence full relief is a long-horizon support-motion obligation, not a local repair inside the current primitive epoch. The latest interface is

`AcceptedResetPDECExclusionOrLongReliefHorizonSupportMotionSAE`.

This quantifies the delayed relief branch; it does not yet exclude the global accepted-reset family. □

Reduction-closed Statement 2.73 (Long relief cycle debt)**.** *The long-relief branch has an explicit* 71-*cycle phase debt. Among the remaining* 35 *nonzero residues, only* 8 *receive prime relief on the first post-reset phase hit; the other* 27 *require later* 71-*period returns.*

Status and reference. The long-relief cycle-debt router is generated by

`experiments/prime_matrix_long_relief_cycle_debt_router.py`

and archived at

`docs/monograph/prime_matrix_long_relief_cycle_debt_router.md.`

It records

$$\texttt{ell=71},\quad \texttt{period\_p=5680},$$
$$\texttt{zero\_cycle\_relief\_count=8},\quad \texttt{positive\_cycle\_debt\_residue\_count=27},$$
$$\texttt{total\_cycle\_debt=101},\quad \texttt{total\_composite\_wait\_count=101},$$
$$\texttt{matches\_full\_horizon\_composite\_missing\_count=true},$$
$$\texttt{max\_cycle\_debt=15},\quad \texttt{max\_cycle\_debt\_residue=67}.$$

Thus the 101 composite formal missing-residue hits in the full-horizon ledger are exactly the sum of the per-residue cycle debts. The worst residue waits from $P = 12847$ to $P = 98047$, a 15-period delay. The latest interface is

$$\texttt{LongReliefCycleDebtPDECExclusionOrSupportMotionSAESummability}.$$

This records the phase cost of long relief; it does not yet prove global summability or global PDEC exclusion. □

Reduction-closed Statement 2.74 (One-period relief deficit)**.** *In the first complete* 71-*residue cycle after the accepted reset, every missing nonzero residue appears formally once, but only* 8 *of the* 35 *formal hits are actual prime relief.*

Status and reference. The one-period relief deficit router is generated by

`experiments/prime_matrix_one_period_relief_deficit_router.py`

and archived at

`docs/monograph/prime_matrix_one_period_relief_deficit_router.md.`

It records

$$\texttt{period\_is\_complete\_residue\_cycle=true},$$
$$\texttt{missing\_nonzero\_required=35},\quad \texttt{formal\_missing\_hit\_count=35},$$
$$\texttt{actual\_relief\_count\_in\_one\_period=8},$$
$$\texttt{composite\_missing\_count\_in\_one\_period=27},$$
$$\texttt{repeat\_prime\_anchor\_count\_in\_one\_period=10},$$
$$\texttt{relief\_deficit\_after\_one\_period=27}.$$

The 27 composite missing hits match exactly the positive cycle-debt residues. Thus one reset-local period is not enough actual load; it forces cycle debt or a support-motion SAE branch. The latest interface is

$$\texttt{OnePeriodReliefDeficitForcesCycleDebtPDECOrSupportMotionSAE}.$$

This is still a reduced hardpoint, not a global theorem closure. □

Reduction-closed Statement 2.75 (Cycle-debt CRT cover pressure)**.** *The* 27 *positive cycle-debt residues are not free local drift. Their composite waits form a CRT cover in the cycle coordinate, using* 24 *distinct blocking prime factors.*

Status and reference. The CRT cover pressure router is generated by

`experiments/prime_matrix_cycle_debt_crt_cover_pressure_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_crt_cover_pressure_router.md`.

It records

$$\texttt{positive\_cycle\_debt\_residue\_count=27},$$
$$\texttt{total\_composite\_waits=101},$$
$$\texttt{global\_unique\_blocker\_factor\_count=24},$$
$$\texttt{global\_blocker\_lcm=337212073559813724487421695331234639247},$$
$$\texttt{max\_row\_residue=67},\quad \texttt{max\_row\_cycle\_debt=15},$$
$$\texttt{max\_row\_blocker\_lcm=55140500775337593}.$$

For a fixed missing residue the same-residue column is

$$P(k) = P_0 + 5680k.$$

Each composite wait imposes one congruence class in the cycle coordinate modulo its least prime blocker. Hence persistent reproduction of the debt pattern requires a large CRT phase package, not merely a local endpoint translation. The latest interface is

$$\texttt{CycleDebtCRTCoverPressurePDECOrGlobalSupportMotionSAE}.$$

This records the global-family pressure, while leaving the PDEC exclusion or SAE summability proof open. □

Reduction-closed Statement 2.76 (Cycle-debt transverse CRT independence)**.** *The cycle-debt CRT cover is transverse to the local period* 5680*. It cannot be absorbed by a local period translation or by a same-slot drift.*

Status and reference. The transverse CRT independence router is generated by

`experiments/prime_matrix_cycle_debt_transverse_crt_independence_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_transverse_crt_independence_router.md`.

It records

$$\texttt{period\_p=5680=2\^{}4*5*71},$$
$$\texttt{global\_unique\_blocker\_factor\_count=24},\quad \texttt{transverse\_blocker\_factor\_count=24},$$
$$\texttt{all\_blocker\_factors\_coprime\_to\_period\_p=true},$$
$$\texttt{gcd\_global\_blocker\_lcm\_with\_period\_p=1},$$
$$\texttt{combined\_period\_equals\_product=true},$$
$$\texttt{all\_row\_shift\_replay\_classes\_zero=true},$$
$$\texttt{all\_row\_lcm\_exceeds\_phase\_support\_width=true}.$$

For a fixed residue column $P(k) = P_0 + 5680k$, every composite wait has a blocking prime q with $\gcd(q, 5680) = 1$. Replaying the same wait after a cycle shift K therefore forces $K \equiv 0 \pmod q$. Rowwise this gives $K \equiv 0 \pmod{\mathrm{lcm}_{\mathrm{row}}}$, and across all rows it gives

$$K \equiv 0 \pmod{337212073559813724487421695331234639247}.$$

The modulus is coprime to 5680, so this is an independent transverse CRT phase package, not a factor of the local period. The latest interface is

`TransverseCRTCoverPDECExclusionOrGlobalSupportMotionSAE`.

Thus local period absorption is closed in this certificate, while global PDEC exclusion or SAE summability remains open. □

Reduction-closed Statement 2.77 (Cycle-debt sparse replay barrier). *If the transverse CRT cover keeps the same blocking graph, exact replay is extremely sparse. A non-sparse persistent branch must move the blocker graph and hence enters a moving-cover PDEC interface.*

Status and reference. The sparse replay barrier router is generated by

`experiments/prime_matrix_cycle_debt_sparse_replay_barrier_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_sparse_replay_barrier_router.md`.

It records

```
global_exact_replay_cycle_modulus= 337212073559813724487421695331234639247,
global_exact_replay_p_gap= 1915364577819741955088555229481412750922960,
max_cycle_debt_support_width=15,
full_relief_cycle_span_ceiling=16,
single_full_debt_copy_per_full_relief_horizon=true,
exact_replay_branch_is_sae_sparse_current_certificate=true,
non_sparse_persistence_forces_moving_blocker_map=true.
```

The full debt word can replay exactly only after a shift by the global transverse lcm. The observed full-relief horizon has length $\lceil 1102/71 \rceil = 16$ residue cycles, whereas the replay modulus is the large integer displayed above. Thus any local or medium support-motion window shorter than that modulus contains at most one exact copy of the full word. The exact branch is therefore a sparse SAE branch. If a counterexample branch requires positive-density or short-window persistence, it must alter the blocking primes, phases, or row combination; this is the moving transverse cover PDEC branch. The latest interface is

`SparseReplaySAEOrMovingTransverseCoverPDEC`.

This is a sharper reduced hardpoint, not a global theorem closure. □

Reduction-closed Statement 2.78 (Cycle-debt near-shift exit boundary). *The moving-cover branch cannot be a small shift of the old debt prefix. Any near shift either hits an exit prime inside the shifted word, or else has no old prefix left to reuse and must build a fresh cover.*

Status and reference. The near-shift exit-boundary router is generated by

`experiments/prime_matrix_cycle_debt_near_shift_exit_boundary_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_near_shift_exit_boundary_router.md.`

It records

near_shift_limit_cycles=16, positive_cycle_debt_residue_count=27,
total_cycle_debt_mass=101, max_cycle_debt=15,
all_near_shifts_close_old_prefix_reuse=true,
shift_1_exit_prime_collision_row_count=27,
shift_1_exit_prime_collision_debt_mass=101,
shift_max_cycle_debt_exit_prime_collision_row_count=1,
shift_near_limit_fresh_cover_required_row_count=27.

For every positive-debt residue, the old composite prefix is immediately followed by the first relief prime in that same residue column. A shift $1 \leq K \leq d$ for a row of debt d pulls that exit prime into the shifted debt word. Hence $K = 1$ breaks all 27 rows, while every $1 \leq K \leq 15$ breaks all rows with $d \geq K$. At $K = 16$ no row has old prefix support left, so all 27 rows require fresh cover material. The latest interface is

FreshMovingCoverPDECOrGlobalSupportMotionSAE.

Thus near-shift reuse of the old CRT cover is closed in this certificate; the remaining branch is fresh moving-cover PDEC or genuine global support-motion SAE. □

Reduction-closed Statement 2.79 (Cycle-debt fresh-cover prime obstacle)**.** *On the same* 27 *positive-debt support rows, every near shifted fresh window contains actual prime obstacles. Same-support fresh cover is therefore closed unless those actual primes are deleted by a PDEC.*

Status and reference. The fresh-cover prime-obstacle router is generated by

`experiments/prime_matrix_cycle_debt_fresh_cover_prime_obstacle_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_fresh_cover_prime_obstacle_router.md.`

It records

total_cycle_debt_mass_per_shift=101, tested_shift_count=16,
total_tested_same_support_slots=1616,
total_prime_obstacles_all_near_shifts=364,
total_new_prime_obstacles_all_near_shifts=263,
all_near_shift_same_support_windows_have_prime_obstacles=true,
min_prime_obstacle_count_per_shift=17,
shift_near_limit_prime_obstacle_count=27,
same_support_fresh_cover_closed_current_certificate=true.

For each $K = 1, \ldots, 16$, the shifted window on the same support rows has the same total length 101 as the original debt word. Direct primality checking of those actual P-values shows at least 17 prime obstacles in every such shifted word; the fully fresh $K = 16$ window still has 27, all new. Hence a same-support fresh moving cover cannot be an all-composite word. The latest interface is

`FreshCoverPrimeObstaclePDECOrSupportRowReplacementSAE`.

The remaining branch must either delete actual prime obstacles, forming a prime-obstacle PDEC, or replace support rows and enter a support-row replacement SAE/PDEC interface. □

Reduction-closed Statement 2.80 (Cycle-debt support-row replacement Hall barrier)**.** *Support-row replacement is a Hall capacity problem over the* 35 *missing residue candidates. Fourteen of the sixteen near phases fail the Hall threshold test; only* $K = 13, 14$ *survive.*

Status and reference. The support-row replacement Hall router is generated by

`experiments/prime_matrix_cycle_debt_support_row_replacement_hall_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_support_row_replacement_hall_router.md`.

It records

```
candidate_missing_residue_count=35,  needed_positive_debt_row_count=27,
total_demand_width=101,  max_required_width=15,
hall_fail_shift_count=14,
hall_survivor_shifts=[13,14],
k13_assigned_immediate_relief_rows=6,
k14_assigned_immediate_relief_rows=8.
```

For a phase K, a candidate row has capacity equal to the longest composite run starting at K, capped at the maximum required width 15. For every threshold t, Hall requires

$$\#\{\text{candidate rows with capacity} \geq t\} \geq \#\{\text{demand rows with debt} \geq t\}.$$

This condition fails for

$$K = 1, 2, \ldots, 12, 15, 16.$$

Only $K = 13$ and $K = 14$ pass, and both are tight survivor phases requiring immediate-relief rows and large-scale row replacement. The latest interface is

`K13K14SupportReplacementSurvivorPDECOrGlobalSAE`.

This is a two-phase survivor hardpoint, not a global theorem closure. □

Reduction-closed Statement 2.81 (K13/K14 tight-Hall survivor island)**.** *The two support-row replacement survivors* $K = 13, 14$ *are not free moving replacement phases. They form a tight Hall island whose zero-slack layers force row replacement, immediate-relief support, and transverse CRT pressure.*

Status and reference. The K13/K14 survivor rigidity router is generated by `experiments/prime_matrix_cycle_debt_k13_k14_survivor_rigidity_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_k13_k14_survivor_rigidity_router.md.`

It records

```
k13_zero_slack_thresholds=[1,4,7,8],
k14_zero_slack_thresholds=[7,8,13,15],
k13_minimum_replacement_count=20, k14_minimum_replacement_count=19,
k13_minimum_immediate_relief_rows=6, k14_minimum_immediate_relief_rows=6,
k13_minimum_overstretch_units=61, k14_minimum_overstretch_units=65,
all_tight_layers_have_transverse_lcm_exceeding_period=true.
```

Here a zero-slack threshold t means that the number of candidate rows with capacity at least t is exactly the number of demand rows with debt at least t. Consequently the whole supply set of that layer is forced: deleting one such row, or moving one of its first t composite slots to a prime obstacle, breaks the Hall inequality at that threshold. For $K = 13$, the forced shells include the disjoint high shell

$$\{1, 13, 19, 30\} \to \{17, 23, 58, 67\}$$

at threshold 8, plus the singleton shell $31 \to 15$ at threshold 7. For $K = 14$, the top shells force $13 \to 67$, $19 \to 23$, and $70 \to 15$ at the corresponding thresholds.

The router then solves exact minimum-cost matching problems rather than relying on the previous greedy witness. Even after optimization, $K = 13$ requires at least 20 row replacements and 6 immediate-relief rows, while $K = 14$ requires at least 19 row replacements and 6 immediate-relief rows. The forced composite slots in every tight layer have transverse CRT least common multiple larger than the local period 5680. The neighboring phases are not survivors: $K = 12$ fails Hall at threshold 1, and $K = 15$ fails at threshold 6. Thus the remaining support-row branch is compressed to

`K13K14TightHallCRTIslandPDECOrMovingSupportSAE.`

This still is not a global theorem closure; it replaces the previous free replacement interpretation with a named tight-Hall CRT island. □

Reduction-closed Statement 2.82 (Shell phase graph obstruction). *The $K = 13, 14$ tight-Hall island cannot be absorbed by a single residue translation of the support. Its forced shells split into multiple incompatible phase deltas.*

Status and reference. The shell phase graph router is generated by

`experiments/prime_matrix_cycle_debt_shell_phase_graph_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_shell_phase_graph_router.md.`

It records

$$\texttt{k13\_forced\_shell\_matching\_product=252829237248000},$$
$$\texttt{k14\_forced\_shell\_matching\_product=2},$$
$$\texttt{k13\_forced\_edge\_count=1}, \quad \texttt{k14\_forced\_edge\_count=3},$$
$$\texttt{k13\_single\_translation\_support\_possible=false},$$
$$\texttt{k14\_single\_translation\_support\_possible=false},$$
$$\texttt{k13\_core\_min\_distinct\_delta\_lower\_bound=7},$$
$$\texttt{k14\_core\_min\_distinct\_delta\_lower\_bound=4}.$$

For each forced Hall shell the router builds the bipartite graph from supply residues to demand residues, keeping only edges whose capacity can cover the demand width. A single translation support motion would require one common $\delta \pmod{71}$ across all forced shells. For $K = 14$ three forced edges already have incompatible deltas:

$$13 \mapsto 67 \quad (\delta = 54), \qquad 19 \mapsto 23 \quad (\delta = 4), \qquad 70 \mapsto 15 \quad (\delta = 16).$$

The remaining 8..12 shell has only two possible matchings; the whole forced core still needs at least four distinct deltas. For $K = 13$, the singleton forced edge is $31 \mapsto 15$ with $\delta = 55$, and the enumerable core shells >= 8, 7..7, and 4..6 already require at least seven distinct deltas. The large 1..3 shell is not used to lower this count; it can only preserve or increase the number of deltas.

Thus the remaining branch is not an ordinary single-phase support translation. It is compressed to

$$\texttt{NonAffineShellPhaseFragmentPDECOrMultiDeltaSupportSAE}.$$

This still is not a global theorem closure; it closes the single-translation support-motion explanation and leaves a non-affine, multi-delta support branch. □

Reduction-closed Statement 2.83 (Multi-delta core CRT load)**.** *The non-affine multi-delta support branch carries a large transverse CRT load on its forced core. It is not a lightweight phase-noise explanation.*

Status and reference. The multi-delta core CRT-load router is generated by

```
experiments/prime_matrix_cycle_debt_multi_delta_core_crt_load_router.py
```

and archived at

```
docs/monograph/prime_matrix_cycle_debt_multi_delta_core_crt_load_router.md.
```

It records

$$\texttt{k13\_enumerated\_core\_matching\_count=96},$$
$$\texttt{k13\_minimum\_delta\_count=7}, \quad \texttt{k13\_minimum\_delta\_matching\_count=2},$$
$$\texttt{k13\_best\_min\_delta\_global\_lcm\_log10=36.165},$$
$$\texttt{k14\_enumerated\_core\_matching\_count=2},$$
$$\texttt{k14\_minimum\_delta\_count=4}, \quad \texttt{k14\_minimum\_delta\_matching\_count=1},$$
$$\texttt{k14\_best\_min\_delta\_global\_lcm\_log10=31.716}.$$

For each core edge $a \mapsto b$, the router expands the source row from the given shift K for the full demand width of b, records the actual composite slots, and multiplies the distinct smallest blocking factors. The resulting lanes are grouped by $\delta = b - a \pmod{71}$. For $K = 14$, the minimum multi-delta core is unique; its four lanes $\delta = 4, 16, 28, 54$ carry total demand width 52 and have global least common multiple about $10^{31.716}$. For $K = 13$, the enumerable rigid core has 96 matchings, only two of which realize the minimum of seven deltas; the best such core has total demand width 71 and global least common multiple about $10^{36.165}$.

Both least common multiples are far larger than the local period 5680. Thus the latest branch is compressed to

`MultiDeltaCoreCRTLoadPDECOrLowShellFullResidueSAE`.

This still is not a global theorem closure: it closes the light multi-delta noise interpretation, while leaving a heavy CRT-load PDEC or the $K = 13$ low-shell full-residue SAE as the next explicit interface. □

Reduction-closed Statement 2.84 (K13 low-shell delta skeleton)**.** *The $K = 13$ low shell does not require a full-residue escape. Relative to the rigid core, it adds only two delta lanes and turns $K = 13$ into a full-debt nine-lane CRT-load branch.*

Status and reference. The low-shell delta skeleton router is generated by

`experiments/prime_matrix_cycle_debt_low_shell_delta_skeleton_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_low_shell_delta_skeleton_router.md`.

It records

```
low_shell_possible_delta_count=71,
low_shell_minimum_delta_count=6,
low_shell_minimum_delta_witness=[23,35,38,58,68,70],
core_delta_count_before_low_shell=7,
low_shell_new_delta_count_over_core=2,
low_shell_new_deltas_over_core=[0,66],
full_k13_delta_count_after_low_shell=9,
full_k13_total_required_width=101,
full_k13_global_lcm_log10=42.095.
```

The router formulates the low shell as a binary matching problem. Edge variables choose the supply-to-demand matching, while delta variables record which residue lanes are used. First, with unit cost on every delta, the low shell alone needs only six deltas despite having all 71 deltas available. Second, after the seven core deltas from the $K = 13$ rigid core are given zero cost, the low shell requires only two new lanes, $\delta = 0$ and $\delta = 66$. Expanding the core and low-shell edges into their actual composite slots gives a full 101-width $K = 13$ load with nine delta lanes and global CRT least common multiple about $10^{42.095}$.

Thus the latest branch is compressed to

`K13FullDebtNineLaneCRTLoadPDECOrResidualSlackTailSAE`.

This still is not a global theorem closure; it closes the low-shell full-residue free-escape interpretation and leaves a full-debt CRT-load PDEC or a residual slack-tail SAE. □

Reduction-closed Statement 2.85 (K13 full-debt nine-lane residual tail)**.** *The residual slack-tail in the $K = 13$ full-debt nine-lane branch is not an artifact of the chosen matching. It is forced by the layer Hall ledger and already carries a transverse CRT load larger than the local period.*

Status and reference. The K13 full-debt nine-lane tail router is generated by

`experiments/prime_matrix_cycle_debt_k13_full_debt_nine_lane_tail_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_k13_full_debt_nine_lane_tail_router.md.`

It records

$$\begin{gathered}\texttt{capacity\_row\_count=27},\quad \texttt{demand\_row\_count=27},\\ \texttt{all\_capacity\_rows\_mandatory=true},\\ \texttt{total\_capacity=119},\quad \texttt{total\_demand\_width=101},\\ \texttt{unavoidable\_tail\_slot\_count=18},\\ \texttt{zero\_slack\_layers=[1,4,7,8]},\\ \texttt{allowed\_deltas=[0,8,22,39,54,55,58,59,66]},\\ \texttt{min\_tail\_factor\_count=8},\quad \texttt{min\_tail\_lcm\_log10=11.488}.\end{gathered}$$

At threshold 1, the Hall count is 27 supply rows against 27 demand rows. Hence every positive-capacity row is mandatory in any full-debt matching. The total capacity is 119, while the full demand is 101; equivalently the sum of the layer slacks is 18. Thus the residual tail contains exactly 18 slots independently of the particular nine-lane matching witness.

The zero-slack layers $1, 4, 7, 8$ are the local rigidity gates. Moving a slot through one of these layers without a compensating arrival would immediately break Hall capacity. Within the fixed nine-lane family, a MILP over all allowed 71 supply-demand edges optimizes the active tail blockers. Even under the minimizing objective, the tail activates at least eight distinct blocking factors,

$$3, 7, 11, 13, 19, 127, 151, 281,$$

whose least common multiple has logarithm about 11.488, already exceeding the local period 5680.

Thus the latest branch is compressed to

$$\texttt{K13LayerSlackTailCRTInvariantPDECOrMovingFamilySAE}.$$

This still is not a global theorem closure. It closes the interpretation that the residual slack-tail is a matching artifact or a cost-free tail; the remaining branch is a layer-invariant tail-CRT PDEC or a genuinely moving family SAE. □

Reduction-closed Statement 2.86 (K13 tail gate drift)**.** *The $K = 13$ layer-tail profile has no near-shift replay. The only other Hall-surviving shift is $K = 14$, and it changes the zero-slack gates and increases the tail balance.*

Status and reference. The K13 tail gate-drift router is generated by

`experiments/prime_matrix_cycle_debt_k13_tail_gate_drift_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_k13_tail_gate_drift_router.md.`

It records

`hall_pass_shifts=[13,14]`,
`same_slack_vector_shifts=[13]`, `same_zero_gate_set_shifts=[13]`,
`k13_tail_balance=18`, `k13_zero_slack_layers=[1,4,7,8]`,
`k14_tail_balance=31`, `k14_tail_increase_over_k13=13`,
`k14_zero_slack_layers=[7,8,13,15]`,
`k14_slack_l1_distance_from_k13=19`.

The router compares every near shift $1 \leq K \leq 16$ with the $K = 13$ slack vector. The complete vector is repeated only at the base shift itself. The same is true for the zero-gate set. Since the Hall-surviving shifts are only 13 and 14, any near moving branch that leaves $K = 13$ must enter $K = 14$.

But $K = 14$ is not a same-profile replay. It preserves only the common gates $7, 8$, loses the low gates $1, 4$, gains the high gates $13, 15$, and raises the tail balance from 18 to 31. Hence a moving family cannot carry the $K = 13$ layer-tail invariant through the near-shift window without changing the gate structure.

Thus the latest branch is compressed to

`K13GateProfileNoNearReplayPDECOrK14HighGateDriftSAE`.

This still is not a global theorem closure. It closes the near same-profile moving replay interpretation; the remaining branch is either a fixed $K = 13$ gate-profile PDEC or a $K = 14$ high-gate drift SAE/PDEC. □

Reduction-closed Statement 2.87 (K14 high-gate low-shell skeleton)**.** *The $K = 14$ high-gate drift branch is not an unregistered moving family. Its tight shells leave only two high-core matchings, and the remaining low shell compresses to a finite delta skeleton.*

Status and reference. The K14 high-gate low-shell skeleton router is generated by

`experiments/prime_matrix_cycle_debt_k14_high_gate_low_shell_skeleton_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_k14_high_gate_low_shell_skeleton_router.md`.

It records

`k14_zero_slack_layers=[7,8,13,15]`,
`k14_high_gate_core_alternative_count=2`,
`k14_best_core_deltas=[4,16,28,54]`,
`k14_low_shell_minimum_delta_count=6`,
`k14_low_shell_new_deltas_over_core=[2,48,58,70]`,
`k14_full_delta_count_after_low_shell=8`,
`k14_full_total_required_width=101`,
`k14_full_global_lcm_log10=48.508`.

The high gates force the singleton shells $13 \mapsto 67$, $19 \mapsto 23$, and $70 \mapsto 15$, together with one of two matchings on the 8..12 shell. For each high-core candidate, the router removes the already-used supply and demand residues and solves the remaining low-shell matching by a binary delta-skeleton

MILP. The best full branch has core deltas $4, 16, 28, 54$, and the low shell adds only the four lanes $2, 48, 58, 70$.

Expanding the resulting twenty-seven edges into actual composite slots covers all 101 demand-width units. The eight-lane load has global CRT least common multiple about $10^{48.508}$, far beyond the local period 5680. Thus the $K = 14$ moving branch returns to a full-debt CRT-load PDEC; it is no longer an unnamed high-gate drift SAE.

Thus the latest branch is compressed to

`K13FixedGateProfilePDECOrK14FullDebtEightLaneCRTLoadPDEC`.

This still is not a global theorem closure. It closes the unregistered $K = 14$ high-gate moving-family explanation and leaves a fixed $K = 13$ gate-profile PDEC or the registered $K = 14$ eight-lane CRT-load PDEC. □

Reduction-closed Statement 2.88 (Two-survivor terminal CRT bifurcation)**.** *The $K = 13$ and $K = 14$ terminal branches are not two views of the same support motion. They form a named terminal CRT bifurcation.*

Status and reference. The two-survivor terminal CRT bifurcation router is generated by

`experiments/prime_matrix_cycle_debt_two_survivor_terminal_crt_bifurcation_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_two_survivor_terminal_crt_bifurcation_router.md`.

It records

```
k13_delta_count=9,  k14_delta_count=8,
common_deltas=[54,58],
delta_union_count=15,  delta_symmetric_difference_count=13,
source_intersection_count=19,  source_symmetric_difference_count=16,
target_intersection_count=27,  pair_intersection_count=1,
union_lcm_log10=57.156,  tail_union_lcm_log10=24.632.
```

Both terminal branches cover the same 27 target demand residues. However, their source supports overlap in only 19 rows, and the actual source-to-target matching overlaps in only one edge, $13 \mapsto 67$. At the lane level, only the two deltas 54 and 58 are common; the symmetric difference contains 13 lanes. Thus $K = 13$ and $K = 14$ cannot be treated as a single support pattern with a small phase perturbation.

The combined blocker set has least common multiple about $10^{57.156}$, and the combined tail blocker set has least common multiple about $10^{24.632}$. Both are far larger than the local period 5680. Therefore the remaining branch is no longer an unnamed support-motion SAE. It is a terminal two-survivor CRT bifurcation PDEC that must be excluded directly.

Thus the latest branch is compressed to

`TwoSurvivorTerminalCRTBifurcationPDECExclusion`.

This still is not a global theorem closure. It closes the interpretation that the two survivor branches can absorb each other as one unnamed moving family. □

Reduction-closed Statement 2.89 (Two-survivor switch-arrival pressure)**.** *The terminal two-survivor bifurcation cannot be absorbed by the common anchor or by the two common delta lanes. Avoiding a fixed-branch PDEC forces a fresh-arrival ColumnCRT load or a branch-exclusive CRT-load exclusion.*

Status and reference. The two-survivor PDEC exclusion router is generated by

```
experiments/prime_matrix_cycle_debt_two_survivor_pdec_exclusion_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-two-survivor-pdec-exclusion-router.md.
```

It records

```
common_pair_count=1,  common_pair_width=15
forced_rematched_target_count=26,  forced_rematched_target_width=86
minimum_branch_exclusive_delta_width=70
minimum_branch_exclusive_delta_lcm_log10=32.582
k14_only_assigned_width=29,  k14_only_capacity_at_k13=0
k14_only_capacity_at_k14=39,  k14_arrival_lcm_log10=20.205
k13_same_support_deficit_at_k14=8,  common_source_changed_count=18
```

The only common actual source-target edge is $13 \mapsto 67$, and it accounts for only 15 units of target width. Hence the remaining 26 targets, of total width 86, must be genuinely rematched. The common delta lanes $54, 58$ do not carry this rematching load: each terminal branch still has at least 70 units of width on branch-exclusive deltas, with least common multiple at least $10^{32.582}$ in the current certificate.

On the switching side, the $K = 14$ branch uses eight source rows that have zero capacity at $K = 13$. These rows acquire 39 units of capacity at $K = 14$ and carry 29 units of assigned demand, with arrival blocker lcm about $10^{20.205}$. Conversely, dragging the $K = 13$ source set to $K = 14$ leaves only 93 units of capacity against the 101-unit demand, a deficit of 8.

Thus the latest branch is compressed to

$$\texttt{TerminalSwitchArrivalColumnCRTPDECOrBranchExclusiveCRTLoadExclusion}.$$

This is still not a global theorem closure. It closes the interpretation that the terminal bifurcation can be repaired by the common anchor or the common delta lanes; the next obligation is to exclude the switch-arrival ColumnCRT load or the persistent branch-exclusive CRT loads. □

Reduction-closed Statement 2.90 (Terminal switch arrival wall)**.** *The $K = 14$ fresh-arrival source is not a smoothly opening capacity source. It is an eight-prime entry wall followed by a post-wall CRT load.*

Status and reference. The terminal switch arrival wall router is generated by

```
experiments/prime_matrix_cycle_debt_terminal_switch_arrival_wall_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-terminal-switch-arrival-wall-router.md.
```

It records

```
arrival_source_count=8,  all_arrival_sources_zero_capacity_at_k13=true
all_entry_wall_cycles_equal_13=true,  all_entry_wall_positions_zero_at_k13=true
entry_wall_lcm_log10=39.482
postwall_capacity_total=39,  postwall_assigned_width_total=29
postwall_assigned_lcm_log10=20.205
entry_plus_assigned_lcm_log10=59.687
entry_plus_postwall_lcm_log10=68.361
```

The eight $K = 14$-only source rows all have zero capacity at $K = 13$. More precisely, each of them is cut at $K = 13$ by a prime obstacle at window-position 0. These eight entry primes have least common multiple about $10^{39.482}$, coprime to the local period 5680.

After crossing this entry wall, the $K = 14$ branch obtains 39 post-wall capacity slots. It uses 29 of them for the actual assignment, with post-wall assigned blocker lcm about $10^{20.205}$. Combining the entry wall with the assigned post-wall load gives lcm about $10^{59.687}$, and combining the entry wall with all post-wall slots gives lcm about $10^{68.361}$, still coprime to 5680.

Thus the latest branch is compressed to

`EightPrimeEntryWallPostWallCRTExclusionOrBranchExclusiveCRTLoadExclusion`.

This is still not a global theorem closure. It closes the interpretation that the $K = 14$ fresh-arrival capacity is an unnamed smooth deformation; the next obligation is to exclude the eight-prime entry-wall/post-wall CRT load, or the parallel branch-exclusive CRT loads. □

Reduction-closed Statement 2.91 (Coupled branch and entry-wall load). *The entry-wall/post-wall load and the branch-exclusive load are not independent terminal exits. They couple on the $K = 14$ branch; the $K = 13$ branch leaves its own branch-exclusive CRT load.*

Status and reference. The coupled branch entry-wall router is generated by

`experiments/prime_matrix_cycle_debt_coupled_branch_entry_wall_router.py`

and archived at

`docs/monograph/prime-matrix-cycle-debt-coupled-branch-entry-wall-router.md.`

It records

```
k13_branch_exclusive_width=73,  k13_branch_exclusive_lcm_log10=36.678
k14_branch_exclusive_width=70,  k14_arrival_assigned_width=29
k14_arrival_branch_overlap_width=21
k14_branch_arrival_union_width=78
k14_branch_plus_entry_plus_postwall_lcm_log10=85.024
both_branches_plus_k14_entry_postwall_lcm_log10=99.349
```

Thus the $K = 14$ terminal branch does not choose between entry-wall load and branch-exclusive load. It carries both. The 29-unit arrival load overlaps the 70-unit branch-exclusive load in 21 units, but their union still forces 78 units of actual load. After adding the entry wall and all post-wall slots, the $K = 14$ coupled lcm is about $10^{85.024}$, coprime to the local period 5680.

The other terminal possibility is $K = 13$. There the remaining branch-exclusive load has width 73 and lcm about $10^{36.678}$. If both branch-exclusive loads are combined with the $K = 14$ entry/post-wall load, the audit lcm is about $10^{99.349}$.

Thus the latest branch is compressed to

`K13BranchExclusiveCRTLoadExclusionOrK14CoupledEntryBranchCRTWallExclusion`.

This is still not a global theorem closure. It replaces the loose entry-wall/branch-exclusive pair by two branch-specific CRT-load exclusions. □

Reduction-closed Statement 2.92 (Branch replay support gap). *The registered terminal branch loads cannot be reproduced by a local moving-slot replay. Any replay of the same blocker package has period at least its CRT lcm, which is already far larger than the phase support width.*

Status and reference. The branch replay support-gap router is generated by

```
experiments/prime_matrix_cycle_debt_branch_replay_support_gap_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-branch-replay-support-gap-router.md.
```

It records

```
period_p=5680,  near_shift_limit_cycles=16
k13_branch_exclusive_width=73
k14_branch_arrival_union_width=78
k14_coupled_audit_slot_count_all_postwall=117
smallest_log10_margin_over_support_width=30.737
all_blocks_coprime_to_period=true
all_nonzero_replay_moduli_exceed_support_width=true
local_moving_slot_replay_excluded_for_registered_blocks=true.
```

The proof is a direct CRT replay calculation. Let $M = 5680$. If a registered blocker package B reappears after T local periods with the same blockers, then for every $q \in B$

$$n_q + TM \equiv 0 \pmod{q}, \qquad n_q \equiv 0 \pmod{q}.$$

All registered q's are coprime to M. Hence $T \equiv 0 \pmod{q}$ for every $q \in B$, and the least nonzero replay period is $\mathrm{lcm}(B)$.

For the $K = 13$ branch-exclusive block this replay modulus is about $10^{36.678}$, while the phase support width is 73. For the $K = 14$ coupled branch-plus-entry-plus-postwall block the replay modulus is about $10^{85.024}$, while the actual support width is 78 and the full audit slot count is 117. Therefore the registered high-factor absorption slots cannot be carried by local moving-slot support motion.

Thus the latest branch is compressed to

$$\texttt{BranchReplayColumnCRTPDECExclusionOrIsolatedTerminalAtomAbsorption}.$$

This is still not a global theorem closure. It closes the local replay interpretation; any far replay is a named ColumnCRT/PDEC package, and any non-replaying survivor must be treated as an isolated finite atom. □

Reduction-closed Statement 2.93 (Global dichotomy for branch replay). *The isolated-atom alternative is not a global structural escape. A persistent registered replay block is a Column-CRT/PDEC family; a nonpersistent registered block is only a finite atom requiring finite base checking.*

Status and reference. The branch replay global dichotomy router is generated by

```
experiments/prime_matrix_cycle_debt_branch_replay_global_dichotomy_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-branch-replay-global-dichotomy-router.md.
```

It records

```
registered_replay_block_count=6
minimum_cycle_replay_modulus_log10=32.582
minimum_p_space_columncrt_modulus_log10=36.337
maximum_p_space_columncrt_modulus_log10=103.103
persistent_registered_replay_routes_to_columncrt_pdec=true
isolated_atoms_cannot_form_infinite_registered_family=true
finite_atom_base_check_required=true.
```

There are only finitely many registered replay blocks in the current terminal certificate. If a global counterexample chain used one of them infinitely often with the same blocker package, the previous replay lemma gives a fixed cycle congruence $T \equiv 0 \pmod{\operatorname{lcm}(B)}$. In the original P-coordinate this is a fixed ColumnCRT class modulo

$$5680 \cdot \operatorname{lcm}(B).$$

The smallest such P-space modulus in the registered list is already about $10^{36.337}$.

If no registered replay block occurs infinitely often, then the registered atoms are finite and cannot be a global structural escape. They still require finite base checking; this router does not erase that obligation. If the blocker package changes, then the object is no longer the same replay block and must return to a PDEC/SAE route or to a new named router.

Thus the latest branch is compressed to

$$\texttt{BranchReplayColumnCRTPDECExclusionOrFiniteAtomBaseCheck}.$$

This remains below an unconditional theorem. The genuine structural hard point is now the exclusion of persistent branch-replay ColumnCRT/PDEC, plus the finite base check for any isolated atoms. □

Reduction-closed Statement 2.94 (Finite atom boundary bridge)**.** *The finite-atom check splits into a covered finite prefix and a post-*100000 *tail packet. The older direct grid verification up to* $P \leq 5000$ *must not be used to absorb the present cycle-debt atoms.*

Status and reference. The finite atom boundary bridge router is generated by

```
experiments/prime_matrix_cycle_debt_finite_atom_boundary_bridge_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-finite-atom-boundary-bridge-router.md.
```

It records

```
direct_grid_verified_max_p=5000
dynamic_finite_bridge_range=[3001,99991]
dynamic_finite_bridge_closed=true
cover_pressure_atom_count=155
cover_pressure_all_atoms_in_dynamic_finite_bridge=true
arrival_wall_atom_count=55
arrival_wall_tail_atom_count=31
tail_atom_p_range=[101087,134047]
finite_atom_base_check_fully_closed=false.
```

The cycle-debt cover-pressure packet has 155 recorded P-coordinates, all lying in the already archived dynamic finite bridge $3001 \leq P < 100000$. These finite-prefix atoms are therefore absorbed by the existing bridge, not by the older direct grid verification up to $P \leq 5000$.

The terminal arrival-wall packet is different. It has 55 recorded P-coordinates. Only 24 lie below 100000; the remaining 31 are post-100000 tail atoms, with P-range 101087 to 134047. They consist of assigned slots, post-wall first primes, and tail slots. Hence the finite-atom alternative is not completely closed by the finite bridge.

Thus the latest branch is compressed to

`BranchReplayColumnCRTPDECExclusionOrPost100000TailAtomRunner`.

This is still not a global theorem closure. It closes the finite-prefix part of the finite-atom check and leaves the post-100000 tail packet to a tail-runner, tail lower-sieve, or ColumnCRT/PDEC route. □

Reduction-closed Statement 2.95 (Post-100000 tail atom exact runner)**.** *The post-*100000 *finite tail atoms in the registered terminal packet are exactly verified. The finite-atom branch is closed for the current registered objects, leaving the persistent branch-replay ColumnCRT/PDEC as the active structural obstruction.*

Status and reference. The post-100000 tail atom exact runner is generated by

```
experiments/prime_matrix_cycle_debt_post100000_tail_atom_exact_runner.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-post100000-tail-atom-exact-runner.md.
```

It records

```
tail_atom_count=31,  tail_atom_p_range=[101087,134047]
kind_counts={assigned_slot:15,postwall_first_prime:8,tail_slot:8}
all_composite_atoms_divisible_by_recorded_factor=true
all_composite_recorded_factors_are_smallest=true
all_postwall_first_primes_verified=true
all_preprime_slots_composite_in_arrival_rows=true
finite_atom_branch_closed_for_registered_atoms=true.
```

The 31 tail atoms split into 23 composite slots and 8 post-wall first-prime boundaries. For every composite slot the recorded factor divides the slot and equals its smallest factor. For every post-wall first-prime boundary, direct trial division confirms primality in the recorded range. Moreover, in each arrival row all slots before the first-prime boundary are verified composite. Thus the finite post-100000 tail packet is no longer a parallel exit for the current registered terminal configuration.

Thus the latest branch is compressed to

`BranchReplayColumnCRTPDECExclusion`.

This still does not prove the row/column theorem. It removes the finite atom alternative and leaves the global persistent replay family as a named ColumnCRT/PDEC obstruction. □

Reduction-closed Statement 2.96 (Fresh-modulus escalation)**.** *A fixed finite ColumnCRT replay class is not a terminal structure for an infinite counterexample chain. New prime layers are coprime to the registered finite modulus and force unbounded modulus escalation, a PDEC/ColumnCRT defect, or a tail-sieve stability contradiction.*

Status and reference. The branch replay fresh-modulus escalation router is generated by

```
experiments/prime_matrix_cycle_debt_branch_replay_fresh_modulus_escalation_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-branch-replay-fresh-modulus-escalation-router.md.
```

It records

```
finite_atom_branch_closed_for_registered_atoms=true
registered_replay_block_count=6
fresh_prime_sample_count_per_block=8
all_registered_blocks_have_coprime_fresh_layers=true
minimum_first_fresh_log10_gain=2.400
minimum_sample_log10_gain=19.435
finite_crt_terminal_description_excluded=true.
```

Let a registered replay class have finite modulus Q. Every prime $\ell \nmid Q$ is a new CRT coordinate independent of the old replay class. When such an ℓ enters the active wheel, the old congruence modulo Q cannot decide the ℓ-layer. A persistent counterexample chain must therefore absorb ℓ into a larger modulus, fail at that layer as a named PDEC/ColumnCRT defect, or prove that the tail sieve absorbs the new coordinate.

The current six registered replay blocks make this explicit. Each has coprime fresh-prime layers. Adding only the first fresh prime increases the modulus by at least $10^{2.400}$, and adding the first eight fresh primes in the recorded sample increases it by at least $10^{19.435}$. Thus a finite registered ColumnCRT class cannot be a final stable object for an infinite counterexample chain.

Thus the latest branch is compressed to

$$\texttt{UnboundedFreshModulusEscalationPDECOrTailSieveStabilityContradiction}.$$

This is still not an unconditional closure. It identifies the remaining global obstruction as unbounded fresh-modulus escalation or a tail-sieve stability contradiction, rather than a fixed finite CRT atom. □

Reduction-closed Statement 2.97 (Fresh-modulus tail-sieve bridge)**.** *After finite ColumnCRT terminal classes have been excluded, a persistent branch replay family has only two remaining branch-replay exits. Either a fresh prime layer creates a PDEC/ColumnCRT defect, or the non-defective unbounded fresh-prime layers form the same one-residue-per-prime tail rough object used in the $B = 3$ lower-sieve ledger.*

Status and reference. The fresh-modulus tail-sieve bridge router is generated by

```
experiments/prime_matrix_cycle_debt_fresh_modulus_tail_sieve_bridge_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-fresh-modulus-tail-sieve-bridge-router.md.
```

It records

```
fresh_modulus_escalation_registered=true
finite_columncrt_terminal_excluded=true
non_pdec_unbounded_fresh_layers_force_tail_rough_object=true
tail_object_interface_closed=true
conditional_external_tail_sieve_closed=true
strict_self_contained_tail_sieve_closed=false
fresh_layer_pdec_excluded=false
row_column_unconditional_closed=false.
```

Let Q be the finite modulus of a registered replay class. Once the fixed finite terminal explanation is removed, every subsequent fresh prime $\ell \nmid Q$ is an independent CRT coordinate. If the ℓ-layer cannot be registered without a projection collision, residue repetition, moving-support escape, or ColumnCRT defect, the branch has entered the named fresh-layer PDEC exit. Otherwise the only information carried by that layer is one forbidden residue class modulo ℓ. Repeating this over the unbounded fresh-prime layers gives the tail object

$$S(P) = \#\{1 \leq k < P : \ k \text{ avoids one prescribed residue class modulo each } q \leq P^{0.43}\}.$$

For each squarefree $d < P$, CRT counting gives

$$A_d(P) = \frac{P-1}{d} + r_d, \qquad |r_d| \leq 1,$$

so this is exactly the same interface as the existing $B = 3$ lower-sieve rough-object ledger. The imported tail ledger has

$$\alpha = 0.43, \qquad \text{model main at } P = 100000 = 4896.256004,$$

and its ten-percent main term still exceeds 401 by 88.625600. Thus, if external explicit Mertens/Dusart estimates or the standard beta-sieve input are accepted, the tail-sieve stability branch is conditionally closed.

The strict self-contained branch is not closed by this router. It still requires the internal Mertens/PNT/Dusart reciprocal-prime tail proof and a separate exclusion of fresh-layer PDEC/ColumnCRT. The current branch-replay frontier is therefore

`FreshLayerPDECColumnCRTExclusionOrSelfContainedTailSieveStabilityClosure`.

This is a compression of the latest branch, not an unconditional proof of the row/column theorem.

□

Reduction-closed Statement 2.98 (Self-contained tail-sieve synchronization). *For the branch replay fresh-modulus branch, the old self-contained tail-sieve opening is no longer an active branch-replay obstruction after the later strict Mertens/PNT and $B = 3$ total-variation synchronization certificates are imported. The remaining branch-replay obstruction is the fresh-layer PDEC/ColumnCRT exit.*

Status and reference. The synchronization router is generated by

```
experiments/prime_matrix_cycle_debt_fresh_modulus_tail
_self_contained_sync_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-fresh-modulus-tail
-self-contained-sync-router.md.
```

It records

```
previous_tail_object_interface_closed=true
previous_non_pdec_unbounded_fresh_layers_force_tail_rough_object=true
strict_self_contained_mertens_tail_proved_latest=true
b3_tv_strict_self_contained_synchronized_latest=true
strict_self_contained_tail_sieve_closed_for_branch_replay=true
fresh_layer_pdec_excluded=false
row_column_unconditional_closed=false.
```

The previous bridge had reduced the non-PDEC unbounded fresh layers to the one-residue-per-prime $B = 3$ tail rough object, but it kept the strict self-contained branch open under the coarse name

`SelfContainedDusartReciprocalPrimeProofAppendixXGe10372.`

Later strict synchronization certificates refine this coarse atom: the theta/PNT envelope and the Meissel–Mertens B_1 interval are imported as self-contained inputs, and the $B = 3$ total-variation ledger is synchronized with that Mertens tail closure. Hence the branch-replay strict remaining basis changes from

```
FreshLayerPDECColumnCRTExclusion
AND SelfContainedDusartReciprocalPrimeProofAppendixXGe10372
```

to

```
FreshLayerPDECColumnCRTExclusion.
```

This still does not close the row/column theorem. It only removes the old tail-sieve analytic opening from this branch; the fresh-layer PDEC/ColumnCRT obstruction remains active. □

Reduction-closed Statement 2.99 (Local fresh-layer collision exclusion)**.** *For the registered branch-replay blocks, a fresh-prime layer cannot create a local projection collision inside the recorded primitive support windows. Any remaining fresh-layer PDEC must therefore be a support-motion escape, a remote P-space ColumnCRT/PDEC recurrence, or a new moving-family block.*

Status and reference. The local collision router is generated by

```
experiments/prime_matrix_cycle_debt_fresh_layer_local_collision_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-fresh-layer-local-collision-router.md.
```

It records

```
period_p=5680
registered_block_count=6
all_sample_fresh_primes_coprime_to_period_p=true
all_first_fresh_primes_exceed_support_width=true
all_first_fresh_primes_exceed_audit_slots=true
all_registered_samples_injective_on_local_windows=true
minimum_first_fresh_minus_support_width=178
minimum_first_fresh_minus_audit_slots=178
local_fresh_layer_projection_collision_excluded=true
row_column_unconditional_closed=false.
```

Let the local period step be $M = 5680$, and let ℓ be a registered fresh prime. In every recorded sample, $\gcd(M, \ell) = 1$. If two slots j_1, j_2 in a local window of length W collide modulo ℓ, then

$$a + j_1 M \equiv a + j_2 M \pmod{\ell},$$

so $\ell \mid (j_1 - j_2)M$. Since M is invertible modulo ℓ, this forces $\ell \mid j_1 - j_2$. But the recorded windows satisfy $W < \ell$, with minimum first-fresh surplus 178 over both support width and audit-slot count. Hence $|j_1 - j_2| < \ell$, and the only possible collision is $j_1 = j_2$.

Thus the local projection-collision subcase of fresh-layer PDEC is excluded for the registered blocks. The remaining branch-replay obstruction is

`FreshLayerSupportMotionEscapeOrRemoteColumnCRTPDECExclusion.`

This is still a named open exit, not a global unconditional proof. □

Reduction-closed Statement 2.100 (Registered support-motion escape removal). *For the registered branch-replay blocks, the support-motion escape is not a remaining local mechanism. After local fresh-layer projection collisions and same-package local support drift are excluded, a persistent registered replay must be a remote P-space ColumnCRT/PDEC class; if the blocker package changes, it is an unregistered moving-family branch.*

Status and reference. The fresh support-motion global router is generated by

```
experiments/prime_matrix_cycle_debt_fresh_support_motion_global_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-fresh-support-motion-global-router.md.
```

It records

```
period_p=5680
registered_block_count=6
local_fresh_layer_projection_collision_excluded=true
registered_local_support_motion_excluded=true
persistent_registered_replay_routes_to_columncrt_pdec=true
registered_support_motion_escape_closed=true
minimum_cycle_log10_margin_over_support_width=30.737
minimum_p_space_columncrt_modulus_log10=36.337
remote_pspace_columncrt_excluded=false
unregistered_moving_family_excluded=false
row_column_unconditional_closed=false.
```

The preceding local-collision certificate removes collisions inside the recorded primitive support windows. The earlier support-gap replay lemma also removes same-package local support drift: a nonzero replay by the same blocker package must move by a multiple of $\mathrm{lcm}(B)$, and the least recorded margin over support width is $10^{30.737}$. Hence a registered block cannot persist through local support motion.

The global dichotomy then leaves only two possibilities. If a registered block recurs infinitely often, it is promoted to a fixed P-space ColumnCRT/PDEC class, whose smallest recorded modulus is about $10^{36.337}$. If the blocker package changes, the branch is not registered support motion but an unregistered moving-family branch requiring a new named router. The current branch-replay frontier is therefore

$$\texttt{RemotePspaceColumnCRTExclusionOrUnregisteredMovingFamilyRouter}.$$

This is a further compression of the branch-replay interface, not a proof of the row/column theorem. □

Reduction-closed Statement 2.101 (Remote ColumnCRT feedback router). *Within the registered branch-replay chain, a remote P-space ColumnCRT class is no longer allowed to remain as a bare fixed-period terminal. Such a bare class is absorbed by the already recorded finite-atom runner, the fresh-modulus escalation router, and the strict tail-sieve synchronization. Therefore the current branch-replay remainder is narrowed to a materialized fresh-layer PDEC/ColumnCRT event or to an unregistered moving-family event.*

Status and reference. The feedback router is generated by

```
experiments/prime_matrix_cycle_debt_remote_columncrt_feedback_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-remote-columncrt-feedback-router.md.
```

It records

```
registered_remote_block_count=6
minimum_remote_pspace_columncrt_modulus_log10=36.337
maximum_remote_pspace_columncrt_modulus_log10=103.103
minimum_first_fresh_log10_gain=2.400
minimum_sample_fresh_log10_gain=19.435
registered_finite_atoms_absorbed=true
fixed_finite_remote_crt_not_terminal=true
non_pdec_fresh_tail_sieve_closed=true
bare_remote_pspace_columncrt_terminal_closed=true
materialized_fresh_layer_pdec_columncrt_excluded=false
unregistered_moving_family_excluded=false
row_column_unconditional_closed=false.
```

The point is only a feedback reduction. If the remote class is an isolated or finite registered atom, it has already been handled by the exact finite runner. If it is a fixed finite CRT replay class, the fresh-modulus escalation router shows that it cannot be a terminal infinite counterexample description. If fresh layers do not materialize a PDEC/ColumnCRT event, the non-PDEC unbounded fresh-layer branch has already been routed to and closed by the strict tail-sieve synchronization. Hence the broad remote ColumnCRT label can be removed as a bare terminal explanation.

The remaining branch-replay interface is consequently

`MaterializedFreshLayerPDECColumnCRTExclusionOrUnregisteredMovingFamilyRouter`.

This still leaves a real open obligation: one must exclude materialized fresh-layer PDEC/ColumnCRT events or unregistered moving-family recurrences. It is not a proof of the row/column theorem. □

Reduction-closed Statement 2.102 (Fresh-layer PDEC admission firewall)**.** *Within the current cycle-debt branch-replay corpus, the materialized fresh-layer PDEC/ColumnCRT label has no unnamed admitted instance. A future fresh-layer PDEC obstruction must be submitted as an explicit primitive same-formal-unit schema; otherwise the branch is an unregistered moving family.*

Status and reference. The firewall router is generated by

```
experiments/prime_matrix_cycle_debt_fresh_layer_pdec_admission_firewall_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-fresh-layer-pdec-admission-firewall-router.md.
```

It records

```
registered_block_count=6
minimum_first_fresh_minus_max_window=178
minimum_remote_plus_first_fresh_log10=38.803
local_registered_materialized_pdec_closed=true
current_materialized_pdec_frontier_closed=true
pdec_family_explicit_input_boundary_closed=true
newlayer_schema_admission_closed=true
newlayer_ranktwo_budget_independent_gate_removed=true
current_corpus_materialized_fresh_layer_pdec_closed=true
future_explicit_primitive_fresh_layer_pdec_schema_submitted=false
unregistered_moving_family_excluded=false
row_column_unconditional_closed=false.
```

For the registered blocks, every sampled first fresh prime is larger than the support and audit windows, and the local affine projection is injective. Thus local fresh-layer phase reuse cannot materialize a PDEC event inside the registered support. If a remote event is to be admitted as PDEC, it must pass the already registered PDEC boundary: the same formal unit, one fixed phase map, at least three physical primitive atoms after quotienting, non-tautology, rank at least two, and cap stability. ColumnCRT or displacement cases are absorbed before admission; sparse fallbacks require their own packet extractor.

Consequently the current branch-replay interface is narrowed to

`FutureExplicitPrimitiveFreshLayerPDECSchemaIfNewOrUnregisteredMovingFamilyRouter`.

This does not prove that no future primitive schema can exist, and it does not exclude unregistered moving families. It only removes the current unnamed materialized fresh-layer PDEC/ColumnCRT label. □

Reduction-closed Statement 2.103 (Cycle-debt branch-replay current frontier zero)**.** *In the present materialized cycle-debt branch-replay corpus, the remaining fresh-layer PDEC/ColumnCRT and unregistered moving-family labels are reduced to future admission firewalls rather than active terminal instances. The branch-replay subfrontier is therefore empty in the current corpus, while the global row/column theorem still depends on the separate final-input gates.*

Status and reference. The current-frontier-zero router is generated by

```
experiments/prime_matrix_cycle_debt_branch_replay_current_frontier_zero_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-branch-replay-current-frontier-zero-router.md.
```

It records

```
current_fresh_layer_pdec_frontier_closed=true
future_explicit_primitive_fresh_layer_pdec_schema_submitted=false
unregistered_moving_family_schema_submitted=false
future_pdec_schema_admission_discipline_closed=true
future_sparse_schema_admission_discipline_closed=true
final_input_firewall_boundary_closed=true
no_hidden_terminal_remaining=true
cycle_debt_branch_replay_current_materialized_frontier_zero=true
future_explicit_schema_global_nonexistence_proved=false
row_column_unconditional_closed=false.
```

The proof is a current-instance compression. A future primitive fresh-layer PDEC schema has not been submitted, so it is not an active mathematical object in the current branch-replay corpus; it is only an admission discipline. The same applies to an unregistered moving family unless it supplies source, shape, phase, persistence, and no-loss ledger fields. The PDEC and sparse schema firewalls specify how such a future object would have to re-enter.

Thus the current materialized cycle-debt branch-replay frontier is

`CycleDebtBranchReplayCurrentMaterializedFrontierZeroWithFutureSchemaFirewall`.

This is still not a proof of the row/column theorem. The global remainder is

`GlobalFinalInputsStillOpen`,

including the noncanonical legal-closure mode and the DStructure/Rankin promotion gate, unless future explicit schemas are submitted and separately excluded. □

Chapter 3

Prime-Matrix Contradiction Fields

Let P be an odd prime and write the $P \times P$ prime matrix entries as

$$n(r,c) = (r-1)P + c, \qquad 1 \le r, c \le P.$$

The row and column programs attempt to show that every nontrivial row or column contains a prime. The current formal appendix reduces a row or non-P column counterexample to a structured bad configuration after removing small-factor locks and tail anchors.

Reduction-closed Statement 3.1 (A/B reduction, recorded form)**.** *A nontrivial full-composite row or column, after small-factor and tail-anchor peeling, induces a Structured-EHPD bad configuration satisfying CRT balance, large-factor non-reuse, and anchor-capacity constraints.*

Reference. See `docs/row-column-reduction-formal-appendix.md` and the theorem list

`docs/monograph/prime-matrix-ab-entrance-theorem-list.md`.

This statement is a reduction interface; it does not by itself prove the Structured-EHPD exclusion theorem. □

3.1 Two-Point Rough-Pair Propositions

A natural next strengthening asks for two simultaneous nonzero-residue constraints in the back half of the prime matrix. Fix an even integer $w > 0$ and ask whether there is an entry x in the back half of the $P \times P$ matrix such that both x and $x - w$ are nonzero modulo every prime $p \le P$, or equivalently both x and $w - x$ are nonzero modulo every prime $p \le P$.

Proposition 3.2 (Equivalence and strength of the two-point rough-pair problem)**.** *For fixed even $w > 0$, the two conditions*

$$p \nmid x(x-w) \quad (p \le P)$$

and

$$p \nmid x(w-x) \quad (p \le P)$$

are identical. In the region $1 < x \le P^2$, any integer nonzero modulo every prime $p \le P$ is prime. Consequently, a proof of this two-point assertion for every sufficiently large P would imply infinitely many prime pairs of gap w; in particular $w = 2$ would imply the twin-prime conjecture.

Proof. For each $p \le P$, the condition $p \nmid x - w$ is $x \not\equiv w \pmod p$, and the condition $p \nmid w - x$ is the same congruence. Both formulations therefore exclude exactly the two residue classes 0 and w modulo each p, with possible collision when $p \mid w$.

If $1 < x \le P^2$ is nonzero modulo every prime $p \le P$ and is composite, then it has a prime factor $q \le \sqrt{x} \le P$, contradicting the nonzero-residue condition. Thus such an x is prime. If the two-point assertion holds for all sufficiently large $P > w$, then x and $x - w$ are both prime in infinitely many growing back-half matrix intervals, giving infinitely many prime pairs of gap w. □

Remark 3.3 (Status)**.** This two-point problem is therefore not a routine corollary of the one-point row/column contradiction field. It is a prime-pair-level strengthening, naturally expressible as a two-point rough Jacobsthal problem. In this monograph it is recorded as a future contradiction-field direction, not as an unconditional theorem.

3.2 Two-Point Secondary-Sieve Contradiction Field

The two-point rough-pair problem admits a more detailed contradiction-field research program. The current working manuscript is recorded in

`docs/monograph/two-point-secondary-sieve-research.md.`

It studies the secondary sieve condition

$$x \not\equiv 0, w \pmod p \qquad (p \le P)$$

inside back-half matrix windows, and tries to exclude a two-point rough-pair counterexample by a hierarchy of local rigidity fields.

Definition 3.4 (Two-point secondary-sieve field)**.** Fix an even integer $w > 0$. A two-point secondary-sieve field is the structured ledger obtained from a hypothetical interval in which every candidate x fails at least one of the two congruence conditions $p \nmid x$ or $p \nmid x - w$ for some prime $p \le P$. Its named channels are small-prime double locks, top-scale large-factor coverage, short-block zero-mass loss, multiplicative near-crossings, Fourier CRT defects, and smoothing boundary terms.

Lemma 3.5 (Fixed-order local crossing complexity)**.** *In the secondary-sieve manuscript, every fixed-order local crossing test is expanded only through truncated hard-skeleton weights*

$$d \le R, \qquad R \le (\log P)^{C_R},$$

with coefficient mass $\sum_{d \le R} |\lambda_d| \le (\log P)^{C_\lambda}$ *and short-block shifts* $|h| < L \asymp (\log\log P)^2$*. For every fixed order k, the effective historical complexity satisfies*

$$Q_{\mathrm{eff},k} \ll_k \left(\sum_{d \le R} |\lambda_d|\right)^k R^k L^{O(k)} \le (\log P)^{C_k}.$$

In particular the required $k = 2, 3$ local crossing tests have polylogarithmic effective complexity.

Proof. After expanding the k truncated weights, only moduli $d_1, \ldots, d_k \le R$ occur. For fixed newline primes p_i, phases, and block shifts, all conditions are one-variable linear congruences. By the Chinese remainder theorem the local system is either inconsistent or occupies one residue class modulo a divisor of

$$\operatorname{lcm}(d_1, \ldots, d_k) \operatorname{lcm}(p_1, \ldots, p_k).$$

The primes p_i are the explicit variables being summed in the crossing estimate; they contribute the usual Y^k scale, not hidden historical complexity. The historical part is bounded by $\mathrm{lcm}(d_1, \dots, d_k) \leq R^k$, the block shifts contribute $L^{O(k)}$ patterns, and the coefficient expansion contributes $(\sum |\lambda_d|)^k$. This proves the displayed bound. The true residual set U_Y is not expanded into a full primorial period. □

Lemma 3.6 (Zero-mass and smoothing discharge)**.** *Assume the SC2/diagonal package supplies the moment bounds*

$$M_1 := \frac{\sum_B X_B R_B}{\sum_B X_B} \leq 0.45, \qquad M_2 := \frac{\sum_B X_B \binom{R_B}{2}}{\sum_B X_B} \geq 0.05,$$

and

$$M_3 := \frac{\sum_B X_B \binom{R_B}{3}_+}{\sum_B X_B} \leq 0.03.$$

Then the weighted zero-hit mass satisfies

$$\frac{\sum_B X_B 1_{R_B=0}}{\sum_B X_B} \geq 0.57.$$

Moreover, if the directional-balance endpoint estimate gives

$$N_{\text{endpoint}}/N_{\text{total}} \leq C_{\text{end}}\Delta + o(1),$$

then choosing $\Delta \leq 0.01/C_{\text{end}}$ gives a smooth-to-sharp loss < 0.03 for all sufficiently large P.

Proof. For every integer $R \geq 0$,

$$1_{R=0} \geq 1 - R + \binom{R}{2} - \binom{R}{3}_+.$$

Multiplying by $X_B \geq 0$ and summing over blocks gives

$$\frac{\sum_B X_B 1_{R_B=0}}{\sum_B X_B} \geq 1 - M_1 + M_2 - M_3 \geq 1 - 0.45 + 0.05 - 0.03 = 0.57.$$

For smoothing, choose even smooth cutoffs $W_- \leq 1_{|t|<1} \leq W_+$ whose difference is supported in $1 - \Delta \leq |t| \leq 1 + \Delta$. The sharp and smoothed counts differ only on this endpoint layer. The displayed directional-balance bound therefore gives loss at most $C_{\text{end}}\Delta + o(1)$, which is < 0.03 after the stated choice of Δ and then taking P large. □

Lemma 3.7 (Adaptive true-hit layering dichotomy)**.** *Let U_Y be the current true residual set and define, for each new prime $p > Y$,*

$$a_p = \frac{1}{|U_Y|}\#\{x \in U_Y : x \equiv 0 \textit{ or } w \pmod{p}\}.$$

Fix $\eta = 0.05$. Either some p has $a_p > \eta$, in which case the configuration is a Single-Prime CRTDefect, or the remaining primes can be greedily partitioned into consecutive layers $\mathcal{P}_j$ satisfying

$$\nu_j^{\text{real}} := \sum_{p \in \mathcal{P}_j} a_p \leq 0.4,$$

and, except for the final layer, $\nu_j^{\text{real}} > 0.4 - \eta \geq 0.35$.

Proof. If $a_p > \eta$ for some p, then a fixed positive proportion of U_Y lies in the two residue classes $0, w$ modulo p. This is exactly the Single-Prime CRTDefect exit, to be excluded by the CRTDefect/Directional Balance part of the SC2 package. Otherwise all $a_p \leq \eta$. Starting from the smallest remaining prime, add primes to the current layer until adding the next one would make the partial sum exceed 0.4. The completed layer has sum at most 0.4, and because the next summand is at most η, every non-final completed layer has sum greater than $0.4 - \eta$. This gives the claimed adaptive true-hit decomposition. □

Proposition 3.8 (Recorded conditional closure chain for the two-point secondary sieve)**.** *Using the three lemmas above, assume the following remaining input holds with the constants recorded in the secondary-sieve research note:*

1. *the I3-Core SC2/CRTDefect package controls the true-residual two-line near-crossing error, the odd-prime* 45° *main-synchronization peak, Single-Prime CRTDefect exits, the normalized* $q = 2$ *parity skeleton, the moment bounds* M_1, M_2, M_3*, and the endpoint Directional Balance used in the zero-mass and smoothing discharge lemma.*

Then the two-point secondary-sieve contradiction field has no remaining top-scale escape channel under the parameter choice

$$\alpha_0 = 0.75, \qquad c_0 = 1/2, \qquad \nu_j^{\mathrm{real}} \leq 0.4$$

for adaptive true-hit layers, subject to the same fixed-w *convention used in the research manuscript.*

Proof sketch and status. The argument is a conditional synthesis, not a final unconditional theorem. A top-scale covering excess is first decomposed into adaptive true-hit layers; any single-prime concentration is routed to the Single-Prime CRTDefect exit. The remaining excess is converted by a TCA linear bridge into a positive short-block load. The block load is forced into the SMC/SC2 near-crossing ledger. The 45° main synchronization is split into large-factor reuse, shared-complement factors, and good synchronized pairs; reuse and shared factors are excluded by short-window rigidity, while good pairs are sent to an auxiliary-prime Fourier CRT-defect. Directional balance excludes the Fourier defect unless the top-scale energy is already below the capacity threshold. Finally, the weighted Bonferroni zero-mass estimate supplies enough zero-hit block mass to make the capacity comparison incompatible.

The current audit is archived under `docs/archive/monograph-reviews/` and classifies this as a research proposition conditional on the single I3-Core package above. The fixed-order complexity input, zero-mass/smoothing discharge, and adaptive layering dichotomy have been inlined as the preceding lemmas. The fact that an unconditional proof would imply fixed-gap prime-pair consequences, including the twin-prime case when $w = 2$, is not a logical obstruction. It only marks the strength of the claim. The reason this chapter remains conditional is that I3-Core has not yet been inlined as an independently verified unconditional proof. □

Proposition 3.9 (Window-compressed conditional variant)**.** *Keep the same fixed even integer* $w > 0$ *and the same remaining I3-Core input as in the preceding proposition. Let* I *be any consecutive back-half window of* $H(P)$ *rows in the* $P \times P$ *matrix, so that* $|I| = H(P)P + O(P)$*. By the fixed-order complexity lemma, write*

$$Q_{\mathrm{eff}} \leq C_Q (\log P)^C.$$

For a prescribed middle-scale error budget $\varepsilon_{\mathrm{SC2}} > 0$*, the conditional row threshold supplied by the current ledger is*

$$H_{\min}^{\mathrm{cond}}(P; \varepsilon_{\mathrm{SC2}}) = \left\lceil \frac{C_Q}{\varepsilon_{\mathrm{SC2}}} P^{1/2} (\log P)^C \right\rceil.$$

Equivalently, one may use the asymptotic form

$$H(P) \geq H^{\mathrm{asym}}_{\min}(P) := \left\lceil P^{1/2}(\log P)^{C_*} \right\rceil, \qquad C_* > C.$$

Under either threshold, the same conditional contradiction-field chain gives an element

$$x \in I, \qquad p \nmid x(x-w) \quad (p \leq P).$$

When $x - w > 1$, both x and $x - w$ are primes.

Recorded scale check. The only scale-sensitive term in the recorded chain is the middle/high-scale $K = 2$ near-crossing error. With top layer $Y = P^{3/4}$ and $Q_{\mathrm{eff}} \leq (\log P)^C$, replacing the original back-half length $\asymp P^2$ by $|I| \asymp H(P)P$ changes the normalized error to

$$\frac{Q_{\mathrm{eff}} Y^2}{|I|} \ll \frac{P^{1/2}(\log P)^C}{H(P)}.$$

The displayed lower bound on $H(P)$ makes this at most $\varepsilon_{\mathrm{SC2}}$ in the explicit form, or $o(1)$ in the asymptotic form. Hence the constants in the TCA, SC2, zero-mass, finite-package, and smoothing ledgers are unchanged. The conclusion is therefore a conditional window-compressed version of the previous proposition. It becomes an unconditional prime-pair theorem only if I3-Core is independently proved and accepted. □

Remark 3.10 (Referee boundary)**.** The two-point secondary-sieve chapter is included to document the contradiction-field architecture and its parameter ledger. The window-compression audit is recorded in `docs/monograph/two-point-window-compression-and-unconditionality.md`. It must not be cited as an unconditional proof of prime pairs, the twin-prime conjecture, or Goldbach-type assertions unless I3-Core is independently proved and accepted.

Remark 3.11 (Current obstruction)**.** The current hard attack reduces I3-Core to a true-residual correlation problem. The local CRT and matrix rigidities do not by themselves exclude a model residual set concentrated on a new prime's bad residue class while still satisfying all previously imposed two-residue exclusions and short-window non-reuse constraints. Thus the remaining task is not another bookkeeping step but a genuine lower-bound/correlation theorem for the true residual set in new prime moduli.

Remark 3.12 (Latest TRC-1G reduction)**.** The single-prime part of the remaining TRC input has been sharpened to a heavy-atom prohibition. If more than 40% of the true residual mass lies in the two bad classes $0, w \pmod{p}$ for some new prime $p > Y$, then Parseval gives a non-zero Fourier coefficient modulo p of size $> 1/5$ after normalization. Equivalently, the three long-direction difference families $0, \pm w \pmod{p}$ carry a fixed positive proportion of the true-residual pair energy. The existing matrix rigidities remove the short-direction, 45°-locked, reuse/shared-complement, and endpoint pieces. The remaining unproved input is therefore a new-modulus long-direction balance theorem for the true residual set, not a finite-template or local CRT bookkeeping issue.

Remark 3.13 (Total large-incidence reduction)**.** The pointwise new-modulus balance can be weakened further. It is enough to prove the total large-incidence inequality

$$\sum_{P^\alpha < p \leq P} \#\{x \in U_{P^\alpha}(I) : p \mid x(x-w)\} < |U_{P^\alpha}(I)|.$$

Indeed, a point not counted on the left is unsieved by every prime up to P, and in the P^2 matrix range this forces both x and $x - w$ to be prime. The CRT model predicts the normalized left side to be $2\log(1/\alpha) + o(1)$; for $\alpha = 3/4$ this is $2\log(4/3) = 0.57536\ldots$, leaving margin $0.42463\ldots$. The unproved part is the true-residual total-incidence estimate with an error below that margin.

Remark 3.14 (RB–TLI audit boundary)**.** The direct attempt to prove the total large-incidence estimate separates cleanly into a numerator and a denominator. The numerator is an upper-sieve problem: after writing $x = pm$ or $x = w + pm$, the variable m again avoids two residue classes modulo each old prime, giving an upper bound of shape

$$A_p \leq 2C_+|I|V_2(P^\alpha)/p + R_p.$$

The denominator requires a model-level lower bound $|U_{P^\alpha}(I)| \geq c_-|I|V_2(P^\alpha)$. For $\alpha = 3/4$, this route would need $C_+/c_- < 1/(2\log(4/3)) = 1.738\ldots$. This relative Buchstab total-incidence input is the remaining unproved prime-pair-strength statement.

Remark 3.15 (Local collapse when $q \mid w$)**.** If an old odd prime q divides the fixed even difference w, then the two forbidden classes $0, w \pmod q$ coalesce into a single forbidden class. The local density is therefore $1-1/q$ instead of $1-2/q$, giving the singular local gain $(q-1)/(q-2)$. This increases the absolute number of admissible residual points and weakens the small-prime synchronization peaks. In the relative RB–TLI ratio, however, the same local factor multiplies both $|U_Y|$ and the model numerator $A_p \sim 2|I|V_w(Y)/p$ for all new primes $p > Y > |w|$. Thus the main relative constant remains $2\log(1/\alpha)$; the collapse improves finite constants but does not by itself close the hardest case $w = 2$.

Remark 3.16 (Buchstab semiprime transition for $w = 2$)**.** The latest numerical audit of the hardest case $w = 2$ shows that the naive constant $2\log(1/\alpha)$ is not the correct conditional main term after the old primes have been sieved. For $\alpha > 2/3$, any new-prime hit $p \mid x$ or $p \mid x - 2$ with $Y = P^\alpha < p \leq P$ forces the complementary quotient to be prime, because it is Y-rough and smaller than Y^2. Thus the large-incidence layer is a Buchstab semiprime transition. The corresponding two-sided main constant is

$$K(\alpha) = \frac{2\log((2-\alpha)/\alpha)}{1 + \log((2-\alpha)/\alpha)}.$$

Exact scans recorded in `docs/rb-tli-w2-scan-large.md` give, for example, $E_U D = 0.691696$ at $P = 10007, \alpha = 3/4$, compared with $K(3/4) = 0.676220$. The remaining target is therefore a two-point stability theorem for this Buchstab semiprime transition, not the earlier naive incidence model.

Remark 3.17 (BST error decomposition)**.** The refined scan decomposes the remaining error. At $P = 10007, \alpha = 3/4$, the shell contributions in the five exponent intervals between 0.75 and 1 track the Buchstab shell constants, and the covariance between the x-side and $(x-2)$-side large-incidence counts is approximately $-9.3 \cdot 10^{-5}$. The zero-incidence set is independent of the intermediate cutoff α, as it is exactly the set unsieved by all primes up to P. Thus the remaining hard input has been reduced to a shifted biprime roughness estimate of the form

$$\#\{p, m : pm \in I,\ pm - 2 \in U_Y\} \leq (B(\alpha) + o(1))|U_Y|.$$

This is the current BST–2 obstruction.

Remark 3.18 (BMD reduction)**.** The BST–2 obstruction can be written as a one-dimensional multiplicative sieve on biprime variables. After setting $x = pm$, the condition $pm - 2$ is unsieved by old primes is, for each $q \leq Y$,

$$pm \not\equiv 2 \pmod q.$$

The prime $q = 2$ is peeled off deterministically: for odd primes p, m, the integer $pm-2$ is odd. Thus the multiplicative sieve is taken over odd q. For each odd $q \leq Y$, since $p, m > Y \geq q$, the forbidden curve has $(q-1)$ points inside $(\mathbb{F}_q^\times)^2$, out of $(q-1)^2$ possible non-zero pairs. The local forbidden density is therefore $1/(q-1)$. The remaining unproved estimate is a biprime multiplicative dispersion bound: weighted sums of the errors in the congruences $pm \equiv 2 \pmod d$, with odd squarefree d, must be $o(|U_Y|)$ for the Buchstab/Rosser weights used in the transition. This is the current BMD input.

Remark 3.19 (Row-column input for BMD)**.** If the prime-matrix row/column theorem is accepted as a conditional input, it can be used only to remove zero-section degeneracies in complete CRT blocks: entire empty rows, columns, or unit residue cells are then not allowed to masquerade as the main BMD error. It does not by itself prove BMD, because BMD is a signed distribution statement. For odd d, expanding

$$1_{pm\equiv 2 \pmod d} = \frac{1}{\varphi(d)} + \frac{1}{\varphi(d)} \sum_{\chi\neq\chi_0} \overline{\chi}(2)\chi(p)\chi(m)$$

shows that the remaining obstruction is a bilinear character-dispersion estimate

$$\sum_{d\le D} \frac{|\lambda_d|}{\varphi(d)} \sum_{\chi\neq\chi_0} \left| \sum_{Y<p\le P} \chi(p) \sum_{\substack{m\in J_p\\ m \text{ prime}}} \chi(m) \right| = o(|U_Y|).$$

Thus the row/column input gives a useful BMD-Zero exclusion, but the genuine remaining hard point is BMD-Char.

Remark 3.20 (Direct BMD attack via an E_2 Bombieri–Vinogradov input)**.** Set $N \asymp P^2$ and

$$a_n = \#\{(p,m) : Y < p \le P,\ m \text{ prime},\ n = pm,\ n \in I\}.$$

For $\alpha > 2/3$ the multiplicity of a_n is bounded. After the deterministic modulus 2 has been peeled off, the BMD remainder for odd squarefree d is exactly the arithmetic-progression discrepancy of this restricted two-prime convolution:

$$R_d = \sum_{\substack{n\in I\\ n\equiv 2 \pmod d}} a_n - \frac{1}{\varphi(d)} \sum_{\substack{n\in I\\ (n,d)=1}} a_n.$$

Thus BMD follows from the standard Bombieri–Vinogradov theorem for restricted E_2 sequences at level

$$d \le P(\log P)^{-B} = N^{1/2}(\log N)^{-B+O(1)}.$$

Indeed, if

$$\sum_{d\le P/\log^B P} \max_{(a,d)=1} |R_d(a)| \ll_A N(\log N)^{-A},$$

then every Rosser/Buchstab weighted remainder is $o(|U_Y|)$ since $|U_Y| \asymp N/(\log P)^2$. The condition $\alpha < 1$ ensures $Y < P/\log^B P$ for large P, so all one-prime forbidden curves $q \le Y$ are included in the available distribution range. Therefore BMD is closed if this standard E_2 BV input is cited; a fully self-contained manuscript must instead prove that input in an appendix.

Remark 3.21 (Self-contained BV–E_2 obligation)**.** The local BV–E_2 appendix records the exact self-contained obligation. A direct multiplicative large-sieve estimate is insufficient: in the balanced dyadic block $p \sim P_1$, $m \sim M_1$, with $P_1 \asymp M_1 \asymp P$ and $Q \asymp P/\log^B P$, Cauchy plus the large sieve gives size roughly $P^3/\log^{2B} P$, while the required Bombieri–Vinogradov error is $P^2/\log^A P$. Thus the only genuinely deep remaining self-contained input is a Type-II dispersion/Kloosterman-cancellation estimate for the balanced convolution blocks. If that standard dispersion theorem is cited, the BMD step is closed as an external-input theorem; if not, this is the next appendix that must be proved line by line.

Remark 3.22 (Weighted BE2–3K core)**.** BMD does not require the full $\max_a$ Bombieri–Vinogradov statement. It only needs the fixed class $2 \bmod d$ against the well-factorable Rosser/Buchstab weights. After Cauchy, the self-contained problem is a variance bound for

$$T_r = \sum_{d \le Q} \lambda_d \left(\sum_{s \equiv 2r^{-1} \bmod d} \beta_s - \frac{1}{\varphi(d)} \sum_{(s,d)=1} \beta_s \right).$$

Opening $\sum_r |T_r|^2$, the off-diagonal conditions

$$rs_1 \equiv 2 \pmod{d_1}, \qquad rs_2 \equiv 2 \pmod{d_2}$$

give, by CRT and Fourier expansion of the interval condition, the Kloosterman phase

$$e\biggl(-\frac{2h\overline{s_1}\overline{d_2}}{d_1} - \frac{2h\overline{s_2}\overline{d_1}}{d_2}\biggr).$$

Thus the true no-black-box core is a weighted bilinear Kloosterman dispersion estimate, denoted BE2–3K in the notes. Proving BE2–3K gives BE2–3, then weighted BV–E_2, then BMD.

Remark 3.23 (Referee status of the no-black-box claim)**.** At the current stage the manuscript is not yet fully no-black-box. The BMD obstruction has been reduced to the single deep BE2–3K Kloosterman-dispersion estimate. If a BFI/Deshouillers–Iwaniec/Kuznetsov type mean-value theorem is cited, BMD is closed as an external-input theorem. If no external input is allowed, BE2–3K remains the unique missing line-by-line proof obligation.

Remark 3.24 (Further reduction of BE2–3K)**.** The BE2–3K core can be narrowed further. Pointwise Weil bounds for individual incomplete reciprocal sums give only single-modulus square-root cancellation and do not provide the arbitrary logarithmic saving required after summing over the d_1, d_2, h families. The common-divisor strata $(d_1, d_2) > 1$ are not the main obstruction: compatibility forces $s_1 \equiv s_2$ modulo the common divisor and yields only polylogarithmic loss. The remaining genuine input is a windowed Kloosterman spectral large-sieve estimate, denoted KLS-window in the notes. Its role is precisely

$$\text{KLS-window} \Rightarrow \text{BE2–3K} \Rightarrow \text{BE2–3} \Rightarrow \text{weighted BV–}E_2 \Rightarrow \text{BMD}.$$

Remark 3.25 (External-theorem closure of KLS-window)**.** The KLS-window input is tied to two classical external sources: the Deshouillers–Iwaniec spectral Kloosterman large-sieve technology, as developed in *Kloosterman sums and Fourier coefficients of cusp forms*, Invent. Math. 70(2), 219–288 (1982), and the Bombieri–Friedlander–Iwaniec dispersion framework with well-factorable weights, as in *Primes in Arithmetic Progressions to Large Moduli. II*, Math. Ann. 277, 361–394 (1987). With these external theorems allowed, the analytic chain closes as

$$\text{DI+BFI} \Rightarrow \text{KLS-window} \Rightarrow \text{BE2–3K} \Rightarrow \text{BE2–3} \Rightarrow \text{weighted BV–}E_2 \Rightarrow \text{BMD}.$$

This is an external-deep-theorem closure, not a fully self-contained reproof of Kuznetsov/dispersion theory.

Chapter 4

Zeta-Zero Contradiction Fields

Assume an off-critical zero $\rho = \beta + i\gamma$ with $\beta > 1/2$. The RH contradiction-field manuscript transfers the resulting smooth prime anomaly into a CRT rough-composite ledger, decomposes it into sparse/dense/tail/internal/global exits, and proves that no named escape channel remains under the recorded local theorems and restricted external inputs.

Theorem 4.1 (RH contradiction-field synthesis, recorded form). *Under the numbered local theorems, controlled-exit lemmas, threshold hierarchy, and restricted external inputs recorded in `paper/rh-proof/rh-contradiction-field.tex`, an off-critical zero has no free escape channel in the contradiction-field event graph.*

Reference. See the RH manuscript in `paper/rh-proof/` and the final audit in `docs/`. The submission warning in the RH manuscript remains in force until independent line-by-line referee verification and final monograph page-number checks are completed. □

Chapter 5

Unified Dependency Map

The two programs share the same logic:

$$\text{counterexample} \Rightarrow \text{structured covering field} \Rightarrow \text{named exits} \Rightarrow \text{capacity/no-cycle contradiction.}$$

The prime-matrix program is more geometric and finite-CRT in nature; the zeta-zero program is analytic and scale-asymptotic. The monograph treats both as projections of the same contradiction-field method.

Layer	Prime-matrix projection	RH projection
Baseline	CRT nonzero class balance	smoothed CRT candidate baseline
Local rigidity	large-factor non-reuse, diagonal locks	sparse/dense/tail routing
Capacity	anchor coverage capacity	projection and square-function capacity
No-cycle	recursive peeling and shell migration	GEE transfer/no-cycle ledger
External inputs	finite verification and Structured-EHPD	explicit formula, KL, Vaaler, BG/Baker, sieve, Vaughan
Final status	reduction package, not final unconditional theorem	verification package with submission warning

Chapter 6

Directory and Theory System

The detailed working directory and theory-system map is maintained in

`docs/monograph/combined-monograph-directory-and-theory-system.md`.

This chapter records the concise version used by the merged monograph.

6.1 Proposed Book Structure

Part	Contents	Status
I	Unified contradiction-field method: entrance, local rigidity, exits, capacity, no-cycle potential	Methodological framework
II	Prime-matrix row/column program: CRT skeleton, A/B reduction, Structured-EHPD interface	Reduction package; not final theorem
III	Two-point secondary sieve: TLI, BST, BMD, WBE2, BE2–3K, KLS-window	External-deep-theorem closure for BMD
IV	Zeta-zero contradiction field: explicit-formula entrance and controlled exits	Verification package; not final RH theorem
V	Dependency map, referee checklist, optimization routes	Audit and future work

6.2 Latest Two-Point Chain

For the $w = 2$ hardest two-point branch, the current analytic chain is

$$\text{DI+BFI} \Rightarrow \text{KLS-window} \Rightarrow \text{BE2–3K} \Rightarrow \text{BE2–3} \Rightarrow \text{WBE2} \Rightarrow \text{BMD} \Rightarrow \text{BST–2} \Rightarrow \text{BST} \Rightarrow \text{TLI}.$$

The chain is closed only in the external-deep-theorem sense: DI and BFI are cited as classical analytic inputs. A fully self-contained version would require reproving the spectral Kloosterman large-sieve and dispersion machinery.

6.3 Prime-Matrix Low-Hole Closure Package

The latest prime-matrix update closes the low-hole bucket submodule as a finite-certificate plus explicit-external-theorem package. The verification is split into five non-overlapping ranges:

$$\begin{aligned} 13 \le P < 61 &\quad:\quad \text{low-range final certificate,}\\ 61 \le P \le 103 &\quad:\quad \text{narrow-band collision-energy certificate,}\\ 107 \le P \le 229 &\quad:\quad \text{exact } P/5 \text{ splitting recursion,}\\ 233 \le P \le 13207 &\quad:\quad \text{integer cross-multiplied product certificate,}\\ P \ge 13208 &\quad:\quad \text{Rosser–Schoenfeld explicit constants.} \end{aligned}$$

The low-range certificate checks 15414 zero buckets: 15282 close by whole-set Hall deficit, 108 close by bridge-critical deletion, and the remaining 24 exceptional phases at $P = 41$ close by exact finite set-cover dynamic programming. The narrow band $61 \le P \le 103$ covers 23100 low phases by fixed ascending high-prime residue blocks. The two finite tail certificates cover 23 and 1520 prime rows respectively, the latter by exact integer cross multiplication rather than floating-point comparison.

For the infinite tail, Rosser–Schoenfeld Corollary 1, formulas (3.5)–(3.6), supplies the prime-counting bounds, while Theorem 7, formula (3.26), supplies the Mertens product bound. Since $P \ge 13208$ implies $\lfloor P/5 \rfloor \ge 2641$, the factor $1 + 1/(2\log^2\lfloor P/5 \rfloor)$ is below 1.009, hence the manuscript's conservative 1.03 Mertens constant is admissible.

This closes the BPN low-hole bucket interface in the stated finite-certificate/external-theorem sense. It does not by itself promote the whole prime-matrix row/column program to a final theorem, because the remaining global exits still require the separately named PDEC/SAE and Rankin-ledger acceptance obligations.

6.4 Prime-Matrix WSH-Hall Phase Certificate

The latest row-program compression keeps the endpoint status honest. The terminal RCI branch has been reduced to the comparison between no-tail primes and balanced two-tail semiprimes. The next local interface is

$$\text{RCI/PDEC} \Rightarrow \text{Distributed-RCI} \Rightarrow \text{WSH-Hall/PDEC}.$$

Here WSH-Hall means that every balanced two-tail semiprime in a row can be injectively matched to a same-row prime through an offset allowed by the small-prime wheel. If such a Hall matching fails, the failing interval must be routed to a named exit: persistent endpoint CRT defect, tail-anchor concentration, or endpoint prime deficit.

A finite phase certificate is archived in

`docs/monograph/prime-matrix-wsh-hall-phase-certificate.md`.

For $17 \le p \le 2000$, $y = \lfloor p/e \rfloor$, and wheel primes $2, 3, 5, 7, 11, 13$, it scans 215074 rows containing balanced two-tail semiprimes. It finds no Hall matching failure, no failure under the candidate radius $3\log^2 q$, maximum minimal matching radius 132, maximum ratio $R/\log^2 q = 2.3771082468335174$, and minimum margin #primes − #semiprimes = 1. The certificate rows list semiprime labels, matched primes, offsets, fixed-offset loads, and wheel-phase loads; each matched offset is checked against the congruence condition $d \not\equiv -b \pmod r$.

This is a computational-certificate update, not a global theorem. The remaining proof obligation is either a uniform WSH-Hall theorem or a proof that every Hall defect necessarily triggers PDEC, Tail-anchor, or Endpoint deficit.

6.5 WSH-Hall Defect Trichotomy

The corresponding defect audit is archived in

`docs/monograph/prime-matrix-wsh-hall-defect-trichotomy.md`.

It converts a possible Hall failure into a consecutive block B_0 of balanced two-tail semiprimes with

$$|N_R(B_0)| < |B_0|.$$

The same block is then examined through five projections: endpoint prime deficit, tail-factor load, small-wheel residue load, fixed-offset capacity, and the terminal mirror $n \mapsto q^2 - n$. The resulting target is the named interface

$$\text{WSH-Expansion-or-Defect.}$$

If Tail-anchor and fixed-offset/PDEC thresholds do not fire, one must prove the expansion inequality

$$|N_R(B_0)| \geq |B_0|$$

for $R = C \log^2 q$; otherwise the failing block must be absorbed by Endpoint/PDEC. A forced-radius audit on the 44 tight certificate rows produces 124 explicit Hall defects and records their tail, phase, offset, and mirror data. This audit fixes the certificate fields and exit geometry; it is not a substitute for the uniform expansion theorem.

6.6 WSH Expansion Margin Ledger

The expansion term itself is audited in

`docs/monograph/prime-matrix-wsh-expansion-margin-ledger.md`.

For $17 \leq p \leq 2000$, $R = \lceil 3 \log^2 q \rceil$, and the same $y = \lfloor p/e \rfloor$, the audit scans every consecutive balanced-semiprime block in every relevant row. It checks 8440419 blocks across 215074 rows and finds

$$\min_B \big(|N_R(B)| - |B|\big) = 1,$$

with no zero-surplus block. Only seven blocks have surplus at most 1.

This strengthens the finite certificate from "a matching exists" to "every interval-Hall block has positive margin" in the audited range. It still does not prove the uniform theorem. The remaining target is:

$$\text{WSH Positive Expansion or Named Defect.}$$

After Tail-anchor, fixed-offset/PDEC, and endpoint-mirror deficit exits are removed, one must prove the positive expansion inequality for all formal blocks.

6.7 Short Critical Block Reduction

The short-critical-block audit is archived in

`docs/monograph/prime-matrix-wsh-short-critical-block-reduction.md`.

It reruns the expansion-margin audit with tight surplus threshold 2. Among the same 8440419 consecutive semiprime blocks, only 45 have surplus at most 2, and every such block has at most three semiprimes. The distribution is:

surplus	$\lvert B\rvert$	count
1	1	4
1	2	3
2	1	24
2	2	10
2	3	4

This suggests the next reduction:

$$\text{SCB–1: } |B| \geq 4 \Rightarrow \text{positive expansion or named defect,}$$

$$\text{SCB–2: } |B| \leq 3 \Rightarrow \text{local exclusion or named defect.}$$

This is still finite evidence and a proof target, not a theorem. It is useful because SCB–2 is a small local problem involving only one, two, or three two-tail semiprimes, where tail labels, fixed offsets, wheel residues, and endpoint mirrors can be written explicitly.

6.8 SCB–2 Local Routing Certificate

The SCB–2 local certificate is archived in

`docs/monograph/prime-matrix-wsh-scb2-local-exclusion-template.md`.

It enumerates all consecutive blocks with $|B| \leq 3$ for $17 \leq p \leq 2000$. The block counts are

$\lvert B\rvert$	count	$\min(\lvert N_R(B)\rvert - \lvert B\rvert)$
1	1496400	1
2	1281326	1
3	1088870	2

There are no negative-surplus or zero-surplus short blocks. The local routing statement is:

$$|B| \leq 3, \quad |N_R(B)| < |B| \Rightarrow \text{Endpoint/PDEC deficit,}$$

with additional Tail-repeat and fixed-offset-full-load checks on tight two- and three-point blocks. Thus SCB–2 is closed as a routing lemma modulo the still-open Endpoint/PDEC exclusion.

6.9 SCB–1 Long-Block Certificate

The complementary long-block certificate is archived in

`docs/monograph/prime-matrix-wsh-scb1-long-block-expansion-template.md`.

It enumerates all consecutive balanced two-tail semiprime blocks with $|B| \geq 4$ for $17 \leq p \leq 2000$ and the same radius $R = \lceil 3 \log^2 q \rceil$. The audit checks 4573823 long blocks and finds

$$\min_{|B|\geq 4} \big(|N_R(B)| - |B|\big) = 3,$$

with no negative-surplus or zero-surplus long block. The four tight long blocks have sizes 4 or 5, and every tight block carries both Endpoint-margin and fixed-offset-full-load labels.

This is a finite certificate, not the uniform SCB–1 theorem. Its proof-theoretic role is to isolate the next target:

$$\text{long-block positive expansion} \quad \text{or} \quad \text{fixed-offset/PDEC absorption.}$$

Together with SCB–2, the present WSH chain is therefore:

$$\text{SCB–2 routing modulo Endpoint/PDEC,} \qquad \text{SCB–1 finite long-block margin,}$$

The remaining global obstructions are fixed-offset/PDEC absorption and Endpoint/PDEC exclusion.

6.10 Fixed-Offset/PDEC Absorption

The fixed-offset absorption interface is archived in

`docs/monograph/prime-matrix-wsh-fixed-offset-pdec-absorption.md`.

It proves the no-third-escape lemma behind the fixed-offset-full-load label. If $p < q$ are consecutive primes and $1 < n < q^2$ is composite, then n has a prime factor at most p. Hence, if a fixed offset d is allowed by the wheel $2, 3, 5, 7, 11, 13$ for every $b \in B$, every missing candidate $b + d$ is explained by some factor in $(13, p]$.

The finite ledger

`docs/monograph/prime-matrix-wsh-fixed-offset-pdec-ledger.md`

checks the tight SCB–1 long blocks. It finds 11 full-load offset rows and 47 candidates, among which 15 are prime and 32 are missing. Every missing candidate has an explaining factor at most p; there is no unexplained candidate.

Thus fixed-offset-full-load is no longer an unnamed escape. It routes to expansion, Tail-anchor, PDEC/low-mod CRTDefect, or SAE/Endpoint. The still-open global target is

$$\text{FO-PDEC: distributed explaining factors} \Rightarrow \text{persistent PDEC or sparse SAE/Endpoint.}$$

6.11 FO-PDEC Low-Mod Equations

The low-mod equation audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-hard-attack.md`.

For every missing candidate $n = b + d = (r - 1)q + c$ and every explaining factor ℓ, the exact CRT equation is

$$r \equiv 1 - cq^{-1} \pmod{\ell}.$$

If $b = uv$ is the corresponding two-tail semiprime, the same missing candidate also satisfies

$$uv + d \equiv 0 \pmod{\ell}.$$

The finite audit checks 43 such low-mod equations in the tight fixed-offset ledger. There are no CRT-equation failures and no bilinear-equation failures.

This closes the equation layer of FO-PDEC. It does not close the global theorem. The remaining target is the quantitative inequality

$$\text{low-mod defect energy} \geq \text{PDEC threshold},$$

unless the non-persistent part is absorbed by SAE/Endpoint.

6.12 FO-PDEC Energy Lower Bound

The energy ledger is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-energy-lemma.md`.

For a multiset E of explaining equations, let t_ℓ be the number of equations using ℓ and let s_ℓ be the number of occupied row residues modulo ℓ. Define

$$\mathcal{E}_\ell(E) = t_\ell - \frac{s_\ell t_\ell}{\ell}, \qquad \mathcal{E}(E) = \sum_\ell \mathcal{E}_\ell(E).$$

For every $0 < \theta < 1$, either some explaining factor has high load $t_\ell > \theta\ell$, which is a Tail/PDEC exit, or

$$\mathcal{E}(E) \geq (1-\theta)|E|.$$

Thus FO-PDEC now produces low-mod energy unconditionally, modulo the named high-load exit.

The finite ledger with $\theta = 0.25$ gives global energy per equation

$$0.9547817296210457$$

and no high-load factor. The remaining obstruction is not energy production; it is the final threshold comparison on the same bad-window set:

$$U_{\mathrm{CRT}}(E) < \mathcal{E}(E) \quad \text{or} \quad L_{\mathrm{PDEC}}(E) \leq (1-\theta)|E|.$$

6.13 FO-PDEC Explicit Threshold

The explicit threshold ledger is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-threshold-comparison.md`.

For each explaining factor ℓ, it computes the nonzero Fourier threshold

$$M_\ell(E) = \max_{1 \leq h < \ell} \left| \sum_{\rho \bmod \ell} g_\ell(\rho) e^{2\pi i h\rho/\ell} \right|.$$

The current best projection is

$$\ell = 199, \qquad h = 95, \qquad M_\ell(E) = 3.959247567099438.$$

Thus the remaining task is now concrete:

$$U_{\mathrm{CRT},199} < 3.959247567099438$$

for the same vector g_{199}. Otherwise this projection must be absorbed by SAE/Endpoint.

6.14 FO-PDEC Dual-Cluster Obstruction

The dual-cluster audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-dual-cluster-route.md`.

For the best projection, the mass is 4 and the Fourier threshold is 3.959247567099438, so the trivial mass bound misses closure by only

$$0.04075243290056196.$$

The best frequency $h = 95$ sends the support to a short dual arc of length 11 modulo 199:

$$40, 126, 61 \mapsto 19, 30, 24.$$

Thus the last obstruction is a concrete dual-cluster exclusion problem: this short dual arc must be ruled out by a PDEC dual certificate, or else absorbed as a non-persistent SAE/Endpoint event.

6.15 FO-PDEC Multiplicity Legitimacy

The primitive-cluster audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-multiplicity-legitimacy.md`.

It checks whether the best projection counts independent bad-window equations or repeated physical candidates. Under the equation or block-local convention, the projection has mass 4 and Fourier value

$$3.959247567099438.$$

Under physical deduplication, it has mass 2 and Fourier value

$$1.9699193446802263.$$

Thus the final PDEC comparison must first fix the multiplicity convention. The lower bound and the upper bound must be computed on the same multiset. Otherwise the argument must switch to the primitive threshold or absorb the repeated events by SAE/Endpoint.

6.16 FO-PDEC Formal Unit Consistency

The formal-unit audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-formal-unit-route.md`.

It checks a stricter requirement: the PDEC vector $g(t)$ must come from one formal bad-window set or multiset with one phase map. The current strong value

$$3.959247567099438$$

is obtained by pooling the finite certificate library. After splitting by formal unit, the audit gives

convention	best Fourier value
global library	3.959247567099438
layer deduplication	2.9698366905785227
physical deduplication	1.9997507790353146
single coordinate-deduplicated formal unit	1.0.

Thus the strong threshold cannot yet be used as a single-branch PDEC lower bound. One must prove FormalUnit-Stitching for the cross-q events and NestedBlock-Independence for exact nested-coordinate duplicates, or route the repeated events to SAE/Endpoint. This is now the narrowest FO-PDEC obstruction before any global unconditional conclusion can be claimed.

6.17 FO-PDEC Branch Separation

The branch-separation theorem and stitching audit are archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-branch-separation-theorem.md`.

They separate two issues. First, events from different matrix widths q cannot be pooled into one PDEC vector without a cross-q persistence theorem. Second, two occurrences with the same

$$(q, row, candidate\_row, column, offset, n, \ell, \rho)$$

give the same CRT equation and must be quotiented unless a weighted Hall-dual independence row is supplied. In the present finite ledger there are 7 exact nested-coordinate duplicates and 9 cross-level physical reuses. After coordinate deduplication inside a single formal unit, the best Fourier value recorded by the audit is only 1.0. Therefore the next narrow obstruction is Weighted Hall Dual Independence, or else SAE/Endpoint absorption of the exact duplicates.

The subsequent dominance audits discharge these finite FO–PDEC subgates without promoting the full theorem. The nested duplicates are laminar repeated coordinates; the fractional weighted Hall differences add one semiprime and one neighboring prime, with zero surplus, so they cannot pay for a second independent unit of mass. The cross-q reuses are coordinate-chart overlaps of the same physical candidate, not independent persistent events. Thus the raw library value 3.959247567099438 and the coordinate-cap value 2.9698366905785227 cannot be used as single-formal-unit lower bounds in the audited branch.

6.18 FO-PDEC Physical Primitive Tautology and SAE Absorption

The physical primitive audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-physical-primitive-tautology.md`.

After physical deduplication the current 199-factor frontier has only two physical atoms. For a prime modulus ℓ and two distinct residues a, b, choosing

$$h \equiv (b-a)^{-1} \pmod{\ell}$$

makes the dual residues adjacent. Hence every two-point support satisfies

$$\max_{h \neq 0} |e(ha/\ell) + e(hb/\ell)| = 2\cos(\pi/\ell).$$

For $\ell = 199$ this is exactly

$$2\cos(\pi/199) = 1.9997507790353146.$$

Therefore the remaining physical primitive value is a two-point Fourier tautology, not a usable PDEC defect threshold.

The two atoms are then routed to the SAE/Endpoint interface and audited in

`docs/monograph/prime-matrix-wsh-fo-pdec-sae-endpoint-absorption.md`.

Each associated fixed-offset fiber has a local prime survivor: the 250541 atom has witness 250543, and the 1664237 atom has witness 1664227. In all four source rows the 199-factor load in the same fiber is 1. Consequently the audited two-atom primitive branch is absorbed by LocalSurvivor witnesses.

This closes only the current finite $\ell = 199$ FO–PDEC sample chain:

$$\begin{gathered}\text{raw library signal} \to \text{nested/weighted rejection,}\\ \text{coordinate-cap signal} \to \text{cross-}q \text{ chart-overlap rejection,}\\ \text{physical two-point signal} \to \text{Fourier tautology,}\\ \text{two physical atoms} \to \text{LocalSurvivor witness absorption.}\end{gathered}$$

It does not close the global prime-matrix theorem. The next honest frontier is the global LocalSurvivor certificate family, or a future non-tautological primitive PDEC family with at least three physical atoms or an extra fixed-frequency constraint. The non-tautological PDEC admission audit records that the current materialized frontier has no such candidate: the raw three-point signal is not admissible as one primitive formal unit, nested and weighted duplicate recovery are blocked, cross-q reuse is only chart overlap, and the physical primitive remnant has two atoms. Those two atoms are already absorbed by LocalSurvivor/SAE witnesses. Thus any future PDEC certificate must introduce a genuinely new same-formal-unit primitive family with at least three physical atoms, or an additional constraint defeating the two-point tautology.

6.19 Materialized LocalSurvivor Packet Ledger

The current LocalSurvivor packet ledger is archived in

`docs/monograph/prime-matrix-local-survivor-materialized-packet-ledger.md`.

It combines the Triad-A1 SparseCap ledger, the FO–PDEC two-atom SAE/Endpoint absorption ledger, and the RPZ Endpoint-SAE finite ledger. The combined current ledger records

quantity	value
materialized packets	9
materialized phase atoms	32
local survivor witnesses	5
finite PDEC atoms	25
open materialized obligations	0.

Thus every LocalSurvivor/SAE packet currently materialized by the proof frontier is either witnessed, vacuous in the current ledger, or routed to a finite PDEC packet beyond the early $P \times P$ rows.

The packet-generation contract is archived in

`docs/monograph/prime-matrix-local-survivor-packet-generation-contract.md`.

It records the structural dichotomy for any future sparse escape. If the window data can be extracted, it is a finite LocalSurvivor packet and must be checked by a witness or blocker-deficit certificate. If the same finite signature persists along an infinite counterexample family, it is no longer a sparse terminal: according to the repeated signature it returns to PDEC, ColumnCRT, endpoint PDEC, TailAnchor, or CofactorAnchor. If the finite signatures do not persist and the layers keep escaping, the branch enters CleanKLS/DLS admission.

Consequently the next narrow frontier is

$$\text{LocalSurvivorPacketGenerationOrNonTautologicalPDEC}.$$

This is still not a final unconditional theorem. It says that the current materialized LocalSurvivor obligations are exhausted and that no unnamed LocalSurvivor terminal remains; the remaining work is packet-extractor completeness, a genuinely non-tautological primitive PDEC family, or the CleanKLS/DLS and referee inputs.

The known-entry extractor audit is archived in

`docs/monograph/prime-matrix-local-survivor-packet-extractor-coverage.md`.

It checks eight currently named LocalSurvivor entry types. Four are covered by machine extractors: Triad-A1 SparseCap, FO–PDEC two-atom SAE/Endpoint, RPZ Endpoint-SAE, and the materialized packet aggregator. Four are covered by contract routes: generic SAE, persistent finite signatures, layer escape to CleanKLS/DLS, and descent/seam return. The audit has no missing or open known entry.

The new-entry admission audit is archived in

`docs/monograph/prime-matrix-new-sparse-entry-admission-audit.md`.

It checks the possible sources of a future sparse entrance: short-window or endpoint sparse packets, PDEC dual sparse caps, endpoint seams, Bohr caps, column-tail/cofactor sparse packets, descent seams, persistent signatures, and layer escape. Each is already admitted to a named route: SAE/LocalSurvivor, PDEC, ColumnCRT, TailAnchor/CofactorAnchor, or CleanKLS/DLS. Hence there is no additional unnamed LocalSurvivor entry route in the current contract system.

After these two audits the LocalSurvivor branch is conditionally exhausted: any future explicitly introduced sparse route must bring its own extractor schema and finite ledger. The current narrow frontier is therefore

$$\text{NonTautologicalPDECOrCleanKLS}.$$

This remains a reduction boundary, not the final prime-matrix theorem.

The clean KLS side is no longer an unnamed black-box exit in the current routing table. The A1 clean-admission router checks that K1–K9 failures return to PDEC/SAE/Multiplicity/Promotion, while a fully admitted clean residual enters the Kuznetsov–LS atom SC-9. The SC-9 frontier router expands this atom into KZ-A–KZ-E: KZ-A–KZ-D are routed by smoothing, trace specialization, Bessel decay, and the spectral-large-sieve/pretrace chain; KZ-E remains the actual WFD block non-concentration input NC-BLK, or else a precisely matched external DI/BFI original dispersion theorem. Thus the updated narrow frontier is

$$\text{NonTautologicalPDECOrNCBLK}.$$

This also remains a reduction boundary, not a final unconditional row/column theorem. The NC-BLK boundary reconciliation audit records the exact reading of this label. On the canonical RIW/Buchstab source branch, the NC-BLK clean-KLS chain is already absorbed by the existing same-set capacity boundary. On the broader full-S non-AP generic branch, NC-BLK remains an exact-source-entropy or external DI/BFI/Kuznetsov route and is not imported into the self-contained theorem. Thus the label no longer denotes an unnamed CleanKLS exit, but it also does not promote the whole row/column theorem.

The subsequent global-terminal-family boundary router updates the frontier one more time. After the NC-BLK reconciliation and the non-tautological PDEC admission audit, the current materialized frontier has no local terminal instance left to eliminate: the non-tautological primitive PDEC candidate count is zero, the materialized LocalSurvivor ledger has no open packet, all known sparse entry routes are admitted, and NC-BLK is reconciled with the canonical boundary. The router records the new narrow label

`CurrentMaterializedFrontierExhausted_GlobalTerminalFamiliesOpen`.

Equivalently, the next required theorem is no longer a fixed numerical improvement for the current $\ell = 199$ sample or a new reading of NC-BLK. It is the global terminal-family exclusion package: every terminal object emitted by a minimal counterexample must be certified by a PDEC-family exclusion, a LocalSurvivor witness/deficit certificate, a CleanKLS/DLS internal large-sieve certificate, or an explicitly accepted external/referee input. This is again a boundary closure, not a final unconditional row/column theorem.

The split router for this global terminal family then removes the remaining ambiguity in the word "family". The non-final reductions are closed: there is no fourth terminal route, the current LocalSurvivor and NC-BLK ledgers are not independent blockers, and the finite-projection terminal dichotomy sends positive-limsup mass to PDEC-CAP and diffuse mass to CleanKLS/DLS. Thus the self-contained mathematical remainder is

`PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve`,

while final promotion also keeps the D-structure/Rankin referee gate. In particular, the frontier is not an unnamed local sample problem; it is the pair of named analytic terminal estimates plus the referee interface.

The self-contained terminal bottleneck router then removes one more apparent branch. Inside the current canonical-source boundary, InternalCleanKLS is not an independent minimal blocker: a clean large-sieve failure returns a dual concentration object to PDEC/SAE, the canonical NC-BLK/CleanKLS branch is already absorbed by the same-set capacity boundary, and the unrestricted generic full-S WFD self-contained theorem is refuted and not claimed. Hence the independent self-contained mathematical bottleneck is

`PDEC_CAP_SameSetGlobalDualCertificate`,

namely a same-bad-window global dual capacity certificate $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$ for the PDEC families. The D-structure/Rankin referee gate remains a separate final-promotion condition, so this is still not a final unconditional row/column theorem.

The next PDEC-CAP router then opens this named bottleneck rather than treating it as a black box. All currently materialized PDEC intermediate gates are routed: the DualCap families have same-M_Q mass sources and closed early P-row exits; the current PersistentCaps route through promotion deletion or NoDeletion–KL/PDEC/CleanKLS; the current ForcedCaps route into multi-bucket ActualPaymentStitching; and LHB projection-stitching is removed by the common full-period completion set. The profinite ActualPaymentStitching router then closes the remaining APS logic by the finite-projection pigeonhole dichotomy. The PDEC-CAP theorem is still not proved. Its current narrow internal frontier is now the single terminal estimate

`PrimitiveMultiAtomSameFormalUnitPDECCertificate`.

Persistent real payment Γ and fixed-shell low-mod persistence are now treated as the same finite-signature formal-unit problem: the former is a phase-bucket/tail-column signature, and the latter is a shell/displacement signature. ColumnCRT displacement is absorbed as displacement PDEC, PDEC dual failure is absorbed as cap refinement, and formal-unit mismatch is absorbed by the multiplicity-stitching normalization. Thus the single persistent terminal is PersistentFiniteSignaturePDECColumnCRT. The flat-clean SC–9 branch has been reconciled with the canonical-source boundary: clean failure returns to PDEC/SAE, canonical NC-BLK is absorbed by the same-set boundary, and generic WFD is not imported into the self-contained claim. Finally, the persistent terminal admission router removes raw ColumnCRT, dual-failure, multiplicity, and two-point-tautology pseudo-terminals. Thus the remaining admitted object must be a primitive multi-atom same-formal-unit PDEC certificate. The intermediate diffuse split is now routed: persistent KL or phase-residue mutual information returns to refined/new-layer PDEC, while only the KL-flat branch may enter CleanKLS/DLS; divergent deletion already implies support exhaustion or a named sparse/PDEC return, and HRO reduces deletion non-divergence to occupancy saturation or TailIndependence. The latter is already NoDeletion/CleanKLS. The occupancy-kernel router then applies the HRO injection bound $|\mathrm{Occ}_t| \le \min(|H_Q(t)|, r)$: sparse old-hole or residue-deficient phases cannot saturate, and any remaining saturation must contain a near-full old-hole selector kernel. The common-variable router writes $c_b = \rho_b + rk_b$, turning every old-prime ban into one forbidden residue for the same shell variable k_b. Hence no legal shell is a capacity/Hall deletion, fixed-shell mass is PDEC/ColumnCRT persistence, and shell-dispersed flat mass is the SC–9 atom. External KLS input closes the flat branch only as an external-theorem version.

6.20 Triad-A1 Self-Contained Theorem-Boundary Closure

The final Triad-A1 review separates the exact theorem that is now closed from a stronger statement that is not true under the recorded hypotheses. In plain terms, the closed result is the following. Once the source of the A1 payment measure is fixed to the canonical RIW/Buchstab decision-tree source, every remaining same-set capacity gate in the full-S terminal chain is internally accounted for. The proof no longer needs to borrow the external FullS–KLS theorem for this canonical-source branch.

Proved-in-text Statement 6.1 (Canonical-source self-contained boundary)**.** *The Triad-A1 same-set capacity / full-S terminal is self-contained on the canonical RIW/Buchstab source branch.*

Recorded proof certificate. The certificate is recorded in

`docs/monograph/prime-matrix-triad-a1-self-contained-theorem-boundary-review.md`.

It checks seven gates: exact theorem statement boundary, final closure certificate, absorption by the same-set frontier, canonical-source provenance, branch coverage without silent generic upgrade, no false generic claim, and separation of the external FullS–KLS contract. The review verdict is

`APPROVE_CANONICAL_SOURCE_SELF_CONTAINED_BOUNDARY`,

with no open review gate. □

Not-claimed Statement 6.2 (Unrestricted generic WFD self-contained theorem)**.** The manuscript does not claim an unrestricted generic full-S well-factorable WFD self-contained theorem.

Reason for non-claim. The generic anti-atom input needed for that broader statement is refuted by the moving-delta model recorded in the A1 no-go audit. Hence the generic WFD version is not an open missing lemma of the canonical-source proof; it is a different and stronger statement that is false under the current formal hypotheses. The external FullS–KLS contract remains available as an external-theorem route, but it is kept separate from the self-contained canonical-source claim. □

Remark 6.3 (Boundary in everyday language)**.** The row-program terminal proof has reached a closed theorem-boundary for the canonical source: the bookkeeping, provenance, and branch split now agree on what is proved. What remains outside this boundary is not a hidden gap in that theorem. It is either a broader generic WFD theorem that the current model refutes, or one of the separate global prime-matrix terminal certificate obligations listed below for promoting the whole row/column program to a final unconditional theorem.

6.21 Claim Status Legend

The monograph uses the following status labels.

Label	Meaning
Proved-in-text	The statement is proved inside the manuscript under the stated hypotheses.
Reduction-closed	The statement has been reduced to a smaller named interface.
External-theorem closed	The statement is closed after citing explicitly named external theorems.
Computational-certificate	The statement is closed over a finite range or certified submodule by reproducible scripts and archived outputs.
Referee-block	A proof package exists, but it still requires line-by-line referee verification.
Not claimed	The statement is not asserted as a theorem of the manuscript.

The status discipline and promotion rules are maintained in

`docs/monograph/claim-status-discipline.md`.

The manuscript provides theorem-like environments with matching names for future editing. A statement may be promoted only when its required proof, reduction endpoint, external theorem template, or certificate package is present.

6.22 External Theorem Index

The external-input index is maintained in

`docs/monograph/external-theorem-index.md`.

For the prime-matrix low-hole tail, the current explicit package is Rosser–Schoenfeld 1962: Corollary 1, formulas (3.5)–(3.6), supplies the required $\pi(x)$ bounds, and Theorem 7, formula (3.26), supplies the required Mertens product bound. These constants close the $P \geq 13208$ tail in the external-theorem sense once the finite certificates below that threshold are accepted.

For the two-point sieve, the critical external package is DI plus BFI. The index records the required checks for the KLS-window input: Kloosterman phase normalization, modulus range, frequency range, inverse-variable support, well-factorable weights, gcd strata, smoothing and sawtooth losses, and the final choice of the logarithmic-loss parameter $B(A)$. The standalone adaptation template is maintained in

`docs/monograph/kls-window-di-bfi-adaptation-template.md`.

It closes the BMD distribution input only in the external-deep-theorem sense; the separate BMD-to-TLI transfer remains a hard obligation.

6.23 Critical Chain Audit

The detailed chain audit is maintained in

`docs/monograph/key-proof-chain-optimization-audit.md`.

Its main purpose is to prevent status mixing. Prime-matrix existence rigidity may support zero-section exclusion, but it does not replace signed spectral distribution estimates such as BMD-Char. Numerical experiments may guide constants and structures, but they are not proof inputs. The RH branch remains a verification package and is not claimed as an unconditional proof.

6.24 Optimization Priorities

1. Separate theorem environments for proved-in-text, reduction-closed, external-theorem closed, and referee-block statements.
2. Replace the overly strong $\max_a$ BV–E_2 formulation by the precise WBE2 well-factorable weighted statement in the main line.
3. Rewrite KLS-window directly in the notation of the cited DI/BFI theorems to reduce translation risk.
4. Compress the prime-matrix program around the single Structured-EHPD exclusion interface.
5. Preserve the RH warning language until every controlled exit has independent referee verification.

6.25 Pre-External-Referee Hard Obligations

The current hard-obligation ledger is maintained in

`docs/monograph/pre-external-referee-remaining-obligations.md`.

It records the author-side status of every terminal interface that cannot honestly be promoted to a final theorem without further proof or independent verification. The decisive interfaces are:

Program	Hard interface	Required closure before theorem promotion

Prime matrix	Structured-EHPD / D-structure exclusion	Prove the exclusion theorem directly, or replace it by a fully proved RHI-type large-prime incidence inequality.
Prime matrix	PDEC / SAE / Rankin exits	Prove persistent endpoint CRT defect exclusion, local isolated-escape exclusion, and complete the formal Rankin certificate ledger.
Prime matrix	RRD / OSPC / Selberg-uniform constants	Put all constants in one convention and prove each budget item over the full asymptotic range, not only in sampled windows.
Two-point sieve	I3-Core true-residual package	Prove true-residual new-modulus balance, TCA, SC2, 45° synchronization peak control, Directional Balance, Zero Mass, and endpoint smoothing.
Two-point sieve	DI/BFI to KLS-window adaptation	State the external theorems in their standard variables and verify phase, modulus, frequency, weights, gcd strata, and logarithmic-loss absorption.
Two-point sieve	BMD to TLI transfer	Prove the Buchstab transfer and total-large-incidence comparison without using a hidden prime-pair lower bound.
RH branch	controlled exits	Prove every sparse, dense, tail, internal, analytic, and global exit under a single load convention.

For the RRD/OSPC constants, the current acceptance matrix is maintained in

`docs/monograph/h5-rrd-ospc-proof-obligation-matrix.md`.

It splits the remaining constant obligation into six independently checkable inequalities and does not claim that those inequalities have already been proved. The first item, RRD-low routing, is recorded in

`docs/monograph/h5-1-rrd-low-exit-theorem.md`.

It proves that an RRD-low excess enters either OSPC* or a weighted CRT-defect exit; excluding those exits remains part of the PDEC/SAE and tail-anchor obligations. The absorption of these two exits into the unified PDEC-or-SAE interface is recorded in

`docs/monograph/h5-4-ospc-weighted-crtdefect-absorption.md`.

This absorption is not an exclusion theorem; the required PDEC and SAE certificates remain open hard obligations. The PDEC certificate format itself is now recorded in

`docs/monograph/h4-pdec-certificate-template.md`.

It fixes the data, Fourier lower bound, admissible CRT upper-bound certificates, and failure-routing rules for persistent defects. This is a certificate-format closure only; the actual PDEC certificates for the formal bad-window families remain to be supplied. The first source lemmas for admissible PDEC constraint rows are recorded in

`docs/monograph/h4-pdec-constraint-source-lemmas.md`.

They justify mass, phase-capacity, mirror-pair, low-hole-bucket, and conditional-routing rows when their stated hypotheses hold. They do not by themselves provide the full constraint table or the final dual upper bound. The first admissibility table for those rows is recorded in

`docs/monograph/h4-pdec-admissible-constraint-table.md`.

It separates tautological, finite-certificate, symbolic, conditional-routing, still-unproved, and rejected rows. In particular, full-cycle CRT balance is not allowed to control an arbitrary bad-window subset. The column-cap source rule is recorded in

`docs/monograph/h4-pdec-column-cap-source-lemma.md`.

It allows column information to enter only through finite column-projection capacity, symbolic column-capacity theorems, or conditional routing to ColumnRadius/ColumnCRT/Tail-anchor exits. The coefficient ledger for those rows remains open. The first coefficient ledger is recorded in

`docs/monograph/h4-pdec-column-cap-coefficient-ledger.md`.

It registers finite column-witness, RCI/CDB, LHB column-residue, and conditional ColumnDefect rows. Most rows still require machine-readable phase blocks before they can become actual rows of the dual certificate matrix. The first materialized phase-block file is

`docs/monograph/h4-pdec-lhb-column-phase-blocks.json`.

It gives finite $Q = 2310$ LHB column rows for $13 \le p \le 47$. Empty anomaly blocks are machine-readable finite rows; diagnostic support blocks still require a projection-capacity proof before entering the final dual system. The support-to-capacity transfer conditions are recorded in

`docs/monograph/h4-pdec-lhb-support-to-capacity-transfer.md`.

They make explicit that support size is not a multiplicity bound for the persistent count vector $g(t)$; a phase-indicator reduction, a multiplicity cap, or an allowed-set projection bound is still required. The multiplicity route is made explicit in

`docs/monograph/h4-pdec-lhb-multiplicity-cap-route.md`.

It states the precise acceptance contract: for the same (p, Q, S, τ) one must prove $g(t) \le M(t)$, and only then may a diagnostic block C be promoted to the capacity row $\sum_{t\in C} g(t) \le \sum_{t\in C} M(t)$. The first finite multiplicity-cap certificate is

`docs/monograph/h4-pdec-lhb-multiplicity-cap-certificate.json`.

It takes $M(t)$ to be the high-layer CRT completion count $C_P(t; Q)$ and verifies, for $Q = 2310$ and $p = 13, 17, 19, 23, 29, 31, 37, 43, 47$, that the WHOLEDEF and BRIDGED support blocks have $\sum_{t\in C} M(t) = 0$. This is an allowed-set projection certificate: it can be used in the LHB branch after proving $S \subset Z_{\mathrm{LHB}}(p, Q)$, but it is not by itself a global PDEC exclusion theorem. The attachment lemma for that branch is

`docs/monograph/h4-pdec-lhb-attachment-lemma.md`.

It proves that an LHB-typed full-row bad window is indeed an element of $Z_{\mathrm{LHB}}(p, Q)$. Thus the remaining global task is no longer this attachment step, but the classification step: every formal PDEC bad window must either be LHB-typed or be routed to a named SAE, ColumnCRT/ColumnRadius, TailAnchor, or Rankin exit. The classification step is recorded in

`docs/monograph/h4-pdec-bad-window-classification-lemma.md`.

It combines the unified PDEC-or-SAE dichotomy, column-conflict routing, tail-anchor persistence, and Rankin access rules to split each named low-mod bad-window family into SAE, LHB-PDEC, or Routed-PDEC. This is still a routing theorem: the non-LHB exits and the final dual $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$ certificate remain open. The homogeneous-splitting rule is recorded in

`docs/monograph/h4-pdec-homogeneous-splitting-lemma.md`.

It closes the bookkeeping case in which one attempted certificate mixes different (Q, τ, F, κ, p) types: such a family must be partitioned into homogeneous subfamilies before applying UPS-1 or the bad-window classification. Thus mixed type is not an additional mathematical exit; the remaining exits are SAE, ColumnCRT/ColumnRadius, TailAnchor, and Rankin. The column-defect routing contract is recorded in

`docs/monograph/h4-pdec-column-defect-routing-contract.md`.

It defines the conditional certificate objects for ColumnRadius and ColumnCRT rows: radius rows use a threshold D_0, a column-witness selector, and phase-compatible weights $W_D(t)$, while displacement rows use a label ℓ, a non-zero displacement residue a, a threshold L_D, and phase-compatible weights $W_{\ell,a}(t)$. Violating such a row routes to the named ColumnRadius or ColumnCRT exit after the zero displacement class has been separated by the same-column prime-witness rigidity. This is a routing-contract closure only; the exits themselves still require materialized weights, thresholds, and exclusion certificates. The first finite materialization of those weights is recorded in

`docs/monograph/h4-pdec-column-defect-weight-certificate.json`.

It uses the refined finite phase $\tau_{\mathrm{fin}} = (p, q, \mathrm{row})$ for $p \leq 1000$ and the five tightest RCI rows for each p. In this finite domain, the blocks $D_{\mathrm{col}} > 81$ and displacement-residue load > 2 are both empty, over 835 phases. This is a finite refined-phase certificate; it does not provide a global D_0, L_D theorem or exclude the ColumnRadius/ColumnCRT exits. The recursive zero-row peeling route is isolated in

`docs/monograph/prime-matrix-recursive-peeling-zero-row-hardpoint.md`.

It proves the exact layer-peeling identity: after a p-sieve zero window is peeled to the previous prime level, any resurrected survivor must be a multiple of the peeled prime with rough cofactor. Thus the recursive object is a punctured zero window, not automatically a lower-level zero row. The accompanying finite audit finds no forced consecutive lower zero rows in the known first-zero-row samples. The useful remaining target is therefore an absorption theorem routing those resurrected points to ColumnCRT, ColumnRadius, or TailAnchor, rather than a direct short-period contradiction. The adjacent-shell refinement is recorded in

`docs/monograph/prime-matrix-adjacent-shell-recursive-descent-route.md`.

For adjacent primes $p < q$, it proves that the only composite old p-sieve survivor in $[1, q^2]$ is q^2. Hence a non-first q-zero row first descends to an old p-sieve q-window, with only the terminal q^2 puncture in the last row. That window either contains a full p-aligned zero row or is a seam window whose two guard intervals have lengths a and $p-(q-p)-a$. The resulting narrow target is SeamGuard-Elimination: guards cannot absorb all forced survivors without entering SAE, PDEC, or ColumnCRT. The multi-level seam descent audit is recorded in

`docs/monograph/prime-matrix-seam-multilevel-descent-route.md`.

In this conditional model, if a seam interval is already an old p-sieve zero window, then after descending to a lower prime $h < p$ the only resurrected points are those with $P^-(n) \in (h, p]$, together with the terminal q^2 puncture in the last q-row. Any full h-aligned row avoiding these resurrected points is therefore a forced h-zero row. The finite ledger checks all 21339 seam windows for $p \leq 500$ and a deterministic $p \leq 2000$ sample of 11488 seam windows, with no persistent blocking cases. This is not a global theorem; it sharpens the remaining gate to the SMD-Global Inequality, or else a proof that persistent revival blocking enters SAE, PDEC, or ColumnCRT. The tail-mirror refinement is recorded in

`docs/monograph/prime-matrix-seam-tail-mirror-descent-route.md`.

For a forced lower h-zero row with row number R, set $N_h = M_h/h$ and $\rho = ((R-1) \bmod N_h)+1$. If $\rho \leq h$ or $N_h - \rho + 1 \leq h$, then periodicity or CRT reflection places a conditional zero row inside the lower $h \times h$ matrix. The finite ledgers above again show full success in the same domains. This motivates the sharper TailMirror-SMD gate: every genuine seam counterexample must hit such a lower head/tail phase, or its persistent non-hit phases must be routed to SAE, PDEC, or ColumnCRT. The all-branch version is recorded in

`docs/monograph/prime-matrix-total-zero-row-recursive-descent-route.md`.

It treats aligned containment, seam windows, and the terminal q^2 puncture by the same descent rule. In the $p \leq 500$ ledger all 21936 non-first q-rows, including 597 aligned rows and 21339 seam rows, force a lower matrix head/tail phase. In the deterministic $p \leq 2000$ sample all 11583 checked rows do the same. The remaining theorem-level statement is TotalDescent-TM: every genuine higher zero row must force a lower $h \times h$ zero row by periodicity or tail reflection, or else its non-hit phase set must enter SAE, PDEC, or ColumnCRT. The fixed $h = 3$ refinement is recorded in

`docs/monograph/prime-matrix-total-descent-h3-margin-route.md`.

Each complete 3-row contains a unique candidate c_R coprime to 6. If $5 \leq P^-(c_R) \leq p$, that row is blocked by a resurrected point when descending from the p-sieve to the 3-sieve; if $P^-(c_R) > p$, then c_R is already a survivor of the old p-sieve and contradicts the assumed p-sieve zero window. Thus the narrowest remaining form of TotalDescent is the six-wheel inequality

$$\#\{R : J_R \subset I_s\} > \#\{R : J_R \subset I_s,\ 5 \leq P^-(c_R) \leq p\}.$$

The audit

`docs/monograph/prime-matrix-total-descent-h3-margin-audit.md` checks this for all 1,552,462 non-first q-rows with $p \leq 5000$, with minimum margin 1. This is a finite certificate and a sharpened interface, not a global theorem: a final proof must establish the six-wheel inequality uniformly, or route full blocking to SAE/PDEC/ColumnCRT. The hard-attack reduction

`docs/monograph/prime-matrix-h3-sixwheel-hard-attack.md`

clarifies the remaining obstruction. The direct six-wheel inequality sits at the Buchstab/linear-sieve boundary $u = 2$, i.e. at square-root interval length near $x = q^2$. Therefore a proof cannot be obtained by a generic short-interval theorem or a finite template. If the six-wheel candidates are all blocked, they admit the first-factor partition $n = \ell m$ with $5 \leq \ell \leq p$, m in an interval of length q/ℓ, and $P^-(m) \geq \ell$. The next theorem-level gate is consequently an H3 full-blocking defect theorem: high first-factor load enters Tail/PDEC, distributed low-load coverage produces low-mod PDEC energy, and endpoint residue concentration is SAE or ColumnCRT. The near-miss audit

`docs/monograph/prime-matrix-h3-full-blocking-defect-audit.md`

checks the same ledger for all $p \leq 5000$. Among 1,552,462 non-first q-rows, only 4015 have margin at most 20, and these occur only up to $p = 317$ and $q = 331$. Their aggregate first-factor load is led by the small-prime skeleton $5, 7, 11, 13, \ldots$. This data does not prove the theorem, but it identifies the correct next split: small-first-factor overload should be absorbed by Tail/PDEC, while low-load many-label blocking should produce low-mod PDEC/ColumnCRT energy. The small-factor envelope route

`docs/monograph/prime-matrix-h3-small-factor-envelope-route.md`

makes this split deterministic. For a cutoff y, let C_y be the load carried by first factors at most y, let $R_y = A - C_y$, and let $\ell_+(y)$ be the next prime. If full blocking holds, then either C_y is already a small-skeleton overload, or at least

$$K_y = \left\lceil \frac{R_y}{\lfloor q/\ell_+(y) \rfloor + 1} \right\rceil$$

middle-tail labels are active. The finite ledger for the near-miss rows gives maximum forced values $K_y = 7$ for $y = 31$ and $K_y = 9$ for $y = 43$. The remaining proof obligation is to convert small-skeleton overload into Tail/PDEC, or many active labels into H3-PDEC energy, with endpoint losses routed to SAE/ColumnCRT. The global scaling audit

`docs/monograph/prime-matrix-h3-global-scaling-law-route.md`

records the underlying scale. The natural six-wheel rough count is

$$\#A_s \prod_{5 \leq \ell \leq p} \left(1 - \frac{1}{\ell}\right) \asymp \frac{q}{\log q}.$$

In the finite ledger, from $p = 331$ onward every prime layer has minimum H3 margin at least 21, and bucket averages track this Mertens scale. Thus a formal counterexample must annihilate a growing $q/\log q$ rough-residue scale. The current proof task is to show that such annihilation forces small-skeleton Tail/PDEC overload, many-label H3-PDEC energy, or endpoint SAE/ColumnCRT. The constant-level formula is recorded in

`docs/monograph/prime-matrix-h3-universal-scaling-inequality.md`.

In the finite ledger, $M_{H3}(p,s) \geq 0.30\,q/\log q$ holds from $p = 113$ onward, and $0.40\,q/\log q$ holds from $p = 2011$ onward. These are not asserted as unconditional short-interval prime theorems. They are target constants for a defect inequality: if the H3 margin falls below $cq/\log q$, then SmallSkeletonOverload, ManyLabel-PDEC, or Endpoint-SAE/ColumnCRT must fire. The global tail-energy lemma

`docs/monograph/prime-matrix-h3-global-tail-energy-lemma.md`

proves the deterministic core of this defect inequality. For any cutoff y, target B, threshold L, and modulus

$$d \geq 2\left(\left\lfloor \frac{q}{\ell_+(y)} \right\rfloor + 1\right),$$

where $\ell_+(y)$ is the next prime after y, the implication

$$M_{H3}(p,s) < B \quad \Longrightarrow \quad C_y > \#A_s - B - 2L \quad \text{or} \quad E_y(d) > L$$

holds for every adjacent pair $p < q$ and every row window $2 \leq s \leq q$. Substituting $B = cq/\log q$ and $L = \lambda q/\log q$ gives a uniform infinite-scale defect formula. This is still not the final H3 lower bound: the remaining theorem-level exits are SmallSkeletonOverload $\Rightarrow$ Tail/PDEC and TailEnergy $\Rightarrow$ H3-PDEC/ColumnCRT. The first exit is bridged in

`docs/monograph/prime-matrix-h3-small-skeleton-pdec-bridge.md`.

Let

$$V_y = \#A_s \prod_{5 \leq r \leq y}\left(1 - \frac{1}{r}\right), \qquad D_y = V_y - R_y,$$

where $R_y = \#\{n \in A_s : P^-(n) > y\}$. The overload inequality $C_y > \#A_s - B - 2L$ gives $R_y < B + 2L$, hence

$$D_y > V_y - B - 2L.$$

Thus, whenever $V_y \geq (c + 2\lambda + \eta)q/\log q$, small-skeleton overload forces a PDEC defect of size $D_y > \eta q/\log q$. The remaining issue is not this bridge, but the exclusion of such persistent PDEC defects and of the tail-energy alternative. The tail alternative is put in Fourier form in

`docs/monograph/prime-matrix-h3-tail-energy-fourier-bridge.md`,

where Parseval gives $\mathcal{F}_y(d) = dE_y(d)$ for the non-zero residue frequencies of the tail first-factor labels. The synthesis

`docs/monograph/prime-matrix-h3-unified-defect-criterion.md`

therefore proves the conditional closure criterion

$$D_y \leq V_y - B - 2L \quad \text{and} \quad \mathcal{F}_y(d) \leq dL \quad \Longrightarrow \quad M_{H3}(p,s) \geq B.$$

With $B = cq/\log q$ and $L = \lambda q/\log q$, this is the current global scale theorem: the empirical $q/\log q$ law is explained by two explicit low-mod defect exclusions. The unconditional row theorem still requires proving those exclusions uniformly. The source of the $q/\log q$ scale is clarified in

`docs/monograph/prime-matrix-h3-first-row-scale-bridge.md`.

Let $\Pi_1(q) = \pi(q)$ be the number of primes in the first row up to q. By the prime number theorem, $\Pi_1(q) \sim q/\log q$. On the other hand, adjacent-shell singleton rigidity says that below q^2 every old p-sieve survivor, apart from the endpoint q^2, is prime. Hence the average H3 margin over the non-first q-rows satisfies

$$\frac{1}{q-1}\sum_{s=2}^{q} M_{H3}(p,s) \sim \frac{q}{2\log q} \sim \frac{1}{2}\Pi_1(q).$$

The factor $1/2$ is simply $\log q/\log(q^2)$. Thus H3 is the square-shell projection of the first-row prime scale; a zero H3 row would have to remove a full first-row-scale mass from one row and therefore must be accounted for by the low-mod defects above. The pointwise boundary is recorded in

`docs/monograph/prime-matrix-h3-pointwise-closure-boundary.md`.

The average identity above is rigorous, but it does not imply pointwise positivity. The missing statement is a pointwise scale-transfer theorem, for example

$$M_{H3}(p,s) \geq \kappa\,\pi(q) \qquad (2 \leq s \leq q)$$

for some fixed $\kappa > 0$, or equivalently the uniform exclusion of the D_y and $\mathcal{F}_y(d)$ defects in the H3 unified criterion. The analytic boundary is recorded in

`docs/monograph/prime-matrix-h3-square-root-short-interval-barrier.md`.

It identifies this pointwise transfer with a prime lower bound in aligned intervals of length $q = \sqrt{q^2}$ near $x = q^2$. This is exactly the square-root short-interval scale, where ordinary average PNT input and the linear sieve at $u = 2$ do not supply a pointwise positive lower bound. The remaining theorem is therefore a genuine square-root defect exclusion, not a bookkeeping consequence of the first-row average. The direct hard attack is archived in

`docs/monograph/prime-matrix-h3-square-root-defect-exclusion-hard-attack.md`.

It shows that the D_y branch can be controlled at a small sieve level by the standard one-dimensional interval-sieve fundamental lemma. The obstruction is the tail branch: for a natural low modulus d, the tail Fourier energy satisfies

$$\frac{1}{2}T_y \leq E_y(d) \leq T_y,$$

so it is essentially the tail-label count. The final hard core is therefore to exclude tail labels and double-rough points as an exact full-row filler, or to force them into ColumnCRT, endpoint phase, or cofactor-structure defects. The local rigidity of such a filler is recorded in

`docs/monograph/prime-matrix-h3-tail-filler-rigidity-hardcore.md`.

Adjacent H3 candidates differ by 2 or 4, and candidates two steps apart differ by 6. If both endpoints are tail-filled, all their large factor families are disjoint. A repeated tail label must be separated by at least $y/4$ candidate steps, and any block of diameter less than y consumes at least two fresh large prime factors per candidate. Thus the remaining obstruction is the global concatenation of these local 2/4/6 rough short-difference units into a whole row. The global concatenation interface is sharpened in

`docs/monograph/prime-matrix-h3-tail-filler-global-chain-capacity.md`.

For any continuous tail-filled block B, full concatenation forces

$$\mathrm{EdgeCap}(B;y) = |B| - 1, \qquad \mathrm{TriCap}(B;y) = |B| - 2.$$

Here EdgeCap counts compatible adjacent 2/4 tail edges and TriCap counts compatible 2/4/6 triples. If $y > \sqrt{q+6}$, a fixed ordered tail-label pair contributes at most once in the row; if $y > q^{2/3}$, every tail-filled point is a double-rough semiprime. Hence the final H3 hard input can now be stated as an explicit capacity exclusion: prove that these edge or triple capacities are strictly below the full-chain requirements, or that full capacity forces PDEC, ColumnCRT, endpoint phase, or cofactor-structure defects. This is a necessary-capacity reduction, not yet an unconditional proof of H3. The macroscopic full-tail branch is then attacked in

`docs/monograph/prime-matrix-h3-tail-edge-selberg-exclusion.md`.

For a fixed even shift $\delta \in \{2,4,6\}$ and an interval I of length H, the classical two-dimensional Selberg upper-bound sieve gives

$$\#\{n \in I : (n(n+\delta), P(y)) = 1\} \le A_\delta(\theta)\frac{H}{(\log y)^2} \qquad (3 \le y \le H^\theta).$$

Every filled adjacent edge gives a y-rough pair with shift 2 or 4, and every filled triple gives a y-rough endpoint pair with shift 6. Hence a continuous tail-filled block B must satisfy

$$|B| \le 2 + A_*(\theta)\frac{q+6}{(\log y)^2}.$$

Thus a macroscopic block of length comparable with q, in particular a whole H3 row of about $q/3$ candidates, cannot be filled only by tail labels at sufficiently large scale. The remaining H3 obstruction is consequently narrower: any surviving bad row must split tail points into many short blocks, and the small-factor cuts or short-block endpoint phases must be routed to PDEC, ColumnCRT, endpoint, or cofactor defects. The short-block routing is made exact in

`docs/monograph/prime-matrix-h3-shortblock-singleton-barrier.md`.

If the tail points are decomposed into maximal consecutive tail blocks, with total tail mass T, number of blocks J, and internal adjacent-tail edges E, then

$$T = J + E.$$

The two-dimensional sieve controls E, so a bad row with T still of order $q/\log y$ would have almost all tail mass in singleton tail blocks. The small-factor cuts themselves are not yet a PDEC contradiction, because the small skeleton has natural size comparable with q. The final local obstruction is therefore the singleton-tail system: a point $a_j = \ell_j m_j$ with $\ell_j > y$ is isolated between two small-skeleton neighbours, so $a_j \equiv 0 \pmod{\ell_j}$ and simultaneously satisfies two non-zero congruences modulo small labels $r_-, r_+ \le y$. One must prove that many such squeezed singleton phases force PDEC, ColumnCRT, endpoint, or cofactor defects. The squeezed singleton phase is converted into a CRT capacity criterion in

`docs/monograph/prime-matrix-h3-singleton-clamp-defect-criterion.md`.

For a fixed clamp datum $c = (\delta_-, \delta_+, r_-, r_+)$, the two small-skeleton congruences are either incompatible or define one class $\rho(c) \pmod{\mathrm{lcm}(r_-, r_+)}$. Adding the tail label $\ell > y$ gives one class modulo $\ell\,\mathrm{lcm}(r_-, r_+)$, hence one row contains at most

$$1 + \left\lfloor \frac{q + O(1)}{\ell\,\mathrm{lcm}(r_-, r_+)} \right\rfloor$$

points from that unit. Consequently many singleton tails must enter one of three branches: tail-label concentration, clamp low-mod concentration, or distributed singleton capacity. The first two are named defect entrances. The final unresolved branch is the distributed signed excess of double-rough semiprimes over the prime survivor channel. This distributed branch is made bilinear in

`docs/monograph/prime-matrix-h3-distributed-singleton-bilinear-obstruction.md`.

For $y > q^{2/3}$, every singleton tail is a product ℓm with $y < \ell \le p$ and m prime. The clamp class $\rho(c) \pmod{R(c)}$ is equivalent to the moving cofactor congruence

$$m \equiv \ell^{-1}\rho(c) \pmod{R(c)}.$$

Thus the final distributed obstruction is a signed bilinear prime-counting defect over the short cofactor intervals I/ℓ. Without tail-label concentration, clamp concentration, endpoint accumulation, or ColumnCRT/cofactor bias, one must prove that this signed bilinear error is only $O(q/\log^2 y)$; this is the current last H3 input. The non-zero-frequency bridge is recorded in

`docs/monograph/prime-matrix-h3-bilinear-large-sieve-defect-bridge.md`.

Expanding the cofactor residue condition modulo $R(c)$ into non-principal characters gives

$$\chi(\ell^{-1}\rho(c)) = \chi(\rho(c))\overline{\chi(\ell)}.$$

Hence a large distributed singleton error forces a non-principal bilinear character sum over the short cofactor intervals. A plain large-sieve bound is still too weak at this short window scale; the final input is a dispersion/Kloosterman-type estimate for this family, or a proof that the corresponding non-zero frequency is already a PDEC, ColumnCRT, or cofactor defect. The additive Kloosterman reduction is

`docs/monograph/prime-matrix-h3-dsb-kloosterman-window-reduction.md`.

The congruence $m \equiv \rho(c)\bar{\ell} \pmod{R(c)}$ gives, after finite additive Fourier expansion, the inverse phase

$$e\left(-\frac{h\rho(c)\bar{\ell}}{R(c)}\right).$$

Thus the final H3 frequency input is a short-window Kloosterman sum with variable ℓ, cofactor window I/ℓ, modulus $R(c)$, and frequency h. The remaining audit splits into four branches: KLS-window coverage of the active parameters, high-lcm clamp escape, high-frequency endpoint tail, and coefficient concentration. The high-lcm branch is routed further in

`docs/monograph/prime-matrix-h3-dsb-high-lcm-clamp-routing.md`.

If $R(c) > R_0$, then one row contributes at most

$$1 + \left\lfloor \frac{q + O(1)}{R_0} \right\rfloor$$

points from a fixed clamp unit. Hence any high-lcm mass of order $q/\log y$ must activate many almost-singleton high-lcm units. Across a persistent family of bad rows this gives a non-zero CRT/Fourier defect, hence a PDEC/ColumnCRT exit; if it is not persistent, it is a sparse single-window escape. This does not yet exclude the branch, but it removes high-lcm mass from the low-modulus KLS-window estimate and routes it to the existing final exits. The common energy source of the two high-lcm exits is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-fourier-energy-clamp.md`.

For a homogeneous high-lcm modulus block, with residue-counting measure μ and total mass U, finite Plancherel gives

$$\sum_{1 \le h < R} |\widehat{\mu}(h)|^2 = R \sum_{a \bmod R} \mu(a)^2 - U^2.$$

In particular, when $U \le R/2$, this non-zero energy is at least $RU/2$. Hence high-lcm escape necessarily injects Fourier energy: persistent energy must be excluded by PDEC/ColumnCRT, and isolated energy by SAE/endpoint analysis. The persistent branch is converted to a PDEC threshold in

`docs/monograph/prime-matrix-h3-dsb-hlc-pdec-threshold-bridge.md`.

For one homogeneous formal unit B with phase-counting vector g_B,

$$L_{\mathrm{HLC}}(B) = \left(\frac{R \sum_a g_B(a)^2 - U_B^2}{R-1} \right)^{1/2}$$

is a forced non-zero-frequency lower bound. Thus a persistent high-lcm unit is excluded by a same-unit certificate $U_{\mathrm{CRT}}(B) < L_{\mathrm{HLC}}(B)$. The case $U_B > R/2$ is not a new escape; it is dense effective-modulus concentration and routes back to KLS-window, PDEC/ColumnCRT, or SAE/endpoint. If this PDEC upper certificate fails, the failure is localized in

`docs/monograph/prime-matrix-h3-dsb-hlc-pdec-failure-localization.md`.

For a failing non-zero frequency h and direction ζ, every cap

$$\mathcal{C}_\alpha = \{a : \Re(\zeta e(ha/R)) \ge \alpha\}, \qquad 0 \le \alpha < L/U,$$

has mass at least $(L - \alpha U)/(1-\alpha)$. Hence failure is a concrete Bohr-cap concentration, which must become a same-unit PDEC constraint or a sparse SAE/endpoint certificate. The cap is further decomposed in

`docs/monograph/prime-matrix-h3-dsb-hlc-bohr-cap-component-route.md`.

With $d = (h, R)$ and $R' = R/d$, the cap is the d-fold lift of one arc in $\mathbb{Z}/R'\mathbb{Z}$. Each lifted component has length at most $d + \omega(\alpha)R$, where $\omega(\alpha) = \arccos(\alpha)/\pi$, and one component carries mass at least $(L - \alpha U)/(d(1-\alpha))$. Thus the remaining cap obstruction is either a large-gcd low-effective-modulus concentration or a short-arc component concentration. The large-gcd branch is descended in

`docs/monograph/prime-matrix-h3-dsb-hlc-high-gcd-descent.md`.

If $d = (h, R) > 1$ and $R' = R/d$, the projected vector $g'(b) = \sum_{a \equiv b \pmod{R'}} g(a)$ satisfies

$$\widehat{g}_R(h) = \widehat{g}_{R'}(h/d),$$

and the Bohr-cap mass is preserved. Hence each high-gcd step strictly lowers the effective modulus and must terminate in a low-effective-modulus PDEC/KLS exit or in the short-arc case. The short-arc case is quantified in

`docs/monograph/prime-matrix-h3-dsb-hlc-short-arc-density-pressure.md`.

If a short arc has length at most $D_0 + \omega(\alpha)R$ and mass forced by the Bohr-cap lower bound, then its density relative to the global density U/R is at least

$$\frac{R(L-\alpha U)}{D_0(1-\alpha)U(D_0+\omega(\alpha)R)}.$$

If this pressure exceeds $1+\epsilon$, a local-density PDEC row or SAE/endpoint certificate is forced; otherwise only the explicit parameter residual remains. The low-pressure residual is optimized in

`docs/monograph/prime-matrix-h3-dsb-hlc-short-arc-pressure-optimizer.md`.

Writing $t = L/U$ and

$$\Psi_{D_0,R}(t) = \sup_{0\le\alpha<t} \frac{t-\alpha}{D_0(1-\alpha)(D_0/R+\omega(\alpha))},$$

the residual is the one-dimensional condition $\Psi_{D_0,R}(t) \le 1+\epsilon$. This implies the same-unit flatness estimate

$$\sum_a g(a)^2 \le \frac{1+(R-1)\tau^2}{R} U^2$$

and the corresponding support lower bound. Thus low-pressure short-arc mass is routed back to the KLS-window coefficient-norm check or to coefficient concentration. The KLS admission step is separated in

`docs/monograph/prime-matrix-h3-dsb-hlc-l2-flat-kls-admission.md`.

It lists six required checks for an HLC L2-flat residual: effective modulus, effective frequency, endpoint smoothing, coefficient L^2 norm, gcd/unit strata, and dyadic or tail-label splitting. The L2-flat estimate supplies the coefficient-norm check; any other failed item is a named exit, and only if all six pass may the KLS-window input be invoked. Clean admission is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-clean-kls-reduction.md`.

A clean HLC formal unit is one in which the six named exits do not occur; then the K1–K6 admission checks pass. The remaining object is the clean HLC Kloosterman window

$$\mathcal{K}_{\mathrm{HLC}}(B),$$

and the remaining external/self-contained input is an estimate $\mathcal{K}_{\mathrm{HLC}}(B) = O(q/\log^2 y)$, denoted HLC–KLS-ext in the notes. The external-theorem adaptation is now separated in

`docs/monograph/prime-matrix-h3-dsb-hlc-kls-external-adaptation.md`.

It matches the clean HLC object to a DI/BFI/Kuznetsov-type windowed Kloosterman input: the inverse phase is $e(-h\rho(c)\overline{\ell}/R(c))$, the modulus and frequency checks are supplied by K1–K2, endpoint smoothing by K3, coefficient L^2 control by K4, gcd/unit strata by K5, and dyadic or tail-label losses by K6. Thus the clean HLC branch is closed in the external-deep-theorem version. A fully self-contained version would still require reproving the corresponding spectral/dispersion Kloosterman theorem inside this manuscript. The self-contained obligation is further compressed in

`docs/monograph/prime-matrix-h3-dsb-hlc-kls-core-reduction.md`.

There the clean HLC window is decomposed into dyadic Type-I/II blocks with standard kernel $\mathfrak{S}(\mathcal{D})$, and the proof of HLC–KLS-ext is reduced to one core estimate $|\mathfrak{S}(\mathcal{D})| \le \mathcal{N}(\mathcal{D})/\log^A y$. Thus the remaining no-black-box line is this single windowed Kloosterman spectral mean estimate. The no-black-box spine is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kls-core-self-contained-spine.md`.

It completes the smooth completion, Kloosterman quadratic-form reduction, coefficient L^2 bookkeeping, and the implication from the Kuznetsov large-sieve atom to the core estimate. It also records why pointwise Weil bounds alone do not yield the required arbitrary logarithmic saving. Hence a fully self-contained version now has one precise remaining analytic line: prove the specialized Kuznetsov large-sieve atom for this window family. That line is expanded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kuznetsov-ls-atom-expansion.md`.

The atom is decomposed into smoothing, the specialized Kuznetsov trace formula, Bessel transform decay, the spectral large sieve including oldforms and Eisenstein spectrum, and the well-factorable dispersion step that supplies the logarithmic saving. The manuscript therefore records the exact no-black-box obligations rather than hiding them under the single phrase "spectral theory". The Bessel-transform part is now treated in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-c-bessel-transform-decay.md`.

It uses the Bessel differential equation, the self-adjoint Bessel operator, and repeated integration by parts to obtain the required rapid transform decay. The remaining no-black-box analytic atoms at this point are the specialized Kuznetsov formula, the spectral large sieve, and the well-factorable dispersion logarithmic saving. The spectral-large-sieve atom is further reduced in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-d-spectral-large-sieve-spine.md`.

This dualizes the spectral large sieve to a spectral projection kernel, accounts for oldforms, Eisenstein spectrum and holomorphic forms as polylogarithmic losses, and reduces KZ–D to a single pre-trace kernel bound with diagonal size T^2 and off-diagonal Schur size N_0. The pre-trace kernel is further decomposed in

`docs/monograph/prime-matrix-h3-dsb-hlc-ptk-d-pretrace-kernel-bound.md`.

There the diagonal term is recorded as the local Weyl volume, and the remaining off-diagonal part is reduced to a spatial lattice-point/correlation row-column bound, denoted LPC–D. The spatial kernel is split in

`docs/monograph/prime-matrix-h3-dsb-hlc-lpc-d-spatial-correlation-split.md`.

The far tail is absorbed by kernel decay, the identity-near sector is empty or parabolic, and the parabolic cusp sector is controlled by divisor bookkeeping. The remaining spatial line is the generic hyperbolic local-correlation bound GHLC–D. That last spatial line is now treated in

`docs/monograph/prime-matrix-h3-dsb-hlc-ghlc-d-local-schur-closure.md`.

The key correction is to estimate the pre-trace object as an integral kernel, not by pointwise hyperbolic neighbour counting. On the local radius $r_* = T^{-1}\log^B y$ one has

$$T^2 \int_0^{r_*} (1+Tr)^{-A} \sinh r \, dr = O_A(1),$$

so the local ball area cancels the T^2 height of the kernel. Combining this local L^1 kernel mass with the Poincare-packet Schur normalization gives GHLC–D, hence LPC–D, PTK–D, and KZ–D. At that point the remaining analytic atoms were KZ–B, the specialized Kuznetsov trace formula, and KZ–E, the well-factorable dispersion logarithmic saving. The KZ–B specialization is treated in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-b-kuznetsov-trace-specialization.md`.

It derives the needed trace formula from the automorphic kernel, Poincare-packet unfolding, the double-coset Kloosterman geometric side, spectral Plancherel, and Bessel-transform normalization. This closes the formula-conversion part; it does not supply the final logarithmic saving. Hence the remaining no-black-box analytic atom in the HLC branch is KZ–E, the well-factorable dispersion logarithmic-saving step. The current KZ–E attack is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-e-well-factorable-dispersion-spine.md`.

It internalizes the well-factorable convolution algebra, the dispersion variance identity, the CRT reduction to a standard Kloosterman phase, and the gcd/endpoint polylogarithmic ledger. This proves that KZ–E would follow from one remaining estimate, denoted WFD-core: a windowed well-factorable Kloosterman dispersion mean estimate. That core is further balanced in

`docs/monograph/prime-matrix-h3-dsb-hlc-wfd-core-balanced-factor-reduction.md`.

Using the square-root well-factorable decomposition $\lambda_c = \sum_{c=uv} \alpha_u \delta_v$, the condition $c \asymp C$ forces $u, v \asymp C^{1/2}$ up to polylogarithmic boundary blocks. The gcd strata are polylogarithmic, and CRT factors the phase modulo uv into coupled phases modulo u and v. At this intermediate stage the remaining atom is BWFD-core, a balanced two-modulus well-factorable Kloosterman dispersion mean estimate. The subsequent completion attack is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-bwfd-core-spectral-completion-attack.md`.

It performs exact finite Fourier completion in the s variable and uses complete Kloosterman multiplicativity modulo uv. This proves that BWFD-core follows from one still narrower atom, BSC-core: balanced complete Kloosterman bilinear correlation with arbitrary logarithmic saving. Ordinary KZ–D spectral large sieve recovers only the raw quadratic norm scale, and pointwise Weil bounds give only single-sum square-root cancellation; neither supplies the required $\log^{-A}$ saving. The current direct attack on BSC-core is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-bsc-core-kloosterman-fraction-attack.md`.

It expands the complete Kloosterman sums term by term and exposes the reciprocal-fraction phase $e(\bar{v}R/u+\bar{u}T/v)$. One-sided degeneracy is localized to quadratic congruences such as $(a_h+\ell)x^2+b_h \equiv 0 \pmod{u}$; the remaining no-black-box atom is KFLS-core, a balanced Kloosterman-fraction large-sieve estimate with arbitrary logarithmic saving. The next square-kernel attack is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kfls-core-square-kernel-attack.md`.

It expands the square of the reciprocal-fraction sum into a four-modulus phase kernel. This also rules out an absolute Schur proof of the full kernel: the exact diagonal recovers only the raw quadratic norm scale. The remaining atom is therefore CFQK-core, a centered four-modulus Kloosterman-fraction correlation estimate covering the semi-diagonal and true off-diagonal layers. The block-centering correction is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-cfqk-core-block-centering-attack.md`.

It shows that the entire same-(u, v) block diagonal is local variance and cannot be forced to be logarithmically small for arbitrary admissible coefficients. The remaining obligations are therefore BD-CEN, the block-centering identity, OSQK-core for one shared modulus, and TFQK-core for the true four-modulus layer. The BD-CEN audit is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-bd-cen-dispersion-centering-audit.md`.

It checks the current KZ–E dispersion spine and finds only the $h = 0$ main-term subtraction, not the same-(u, v) block projection required by BD-CEN. Thus the first blocking obligation is the identity (BDC-5); OSQK-core and TFQK-core are conditional downstream obligations until that identity is proved. The subsequent no-go note is

`docs/monograph/prime-matrix-h3-dsb-hlc-bd-cen-no-go-route-fork.md`.

It shows that, for the current unblocked WFD/KZ–E object, BD-CEN is false rather than merely unproved: a single nonzero block with nonzero frequency survives the $h = 0$ subtraction but has no cross-block counterpart. The remaining choices are therefore SOURCE-CEN, BLK-energy-core, or an explicit external DI/BFI input. The SOURCE-CEN no-go note

`docs/monograph/prime-matrix-h3-dsb-hlc-source-cen-no-go.md`

checks the first of these choices. It shows that source block-centering would replace the unblocked linear WFD target in (KE-13) by a different centered object and would leave a block mean term which still has to be estimated. Thus SOURCE-CEN is not an identity for the present target. The subsequent BLK-energy audit

`docs/monograph/prime-matrix-h3-dsb-hlc-blk-energy-core-obstruction.md`

checks the second choice. As an arbitrary coefficient-array theorem at the square-kernel level, a one-block one-atom test makes $\sum_b |S_b|^2$ comparable with the raw norm, so a bare arbitrary logarithmic saving for BLK-energy-core is false. The remaining self-contained obligation is therefore the sharper NC-BLK input: prove block non-concentration for the actual WFD coefficients produced by the Type-I/II, Fourier, and well-factorable structure. Without that new input, this branch is only closed in an external-theorem version using a precisely matched DI/BFI dispersion theorem. The CRT lift-parity variant is audited in

`docs/monograph/prime-matrix-crt-lift-parity-audit.md`.

For adjacent primes $p < q$, the non-trivial q-columns are not covered by the new prime q; therefore a hypothetical q-row zero does descend to an old p-sieve zero window of length q. However, the old p-sieve row period already has an even number of zero rows by reflection symmetry, so multiplying the period by the odd factor q preserves evenness and does not create a parity contradiction. The retained conclusion is only the early-window dichotomy: an aligned lower zero row or a seam window routed to PDEC/SAE. The attempted recursive continuation is audited in

`docs/monograph/prime-matrix-recursive-mirror-descent-audit.md`.

It separates three objects: zero rows in a full CRT period at fixed width q, zero rows after peeling the largest base prime, and zero rows in an actual smaller-width matrix. These objects do not imply one another. In the finite audit, full-period zero rows occur for several adjacent pairs, but no early q^2 zero row and no actual smaller matrix zero row is forced. The next usable gate remains seam-window exclusion, not recursive parity. The annulus-mirror localization audit

`docs/monograph/prime-matrix-annulus-mirror-localization-audit.md`

separates two reflections. The CRT reflection preserving zero residue classes is $n \mapsto -n \pmod{M_p}$ and sends a small row to the tail of the CRT period. The terminal reflection $n \mapsto q^2 - n$ does land in an early interval, but it changes the forbidden classes from 0 to $q^2 \bmod \ell$. It is therefore a terminal non-zero-class block, not a smaller prime-matrix zero row. The scaled-row and half-width version is audited in

`docs/monograph/prime-matrix-scaled-peeling-halfwidth-audit.md`.

It shows that the scaled height nP/p is only an approximate locator: a lower aligned zero row appears only when the upper window fully contains a lower row. In the same samples, passing to a prime near half the upper width gives no automatic pair of consecutive lower zero rows, because the half-level sieve resurrects points whose least factor lies above the half-width level. The resurrected-point absorption route is now recorded in

`docs/monograph/prime-matrix-rpz-absorption-defect-route.md`

with the finite audit

`docs/monograph/prime-matrix-rpz-absorption-defect-audit.md`.

It routes complete absorption into five alternatives: a survivor remains, TailAnchor load, ColumnCRT displacement load, ColumnRadius, or a residual Distributed-RPZ branch. In the known first-zero-row samples the half-level has 15 resurrected points, no missing absorber, and maximum one-window label and top-column loads equal to 1. Thus the route does not justify a single-window TailAnchor exclusion; the next theorem-level gate is the Distributed-RPZ capacity barrier, or a proof that distributed absorption necessarily returns to one of the named column or tail exits. The first attack on this gate is recorded in

`docs/monograph/prime-matrix-rpz-sliding-plateau-barrier.md`.

It proves a sliding-platform multiplicity formula: if a zero window remains zero under a consecutive block of shifts, then a resurrected source in the common core repeats with the full platform multiplicity. Hence distributed absorption on such a platform is either a persistent TailAnchor load or, after source deletion, a boundary-compressed branch. The audit in

`docs/monograph/prime-matrix-rpz-sliding-plateau-audit.md`

finds platform length 5 in the known samples and TailAnchor triggering at $T_0 = 4$ in all five cases. The remaining gate is therefore the boundary-compressed RPZ branch, not unrestricted distributed absorption. The boundary-compressed branch is further routed in

`docs/monograph/prime-matrix-rpz-boundary-compressed-core-route.md`.

It proves that, if TailAnchor is excluded, the center interval $J_{T_0} = [L + A + T_0, R + B - T_0]$ is an h-sieve zero interval. The audit

`docs/monograph/prime-matrix-rpz-bcb-core-audit.md` finds that all 11 center survivors in the known samples are TailAnchor core sources; after conditional TailAnchor deletion, all five center intervals are clean and contain an aligned half-width row. The next gate is therefore the BCB grid/endpoint exclusion: either the center contains an aligned lower zero row, or the endpoint seam enters SAE, PDEC, or ColumnCRT. The exact grid criterion is recorded in

`docs/monograph/prime-matrix-rpz-bcb-grid-endpoint-criterion.md`.

For $J = [u, v]$, $N = v - u + 1$, and $\delta_h(u) \equiv 1 - u \pmod h$, the interval contains an aligned h-row if and only if $N \geq \delta_h(u) + h$. Failure is therefore a finite endpoint seam phase. The audit in

`docs/monograph/prime-matrix-rpz-bcb-grid-endpoint-audit.md`

matches this criterion in all five samples and finds no endpoint-defect sample. The remaining gate is endpoint persistence exclusion: a failed grid phase must be isolated as SAE or persistent as PDEC/ColumnCRT. This endpoint gate is routed in

`docs/monograph/prime-matrix-rpz-bcb-endpoint-persistence-route.md`

and audited in

`docs/monograph/prime-matrix-rpz-bcb-endpoint-phase-ledger.md`.

For fixed (h, N), endpoint failure is a finite residue ledger in $u \bmod h$. In the known samples there are no actual endpoint failures and only two possible failure phases. Sparse loads are SAE obligations; persistent same-phase loads are PDEC/ColumnCRT objects. Thus the RPZ endpoint branch is now named, not closed: the remaining work is either certificate closure for SAE/PDEC/ColumnCRT or recursive descent from the lower h-zero row. The two remaining RPZ attacks are now advanced together in

`docs/monograph/prime-matrix-rpz-dual-track-closure-route.md`.

Track A fixes the certificate interfaces for RPZ-SAE, RPZ-PDEC, and RPZ-ColumnCRT. Track B uses the lower-zero descent audit

`docs/monograph/prime-matrix-rpz-lower-zero-descent-audit.md`

which shows that, in the known BCB samples, all six conditional lower zero rows descend to a $p = 2$ impossible zero row with no endpoint/grid obstruction. This is still a sample-level descent audit; the global task is to prove descent-grid persistence or route every obstruction to the named certificate interfaces. The obstruction side is now made explicit in

`docs/monograph/prime-matrix-rpz-lower-descent-obstruction-ledger.md`

and

`docs/monograph/prime-matrix-rpz-certificate-materialization-interface.md`.

For each adjacent descent $p \to r$, obstruction is a finite phase condition modulo $\prod_{\ell \leq r} \ell$ and is classified as either grid failure or endpoint-puncture blocking. The current descent tree has twenty actual transition nodes, all successful, with no actual blocked node. This does not close the global theorem; it reduces any future obstruction to the SAE finite package, PDEC endpoint phase row, or ColumnCRT displacement row. The certificate skeleton builder

`docs/monograph/prime-matrix-rpz-certificate-skeleton-package.md`

materializes these rows: two endpoint SAE candidates, two endpoint PDEC/ColumnCRT rows, and three lower-descent grid-failure PDEC/ColumnCRT rows. These are certificate obligations, not completed exclusions. The endpoint SAE finite certificate

`docs/monograph/prime-matrix-rpz-endpoint-sae-finite-certificate.md`

checks that the two endpoint candidates have possible load two but actual load zero in the current BCB ledger. Hence the current finite endpoint SAE line is vacuously closed; global SAE exclusion is still not proved. The lower grid-failure certificate

`docs/monograph/prime-matrix-rpz-lower-grid-fail-avoidance-certificate.md`

reduces every adjacent descent $p \to r$ to the exact test $\delta = -(a-1)(p-r) \bmod r$ and grid success iff $\delta \le p-r$. The current descent tree has twenty actual transition nodes and none hits grid failure. The remaining global task is to prove this phase inequality for formal descent paths, or route its failure to PDEC/ColumnCRT. The formal path note

`docs/monograph/prime-matrix-rpz-formal-descent-phase-inequality.md`

sharpens this statement: once a formal descent path exists, the phase inequality is automatic from interval containment. The true remaining task is path existence for every formal counterexample branch, or exclusion of the first obstruction phase through SAE/PDEC/ColumnCRT. The first-obstruction dichotomy

`docs/monograph/prime-matrix-rpz-first-obstruction-dichotomy.md`

removes endpoint puncture as a genuine obstruction: a contained lower row containing the endpoint ap would force $r \mid a$, contradicting the roughness condition needed for the endpoint to revive. Hence every non-descending branch has a first grid-failure seam phase, which must be discharged by SAE/PDEC/ColumnCRT certificates. The first-grid-failure seam certificate

`docs/monograph/prime-matrix-rpz-first-grid-fail-seam-certificate.md`

puts this obstruction in normal form. If $g = p - r$ and $g < \delta < r$, the bad row is the union of two adjacent lower-row caps of lengths δ and $r + g - \delta$, with conserved missing mass $r - g$ and at most the right endpoint ap as an r-rough survivor. The certificate produces twelve seam phase rows covering all 1752 grid-failure phases in the current ledger. It now also records the exact PDEC support row for each seam:

$$S_\tau = \{a \bmod Q : \ a \equiv \rho \pmod r\}, \qquad F_\tau = \mathbf{1}_{a\equiv\rho \pmod r} - 1/r,$$

whose non-zero Fourier support is contained in the frequencies jQ/r, $1 \le j < r$. The same certificate splits the endpoint ap into 1348 phases already killed by lower labels and 404 Q-unit endpoint phases. Thus the next narrow gate is no longer support extraction, but either an endpoint-PDEC upper bound $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$ for these same bad-window families or a ColumnCRT displacement certificate with explicit selectors Π, λ and threshold L_D. The unit-endpoint gate certificate

`docs/monograph/prime-matrix-rpz-unit-endpoint-columncrt-gate.md` pushes this one step further. For a unit endpoint seam,

$$c \equiv ap \equiv p\rho \equiv r + (p - r) - \delta \pmod r,$$

so the endpoint lies in a fixed non-trivial lower column. If $\pi = hr + c$ is a same-column prime witness, then the endpoint lower row satisfies $H \equiv -cr^{-1} \pmod p$, and the displacement $d = h - H \pmod p$ is independent of the row phase and non-zero. In the current ledger all twelve gate rows have explicit witnesses and non-zero displacements, covering all 404 unit endpoint phases. This routes persistent unit endpoint seams to fixed non-zero ColumnCRT candidates; it still does not exclude the ColumnCRT threshold L_D or prove that formal counterexample families avoid these gates. The threshold obstruction audit

`docs/monograph/prime-matrix-rpz-columncrt-threshold-obstruction.md`

then shows why this cannot be closed by merely lowering L_D. Inside a fixed unit gate, all unit endpoint residues already have the same label p and the same non-zero displacement class $d \bmod p$, so the intrinsic load of that class is

$$R_{p,d} \geq \prod_{\ell < r} (\ell - 1).$$

In the present ledger the maximum intrinsic load is 48, and the test value $L_D = 2$ would send ten gate rows, covering 400 phases, into ColumnCRTDefect rather than exclude them. Thus the remaining legitimate closures are a formal-family avoidance theorem, an endpoint-PDEC upper bound, or an independent exclusion theorem for the resulting ColumnCRTDefect. The three-route audit

`docs/monograph/prime-matrix-rpz-three-route-closure-audit.md`

compares these options under the same ledger. It finds no closed route yet, but ranks formal-family avoidance first: the present lower-descent ledger has twenty actual transition nodes and no grid-failure node, while the endpoint-PDEC route still lacks an admissible U_{CRT} upper bound and the ColumnCRT route has been reduced to an independent defect-exclusion theorem. The formal phase automaton

`docs/monograph/prime-matrix-rpz-formal-phase-automaton.md`

turns the first route into an explicit start-phase admission problem. For each prime level p it defines an accepted set $A_p \bmod P(p)$: phases in A_p descend canonically through successive prime levels to $p = 2$, while phases outside A_p hit a first-grid-fail seam and return to the already materialized exits. In the current ledger all six BCB-Core start rows belong to their accepted sets. The global missing theorem is that every formal counterexample-family start phase lies in A_p, or else is captured by the seam exits. The rejected-phase absorption certificate

`docs/monograph/prime-matrix-rpz-rejected-phase-absorption.md`

removes the second ambiguity within the present automaton range. It checks all 27924 rejected phases, finds twelve distinct first-fail seam rows, and verifies that every one of them is already materialized by the first-grid-fail seam certificate. Hence the RPZ formal route is now a strict two-way routing statement: an accepted start phase descends to $p = 2$, while a rejected start phase enters the named seam/PDEC/ColumnCRT certificate chain. This is an absorption theorem, not an exclusion theorem; the remaining hard point is the actual discharge of those seam/PDEC/ColumnCRT exits, or a proof that formal counterexample starts always lie in A_p. The seam exit pressure ledger

`docs/monograph/prime-matrix-rpz-seam-exit-pressure-ledger.md` then compresses this remaining exit. The 27924 rejected phases reduce to twelve seam rows; within those rows 1348 endpoint phases are already killed by lower labels, while the remaining 404 unit endpoint phases fall into ten fixed non-zero ColumnCRT displacement classes, with maximum aggregated load 96. Thus the next gate is no longer a phase search: one must either prove formal-family avoidance of the twelve seam rows, provide $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$ for their single-residue PDEC supports, or prove an independent exclusion theorem for the ten ColumnCRT displacement classes. The symbolic ladder certificate

`docs/monograph/prime-matrix-rpz-symbolic-ladder-certificate.md`

rewrites the preferred avoidance route as local adjacent-prime digits. For each adjacent pair $p > r$, with $g = p - r$, it uses

$$\delta_p(a) = -(a-1)g \pmod r, \qquad \delta_p(a) \le g$$

as the exact success digit. Failure of this inequality is precisely the first-grid-fail seam already materialized above. The certificate verifies the resulting counting formula by full enumeration through modulus $P(19) = 9699690$ and symbolically records the same ladder through $p = 97$. The remaining theorem is now sharply stated: derive these digit inequalities from the formal counterexample construction itself, not from a finite phase enumeration. The BCB start-digit ledger

`docs/monograph/prime-matrix-rpz-bcb-start-digit-ledger.md`

extracts these digits for the current BCB-Core starts. All twenty digit nodes are safe, but ten of them have zero margin $\delta_p(a) = p - r$. This is a useful negative result for the proof strategy: the desired avoidance theorem cannot be closed by a coarse positive margin estimate. It must use exact congruence information for the start row modulo each lower prime r. The accepted lower-row selector ledger

`docs/monograph/prime-matrix-rpz-bcb-accepted-row-selector.md`

weakens the required positive statement to the exact selector condition

$$C_h(J) \cap A_h \neq \varnothing, \qquad C_h(J) = \{m : \ J \supset [(m-1)h+1, mh]\}.$$

For the current BCB-Core samples all five core intervals have such a selector, and all six candidate lower rows are accepted. This is still only a current-ledger certificate; the global theorem must prove selector existence for every formal BCB core interval, or else route the all-rejected selector failure back to the seam/PDEC/ColumnCRT exits. Finally, the selector gap ledger

`docs/monograph/prime-matrix-rpz-selector-gap-threshold.md`

splits selector existence into two verifiable mechanisms. If the number of complete candidate rows in $C_h(J)$ exceeds the maximum cyclic rejected gap of A_h, length alone forces a selector. In the current samples this explains two of the five BCB cores; the remaining three are short-candidate cases and require exact endpoint phase information. Thus the next proof obligation is not merely to enlarge J, but to control the endpoint phase of short cores relative to the rejected gaps of A_h. The short-candidate endpoint phase ledger

`docs/monograph/prime-matrix-rpz-short-candidate-phase-ledger.md`

performs this refinement at modulus $hP(h)$ rather than just modulo h. In the current ledger the three short-candidate BCB records all hit an accepted selector. However, the full (h, ℓ) phase families still contain all-rejected endpoint phases: 588 for $(h, \ell) = (7, 13)$, 11880 for $(11, 19)$, and 344760 for $(13, 25)$. These rejected phases are not unnamed escapes; each carries a first-failure seam key and therefore returns to the seam/PDEC/ColumnCRT exits. The remaining theorem-level gate is consequently exact: prove that the formal BCB construction avoids these all-rejected $hP(h)$ endpoint phases, or close the corresponding seam/PDEC/ColumnCRT certificates. The follow-up forbidden-block ledger

`docs/monograph/prime-matrix-rpz-short-phase-block-formula.md`

removes another layer of enumeration. Since the relevant short cores have length $\ell < 2h$, a starting endpoint u contains at most one complete h-row. For a lower row m, the endpoint interval that contains it is exactly

$$mh - \ell + 1 \leq u \leq (m-1)h + 1,$$

of width $\ell - h + 1$. Hence the all-rejected endpoint set is the disjoint union of such blocks over rejected row phases $m \bmod P(h)$. The formula matches enumeration in all three short families, and the current endpoints have positive distance from the rejected endpoint union. The remaining non-enumerative input is therefore narrower still: derive from the formal BCB construction that the induced candidate row phase belongs to A_h, or else close its named first-failure exit. The BCB candidate phase identity ledger

`docs/monograph/prime-matrix-rpz-bcb-candidate-phase-identity.md`

then makes the induced row phase explicit. If the top zero row is R, the top prime is P, the half-prime is h, the plateau extremes are s_-, s_+, and the tail threshold is T, the forced core is

$$L = (R-1)P + 1 + s_- + T, \qquad U = RP + s_+ - T.$$

The complete lower rows inside this core are exactly

$$m_{\min} = \left\lfloor \frac{L + h - 2}{h} \right\rfloor + 1, \qquad m_{\max} = \left\lfloor \frac{U}{h} \right\rfloor.$$

Equivalently, after writing $P = Qh + d$, these phases depend only on $R \bmod hP(h)$ and the plateau parameters. In the current ledger the formula matches all five BCB samples and all six induced lower row phases lie in A_h. This is the sharp remaining input: prove the same residue implication for every formal BCB counterexample, or send any rejected induced phase to its first-failure exit. The accepted-preimage ledger

`docs/monograph/prime-matrix-rpz-bcb-accepted-preimage-ledger.md`

enumerates this final residue condition for the current BCB parameter families. The $P = 13, h = 5$ family has no bad residue. The other four families still have bad residues, split between no-candidate endpoint failures and all-rejected candidate phases. The actual top-row residues in the ledger all lie in the selector preimage, with positive distance from the bad set where the bad set is nonempty. Thus the residue preimage computation is now explicit, but the proof is still not global: the formal counterexample construction must be shown to land in the selector preimage, or the bad residue must be discharged through its named BCB endpoint or first-failure seam/PDEC/ColumnCRT exit. The core-run obstruction ledger

`docs/monograph/prime-matrix-rpz-bcb-core-run-obstruction.md` adds a stronger upstream test for the no-TailAnchor branch. In that branch BCB-Core forces J_T itself to be an h-sieve zero interval, so its length cannot exceed the maximum cyclic run of integers divisible by primes $\leq h$ modulo $P(h)$. For the current layers these exact maxima are

$$h = 5 : 5, \qquad h = 7 : 9, \qquad h = 11 : 13, \qquad h = 13 : 21.$$

The five current BCB cores have lengths $9, 13, 15, 19, 25$, respectively, so every one of them exceeds the corresponding low-sieve run bound, with minimum margin 4. Thus the current finite BCB no-TailAnchor samples are obstructed before the accepted-preimage question is reached. The global theorem-level input is now a Jacobsthal-type low-sieve run bound for the formal layers; without such a bound, the remaining bad cases still route to the named endpoint and first-failure exits. This input is isolated in

`docs/monograph/prime-matrix-rpz-bcb-jacobsthal-closure-interface.md`.

Writing $G(h)$ for the maximum length of a consecutive interval all of whose points are divisible by some prime $\leq h$, the exact BCB closure condition is

$$G(h) < P + m - 1 - 2T,$$

where m is the sliding platform length. Under this inequality the no-TailAnchor BCB branch closes immediately; if it fails, the proof must keep the TailAnchor, endpoint, first-failure, PDEC, and ColumnCRT exits active. The risk scan

`docs/monograph/prime-matrix-rpz-bcb-jacobsthal-risk-scan.md`

checks this linear closure route against the Ziller–Morack primorial Jacobsthal data. It shows that the current finite samples are safe, but the naive global specialization $m = 5$, $T = 4$, and only $P > 2h$ fails already at $h = 43$: there $G(h) = 89$, the minimal top prime is $P = 89$, and the core length is only 85. Thus the finite core-run obstruction cannot be promoted to a global theorem by a blanket linear bound. A global proof must either force longer platforms m, prove stronger restrictions on the formal BCB parameter set, or close the TailAnchor/endpoint/first-failure exits. The platform-threshold ledger

`docs/monograph/prime-matrix-rpz-bcb-platform-length-threshold.md`

rewrites this requirement as the exact integer condition

$$m \geq G(h) - P + 2 + 2T.$$

For $T = 4$, the first substantive unsafe layer with $h \geq 5$ is again $h = 43$, where the required platform length is 10 rather than 5. The next attack must therefore prove platform-length growth or prove that short platforms trigger one of the named exits. The short-platform embedding route

`docs/monograph/prime-matrix-rpz-bcb-short-platform-embedding-route.md`

handles the second alternative. If $N = P + m - 1 - 2T \leq G(h)$ and no TailAnchor is present, then the forced core J_T is an h-sieve zero interval of length N and hence must be embedded inside an extremal low-sieve covered block modulo $P(h)$. Therefore its left endpoint lies in a finite Jacobsthal endpoint set $E_{h,N}$. Persistent occurrence of such an endpoint phase is no longer an unnamed escape: it is routed to SAE if sparse, or to PDEC/ColumnCRT if persistent. The accompanying certificate

`docs/monograph/prime-matrix-rpz-bcb-short-platform-embedding-certificate.md` materializes the max-block subfamily $E^{\max}_{h,N}$ by reconstructing CRT left endpoints from the Ziller–Morack modulus representation. For the current $m = 5, T = 4$ interface, the first non-trivial short platform is $h = 43$, $P = 89$, $N = 85$, with 240 distinct max-block endpoint phases; across the scanned table, 1197 reconstructed longest blocks pass the coverage and boundary checks. This remains a certificate for the longest-block subfamily, not a proof that the full endpoint set $E_{h,N}$ has been exhausted.

These obligations are not typographical tasks. They are the remaining theorem-level gates. Until they are closed, the monograph may be submitted only as a synthesis, dependency, and verification manuscript, not as a final unconditional proof of the prime-matrix theorem, prime-pair assertions, or RH.

Chapter 7

Referee Closure Audit

This chapter records the strict closure standard used for the whole monograph. A contradiction-field chain is considered closed only if its entrance, decomposition, exits, and global contradiction are all verified under the same load convention.

Definition 7.1 (Closure certificate)**.** A closure certificate for a contradiction field consists of four data:

1. an entrance theorem sending every counterexample into the field;
2. a disjoint or source-deleting decomposition into named exits;
3. for each exit, an absorption, descent, transfer, external theorem, or finite certificate;
4. a global comparison showing that final load is simultaneously bounded below and above in incompatible ways.

If any item is replaced by an unverified structural input, the conclusion is conditional on that input.

Theorem 7.2 (Current monograph closure)**.** *The present manuscript closes the audit and dependency structure of the two contradiction-field programs: all known entrances, decompositions, exits, external inputs, finite certificates, and referee obligations are explicitly named. It does not close the final unconditional theorem status of either program.*

Proof. For the prime-matrix program, the A/B appendix gives the entrance reduction from a row or column counterexample to a Structured-EHPD bad configuration. Inside the BPN branch, the low-hole bucket submodule is now closed by the low-range certificate, the narrow-band collision certificate, the two finite tail certificates, and the Rosser–Schoenfeld explicit tail package recorded above. Inside Triad-A1, the same-set capacity / full-S terminal is now closed as a canonical-source self-contained theorem-boundary: the source provenance is canonical RIW/Buchstab, the final closure certificate has no open gates, and the unrestricted generic WFD self-contained statement is explicitly refuted rather than claimed. These are genuine submodule and theorem-boundary closures. The D-structure exclusion and the remaining global PDEC/SAE/Rankin or LocalSurvivor/CleanKLS certificates remain decisive inputs requiring independent line-by-line review. Therefore the whole chain is closed as a reduction, certificate, and boundary package but not as a final unconditional prime-matrix theorem.

For the RH program, the RH manuscript has completed U1–U5: statement wording, AEX acceptance, EXT localization, manuscript wording, BibTeX/PDF compilation, and warning-free LaTeX logs. The final audit still requires independent referee verification of every local theorem and controlled exit. Therefore the chain is closed as a verification package but not as a final unconditional RH theorem.

The status table and the referee audit in `docs/monograph/` record exactly which pieces are proved, reduced, external, computational, or awaiting review. Thus the monograph's logical boundary is explicit and closed, while promotion to final theorem status remains conditional on external verification. □

Theorem 7.3 (Prime-matrix frontier). *The current prime-matrix row/column unconditional theorem is routed to named terminal gates but is not closed. The canonical-source Triad-A1 self-contained boundary is closed, and the unrestricted generic WFD self-contained theorem is refuted and not claimed. After the global terminal-family split and the self-contained bottleneck router, the independent self-contained mathematical bottleneck is the same-set PDEC capacity certificate*

$$\texttt{PDEC\_CAP\_SameSetGlobalDualCertificate}.$$

Inside that bottleneck, the transverse formal-unit embedding and the downstream canonical layer admission/nonzero-transfer/thin-return gate are now closed on the canonical-source self-contained branch. The remaining materialized open gate is the generic/external original DI/BFI route

$$\begin{matrix}\texttt{DIBFIQuantifiedNoProjection} \\ \texttt{WindowCertificate}\end{matrix},$$

after the APS profinite dichotomy, the persistent finite-signature unification, and the SC–9 boundary reconciliation are closed as routing laws, and after the persistent terminal admission and finite-arc routers remove raw, low-rank, continuous-cap, and unnamed finite-arc pseudo-terminals. The previous transverse clean large-sieve atom is now routed to the A1 CleanKLS/SC–9 frontier, and its self-contained source-support branch is further embedded into the canonical A1/KZ-E source; the canonical layer closure then returns to the already closed A1 selector/decision-tree/source-provenance boundary. The PDEC-CAP theorem itself remains open in the broader global/external sense. Final theorem promotion still separately requires the D-structure/Tail-log4/finite Rankin referee interface. The boundary-lift router records the canonical-source branch as

$$\texttt{NoFurtherCanonicalSourceSelfContainedPDECCapGap},$$

while keeping the generic/external DI/BFI route and the final D-structure/Rankin referee interface outside that self-contained closure. The subsequent terminal-promotion closure router sharpens the canonical-source self-contained boundary to

$$\texttt{NoFurtherCanonicalSourceTerminalPromotionGap}.$$

Proof. The frontier router

`docs/monograph/prime-matrix-row-column-unconditional-frontier-router.md`

checks the merged boundary audit, the Triad-A1 terminal confluence router, the terminal triad contract, the formal-unit audit, the stitching feasibility audit, the nested-duplicate dominance audit, the line-by-line referee matrix, and the claim-status table. It records

`row_column_unconditional_closed=false`,

while preserving the closed canonical-source boundary and the refuted generic branch. The terminal triad has no fourth exit, so any final counterexample must be absorbed by PDEC, LocalSurvivor, or CleanKLS/DLS certificates, together with the still guarded D-structure/Tail-log4/Rankin interfaces.

The global terminal-family boundary router and its split router then show that the current materialized frontier is exhausted and that all non-final reductions are closed. The open terminal family gates reduce to PDEC-CAP, KLS/external-or-internal large sieve, and the D-structure/Rankin referee interface. For the fully self-contained route, the remaining choice was

`PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve`.

The self-contained bottleneck router closes the non-final CleanKLS ambiguity. CleanKLS failure returns to PDEC/SAE, the canonical clean branch is absorbed by the closed same-set boundary, the generic full-S self-contained strengthening is refuted and not claimed, and the SC-9 route has already been accounted for. Therefore InternalCleanKLS is no longer an independent self-contained blocker. What remains is exactly the global same-set PDEC dual capacity certificate $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$, plus the independent referee-promotion gate. This proves the stated frontier and also explains why it is not yet a final unconditional theorem.

The PDEC-CAP same-set global-dual router refines that frontier once more. It records

`closed_current_materialized_pdec_gates=true`

and

`pdec_cap_same_set_global_dual_closed=false`.

Thus the current local PDEC materialization is exhausted, but the global proof is not. The profinite APS router closes the missing dichotomy: along the inverse tower of signatures, the real payment graph Γ_n either has a positive-limsup persistent finite signature, which becomes a multi-bucket PDEC formal unit, or avoids every finite signature, which is a distributed CleanKLS/DLS input unless FiberDeletion or NoDeletion–KL has already routed it back to PDEC. The first persistent narrow hardpoint is therefore the terminal estimate

`PrimitiveMultiAtomSameFormalUnitPDECCertificate`.

The diffuse terminal split router first sharpens the nonpersistent side. No unnamed FiberDeletion, NoDeletion–KL, or CleanKLS exit remains; the side is reduced to either SC–9 or a finite-shell low-mod persistence object. The latter is a PDEC/ColumnCRT input. The persistent-signature unification router then identifies persistent MFU and fixed-shell persistence as the same finite-signature formal-unit grammar; ColumnCRT displacement, PDEC cap failure, and formal-unit mismatch all route back into the PDEC/ColumnCRT family rather than creating a parallel terminal. The SC–9 boundary reconciliation then removes the flat-clean SC–9 branch as an independent canonical-source blocker: clean failure returns to PDEC/SAE, canonical NC-BLK is absorbed by the same-set boundary, and generic WFD is not claimed as self-contained. The persistent terminal admission router then forces any remaining PDEC object to be primitive, multi-atom, same-formal-unit, non-tautological, and not already absorbed by SAE/Endpoint. The deletion support-exhaustion bridge closes the implication from divergent deletion to exhausted support or named sparse/PDEC return; the HRO router then shows that if deletion does not diverge, either TailIndependence returns to NoDeletion/CleanKLS or old-hole residue occupancy saturates. The occupancy-kernel router uses $|\mathrm{Occ}_t| \le \min(|H_Q(t)|, r)$ to remove the sparse-old-hole subcase. Therefore the deletion-side hardpoint first becomes the dense old-hole selector kernel. The common-variable router then rewrites each selected column as $c_b = \rho_b + rk_b$; every old-prime condition forbids exactly one residue of the same shell variable k_b. The resulting alternatives are named: no legal shell gives capacity/Hall deletion, positive mass on a fixed shell or finite shell packet gives PDEC/ColumnCRT persistence, and genuine multishell flatness enters the already reconciled SC–9/NC-BLK boundary. An external KLS/DI/BFI/Kuznetsov citation would close only the external-input version.

The primitive terminal is then reduced further. The primitive multi-atom rank router removes rank-zero and rank-one pseudo-terminals: after physical deduplication, such remnants are reuse defects, two-point Fourier tautologies, fixed-shell PDEC/ColumnCRT objects, or SAE objects. What remains is a rank-at-least-two cap-stable primitive kernel. The cap-stable kernel router rewrites its inequality by the contrapositive of cap localization: if a direction reaches L_{PDEC}, then a direction cap has mass at least $(L_{\mathrm{PDEC}} - \alpha M)/(1-\alpha)$ and must return to SAE, refined PDEC, ColumnCRT, or multiplicity normalization. Hence a genuinely cap-stable kernel only has to certify that all legal direction caps stay below their thresholds. The uniform-cap finite-basis router then removes the continuum in (ζ, α): for a finite formal unit, every such cap is the preimage of a finite cyclic arc under a nontrivial character. Finally, the finite-arc transverse router splits a high arc into low transverse support, persistent transverse bias, or transverse-flat dispersion. These route respectively to SAE/ColumnCRT/fixed-shell PDEC, refined PDEC, or CleanKLS/DLS or an external large-sieve input. The current narrow PDEC-CAP hardpoint is thereby reduced to transverse clean analysis. The transverse clean reduction router observes that a finite arc fixes only one character direction; a rank-at-least-two primitive kernel still leaves a transverse quotient. Nonflat transverse defects are named returns to SAE, refined PDEC, or ColumnCRT. If none occurs, the residual is exactly a transverse L^2-flat clean large-sieve atom. The current narrow PDEC-CAP hardpoint is therefore first reduced to

`TransverseQuotientCleanLargeSieveAtom`.

The transverse clean-atom frontier router then imports the already registered A1 CleanKLS/SC–9 grammar. K1–K9 admission failure returns to PDEC, SAE, ColumnCRT, or multiplicity normalization. If admission succeeds, the object enters the SC–9 frontier: self-contained actual-coefficient NC-BLK or external DI/BFI. The canonical NC-BLK absorption cannot be used for a generic transverse quotient until the quotient coefficients are shown to have the canonical RIW/Buchstab source support, or until actual transverse block non-concentration is proved directly. The naive factor-residue incidence bridge is blocked by the internal-fiber obstruction. Thus the current self-contained hardpoint is

`TransverseSourceSupportNonconcentrationCertificate`.

This certificate is then split once more. The A1/KZ-E actual-source provenance ledger is closed for the canonical RIW/Buchstab pre-Cauchy source, the canonical RIW support route is reduced to Buchstab-layer support, and the raw thick squarefree support count is closed. The transverse embedding router then proves that the transverse quotient formal unit is a legal restriction, quotient, or conditioning of that canonical A1/KZ-E source without changing its coefficients: actual payment is a deterministic pushforward, finite signatures are finite projections, direction arcs are preimage restrictions, and the transverse quotient is a finite factor. The canonical layer closure router then observes that the downstream layer-admission/nonzero-transfer/thin-return gate is exactly the already routed A1 chain: selector retention, finite signatures, RIW/Buchstab decision tree, actual source provenance, and canonical branch boundary. Hence the canonical-source self-contained PDEC-CAP branch has no remaining layer-transfer gate. The original external DI/BFI route is still separate and requires

`DIBFIQuantifiedNoProjectionWindowCertificate`,

unless one explicitly accepts a windowed KLS external theorem as an external input. This does not promote the full row/column statement to an unconditional theorem.

Finally, the boundary-lift router

`prime-matrix-self-contained-pdec-cap-boundary-lift-router`

propagates this conclusion back to the upper self-contained bottleneck. It records

`canonical_source_self_contained_pdec_bottleneck_closed=true`

and

`narrowest_self_contained_boundary=NoFurtherCanonicalSourceSelfContainedPDECCapGap`.

The same audit keeps

`pdec_cap_same_set_global_dual_closed=false`

and

`row_column_unconditional_closed=false`.

Thus the PDEC-CAP gap has been removed only from the canonical-source self-contained branch. The broader generic/external branch still needs the quantified no-projection DI/BFI certificate, and the whole prime-matrix theorem still needs the global terminal family promotion review and the D-structure/Rankin referee interface.

The canonical terminal-promotion closure router then reconciles the older global split with the boundary lift. It observes that the current materialized terminal frontier is exhausted, the terminal triad has no fourth exit, PDEC-CAP promotion is closed on the canonical-source branch, canonical CleanKLS/NC-BLK is absorbed or returns to PDEC/SAE, and all known LocalSurvivor entrances are covered by witness, extractor, or admission contracts. It records

`latest_self_contained_hardpoint_closed=true`

and

`open_self_contained_gates=[]`.

Thus the latest self-contained boundary is

`NoFurtherCanonicalSourceTerminalPromotionGap`.

This is still a canonical-source theorem-boundary closure, not an unrestricted global terminal-family exclusion theorem. The external DI/BFI route and the D-structure/Rankin referee gate remain outside the closure.

The final self-contained theorem router then packages the exact statement boundary. The approved self-contained statement is the Prime Matrix canonical-source terminal theorem boundary: the Triad-A1 same-set/full-S terminal together with canonical terminal promotion on the canonical RIW/Buchstab source branch. Under that exact boundary it records

`canonical_source_self_contained_theorem_closed=true`

and

`open_self_contained_gates=[]`.

Its terminal is

`NoFurtherCanonicalSourceSelfContainedTheoremBoundaryGap`.

The router also records the non-claims: the unrestricted generic full-S WFD self-contained theorem is refuted and not claimed, and the unrestricted/global Prime Matrix row-column unconditional theorem is not claimed. The external DI/BFI route and the D-structure/Rankin referee promotion gate remain outside this self-contained theorem boundary.

The global-unconditional obstruction router makes the last point explicit. It records

`current_corpus_global_unconditional_self_contained_closure_possible=false`

and

`row_column_unconditional_closed=false`.

The named blockers are the refuted unrestricted generic WFD route, the still open actual full-S source bridge or strengthened source anti-atom theorem, the DI/BFI no-projection window certificate, and the D-structure/Tail-log4/finite Rankin promotion gate. Thus the current manuscript closes the exact canonical-source self-contained theorem boundary, but does not close the unrestricted global row-column theorem without new inputs.

The actual-source bridge reconciliation router sharpens this last obstruction list. It checks the canonical provenance ledger, the source-lock branch split, and the final canonical-source theorem certificate together. On the canonical RIW/Buchstab branch it records

`actual_source_bridge_closed_for_canonical_branch=true`.

Hence the former broad blocker

`ActualFullSSourceBridge`

is superseded inside the canonical self-contained theorem boundary. The global complement is still open, however:

`actual_source_bridge_closes_global_unrestricted=false`.

The updated global blockers are

`UnrestrictedGenericWFD`,
`NoncanonicalFullSComplementAntiAtomOrExternalDIBFI`,
`FullSNonAPStrengthenedSourceAntiAtom`,
`DIBFIQuantifiedNoProjectionWindowCertificate`,
`DStructureTailLog4FiniteRankinPromotion`.

Thus the actual-source bridge is closed only for the canonical branch. Full global unrestricted closure still requires a strengthened anti-atom for the noncanonical full-S complement, or an external quantified DI/BFI no-projection certificate, together with the final D-structure/Rankin promotion gate.

The noncanonical-complement input-contract router then fixes the remaining input boundary. It records

`contract_boundary_closed=true`, `canonical_branch_removed_from_remainder=true`,

but also

`generic_wfd_template_available=false`,
`noncanonical_complement_closed=false` for the current corpus.

Thus the remaining full-S complement is not an unrestricted generic WFD lemma. The moving-delta separation blocks that template, while the canonical branch has already been subtracted. Under the current terminal map there is no hidden fourth route: one must prove actual source identity with the canonical RIW/Buchstab source, prove a strengthened anti-atom for the actual noncanonical source, or provide the quantified no-projection DI/BFI route. Final unconditional row-column promotion still additionally requires the D-structure/Tail-log4/Rankin acceptance gate.

The noncanonical-complement trilemma router sharpens this boundary. It records

$$\texttt{trilemma\_boundary\_closed=true},$$

but also

$$\texttt{self\_contained\_noncanonical\_package\_closed=false},$$
$$\texttt{external\_contract\_package\_closed\_if\_fulls\_kls\_ext\_accepted=true}.$$

The router separates the only legal closure modes after the canonical branch has been subtracted:

actual source identity,
strengthened actual-source anti-atom,
or acceptance/proof of FullS-KLS-ext.

Its negative content is important: the generic full-S self-contained anti-atom route is refuted by the moving-delta capacity model under the recorded formal WFD, Type/Fourier, K4/K6, and naive-incidence hypotheses. Hence the noncanonical package cannot be closed by returning to the generic WFD template. Without the external FullS-KLS-ext input, one must prove actual source identity or a strengthened actual-source anti-atom; with the external input, the remaining primary-source issue is the line-by-line `DIBFIPrimarySourceSpecializationProof`.

The closure-input atlas finally reviews the explored models together. The CRT/MinRep equivalence, cylindrical line covering, the P-column anchor wheel, the layered $P^2 \pm k$ wheel, far-tail cofactor inversion, SN recursive peeling, named-exit absorption, and PDEC cap refinement no longer point to independent unnamed branches. They all feed the same terminal certificate package:

$$\texttt{SAE-Cert},\quad \texttt{PDEC-Cert},\quad \texttt{ColumnCRT-Cert},$$
$$\texttt{CleanMultishellKLS},\quad \texttt{TotalDescent}.$$

Together with the noncanonical full-S complement package and the D-structure/Rankin promotion package, these are the minimal remaining input basis. Thus future work should prove these packages directly rather than search for another generic WFD template or a fixed numerical constant.

The D-structure/Rankin promotion package is now fixed as an acceptance boundary rather than a hidden terminal. The corresponding router records

$$\texttt{promotion\_package\_boundary\_closed=true},$$
$$\texttt{promotion\_package\_independently\_accepted=false},$$
$$\texttt{rankin\_sample\_pass=true}.$$

It checks that the package is listed in the minimal input basis, that it is separated from the canonical-source self-contained boundary, that PM-16 remains a BLOCK-REFEREE item in the line-by-line matrix, and that the Rankin ledger has a checkable certificate format. The current sample has exact smooth-core count 39, allowed budget 40.0, and both the Rankin and exact budget tests passing.

This is a format and sample verification, not a global acceptance of the whole promotion package. Final promotion still requires independent acceptance of the D-structure/Structured-EHPD entrance reductions, Tail-log4 BG/RKS theorem-number and adaptation checks, reproducible finite-verification hashes, and Rankin certificates for all formal colored corridors, with failures routed to PDEC/SAE.

The terminal-certificate package has a further compression. The no-unnamed-escape machine and the named-exit absorption contract are already closed at the routing level. The canonical-source clean KLS branch is absorbed by the final canonical boundary, and the noncanonical clean branch belongs to the noncanonical complement or to an external KLS route. TotalDescent is also not a separate terminal: a formal descent path reaches $p = 2$, while a first failure is a first-grid-fail seam already routed to PDEC, ColumnCRT, or SAE. Therefore the independent terminal certificate work is reduced to

$$\texttt{SAE-Cert}, \quad \texttt{PDEC-Cert}, \quad \texttt{ColumnCRT-Cert}.$$

This is a structural compression of the first package, not yet a proof of those three certificate families.

The ColumnCRT branch is then absorbed once more. A persistent nonzero column displacement overload is a displacement PDEC after adjoining the displacement signature to the finite phase space; a sparse displacement overload is an SAE/endpoint event; and balanced displacement loads become admissible constraints in the PDEC dual problem. RPZ unit endpoint gates show that tuning the displacement threshold alone cannot prove exclusion, but they do not create a third terminal family. Hence the independent terminal remainder is reduced to

$$\texttt{SAE-Cert}, \quad \texttt{PDEC-Cert} \quad \text{including endpoint, displacement, cofactor, and primitive variants.}$$

This is still not a proof of the terminal certificate package; it is the removal of ColumnCRT as an independent terminal.

The SAE branch is also removed as an independent terminal by the SAE absorption router. The router checks the SAE local-reduction lemma, the materialized LocalSurvivor packet ledger, the known extractor coverage audit, the new-sparse-entry admission audit, the packet-generation dichotomy, and the persistent-terminal admission boundary. It records

$$\texttt{sae\_independent\_terminal\_removed=true}.$$

The absorption law is the following. A sparse or single-window escape must either emit a finite LocalSurvivor packet with a witness or blocker-deficit certificate, or else a finite signature persists and returns to PDEC, ColumnCRT, TailAnchor, or CofactorAnchor; if the signature layers keep escaping, the branch enters CleanKLS/DLS admission or an external input package. The currently materialized LocalSurvivor/SAE packets are exhausted, the known sparse entries have extractors or contract fallbacks, and no additional unnamed sparse entry is admitted by the current contract system. Hence SAE is not a third terminal family. The first terminal package is now reduced to

$$\begin{gathered}\texttt{PDEC-Cert} \quad \text{including endpoint, displacement, cofactor, primitive, and} \\ \text{non-tautological variants,}\end{gathered}$$

together with the hygiene rule that any genuinely new sparse route must bring an explicit packet-extractor schema. This still does not prove the terminal package or the unrestricted row-column theorem; it deletes SAE as an independent terminal and pushes the remaining structural burden back to PDEC or to a future explicit sparse schema.

The broad PDEC family is then turned into an explicit input boundary. The PDEC-family router records

`pdec_family_explicit_input_boundary_closed=true`,
`current_materialized_pdec_frontier_closed=true`,
`canonical_source_pdec_cap_closed=true`,
`global_pdec_family_unconditional_closed=false`.

It checks the persistent-terminal admission gate, the non-tautological PDEC admission audit, the primitive-rank boundary, the rank-two cap-stable inverse step, the finite cyclic-arc reduction, the transverse clean reduction, the clean-frontier route, and the canonical-layer closure. Thus the currently materialized non-tautological primitive PDEC candidate count is zero, and the canonical-source PDEC-CAP route has returned to the final self-contained boundary. A future PDEC obstruction is not admitted as a label: it must come with a same-formal-unit phase map, at least three physical primitive atoms after deduplication, no two-point tautology, rank at least two after shell and column quotients, cap stability under finite cyclic-arc localization, and no sparse, persistent-biased, or clean-externalized transverse return. Hence the first package is now reduced to future explicit primitive PDEC schemas and future explicit sparse packet schemas, while generic branches remain in the noncanonical trilemma and final theorem promotion remains in the D-structure/Rankin acceptance package.

The future sparse branch is also turned from a hygiene rule into an explicit admission boundary. The sparse-schema router records

`future_sparse_packet_schema_boundary_closed=true`,
`current_materialized_sparse_frontier_closed=true`,
`global_sparse_family_unconditional_closed=false`.

It uses the materialized LocalSurvivor ledger, the known packet-extractor coverage audit, the new-sparse-entry admission audit, the SAE absorption router, the packet-generation dichotomy, and the PDEC-family boundary. Thus a future sparse obstruction is not admitted as a bare terminal label. It must specify the source class, a finite window or fixed-offset fiber, the candidate set, the blocker families and their incidence projection, a witness or a strict blocker-deficit inequality, phase and formal-unit normalization with deduplication, a finite-signature persistence test, a layer-escape test, and a reproducible ledger with no open local obligation. If the finite signature persists, the branch returns to explicit PDEC/ColumnCRT/Tail/Cofactor schemas; if the packet layers escape, it enters CleanKLS/DLS or an explicit external large-sieve input. Hence the current materialized sparse frontier is zero, but the global sparse family is not unconditionally excluded.

Combining these items gives the final input firewall. The firewall router records

`final_input_firewall_boundary_closed=true`,
`current_materialized_terminal_frontier_closed=true`,
`no_hidden_terminal_remaining=true`,
`all_final_inputs_independently_accepted=false`,
`row_column_unconditional_closed=false`.

The remaining named inputs are exactly future explicit primitive PDEC schemas, future explicit sparse packet-extractor schemas, one legal branch of the noncanonical full-S trilemma, and independent D-structure/Rankin promotion acceptance. Thus the author-side boundary and naming problem is closed: there is no hidden terminal left in the current material. The unrestricted row-column theorem is still not promoted, because the named inputs have not all been proved or independently accepted.

The four named inputs are then attacked one by one. The attack router records

`current_materialized_frontier_zero=true`, `no_hidden_terminal_remaining=true`, `conditional_closure_chain_complete=true`, `all_current_obligations_closed=false`, `unconditional_closure_possible_from_current_corpus=false`.

The future primitive PDEC and future sparse packet-extractor inputs have no present materialized obligation: they are admission firewalls for future routes. The real current obligations are the noncanonical full-S complement and D-structure/Rankin promotion. The canonical actual-source branch is closed, but the unrestricted noncanonical complement is not; the recorded primary-source audit rejects deriving the required full-S non-AP WFD Kloosterman large-sieve estimate from the existing DI/BFI sources. The D-structure/Rankin boundary is named and the sample Rankin certificate passes, but the full ledger and independent acceptance are not complete. Therefore the logical chain is conditionally complete: if future PDEC/sparse routes are discharged by their schemas, one legal noncanonical branch is proved or accepted, and the D-structure/Rankin package is independently accepted, then the no-hidden-terminal chain promotes the theorem. Without those new inputs, the present corpus cannot honestly claim the unrestricted unconditional row-column theorem.

The noncanonical input is finally narrowed further. The noncanonical-narrowing router records

`noncanonical_narrowing_boundary_closed=true`, `ap_source_lift_rejected=true`, `generic_self_contained_antiatom_refuted=true`, `exact_source_entropy_closed=false`, `external_full_s_contract_closed_if_accepted=true`, `self_contained_noncanonical_closed=false`.

The AP-source lift is rejected by the AP/non-AP branch definition and by the object ledger; the generic self-contained anti-atom is refuted by the moving-delta capacity model; the canonical branch has already been removed. Therefore the noncanonical complement has only two real remaining inputs: an internal theorem proving exact source entropy, equivalently a strengthened source anti-atom for the actual full-S non-AP source, or an external/new proof of the full-S non-AP WFD Kloosterman large-sieve input.

The last-remaining-atoms router packages this boundary into the final open atoms. It records

`last_remaining_atom_boundaries_closed=true`,
`conditional_logic_chain_complete=true`,
`all_last_atoms_proved_or_accepted=false`,
`row_column_unconditional_closed=false`.

The final atoms are

`ActualFullSNonAPExactSupportAtom`,
`ModulusDependentCompletedFullSKLSInput`,
`DStructureRankinIndependentAcceptance`.

The first two are alternative closures for the noncanonical full-S complement: either an internal exact support/source-entropy theorem for the actual full-S non-AP source, or an external/new modulus-dependent completed full-S Kloosterman large-sieve input. The third is the independent promotion acceptance for the D-structure/Tail-log4/finite Rankin package. Thus the no-hidden-terminal logic chain is conditionally complete: one of the first two atoms, together with the D-structure/Rankin acceptance atom, promotes the row-column theorem. Since the present corpus has not proved or independently accepted these atoms, the unrestricted unconditional row-column theorem remains open.

The three-final-atoms hard-attack router then pushes this boundary one step further. It records

$$\texttt{three\_atom\_attack\_boundary\_closed=true},$$
$$\texttt{conditional\_logic\_chain\_complete=true},$$
$$\texttt{all\_three\_atoms\_proved\_or\_accepted=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The final mathematical choice is:

```
ActualFullSNonAPSourceCapacityAntiAtomForActualSource
or
CDependentResidueWeightSpectralCancellationInput
```

The first input is a theorem that the actual full-S non-AP source capacity measure has no moving same-(u, v) atom. The generic version is refuted by the moving-delta model, and it is not forced by K4/K6 or naive incidence. The second input is a spectral/dispersion cancellation theorem for the completed c-dependent residue weights $B_{c,x} = \sum_k \beta_{x+kc}$; pointwise Weil, L^2 budgeting, the ordinary large sieve, and flat-residue shortcuts do not supply it. The final promotion input is:

```
DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
```

Thus the minimal unconditional input basis is

```
(ActualFullSNonAPSourceCapacityAntiAtomForActualSource
or CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
```

The current corpus has not proved this mathematical alternative and has not completed the independent promotion acceptance, so the unrestricted unconditional row-column theorem is not yet a proved theorem of the manuscript.

The final unconditional-closure attempt router then checks whether the mathematical alternative can be closed from the current corpus. It records

$$\texttt{final\_attempt\_boundary\_closed=true},$$
$$\texttt{math\_lanes\_collapsed\_to\_common\_core=true},$$
$$\texttt{internal\_math\_proof\_found\_in\_current\_corpus=false},$$
$$\texttt{external\_math\_match\_found\_in\_current\_corpus=false},$$
$$\texttt{independent\_promotion\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The c-dependent completed-residue route is not an independent terminal: finite Fourier inversion connects it to the existing BWFD–BSC–KFLS chain, whose self-contained remaining duty is actual same-(u, v) block non-concentration, unless a precisely matched external DI/BFI/Kuznetsov dispersion theorem is supplied. The actual-source anti-atom is the same moving-block spread/source-entropy duty. Therefore the irreducible final input basis is

```
(MovingBlockSpreadNCBLKForActualFullSNonAPWFDCoefficients
OR PreciselyMatchedExternalDIBFIKuznetsovDispersionTheorem)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
```

The current corpus does not prove moving-block spread, does not contain a line-by-line external theorem match for the current full-S, non-AP, uncentered, no-projection object, and has not completed independent D-structure/Rankin promotion acceptance. Thus the final input boundary is closed, but the unrestricted unconditional theorem is still not proved.

The irreducible-input refinement router then sharpens the external label. The phrase "precisely matched external DI/BFI/Kuznetsov dispersion theorem" cannot be read as a claim that the existing primary DI/BFI sources have already been matched to this object: the BFI AP theorem, the DI/Maynard J-scale, and the AP-source lift route are all blocked for the present full-S, non-AP, uncentered, no-projection WFD window. The refined basis is

```
(MovingBlockSpreadNCBLKForActualFullSNonAPWFDCoefficients
OR FullSNonAPWFDKLSTheoremInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Here FullSNonAPWFDKLSTheoremInput means a new proof or an independently cited theorem which directly estimates the current full-S non-AP WFD Kloosterman/dispersion object. The router records

`refinement_boundary_closed=true`,

`row_column_unconditional_closed=false`.

The subsequent joint attack compares the two remaining mathematical lanes directly. The canonical RIW/Buchstab source branch has already been closed and subtracted from the noncanonical full-S complement, so its support theorem cannot be imported as a proof of the noncanonical moving-block anti-atom. The internal lane is therefore narrowed to an exact support package for the actual noncanonical source:

```
FullSNonAPExactFactorSupportPackageForActualNoncanonicalSource.
```

The package has four subclauses:

exact factor-support lower bound,
balanced-range thresholds,
Type/Fourier capacity compatibility,
factor-residue incidence bridge or direct support proof.

The external lane remains

```
FullSNonAPWFDKLSTheoremInput.
```

The BFI, DI, and Maynard sources provide the AP/dispersion/Kuznetsov technology class, but the current audit finds no ready theorem covering the present c-dependent completed residue weights, uncentered no-projection full-S non-AP object, with arbitrary logarithmic saving. Thus the latest basis is

```
(FullSNonAPExactFactorSupportPackageForActualNoncanonicalSource
OR FullSNonAPWFDKLSTheoremInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The two lanes can be narrowed once more. On the internal side, the full-S balanced-range threshold is closed, since $C \asymp P/\log^{O(1)} P$ and the balanced factors of $c = uv$ dominate every fixed logarithmic threshold. The remaining exact-support and Type/Fourier-capacity clauses are therefore a single source anti-atom contract:

```
FullSNonAPStrengthenedSourceAntiAtomContractForActualNoncanonicalSource.
```

On the external side, the full-S window may be completed modulo c; the real residue is the c-dependent completed weight

$$B_{c,x} = \sum_k \beta_{x+kc}.$$

Thus the external theorem input is narrowed to

```
CDependentResidueWeightSpectralCancellationInput.
```

The refined terminal basis is

```
(FullSNonAPStrengthenedSourceAntiAtomContractForActualNoncanonicalSource
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The dual-lane common-core reconciliation router then checks whether the internal and external lanes are still independent self-contained obstacles. It records

```
common_core_reconciliation_closed=true,
self_contained_common_core_proved=false,
external_contract_accepted_as_final_input=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The reconciliation law is this: if `CDependentResidueWeightSpectralCancellationInput` is accepted as a FullS–KLS or c-dependent spectral theorem, it is an external theorem input. If it is proved internally from the present corpus instead, finite Fourier completion sends it back to the BWFD–BSC–KFLS chain, and then to actual same-(u, v) block non-concentration. That is the same moving-block source-entropy core as the internal source anti-atom lane. Consequently the reconciled basis is

```
((ActualA1FullSSourceLockOrNewFullSNonAPSourceAntiAtomTheoremInput)
OR AcceptedFullSKLSExtOrCDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis, without an external black box, is

```
ActualA1FullSSourceLockOrNewFullSNonAPSourceAntiAtomTheoremInput
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Thus the independence gap between the two mathematical lanes is closed. What remains is not two separate self-contained obstacles, but one actual-source anti-atom/source-lock input plus the independent D-structure/Rankin promotion acceptance. The unrestricted unconditional row-column theorem is therefore still not proved.

The self-contained narrowest-core router then removes one more ambiguity from the no-external-black-box basis. It records

`narrowest_core_reduction_closed=true`,
`canonical_source_lock_absorbed_for_canonical_branch=true`,
`source_lock_option_removed_from_global_remainder=true`,
`external_black_box_used=false`,
`generic_self_contained_antiatom_available=false`,
`noncanonical_actual_source_core_proved=false`,
`dstructure_rankin_independent_acceptance_completed=false`,
`row_column_unconditional_closed=false`.

The source-lock option has already been absorbed by the canonical RIW/Buchstab branch and the actual-source provenance ledger. It closes the canonical-source self-contained theorem, but it does not close the unrestricted global noncanonical full-S complement. The generic self-contained anti-atom is also unavailable, since the moving-delta model refutes it under the recorded formal WFD/Type/Fourier hypotheses. Therefore the latest fully self-contained global basis is

```
ActualNoncanonicalFullSSourceEntropyOrStrengthenedAntiAtomTheoremInput
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The first input means exact source entropy, equivalently no moving same-(u, v) atom, for the actual noncanonical full-S non-AP WFD coefficients. It is not an unrestricted generic WFD template. This is a sharper boundary, not a proof of the remaining source core or of the independent D-structure/Rankin promotion gate.

The noncanonical source-core atomization router then removes the remaining naming duality between "source entropy" and "strengthened anti-atom." It records

`source_core_atomization_closed=true`,
`entropy_antiatom_duality_removed=true`,
`balanced_range_threshold_closed=true`,
`k4_k6_or_naive_incidence_suffices=false`,
`canonical_import_allowed=false`,
`actual_support_capacity_core_proved=false`,
`dstructure_rankin_independent_acceptance_completed=false`,
`row_column_unconditional_closed=false`.

Exact source entropy and the strengthened anti-atom are two formulations of the same actual noncanonical source-capacity core. The balanced range threshold is already closed in the full-S regime. K4/K6 residue and dyadic controls do not imply moving factor-pair support, naive factor-residue incidence is blocked by the large internal fiber inside one (u, v) block, and canonical RIW/Buchstab support cannot be imported into the noncanonical complement. Hence the latest self-contained basis is

```
ActualNoncanonicalFullSFactorSupportCapacityTheoremInput
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The first input asks for log-power lower absolute support of the exact u- and v-factors in every surviving actual noncanonical full-S non-AP balanced block, together with Type/Fourier capacity compatibility so that no moving (u, v) pair receives an unrecorded capacity multiplier. This closes a naming and shortcut boundary, but it is not yet a proof of that support-capacity theorem.

The actual capacity-ledger microatom router then checks one last possible shortcut inside that support-capacity statement. It records

$$\texttt{microatom\_boundary\_closed=true},$$
$$\texttt{support\_only\_suffices=false},$$
$$\texttt{registered\_multiplier\_discipline\_would\_suffice\_with\_support=true},$$
$$\texttt{exact\_uv\_support\_proved=false},$$
$$\texttt{registered\_multiplier\_discipline\_proved=false},$$
$$\texttt{actual\_final\_capacity\_antiatom\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The correction is that raw support width is not itself a source anti-atom. A model with many raw (u, v) pairs but one unrecorded Type/Fourier/fiber multiplier can still concentrate the final capacity measure on a single moving pair. Conversely, if all such multipliers are registered in the same formal unit and bounded by a fixed logarithmic power, then exact support does imply the final anti-atom: with $L = \log y$, $|\alpha_u|, |\delta_v| \leq L^C$, registered multipliers $W_{u,v} \leq L^E$, and $S_u S_v \geq L^{2A+4C+E}$, one obtains

$$\max_{u,v} M_{u,v} / \sum M_{u,v} \leq L^{-2A}.$$

Thus the sharp self-contained input is best stated as

```
ActualFinalCapacityAntiAtomLedgerForNoncanonicalFullS
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

A proof may supply the package

```
ActualNoncanonicalExactUVSupportLowerBound
AND ActualTypeFourierRegisteredCapacityMultiplierDiscipline.
```

This closes the support-only shortcut and fixes the conditional implication, but it still does not prove the final actual capacity anti-atom ledger or the independent D-structure/Rankin acceptance.

The registered capacity-multiplier discipline router then attacks the second micro-input in that package. It records

$$\texttt{registered\_capacity\_multiplier\_discipline\_closed=true},$$
$$\texttt{exact\_uv\_support\_proved=false},$$
$$\texttt{actual\_final\_capacity\_antiatom\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

This is a bookkeeping closure, not a new cancellation theorem. The Type decomposition, dyadic summation, CRT phase normalization, the Fourier h-window and tail, the coefficient, gcd, endpoint and smoothing losses, the full-S completion fiber, and tail-label bookkeeping are all registered inside the same actual formal unit and cost only fixed logarithmic powers. These costs are absorbed into the final capacity measure $M_{u,v}$, or equivalently into a registered multiplier $W_{u,v} \leq L^E$. The router does not use the external DI/BFI no-projection theorem-match branch and does not prove factor support.

Consequently the fully self-contained source-side basis is reduced from

```
ActualNoncanonicalExactUVSupportLowerBound
AND ActualTypeFourierRegisteredCapacityMultiplierDiscipline
```

to the single source micro-input

```
ActualNoncanonicalExactUVSupportLowerBound.
```

Together with the independent promotion gate, the latest basis is

```
ActualNoncanonicalExactUVSupportLowerBound
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the multiplier-discipline item, but the exact u, v-support theorem and the independent D-structure/Rankin acceptance remain open.

The ExactUVSupport terminal-attack router then audits the remaining source micro-input. It records

$$\texttt{exact\_uv\_support\_terminal\_boundary\_closed=true},$$
$$\texttt{registered\_capacity\_multiplier\_discipline\_closed=true},$$
$$\texttt{exact\_uv\_support\_proved=false},$$
$$\texttt{actual\_final\_capacity\_antiatom\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The audit removes the remaining shortcut routes. Formal WFD hypotheses allow point-supported factors, so they do not force actual u, v-support. K4/K6 and naive factor-residue incidence still live on the wrong projections. Raw Buchstab/Mertens counting gives enough squarefree products only after the canonical layer is admitted, and the canonical RIW/Buchstab support chain closes only the canonical-source branch. It cannot be imported into the noncanonical full-S complement.

Thus the current fully self-contained basis is still

```
ActualNoncanonicalExactUVSupportLowerBound
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

If the exact-support theorem is supplied, the already closed multiplier discipline gives the final capacity anti-atom by the recorded inequality. Without that theorem, and without independent D-structure/Rankin acceptance, the unrestricted row-column theorem cannot yet be stated as unconditional.

The ExactUVSupport failure-packetization router then rewrites the remaining source input in its contrapositive finite form. It records

$$\texttt{failure\_packetization\_closed=true},$$
$$\texttt{exact\_uv\_support\_proved=false},$$
$$\texttt{actual\_support\_failure\_packet\_exclusion\_proved=false},$$
$$\texttt{actual\_final\_capacity\_antiatom\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

After the actual noncanonical clean block, exact u, v-support notion, and registered multiplier threshold are fixed, the positive lower-bound input

```
ActualNoncanonicalExactUVSupportLowerBound
```

is equivalent to the absence of a positive-mass support-failure packet:

```
ActualNoncanonicalSupportFailurePacketExclusion.
```

Such a packet must carry the source class, formal-unit id, block key, exact u- and v-support sets, the $L^{2A+4C+E}$ threshold, the registered capacity profile, return tests to the named exits, and a reproducible certificate. Hence support failure can no longer remain an unnamed obstruction. This packetization is an equivalent boundary reformulation, not a proof of packet exclusion. The latest self-contained input basis can therefore be written as

```
ActualNoncanonicalSupportFailurePacketExclusion
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The manuscript still lacks a proof that all such packets are absent or must return to the PDEC, SAE, ColumnCRT, CleanKLS, or external KLS exits, and it still lacks independent D-structure/Rankin acceptance.

The support-failure packet return-dichotomy router then closes the return alphabet for these packets. It records

```
support_failure_packet_return_dichotomy_closed=true,
clean_core_packet_exclusion_proved=false,
actual_final_capacity_antiatom_proved=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

An actual noncanonical support-failure packet cannot be a fifth terminal family. If it is not a clean-core packet, then it must return through one of the existing named interfaces: an isolated finite packet gives a LocalSurvivor/SAE certificate; a persistent finite signature gives an explicit primitive PDEC schema; a column, displacement, endpoint or cofactor load returns through the ColumnCRT-to-PDEC/SAE absorption; and a drifting moving block enters CleanKLS/DLS, exact source entropy, or an explicit external KLS input. The canonical and generic-template escapes have already been blocked by the ExactUVSupport audit.

Thus the latest source-side micro-input is

```
ActualNoncanonicalCleanCoreSupportFailurePacketExclusion.
```

A clean-core packet is a positive-mass actual noncanonical full-S non-AP balanced block, in the same formal unit, below the exact u, v-support threshold, with no canonical import, no finite sparse witness, no persistent PDEC signature, no column or displacement defect, and no external or generic CleanKLS exit. The latest self-contained input basis is therefore

```
ActualNoncanonicalCleanCoreSupportFailurePacketExclusion
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes return completeness, not clean-core exclusion. The manuscript still lacks a proof that such clean-core packets are impossible, or a direct proof of the final capacity anti-atom.

The clean-core moving-atom sharp-input router then corrects one final over-strong formulation. It records

```
clean_core_moving_atom_sharp_boundary_closed=true,
clean_core_moving_atom_exclusion_proved=false,
actual_final_capacity_antiatom_proved=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

Excluding every clean-core low-support packet is sufficient, but it is stronger than the final capacity anti-atom requires. Once the registered multiplier discipline is closed, the contrapositive of the recorded capacity inequality says that any final $M_{u,v}$-atom produces a low-support packet. The converse is not needed: a low-support packet that does not concentrate final capacity is not a terminal obstruction. Therefore the sharp source-side input is

```
ActualNoncanonicalCleanCoreMovingAtomExclusion.
```

Here a clean-core moving atom is a positive-mass actual noncanonical full-S non-AP balanced block, after all return tests, with some pair (u, v) satisfying

$$M_{u,v} / \sum_{u,v} M_{u,v} > \log^{-2A} .$$

The latest self-contained input basis is

```
ActualNoncanonicalCleanCoreMovingAtomExclusion
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This is only a sharp reformulation. It is not implied by formal WFD, Type decomposition, Fourier smoothing, or fixed-projection diffuse; the moving-delta obstruction remains.

The clean-core terminal normal-form router then removes the last naming ambiguity around that sharp input. It records

$$\texttt{clean\_core\_terminal\_normal\_form\_closed=true},$$
$$\texttt{internal\_exact\_entropy\_proved=false},$$
$$\texttt{external\_completed\_kls\_accepted=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The internal normal form of `ActualNoncanonicalCleanCoreMovingAtomExclusion` is exact clean-core source entropy: for every post-return clean-core moving block $b = (u, v)$,

$$\max_b M_b / \sum_b M_b \le \log^{-2A} .$$

The external alternative is not a generic DI/BFI or Kuznetsov citation; it is the completed, modulus-dependent full-S input `ModulusDependentCompletedFullSKLSInput`. Thus the latest conditional terminal input basis is

```
(ExactCleanCoreFullSNonAPWFDSourceEntropy
OR ModulusDependentCompletedFullSKLSInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis is

```
ExactCleanCoreFullSNonAPWFDSourceEntropy
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the terminal input normal form only. It does not prove exact entropy, accept the completed KLS input, or complete the independent D-structure/Rankin gate.

Finally, the clean-core exact-entropy atom router rewrites the failure mode of that self-contained input. It records

$$\texttt{clean\_core\_exact\_entropy\_atom\_boundary\_closed=true},$$
$$\texttt{clean\_core\_terminal\_support\_incidence\_proved=false},$$
$$\texttt{exact\_clean\_core\_entropy\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

If exact clean-core entropy fails, there is a clean-core moving atom. The registered multiplier discipline removes unrecorded Type/Fourier/fiber capacity as an explanation; packetization forces any support failure to be reproducible; and the return dichotomy routes every non-clean-core packet back to the named exits. The only remaining self-contained failure mode is a clean-core terminal support atom. A sufficient actionable proof package is therefore

```
CleanCoreTerminalSupportIncidenceTheorem
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Here `CleanCoreTerminalSupportIncidenceTheorem` means that every positive-mass post-return actual clean-core full-S non-AP WFD block has enough exact u/v support, inside the same formal unit, to absorb the divisor bound and all registered capacity multipliers. This theorem is not proved in the current corpus.

The clean-core support-incidence attack router then audits whether this theorem follows from the already closed ingredients. It records

$$\texttt{clean\_core\_support\_incidence\_attack\_boundary\_closed=true},$$
$$\texttt{clean\_core\_exact\_layer\_transfer\_proved=false},$$
$$\texttt{clean\_core\_terminal\_support\_incidence\_proved=false},$$
$$\texttt{external\_completed\_kls\_accepted=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The range threshold, registered capacity budget, and raw squarefree/Buchstab counting in thick blocks are no longer the obstruction. The naive factor-residue incidence bridge is still blocked by the internal h, ℓ, x, z fiber, and canonical RIW/Buchstab support cannot be imported into the noncanonical clean-core branch. The remaining internal input is

```
CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn.
```

The latest conditional basis is therefore

```
(CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn
OR ModulusDependentCompletedFullSKLSInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis replaces the parenthesized alternative by

```
CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn.
```

This input requires exact clean-core layer admission, nonzero transfer to the actual α/δ coefficients, and return of thin or rejected blocks to the named exits. It is still open.

The clean-core layer-transfer path router then separates the available canonical template from what is needed for the noncanonical clean-core residual. It records

```
clean_core_layer_transfer_path_boundary_closed=true,
clean_core_path_partition_proved=false,
clean_core_exact_layer_transfer_proved=false,
external_completed_kls_accepted=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The canonical RIW/Buchstab decision tree and source-provenance ledger are closed only on the canonical-source branch. The noncanonical clean-core residual must supply its own actual coefficient path partition, or leave through the completed KLS or named-return interfaces. The newest internal input is

```
CleanCoreExactCoefficientPathPartitionNoCancellationAndThinReturn.
```

It asks for a polylogarithmic partition of the actual clean-core α/δ coefficients into exact path signatures, nonzero same-path contribution without cancellation, and return of thin, over-budget, cancelling, or unidentified-source blocks to the named exits.

The clean-core path-source firewall router then moves this input to the coefficient source level. It records

```
clean_core_path_source_firewall_boundary_closed=true,
clean_core_precauchy_source_law_proved=false,
clean_core_path_partition_proved=false,
external_spectral_atom_accepted=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The path partition must be built on the pre-Cauchy actual clean-core α/δ coefficient formula. The canonical RIW/Buchstab decision tree is only a canonical-source template, and generic WFD well-factorability is blocked by the moving-delta model. The latest conditional basis is

```
(CleanCorePreCauchyCoefficientSourceLawAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The self-contained source-side input is

```
CleanCorePreCauchyCoefficientSourceLawAndReturn.
```

It requires an exact source formula before Cauchy, Type/Fourier and completion, plus polylog path signatures, same-path nonzero contribution, and named return for source-law failure, over-budget paths, or thin/rejected blocks.

The clean-core pre-Cauchy source-law atom router then compresses this source-law input to an origin-generation ledger. It records

$$\texttt{clean\_core\_precauchy\_source\_law\_atom\_boundary\_closed=true},$$
$$\texttt{origin\_generation\_ledger\_implication\_closed=true},$$
$$\texttt{clean\_core\_original\_coefficient\_generation\_ledger\_proved=false},$$
$$\texttt{clean\_core\_precauchy\_source\_law\_proved=false},$$
$$\texttt{external\_spectral\_atom\_accepted=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The compression theorem is that a pre-Cauchy source law is exactly an origin ledger for the actual clean-core α/δ coefficients in one formal unit before Cauchy, dispersion, Type/Fourier and completion. If that ledger lists every summand, branch key, u/v map, sign and local factor, then the branch keys give the exact path signatures, the registered K6/tail-label costs give the polylog path budget, and same-key nonzero or sign-split refinement gives no cancellation. Missing source entries, over-budget paths, thin/rejected blocks or unresolved cancellation must return to the named exits or to the external spectral input. The latest conditional basis is

```
(CleanCoreOriginalCoefficientGenerationLedgerAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The self-contained source-side input is

```
CleanCoreOriginalCoefficientGenerationLedgerAndReturn.
```

This input is not proved here. The router only shows that the former source-law input has no smaller honest self-contained evidence than the actual origin ledger; the canonical RIW/Buchstab ledger remains scoped to the canonical-source branch, and generic WFD well-factorability cannot replace it.

The clean-core origin-source admission router then compresses the origin ledger to a primitive source-constructor admission gate. It records

$$\texttt{clean\_core\_origin\_source\_admission\_boundary\_closed=true},$$
$$\texttt{constructor\_admission\_implies\_origin\_ledger=true},$$
$$\texttt{unregistered\_source\_return\_absorbed=false},$$
$$\texttt{clean\_core\_primitive\_source\_constructor\_admission\_proved=false},$$
$$\texttt{clean\_core\_original\_coefficient\_generation\_ledger\_proved=false},$$
$$\texttt{external\_spectral\_atom\_accepted=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

An origin ledger is not an estimate; it first needs a source constructor. If the actual clean-core α/δ coefficients are emitted by a pre-Cauchy primitive constructor inside one formal unit, then the emitted-summand schema expands into the origin ledger. If no such constructor is registered, the object must return as an unregistered-source failure rather than remain a clean-core terminal. The latest conditional basis is

```
(CleanCorePrimitiveSourceConstructorAdmissionAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The self-contained source-side input is

`CleanCorePrimitiveSourceConstructorAdmissionAndReturn.`

This input is still open: the current material proves the canonical constructor only on the canonical-source branch and blocks generic WFD as a constructor, but it does not prove noncanonical clean-core constructor admission or absorption of every unregistered source.

The clean-core constructor source-class firewall then separates this admission gate by source class. It records

```
constructor_source_class_firewall_boundary_closed=true,
source_class_partition_closed=true,
unregistered_source_return_absorbed=true,
actual_noncanonical_primitive_constructor_formula_proved=false,
clean_core_primitive_source_constructor_admission_proved=false,
external_spectral_atom_accepted=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The firewall has five classes. The canonical constructor is already closed but scoped to the canonical-source branch. A generic well-factorable template is not a constructor. Unregistered or mixed formal-unit sources return through Multiplicity/Stitching or the K7 formal-unit failure. The external spectral class remains the completed c-dependent residue-weight input. Thus the only self-contained source-side atom is

`ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn.`

Equivalently, the latest conditional basis is

```
(ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The source-side input asks for a pre-Cauchy definition of the actual noncanonical clean-core coefficients, a deterministic summand emitter for α/δ, branch keys, u/v maps, signs and local factors, and compatibility with the same formal unit used by the support, capacity and return ledgers. This formula is not proved here.

Finally, the external-lemma parameter-match router compares this remaining self-contained atom with the available DI/BFI/Kuznetsov tools. It records

```
external_lemma_parameter_match_boundary_closed=true,
external_lemmas_match_constructor_formula=false,
external_lemmas_close_self_contained_remainder=false,
external_spectral_atom_accepted=false,
actual_noncanonical_primitive_constructor_formula_proved=false,
row_column_unconditional_closed=false.
```

The comparison is structural. DI/Kuznetsov and BFI act after coefficients, residue weights, or Kloosterman phases have already been produced: they give spectral or AP-distribution control. The current self-contained atom is earlier in the chain: it asks for the pre-Cauchy actual noncanonical source constructor and summand emitter. Therefore those external lemmas cannot replace the constructor formula. They remain relevant only to the external spectral branch, whose narrow target is

```
CDependentResidueWeightSpectralCancellationInput.
```

The conditional basis is unchanged:

```
(ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis remains

```
ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The clean-core reverse-provenance functor router then audits whether the older geometric and payment models can be used in the opposite direction. It records

$$\texttt{reverse\_provenance\_functor\_boundary\_closed=true},$$
$$\texttt{payment\_pushforward\_functoriality\_available=true},$$
$$\texttt{pushforward\_reverse\_uniqueness\_rejected=true},$$
$$\texttt{finite\_projection\_recovers\_gamma\_not\_source=true},$$
$$\texttt{constructor\_formula\_equivalent\_to\_registered\_fiber\_emitter=true},$$
$$\texttt{registered\_primitive\_prepushforward\_fiber\_emitter\_proved=false},$$
$$\texttt{actual\_noncanonical\_primitive\_constructor\_formula\_proved=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The point is functorial rather than numerical. The actual payment graph Γ, its finite projections, cyclic arc restrictions, and transverse quotients are forward pushforwards or finite factors of a pre-Cauchy source. They preserve the registered formal unit and prevent source replacement, but they do not invert the pushforward: several primitive source disintegrations may have the same Γ. Thus one cannot recover a primitive constructor merely from the downstream payment graph.

Modulo the already closed return discipline, however, the constructor formula is equivalent to a more auditable input:

```
RegisteredPrimitivePrePushforwardFiberEmitterAndReturn.
```

Such an emitter must list, before Cauchy/dispersion, finite or polylogarithmically many primitive preimage summands over each positive-mass clean-core payment atom, with α/δ, branch keys, u/v maps, signs, local factors, formal-unit registration, and coefficient identities. Empty fibers, incompatible registrations, over-budget fibers, thin blocks, or unregistered sources must return through named exits. The latest conditional basis is therefore

```
(RegisteredPrimitivePrePushforwardFiberEmitterAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis becomes

```
RegisteredPrimitivePrePushforwardFiberEmitterAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The clean-core fiber-emitter field audit then separates the already available payment skeleton from the still missing signed coefficient lift. It records

`fiber_emitter_field_audit_boundary_closed=true`,
`payment_fiber_skeleton_closed=true`,
`first_cover_payment_map_closed=true`,
`payment_count_identity_closed=true`,
`alpha_delta_coefficient_lift_proved=false`,
`polylog_branch_schema_proved=false`,
`registered_primitive_prepushforward_fiber_emitter_proved=false`,
`row_column_unconditional_closed=false`.

The payment-level skeleton is now explicit: a completion y and a low hole c are sent by the first-cover rule $\mathrm{pay}(c, y)$ to a payment atom in Γ, and the count identity

$$\text{payment\_count} = \sum_{\text{phase}} M(\text{phase})\,|H_{\text{low}}(\text{phase})|$$

has been checked. ActualPaymentStitching also leaves no fourth payment-layer exit. What is not supplied by this skeleton is the signed clean-core primitive coefficient table. Thus the newest self-contained input is

```
ExactAlphaDeltaLiftForRegisteredPaymentFiberSkeletonAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
(ExactAlphaDeltaLiftForRegisteredPaymentFiberSkeletonAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The alpha/delta disintegration router then rewrites this lift in its intrinsic form. It records

`alpha_delta_disintegration_boundary_closed=true`,
`payment_base_map_closed=true`,
`lift_equivalent_to_signed_disintegration_dictionary=true`,
`canonical_decision_tree_template_scoped=true`,
`generic_or_unregistered_dictionary_blocked=true`,
`registered_alpha_delta_disintegration_dictionary_proved=false`,
`exact_alpha_delta_lift_proved=false`,
`row_column_unconditional_closed=false`.

Let Φ denote the first-cover payment map from primitive source summands to the completion-hole payment skeleton. An exact alpha/delta lift is equivalent to a registered signed disintegration dictionary for the actual noncanonical source measure over Φ: one must list the signed preimage summands, branch keys, u/v maps, signs, local factors, variation/support budgets, and the push-forward identity. The canonical RIW/Buchstab decision tree is a model for such a dictionary only on the canonical branch; generic well-factorability or unregistered sources do not provide it. The latest self-contained basis is therefore

```
RegisteredAlphaDeltaDisintegrationDictionaryAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
(RegisteredAlphaDeltaDisintegrationDictionaryAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Finally, the disintegration automaticity router removes a false hard point. It records

$$\texttt{disintegration\_automaticity\_boundary\_closed=true},$$
$$\texttt{discrete\_payment\_base\_map\_closed=true},$$
$$\texttt{signed\_fiber\_disintegration\_formal=true},$$
$$\texttt{pushforward\_identity\_is\_real\_gate=true},$$
$$\texttt{actual\_signed\_source\_measure\_phi\_compatibility\_budget\_proved=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

Once a signed source measure ν and the first-cover map Φ are given, the fiber decomposition

$$\nu = \sum_a \nu|_{\Phi^{-1}(a)}$$

is formal on the discrete payment skeleton. The real compatibility gate is not disintegration itself, but the proof that ν is the actual noncanonical alpha/delta source measure in the same formal unit, that $\Phi_*\nu$ equals the target payment-side coefficient, and that variation, absolute support, and branch-key budgets are registered. The latest self-contained basis is

```
ActualSignedSourceMeasurePhiCompatibilityBudgetAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
(ActualSignedSourceMeasurePhiCompatibilityBudgetAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This is a boundary refinement, not a proof of the signed source or budget identity.

The geometric Phi/budget bridge router then returns the line-covering, cylindrical, P-column anchor, layered wheel, and dynamic-capacity models to this same interface. It records

$$\texttt{geometric\_phi\_budget\_bridge\_boundary\_closed=true},$$
$$\texttt{geometric\_payment\_base\_available=true},$$
$$\texttt{geometry\_defines\_signed\_source\_measure=false},$$
$$\texttt{geometry\_proves\_phi\_pushforward\_identity=false},$$
$$\texttt{geometry\_supplies\_budget\_return\_shape=true},$$
$$\texttt{actual\_signed\_source\_measure\_phi\_compatibility\_budget\_proved=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The geometric models provide the unsigned payment base map, candidate variation/support budgets, and named returns such as PDEC, SAE, ColumnCRT, and CleanKLS. They do not define the pre-Cauchy signed α/δ source measure, nor do they prove the coefficient identity $\Phi_*\nu =$ target payment-side coefficient. Hence the previous single input splits into two atoms:

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND GeometricVariationBranchBudgetCertificateOrNamedReturn.
```

The latest conditional basis is therefore

```
((ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND GeometricVariationBranchBudgetCertificateOrNamedReturn)
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND GeometricVariationBranchBudgetCertificateOrNamedReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This again refines the boundary rather than proving the two remaining atoms.

The geometric variation/branch budget router then attacks the second atom. It records

$$\texttt{geometric\_budget\_attack\_boundary\_closed=true},$$
$$\texttt{geometry\_ledger\_alphabet\_closed=true},$$
$$\texttt{dprc\_analytic\_capacity\_bound\_proved=false},$$
$$\texttt{signed\_variation\_branch\_lift\_proved=false},$$
$$\texttt{geometric\_variation\_branch\_budget\_certificate\_proved=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The line-cylinder, bottom-deficit, P-column anchor, dynamic promoted wheel, layered wheel, and input-atlas ledgers close the unsigned support and named-return alphabet. They do not control the total variation of the actual signed source, since mass may cancel after pushforward along Φ. The budget atom therefore splits as

```
GeometricVariationBranchBudgetCertificateOrNamedReturn
=> DPRCAlpha043CenteredDiscrepancyOrNamedLayerReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn.
```

The latest self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND DPRCAlpha043CenteredDiscrepancyOrNamedLayerReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
((ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND DPRCAlpha043CenteredDiscrepancyOrNamedLayerReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn)
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the budget-side interface, not the two new inputs.

The DPRC centered-discrepancy router then attacks the DPRC atom itself. It records

$$\texttt{dprc\_centered\_discrepancy\_boundary\_closed=true},$$
$$\texttt{rsm\_identity\_closed=true},$$
$$\texttt{bes\_compression\_closed=true},$$
$$\texttt{dprc\_centered\_discrepancy\_input\_proved=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The relative-sieve margin identity reduces $T_Y < S_Y$ to an explicit model-gap/finite ledger and a centered positive-discrepancy bound. The BES compression reduces that bound to the exclusion of simultaneous high L^1 and high L^2 positive beta-bucket synchronization. If such a danger intersection is not directly excluded, the DLS route requires a named return:

```
PointLoad/ColumnCRT,  ShortWindow/SAE,  LowPhase/PDEC.
```

Thus

```
DPRCAlpha043CenteredDiscrepancyOrNamedLayerReturn
=> ExplicitModelGapAndFiniteDPRCLedger
AND BESDangerIntersectionExclusionOrDLSNamedReturn.
```

The latest self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND BESDangerIntersectionExclusionOrDLSNamedReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
((ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND BESDangerIntersectionExclusionOrDLSNamedReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn)
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the DPRC interface, not the finite/model-gap or BES-DLS proofs.

The BES-DLS named-return router then attacks the BES-DLS atom. It records

```
bes_dls_named_return_boundary_closed=true,
dls_kernel_and_danger_algebra_closed=true,
bes_dls_named_return_input_proved=false,
row_column_unconditional_closed=false.
```

The centered kernel, danger intersection, spike-bucket pigeonhole, and quadratic energy expansion are now registered. If the BES danger intersection occurs, it may no longer remain an unnamed synchronization failure. It must enter one of three exits:

```
PointLoad -> ColumnCRT/tail-anchor/displacement PDEC,
ShortWindow -> SAE/LocalSurvivor/persistent sparse PDEC,
LowPhase -> W-unit PDEC/new-layer PDEC/flat DLS.
```

Thus

```
BESDangerIntersectionExclusionOrDLSNamedReturn
=> DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSLowPhasePDECNewLayerOrFlatDLSBound.
```

The latest self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSLowPhasePDECNewLayerOrFlatDLSBound
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the named-return boundary, not the three microinputs.

The DLS LowPhase PDEC/flat-DLS router then attacks the LowPhase atom. It records

$$\texttt{dls\_lowphase\_boundary\_closed=true},$$
$$\texttt{lowphase\_input\_proved=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

LowPhase is not a single fixed-modulus law. If a fixed-wheel unit-class peak persists, it must return a W-unit PDEC. If a new Fourier layer concentrates in low dimension, it must return a new-layer PDEC. If both fixed-wheel and new-layer peaks dilute, the remaining mass is genuinely high-modulus and flat, so it must be absorbed by flat DLS/KLS. Thus

```
DLSLowPhasePDECNewLayerOrFlatDLSBound
=> DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND DLSNewLayerFourierConcentrationPDECReturn
AND DLSFlatHighModLargeSieveAbsorption.
```

The latest self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND DLSNewLayerFourierConcentrationPDECReturn
AND DLSFlatHighModLargeSieveAbsorption
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the LowPhase boundary, not the three new microinputs.

The new-layer external-lemma matching router then compares the most structural LowPhase microinput with the already registered external FullS–KLS and CleanKLS/DLS interfaces. It records

$$\texttt{direct\_external\_lemma\_closes\_newlayer=false},$$
$$\texttt{self\_contained\_newlayer\_closed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The comparison is not a new numerical experiment. It is a parameter-level boundary check: external KLS/FullS lemmas estimate a prepared, completed, coefficient-given flat spectral block. The current new-layer atom sits one layer earlier. It must first prove that low-dimensional concentration on the added wheel fibre is an actual PDEC certificate, or that after deleting all such fibres the remaining object satisfies the clean flat-admission hypotheses. Hence

```
DLSNewLayerFourierConcentrationPDECReturn
=> ExactNewLayerFiberPDECProjectionMorphism
AND NewLayerNoConcentrationImpliesFlatAdmission.
```

The latest self-contained basis becomes

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND ExactNewLayerFiberPDECProjectionMorphism
AND NewLayerNoConcentrationImpliesFlatAdmission
AND DLSFlatHighModLargeSieveAbsorption
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Thus the external theorem can attach only after a new-layer projection/admission morphism; it cannot be used directly to erase the self-contained new-layer obligation.

The new-layer PDEC projection router then attacks the first of these two atoms. It records

$$\texttt{newlayer\_projection\_slicer\_closed\_schema\_admission\_open},$$
$$\texttt{exact\_newlayer\_projection\_morphism\_closed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

Two deterministic sublemmas are closed. First, on the same full-period completion set C_P, lifting Q to rQ cannot create support outside the old projection. Second, a large $r \nmid h$ Fourier coefficient on the new fibre can be rotated and sliced by a layer-cake argument to produce a cyclic-arc or half-plane cap with comparable signed mass discrepancy. Hence

```
ExactNewLayerFiberPDECProjectionMorphism
=> RegisteredNewLayerPDECFormalUnitAndCapStableSchema.
```

The latest self-contained basis is therefore

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND RegisteredNewLayerPDECFormalUnitAndCapStableSchema
AND NewLayerNoConcentrationImpliesFlatAdmission
AND DLSFlatHighModLargeSieveAbsorption
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The remaining obstruction is formal PDEC admission for the sliced cap: the same Ω, τ, w and phase map must be fixed, the cap must be non-tautological and cap-stable, and its lower and upper budgets must be computed in that same formal unit. □

Theorem 7.4 (Early-zero gap carrier and CRT-asymmetry router). *The early-zero gap-carrier observation is a valid entrance reduction but not by itself a final contradiction. If the k-th P-row*

$$I_{P,k} = [(k-1)P + 1, kP]$$

contains no prime, and if a and b are the nearest primes respectively to the left and to the right of $I_{P,k}$, then a, b are adjacent primes and $b - a > P$. The CRT period M_P repeats the small-factor covering pattern, but it does not repeat the primality of the carrier endpoints. Consequently a persistent endpoint phase is a PDEC/ColumnCRT object, an isolated endpoint phase is an SAE object, and the non-periodic prime endpoint case returns to the H3-DSB/KLS branch, whose current self-contained hard core is NC-BLK unless an external DI/BFI dispersion input is accepted.

Proof. The finite certificate is recorded in

`docs/monograph/prime-matrix-early-zero-gap-crt-asymmetry-router.md.`

If $I_{P,k}$ contains no prime, then every prime between a and b would have to lie inside $I_{P,k}$, contradicting the hypothesis. Hence a, b are adjacent. Since $a \le (k-1)P$ and $b \ge kP+1$,

$$b - a \ge (kP+1) - (k-1)P = P + 1 > P.$$

This proves the carrier-gap lemma.

The second assertion is a boundary statement. Small-factor coverage of a row is periodic modulo M_P; primality of the carrier endpoints is not periodic modulo M_P. Therefore the carrier cannot be used as a naked CRT-period contradiction. If its endpoint phase recurs as a fixed formal unit, it is a PDEC/ColumnCRT candidate. If it is isolated, it is SAE. If the endpoint primality cannot be made periodic while the covering pressure persists, the already established H3 singleton-tail reduction sends the obstruction to the H3-DSB/KLS chain. The present document has not closed NC-BLK internally and has not replaced it by an accepted external DI/BFI theorem, so the global row/column theorem remains open. □

Theorem 7.5 (Period-lift carrier drift router)**.** *Let*

$$L_P = \prod_{\substack{q<P \\ q \text{ prime}}} q.$$

If x is a P-zero-row multiplier, then $x + tL_P$ is again a P-zero-row multiplier for every $t \ge 0$. The corresponding intervals are shifted by PL_P. However, the adjacent prime carriers around these intervals are not forced to translate by PL_P. Thus the period-lifted counterexample cover and the true adjacent-prime endpoint chain have an intrinsic asymmetry: persistent endpoint drift is a PDEC/ColumnCRT object, sparse drift is SAE, and non-periodic drift with persistent covering pressure returns to the H3-DSB/KLS NC-BLK or external DI/BFI branch.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-early-zero-period-lift-carrier-drift-router.md.`

For each non-trivial column $1 \le c < P$, the zero-row hypothesis gives a prime $q < P$ with $q \mid Px + c$. Since $q \mid L_P$,

$$P(x + tL_P) + c \equiv Px + c \pmod{q},$$

so the same small prime q covers the lifted entry. This proves exact period lift of the covering chain.

Let a_t, b_t be the adjacent primes immediately outside the lifted interval $[P(x+tL_P)+1, P(x+tL_P)+P]$. The congruence calculation above contains no condition forcing $a_t = a_0 + tPL_P$ or $b_t = b_0 + tPL_P$. It only proves absence of primes inside the lifted interval. Hence endpoint translation is not a CRT consequence. Any infinite structured use of the endpoint drift must therefore be named: fixed recurring drift gives a PDEC/ColumnCRT formal unit; isolated drift gives SAE; and non-periodic endpoint drift leaves the tail filler/short-window dispersion problem already registered as H3-DSB/KLS with NC-BLK or external DI/BFI as the remaining input. This proves the router but not the final row/column theorem. □

Theorem 7.6 (Q2-stage endpoint inversion router)**.** *Let an early zero row be straddled by adjacent primes $Q_1 < Q_2$. If $Q_2 < P^2$, then every composite in the open carrier gap (Q_1, Q_2) already has* *a prime factor below P; otherwise the branch enters the square-anchor endpoint problem. At the Q_2-stage, the full wheel*

$$M_{\leq Q_2} = \prod_{\substack{\ell \leq Q_2 \\ \ell \text{ prime}}} \ell$$

does not reproduce the two prime endpoints. For every $t \geq 1$,

$$Q_1 + tM_{\leq Q_2} \equiv 0 \pmod{Q_1}, \qquad Q_2 + tM_{\leq Q_2} \equiv 0 \pmod{Q_2},$$

so the endpoint-stable full-wheel replay is impossible. The surviving branches are: a persistent closed carrier block, which is a ColumnCRT/PDEC object; a sparse carrier block, which is SAE; or moving endpoints/support, which returns to the moving-family H3-DSB/KLS branch.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-q2-carrier-stage-crt-asymmetry-router.md.`

The below-square assertion follows because a composite $n < Q_2 < P^2$ has a least prime factor at most $\sqrt{n} < P$, while adjacency of Q_1, Q_2 excludes primes in the open gap. The endpoint inversion is the displayed congruence: both endpoints divide the full Q_2-wheel modulus, so any positive full-wheel copy of either endpoint is composite. If one uses only $M_{<Q_2}$, the left endpoint Q_1 is still killed by its own zero class, and the right endpoint Q_2 is not forced to remain prime by CRT data. Thus the full Q_2-stage gives a genuine local contradiction only to endpoint-stable replay. Without endpoint-stable replay, the obstruction is precisely one of the named ColumnCRT/PDEC, SAE, or moving-family exits. Hence the router is closed, but the global row/column theorem is not. □

Theorem 7.7 (Q2 endpoint replacement and aperture growth). *In the Q_2-stage of the preceding theorem, full-wheel replay creates a replacement debt for the true adjacent-prime endpoints. More precisely, in the below-square branch the full Q_2-wheel copy of $[Q_1, Q_2]$ is a closed composite block. Hence if $A_t < B_t$ are the adjacent primes straddling that copied block, then*

$$B_t - A_t \geq Q_2 - Q_1 + 2.$$

Consequently a fixed bounded carrier aperture cannot support infinitely many endpoint-stable replays. A fixed aperture of width W, starting from closed carrier width $W_0 = Q_2 - Q_1 + 1$, permits at most

$$\left\lfloor \frac{W - W_0}{2} \right\rfloor$$

such replacements. Any infinite continuation must move or enlarge the aperture and is therefore routed to moving-aperture ColumnCRT/PDEC, SAE, or the H3-DSB/KLS moving-family branch.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-q2-endpoint-replacement-aperture-growth-router.md.`

For an early zero row $[(k-1)P+1, kP]$, the left endpoint satisfies $Q_1 \leq (k-1)P = kP - P$, with equality possible only in the $k = 2$ boundary case $Q_1 = P$; the right endpoint satisfies $Q_2 > kP$. In the $Q_2 < P^2$ branch, every composite in the open gap has least prime factor below P, and the full Q_2-wheel kills Q_1 and Q_2 themselves. Thus every point of the copied closed interval

$$[Q_1 + tM_{\leq Q_2}, Q_2 + tM_{\leq Q_2}]$$

is composite for $t \geq 1$. The nearest true primes around this copied block must lie outside it, giving the displayed $+2$ lower bound. Iterating this lower bound inside a fixed aperture gives the finite bound $\lfloor (W - W_0)/2 \rfloor$. Therefore the bounded-aperture branch is closed; the remaining branches are exactly the named moving, PDEC, SAE, and H3 exits. □

Theorem 7.8 (Fresh endpoint layer cascade). *After the bounded-aperture branch is removed, any continued full-wheel endpoint replay must introduce an unbounded sequence of fresh endpoint prime layers. Let M_j be the full-wheel modulus at stage j, and let B_{j+1} be the new right adjacent-prime endpoint forced after the copied closed composite block. Then*

$$B_{j+1} > M_j.$$

If $L_j = \log M_j$, the next full wheel satisfies

$$L_{j+1} \geq L_j + \log B_{j+1} > 2L_j.$$

Thus the endpoint cascade cannot terminate in a fixed finite CRT period. A persistent fresh endpoint phase is a ColumnCRT/PDEC object; a sparse cascade is SAE; and a non-PDEC unbounded fresh-layer cascade is routed to the one-residue per fresh prime tail-sieve/H3-DSB/KLS branch.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-q2-endpoint-fresh-layer-cascade-router.md.`

At stage j, full-wheel replay moves the current closed carrier by at least M_j and turns it into a closed composite block. The true right adjacent prime around that copied block must therefore lie to the right of the copied block, hence beyond M_j. This prime is not contained in any fixed finite endpoint layer below the old bound. When the next wheel includes it, its next copy is divisible by itself, so the same replacement mechanism repeats. The displayed logarithmic inequality follows immediately. Since a finite CRT period contains only finitely many prime layers, it cannot support this infinite cascade. The only remaining possibilities are the named PDEC, SAE, or tail-sieve/H3 exits. □

Theorem 7.9 (Fresh-layer tail-mass dichotomy). *In the non-PDEC branch of the fresh endpoint layer cascade, assume the local aperture at stage j is W_j and the fresh endpoint prime is B_j. If the fresh layer does not create a persistent ColumnCRT/PDEC correlation, then its local one-residue contribution is bounded by*

$$\frac{W_j}{B_j}.$$

Under the controlled-aperture condition $\log W_j = o(\log B_j)$, and in particular under the endpoint-replacement model $W_j = W_0 + 2j$, the total fresh-tail mass is summable:

$$\sum_j \frac{W_j}{B_j} < \infty.$$

Thus the controlled non-PDEC fresh-tail branch is SAE. If instead the aperture repeatedly grows fast enough to track the fresh modulus, the branch is no longer local endpoint replacement; it is aperture explosion or global support motion and must be routed to H3-DSB or moving-support PDEC.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-q2-fresh-layer-tail-mass-dichotomy-router.md.`

The previous cascade theorem gives $L_{j+1} > 2L_j$, where $L_j = \log M_j$ is the logarithm of the full-wheel modulus, and also gives $B_{j+1} > M_j$. Hence the denominators B_{j+1} have lower bounds $\exp(L_j)$

whose logarithms grow at least double-exponentially in the stage index. If $\log W_j = o(\log B_j)$, then for all sufficiently large j one has $W_j \le B_j^{1/2}$, so

$$\frac{W_j}{B_j} \le B_j^{-1/2},$$

and the right-hand series converges by the same double-exponential lower growth. The linear endpoint-replacement aperture $W_j = W_0 + 2j$ is a special case.

The only ways to avoid this SAE absorption are exactly the two named failures of the hypotheses. A persistent concentration of fresh-layer hits is, by definition, a ColumnCRT/PDEC formal unit. A failure of controlled aperture is an aperture explosion or support-motion event, not the local endpoint replay branch. Therefore the router closes the controlled fresh-tail outlet but not the global row/column theorem. □

Theorem 7.10 (Q2 aperture-explosion schema firewall)**.** *After the controlled fresh-tail branch has been absorbed into SAE, any remaining aperture-explosion branch must present an explicit persistent structure: support motion, a blocker-package change, a fresh-layer ColumnCRT/PDEC unit, or a moving-family schema. In the current materialized corpus, unnamed aperture explosion is not an admissible terminal. Registered support motion has already been routed away, current materialized fresh-layer PDEC/ColumnCRT has already been blocked by the admission firewall, and the branch-replay current frontier is zero. Hence the only surviving form is a future explicit schema, which must reopen under the stated firewall; the global row/column theorem is still not proved.*

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-q2-aperture-explosion-schema-firewall-router.md.`

The previous theorem leaves only two ways out of SAE: persistent fresh-layer correlation, which is PDEC/ColumnCRT, or failure of controlled aperture, which is support motion or aperture explosion. The current registered support-motion certificate eliminates local support drift for the registered blocks and routes persistent registered replay to named ColumnCRT/PDEC objects. The fresh-layer admission firewall eliminates unnamed materialized PDEC/ColumnCRT in the current corpus. The current-frontier-zero certificate then states that no active branch-replay terminal instance remains in the materialized ledger.

Therefore an aperture-explosion claim cannot be kept as an unnamed terminal. It must specify its source family, blocker package, shape, phase map and persistence test, with a reproducible ledger. Such a future schema may reopen the branch, but it is not a present closed counterexample chain. This proves the schema-firewall synchronization only, not the final theorem. □

Theorem 7.11 (Q2 CRT ladder to final exact-source alignment)**.** *The Q2-stage CRT ladder has no remaining current unnamed local terminal: the controlled fresh tail is SAE, and aperture explosion must enter an explicit schema firewall. However, this does not by itself prove the row/column theorem. The CRT ladder controls positions and phases of allowed and forbidden classes; it does not produce a mass-dispersion estimate for the actual pre-Cauchy source over exact (u, v)-fibers. Therefore the Q2 branch can promote only through one of the final source inputs:*

$$\begin{gathered} \mathit{NonterminalExactUVFiberAperiodicityEstimateForActualPreCauchySource} \\ \text{or } \mathit{ExternalDIBFIKuznetsovDispersionTheoremMatch}, \end{gathered}$$

together with independent DStructure/Rankin promotion acceptance.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-q2-to-final-exact-source-alignment-router.md.`

The preceding Q2 routers prove that a full Q_2-wheel repeats the composite cover but destroys endpoint-prime replay, that continued endpoint replacement forces fresh endpoint layers, that controlled fresh tails are summable, and that current unnamed aperture explosion is barred by the schema firewall. Thus there is no current Q2-local terminal left to carry an infinite counterexample chain.

The remaining obstruction is of a different type. The final source anti-atom concerns the distribution of the actual signed pre-Cauchy source among exact (u, v)-fibers before Cauchy, pair energy and terminal extraction. CRT wheel rigidity gives congruence positions, not a bound on that source mass. Hence the strict self-contained branch must prove the nonterminal exact-UV fiber aperiodicity estimate, equivalently the already named source-domain rank/no-collapse package. The alternative is an accepted external DI/BFI/Kuznetsov dispersion theorem. In either case the separate DStructure/Rankin promotion gate remains open. This proves alignment of the remaining Q2 branch, not unconditional closure. □

Theorem 7.12 (Global CRT homogeneity frontier)**.** *The Q1/Q2 adjacent-prime carrier gives a genuine endpoint obstruction: at the full Q_2-wheel the two endpoint primes cannot replay as primes. Nevertheless this obstruction is not, by itself, a global CRT phase contradiction. For any squarefree wheel modulus M_Y and any fresh prime $r \nmid M_Y$, the lifts $a + tM_Y$, $0 \le t < r$, of a fixed old residue a run through all residue classes modulo r. Hence the fresh layer deletes exactly one class in each old fiber. Finite CRT growth is therefore a homogeneous one-class deletion mechanism, not a deterministic short-interval occupancy theorem, unless the additional phase equations form a registered PDEC/ColumnCRT collision.*

Consequently the pure global-CRT phase-contradiction outlet is closed as a final proof route. The remaining strict self-contained branch must pass through actual-source exact-(u, v) fiber dispersion and then the pointwise primitive alpha/delta kernel table. Its first active atom is

$$\mathit{AlphaRowAnchorPhaseEmissionFormulaLedger},$$

together with the independent pre-Cauchy arithmetic identity, the same-unit rank/multiplicity certificate, and independent DStructure/Rankin promotion acceptance.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-global-crt-homogeneity-frontier-router.md.`

The previous Q2 routers already prove endpoint inversion, fixed-period removal, controlled fresh-tail SAE absorption, and the aperture-explosion schema firewall. These remove the local endpoint-replay and unnamed moving outlets. The remaining question is whether the infinite CRT wheel itself forces a global phase contradiction.

The lift calculation above is exact because M_Y is invertible modulo the fresh prime r. As t varies modulo r, the class $a + tM_Y$ also varies through all classes modulo r; precisely one value is 0. Thus each finite wheel extension applies the same relative deletion to every old fiber. The Euler product explains the critical density scale, but it does not supply a lower bound for the actual mass in a moving interval of length P near P^2.

Therefore CRT position and phase rigidity must be supplemented by actual source information. The already archived strict descent identifies that source information with exact-(u, v) fiber aperiodicity and then with the pointwise primitive alpha/delta kernel table. Since the alpha-row emission formula, the signed pre-Cauchy identity, the rank/multiplicity certificate, and DStructure/Rankin promotion acceptance remain open, this theorem is a frontier synchronization, not a proof of the row/column theorem. □

Theorem 7.13 (Global CRT route to terminal saturation). *The global CRT/Q1-Q2 route does not stop at the coarse alpha-row atom. After the pure CRT homogeneity firewall is imported, the local alpha-row anchor/phase formula has already been synchronized to the terminal PDEC/CleanKLS gate. The PDEC/CleanKLS gate has then been attacked to the current internal saturation frontier: the CleanKLS/Kuznetsov-DLS arm returns to the strict acyclic terminal family, the nonrecursive constructor/signed-lift breaker returns to the signed coordinate-source cycle, and the seed-cycle-cut branch is saturated. Therefore the current non-cyclic strict inputs are*

$$\begin{gathered} \mathit{AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate} \\ \text{or } \mathit{NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact}, \end{gathered}$$

together with the model-gap ledger, rate preservation for the moving-atom packet, and independent DStructure/Rankin promotion acceptance.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-global-crt-terminal-saturation-sync-router.md.`

The previous theorem leaves the strict route at the pointwise alpha/delta kernel table. The archived alpha-row terminal synchronization shows that the unsigned carry-shell, source-tuple and phase-wheel parts of the alpha formula are local geometric data, while the signed-lift and anchor-overload parts return to the global PDEC/CleanKLS terminal gate and the model-gap ledger.

The archived terminal split then separates the terminal gate into a direct same-set PDEC scope arm and a clean Kloosterman large-sieve arm. Continuing the clean arm through the Kuznetsov/DLS files returns to the strict acyclic terminal family rather than producing a self-contained log-saving theorem. Continuing the nonrecursive breaker returns to the signed source-coordinate cycle; continuing seed-cycle-cut returns to the same row-level fixed point unless a new joint emitter formula is supplied.

Thus the CRT route, the alpha-row route and the current terminal-family route all meet at the same saturated frontier. A proof must either supply the same-set PDEC scope match or a new explicit joint alpha/delta constructor formula, while also paying the model, rate and DStructure/Rankin gates. Since none of these has been proved here, this is a sharpening of the frontier, not unconditional closure. □

Theorem 7.14 (Global CRT route to branch-trace frontier). *The non-PDEC side of the current global CRT frontier can be sharpened further. The alternative "new explicit joint alpha/delta formula" is not a single undifferentiated atom. If it is split through the existing alpha-side, same-row, row-level and signed-source routes, it returns to the already registered source-coordinate fixed point. Therefore a genuinely new formula must be anti-split: it must produce atomic joint rows before alpha-side projection. The ordinary anti-split declaration still degrades to the old split cycle unless it contains built-in word/coefficient pairing, and that built-in pairing further requires an exact atomic joint branch trace. Thus the current strict frontier is*

$$\begin{gathered} \mathit{AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate} \\ \text{or } \mathit{ExactAtomicJointBranchTraceSignedCoefficientFormulaOrReturn}, \end{gathered}$$

with the ExactUV, model-gap, rate-preservation and DStructure/Rankin gates still present.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-global-crt-branch-trace-frontier-router.md.`

The same-set PDEC scope arm remains an independent possible new certificate, but it is not proved in the present strict corpus. On the other arm, the archived joint-formula obligation says that the old joint declaration, alpha-side primitive rule and same-row origin identity return to the signed-source fixed point. The anti-split router therefore requires a non-split joint primitive word/coefficient formula. The declaration firewall then sharpens this to atomic pre-Cauchy joint rows with built-in pairing, and the atomic pairing router sharpens it to a built-in signed coefficient pairing closed form.

The signed coefficient is odd branch data: it depends on orientation and local factors, not only on the unsigned CRT skeleton. Hence the closed-form frontier is an exact atomic branch trace that, before Cauchy, Phi and payment pushforward, gives the basis word, signed coefficient, alpha/delta payload, orientation/local factor, exact (u, v), and named returns in the same formal unit. The current corpus has not supplied that trace formula. This proves the frontier sharpening only, not the row/column theorem. ☐

Theorem 7.15 (Global CRT route to signed-payload frontier)**.** *After importing the strict atomic-payload audit, the branch-trace side of the current global CRT frontier sharpens from*

$$\mathit{ExactAtomicJointBranchTraceSignedCoefficientFormulaOrReturn}$$

to

$$\mathit{AtomicSignedPayloadTraceConstructorBeforeAssignmentOrReturn}.$$

Moreover, in the present self-contained internal corpus the PDEC same-set arm is saturated: unless a new acyclic same-set scope certificate, or an external no-projection DI/BFI input, is supplied, that arm returns to the new-joint formula line, and the new-joint line has already been routed through branch trace to signed payload. Thus the internal self-contained frontier is

$$\begin{aligned}&\mathit{AtomicSignedPayloadTraceConstructorBeforeAssignmentOrReturn}\\&\quad\text{and } \mathit{ActualEmitterExactUVBoundedMultiplicityIncidenceTheorem}\\&\quad\text{and } \mathit{ExplicitModelGapAndFiniteDPRCLedger}\\&\quad\text{and } \mathit{RatePreservationLedger\_FOR\_moving\_atom\_packet}\\&\quad\text{and } \mathit{DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance}.\end{aligned}$$

If a new PDEC scope input is retained, the global activity basis is instead

$$\begin{aligned}&\mathit{AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate}\\&\quad\text{or } \mathit{AtomicSignedPayloadTraceConstructorBeforeAssignmentOrReturn},\end{aligned}$$

with the same ExactUV, model, rate and DStructure/Rankin gates.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-global-crt-signed-payload-sync-router.md.`

The preceding theorem left the non-PDEC side at an exact atomic branch trace. The archived signed-payload audit separates such a trace into its visible coordinate part and its signed payload part. The coordinate part follows the CRT anchor, phase and word-coordinate data. The payload part consists of the orientation bit, local-factor product, signed coefficient, same-row word/coefficient identity, alpha/delta prepushforward identity and exact (u, v) key. These are not consequences of the unsigned CRT skeleton.

The PDEC arm is treated separately. The archived scope-saturation audit says that current internal PDEC arguments either require a new same-set scope certificate, require an external no-projection DI/BFI input, or return to the new-joint formula line. The branch-trace synchronization has already routed that new-joint line to exact atomic branch trace, and the payload audit routes exact atomic branch trace to signed payload. Therefore the self-contained internal route has the stated payload frontier, while the new PDEC scope route is retained as an independent unproved input. No signed payload constructor, PDEC scope certificate, ExactUV incidence theorem, model-gap ledger, rate preservation ledger, or DStructure/Rankin promotion is proved here, so this is a frontier sharpening rather than unconditional closure. □

Theorem 7.16 (Predecessor-gap P-CRT uniformity route)**.** *Let $p^- < P$ be consecutive primes. A large gap $P - p^-$ creates a local first-row actual-prime deficit in the columns $p^- + 1, \ldots, P - 1$, but it does not create an imbalance in the complete P-level CRT wheel. In the full period $M_{\le P} = PM_{<P}$, before imposing the P-zero class, each residue class modulo P contains exactly $\varphi(M_{<P})$ residues coprime to $M_{<P}$. After imposing the P-zero class, the P-column is deleted and every nonzero column $c \in \mathbb{F}_P^*$ still contains exactly $\varphi(M_{<P})$ surviving CRT classes. The reflection $n \mapsto -n \pmod{M_{\le P}}$ also pairs c with $P - c$ exactly. Therefore a predecessor-gap argument can close the row/column theorem only if it supplies a localization transfer from this complete-wheel identity to the initial $P \times P$ square, or else a new PDEC scope or signed-payload input.*

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-predecessor-gap-pcrt-uniformity-router.md.`

The complete-wheel count is a direct CRT bijection. For every unit $a \bmod M_{<P}$ and every $c \bmod P$, there is a unique residue $n \bmod PM_{<P}$ satisfying $n \equiv a \bmod M_{<P}$ and $n \equiv c \bmod P$. Hence each P-column has $\varphi(M_{<P})$ such classes before the P-sieve. The P-sieve removes exactly the class $c = 0$, leaving all nonzero columns equal. Negation gives the stated reflection symmetry.

The gap $P - p^-$ is different in kind. It says that several first-row columns contain no prime atom, not that the full wheel has unequal nonzero columns. Passing from the full-period identity to the initial $P \times P$ arc is precisely a short localization problem. The archived scan for $P \le 5000$ records the largest predecessor gaps and observes that the first-row missing columns are repaired later in those finite squares, but this empirical repair is not used as proof. Without a localization transfer, the route returns to the current frontier: same-set PDEC scope, signed payload, ExactUV, model, rate and DStructure/Rankin gates. Thus this theorem closes a false shortcut, not the row/column theorem. □

Theorem 7.17 (Localized P-CRT transfer and the Linnik–2 barrier)**.** *The column side of*

$$LocalizedPCRTColumnUniformityTransferToInitialPxPSquare$$

is equivalent to the following pointwise least-prime-in-progressions assertion for prime moduli:

$$\forall c \in \mathbb{F}_P^* \quad \exists\, \ell \le P^2 \quad \textit{prime with} \quad \ell \equiv c \pmod{P}.$$

Thus any proof that localizes complete P-wheel nonzero-column uniformity to the initial $P \times P$ square must either prove this Linnik–2 type pointwise AP assertion, or else introduce a genuinely new structural input such as a same-set PDEC scope certificate or a signed-payload constructor.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-localized-pcrt-transfer-linnik2-barrier-router.md.`

The c-th non-P column of the initial square is exactly the finite progression

$$c,\ c+P,\ c+2P,\dots,c+(P-1)P,$$

intersected with the prime numbers. Since $1 \le c < P$, every term is congruent to $c \bmod P$ and is at most P^2. Therefore the column has a prime if and only if the displayed pointwise AP assertion holds for that residue c. Requiring this for every nonzero column gives the asserted equivalence.

The complete CRT wheel does not contain this pointwise first-prime information. It says that the nonzero columns have equal total mass over a full period $M_{\le P}$, while the column theorem asks for a hit in the first P terms of each residue class. Mean AP estimates or complete-period symmetry can at best control averaged exceptions; they do not remove a single bad residue class. The archived finite scan up to $P \le 3000$ records no missing class and a largest first-prime row 108, but this is only diagnostic. In the absence of a new pointwise AP theorem, the route returns to the active frontier consisting of PDEC scope, signed payload, ExactUV, model, rate and DStructure/Rankin gates. Hence this is a barrier identification, not an unconditional proof. □

Theorem 7.18 (Terminal-row CRT atoms and the square-root gate)**.** *For $P = 5$, the terminal-row and next-row atoms 23 and 29 are units modulo $M_{\le 5} = 2\cdot 3\cdot 5$. For $P = 7$, the atoms $43, 47$ and 53 are units modulo $M_{\le 7} = 2\cdot 3\cdot 5\cdot 7$. Hence none of these atoms can be absorbed by a prime factor $\le P$. More generally, if p_+ is the next prime after P and*

$$1 < n < p_+^2, \qquad (n, M_{\le P}) = 1,$$

then n is prime.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-terminal-row-crt-atom-router.md.`

For $P = 5$,

$$23 \equiv (1,2,3) \pmod{(2,3,5)}, \qquad 29 \equiv (1,2,4) \pmod{(2,3,5)}.$$

For $P = 7$,

$$\begin{aligned} 43 &\equiv (1,1,3,1) \pmod{(2,3,5,7)},\\ 47 &\equiv (1,2,2,5) \pmod{(2,3,5,7)},\\ 53 &\equiv (1,2,3,4) \pmod{(2,3,5,7)}. \end{aligned}$$

Every displayed coordinate is nonzero, so the hypothesis that any of these numbers is divisible by a prime $\le P$ contradicts its CRT vector.

For the general square-root gate, suppose n is composite and $(n, M_{\le P}) = 1$. Then the least prime factor of n is strictly larger than P, hence at least p_+. A composite integer all of whose prime factors are at least p_+ is at least p_+^2, contradicting $n < p_+^2$. Thus n is prime.

This proves an atom-level principle: a reduced CRT atom in the terminal row or the immediate post-square row is automatically a true prime. It does not prove that every prime P has such an atom in a prescribed short row. Complete CRT-period symmetry supplies full-period uniformity and reflection, but not the local pointwise existence needed in the initial square. The remaining global input is therefore a terminal-row reduced-residue existence theorem, the localized P-CRT/Linnik–2 route, or the already registered AP zero-packet, Page-sparsity, signed-payload, or PDEC frontiers. □

Theorem 7.19 (Terminal rows as square-phase Jacobsthal special phases)**.** *For $1 \le r < P$, the next-row and last-row reduced atoms are exactly*

$$(P^2 + r, M_{<P}) = 1, \qquad (P^2 - r, M_{<P}) = 1.$$

Equivalently, the next-row full-cover failure is the covering of $r = 1, \dots, P-1$ by the forbidden classes

$$r \equiv -P^2 \pmod q, \qquad q < P,$$

and the last-row full-cover failure is the covering by

$$r \equiv P^2 \pmod q, \qquad q < P.$$

Thus either full cover forces the square phase P^2 to start a covered block of length $P-1$ in the primorial period $M_{<P}$.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-terminal-row-square-phase-bridge-router.md.`

For every prime $q < P$, since P is a unit modulo q,

$$q \mid P^2 + r \iff r \equiv -P^2 \pmod q,$$

and

$$q \mid P^2 - r \iff r \equiv P^2 \pmod q.$$

Therefore the absence of a reduced atom on one side is exactly the statement that these one-residue-per-q forbidden classes cover the whole interval $[1, P-1]$. In the primorial period this is precisely a consecutive covered block of length $P-1$ beginning at the corresponding square phase.

The full-period Jacobsthal shortcut is not available in the present corpus: known and archived finite data already show primorial periods with covered blocks of length at least $P-1$. The useful remaining target is not a uniform period-wide maximum-run bound, but the special-phase assertion that the particular phase P^2 cannot align with such a long block, or that such alignment is a PDEC/SAE/ColumnCRT defect. Hence this is a route sharpening, not an unconditional row/column proof. □

Theorem 7.20 (Linnik–2 defects and nonprincipal character obstruction)**.** *For a prime modulus P, put*

$$\vartheta_a(P) = \sum_{\substack{\ell \le P^2 \\ \ell \text{ prime} \\ \ell \equiv a \pmod P}} \log \ell, \qquad a \in \mathbb{F}_P^*,$$

and for each Dirichlet character χ mod P put

$$T_\chi(P) = \sum_{a \in \mathbb{F}_P^*} \chi(a) \vartheta_a(P).$$

Let $T_0(P)$ denote the principal-character mass and let

$$N_a(P) = \sum_{\chi \ne \chi_0} \overline{\chi(a)} T_\chi(P)$$

be the nonprincipal projection at the class a. Then

$$\vartheta_a(P) = \frac{T_0(P) + N_a(P)}{P-1}.$$

Consequently a Linnik–2 zero-column defect, namely $\vartheta_a(P) = 0$ for some $a \in \mathbb{F}_P^*$, forces the exact negative phase alignment

$$N_a(P) = -T_0(P)$$

and hence

$$\sum_{\chi \neq \chi_0} |T_\chi(P)|^2 \geq \frac{T_0(P)^2}{P-2}.$$

Thus the remaining AP route is equivalent to excluding this pointwise nonprincipal negative projection at $x = P^2$, or else returning to the same-set PDEC scope or signed-payload structural frontier.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-linnik2-nonprincipal-character-obstruction-router.md.`

All identities are finite orthogonality identities on the group $\mathbb{F}_P^*$. Since

$$T_\chi(P) = \sum_{b \in \mathbb{F}_P^*} \chi(b)\vartheta_b(P),$$

character inversion gives the displayed formula for $\vartheta_a(P)$. If a nonzero column has no prime below P^2, then $\vartheta_a(P) = 0$, so the nonprincipal projection must cancel the entire principal mass. Applying Cauchy's inequality to

$$T_0(P) = \left| \sum_{\chi \neq \chi_0} \overline{\chi(a)} T_\chi(P) \right|$$

and using $\sum_{\chi \neq \chi_0} |\chi(a)|^2 = P - 2$ gives the stated energy lower bound.

The archived finite scan up to $P \leq 3000$ finds no theta-zero residue class, with largest normalized negative theta deficit about 0.686688 at $P = 73$, but this empirical fact is not used in the proof. The theorem only sharpens the hard point: complete CRT symmetry and averaged AP estimates do not by themselves exclude the required negative nonprincipal projection. No pointwise projection bound, PDEC scope certificate, signed-payload constructor, ExactUV incidence theorem, model-gap ledger, rate-preservation ledger, or DStructure/Rankin promotion is proved here. □

Theorem 7.21 (Energy capacity and rank-one phase for the Linnik–2 defect)**.** *In the notation of the preceding theorem, set*

$$\rho_a(P) = -\frac{N_a(P)}{T_0(P)}.$$

Then

$$\rho_a(P) = 1 - \frac{\vartheta_a(P)}{T_0(P)/(P-1)}.$$

Thus a zero-column defect is exactly the rank-one condition $\rho_a(P) = 1$. The energy condition

$$\sum_{\chi \neq \chi_0} |T_\chi(P)|^2 \geq \frac{T_0(P)^2}{P-2}$$

is only a necessary condition for such a defect. Therefore an L^2 capacity argument can close this route only if it proves the strict reverse inequality. Otherwise the remaining obligation is a pointwise phase statement: exclude a negative projection of size $T_0(P)$ in one evaluation direction.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-linnik2-rankone-phase-capacity-router.md.`

The first identity follows directly from $\vartheta_a(P) = (T_0(P) + N_a(P))/(P-1)$. Hence $\vartheta_a(P) = 0$ if and only if $\rho_a(P) = 1$. The displayed energy lower bound is exactly the Cauchy consequence proved in the preceding theorem. Its converse is false as a matter of logic: total nonprincipal energy above the floor does not force the energy to point in the single evaluation direction attached to a.

The evaluation vectors $v_a = (\overline{\chi(a)})_{\chi \neq \chi_0}$ satisfy

$$\langle v_a, v_b \rangle = \begin{cases} P-2, & a = b, \\ -1, & a \neq b, \end{cases}$$

so the residue tests are simplex directions in the nonprincipal character space. The archived scan up to $P \le 3000$ records 423 prime moduli for which the total-energy floor is already exceeded while no zero residue class exists. This diagnostic is not a proof input; it identifies why the capacity route is too coarse. The remaining named input is the rank-one negative evaluation projection exclusion at $x = P^2$, or else a return to same-set PDEC scope or signed payload. □

Theorem 7.22 (Three-claim frontier and the sharp AP zero-packet barrier)**.** *The three main claims currently have distinct frontiers. The row/column chain has been sharpened to the rank-one condition*

$$\rho_a(P) < 1 \qquad (a \in \mathbb{F}_P^*),$$

which is equivalent to

$$\vartheta_a(P) > 0 \qquad (a \in \mathbb{F}_P^*).$$

The two-point secondary-sieve chain remains conditional on the I3-Core true residual total-incidence input, or on the external DI/BFI–Kloosterman large-sieve route. The RH contradiction-field chain remains a verification package awaiting independent acceptance of its controlled exits.

For the row/column AP branch, the rank-one condition is therefore precisely the sharp positivity statement

$$\theta(P^2; P, a) > 0 \quad \textit{for every } a \in \mathbb{F}_P^*.$$

In explicit-formula language this requires the signed nonprincipal zero packet in each residue class to be strictly smaller than the main term, which is of order P. Complete CRT averages, Bombieri–Vinogradov type mean estimates, standard Linnik bounds with exponent > 2, and GRH-shape errors $O(P \log^2 P)$ do not by themselves imply this $x = P^2$ pointwise positivity.

Proof. The certificate is recorded in

`docs/monograph/three-claims-frontier-rankone-explicit-formula-router.md.`

The equivalence $\rho_a(P) < 1 \iff \vartheta_a(P) > 0$ follows from the identity

$$\rho_a(P) = 1 - \frac{\vartheta_a(P)}{T_0(P)/(P-1)}$$

proved in the preceding theorem. Since $\vartheta_a(P)$ is exactly the Chebyshev prime weight in the class a mod P up to P^2, positivity is equivalent to a prime in that nonzero class below the square barrier.

The explicit formula for $\theta(P^2; P, a)$ has a principal term of size about P, plus signed nonprincipal zero packets and controlled trivial terms. Therefore the sharp remaining analytic input is a pointwise bound on those signed packets for every a. Mean estimates and full-period CRT identities only control averages, not a single exceptional simplex direction. GRH-shape errors at $x = P^2$ are of size $P \log^2 P$, too large to beat a main term of size P without additional sharp constants or cancellation. Thus this theorem is a frontier separation and barrier identification, not a promotion of any of the three claims to final unconditional theorem status. □

Theorem 7.23 (Explicit AP zero-packet Siegel split)**.** *The remaining row/column AP input*

`ExplicitAPZeroPacketBoundBeatingMainTermAtXEqualsP2`

splits into two genuine analytic branches. First, any exceptional real zero β of a real nonprincipal character modulo P contributes at $x = P^2$ on the scale

$$P^{2\beta-1}/\beta.$$

If $\beta = 1 - \lambda/\log P$, this is $P\exp(-2\lambda)/\beta$, hence still of the same order as the principal term for bounded λ. This is the square-scale Siegel-bias branch. Second, after extracting any real exceptional packet, the nonreal zeros leave a signed residue packet; closing it requires pointwise phase cancellation in every evaluation direction $a \bmod P$, not just average energy or complete CRT uniformity.

Consequently the current AP branch is not a single unnamed zero-packet error. It is the conjunction

```
SiegelExceptionalBiasExclusionAtSquareScale
AND NonrealZeroPacketResiduePhaseCancellationAtXEqualsP2,
```

with the trivial and finite explicit-formula terms kept as a downstream constant ledger once a positive main-term margin is available.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-explicit-ap-zero-packet-siegel-split-router.md.`

The character explicit formula for $\theta(P^2; P, a)$ has a principal term of size P and a signed sum of nonprincipal zero terms. A real zero close to 1 cannot be dismissed by a mean theorem, because its square-scale contribution remains comparable to the main term on one of the two real character half-classes. Deuring–Heilbronn repulsion organizes the remaining zeros but does not itself prove that this biased half-class is positive before P^2.

After this real branch is separated, the nonreal packet is still pointwise in the residue a. Bombieri–Vinogradov, complete CRT identities, total nonprincipal energy bounds, and GRH-shape estimates do not exclude a single rank-one evaluation direction from carrying too much signed mass at the square barrier. Therefore the latest analytic AP route remains open and must either prove the two displayed branches or return to the already named structural inputs: same-set PDEC scope, atomic signed payload, exact-(u, v) multiplicity, model/rate preservation, and the D-structure Rankin promotion gate. No unconditional row/column theorem is claimed here. □

Theorem 7.24 (Siegel branch as a quadratic half-class margin)**.** *For prime modulus P, let χ_P be the quadratic character and define*

$$T_\chi(P) = \sum_{\substack{\ell \le P^2 \\ \ell \text{ prime},\ \ell \ne P}} \chi_P(\ell) \log \ell, \qquad T_0(P) = \sum_{\substack{\ell \le P^2 \\ \ell \text{ prime},\ \ell \ne P}} \log \ell.$$

Then the real-character part of the Siegel branch is measured by

$$r_{\text{quad}}(P) = \frac{|T_\chi(P)|}{T_0(P)}.$$

The dangerous quadratic half-class retains an average main-term margin proportional to $1-r_{\mathrm{quad}}(P)$*. Thus the square-scale Siegel branch is reduced to either proving the margin theorem*

`QuadraticHalfClassSquareScaleBiasMarginTheorem`,

or accepting an independently certified effective no-Siegel-zero or β*-gap input strong enough at* $x = P^2$*.*

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-siegel-quadratic-halfclass-margin-router.md`.

The identity is the specialization of the nonprincipal character expansion to the unique real quadratic character modulo P. The sum T_χ is the difference between the Chebyshev weights falling in the quadratic-residue and quadratic-nonresidue half-classes. If the real projection almost equals the principal mass in magnitude, one half-class may have only a small average margin left before the nonreal packet and finite terms are charged. Conversely, any explicit bound $r_{\mathrm{quad}}(P) \le 1 - \eta(P)$ supplies exactly that half-class margin.

The archived finite scan $7 \le P \le 1000$ records a maximum diagnostic ratio about 0.13027, attained at $P = 7$. This scan is not used as an infinite proof. The theorem only identifies the exact remaining self-contained target for the real branch. The nonreal zero-packet phase-cancellation branch and the structural PDEC/signed-payload/ExactUV/D-structure inputs remain open. □

Theorem 7.25 (Nonreal packet as two half-class simplexes)**.** *After removing the principal character and the quadratic character, define*

$$R_a = (P-1)\theta(P^2; P, a) - T_0(P) - \chi_P(a)T_\chi(P).$$

Let w_a *be the evaluation vector over the remaining nonreal characters. Then*

$$\langle w_a, w_b\rangle = \begin{cases} P-3, & a = b, \\ -2, & a \neq b \text{ and } \chi_P(a) = \chi_P(b), \\ 0, & \chi_P(a) \neq \chi_P(b). \end{cases}$$

Thus the nonreal packet splits into two orthogonal regular simplexes indexed by the quadratic-residue and quadratic-nonresidue half-classes. A zero column in class a *would force*

$$R_a = -T_0(P) - \chi_P(a)T_\chi(P),$$

so the nonreal packet must supply a rank-one negative projection of size at least

$$(1 - |T_\chi(P)|/T_0(P))T_0(P).$$

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-nonreal-halfclass-simplex-phase-router.md`.

The character orthogonality relation over $\mathbb{F}_P^*$ gives $\sum_\chi \overline{\chi(a)}\chi(b) = P - 1$ if $a = b$, and 0 otherwise. Removing the principal character subtracts 1, and removing the quadratic character subtracts $\chi_P(a)\chi_P(b)$. This gives the displayed inner products. The identity for R_a is the inverse character expansion with those two components removed. Setting $\theta(P^2; P, a) = 0$ gives the displayed zero-column requirement.

Cauchy's inequality gives a necessary energy floor

$$E_{\mathrm{nonreal}} \geq \frac{T_0(P)^2(1 - |T_\chi(P)|/T_0(P))^2}{P-3},$$

but this is still only a capacity condition. The archived scan $7 \leq P \leq 1000$ records many cases where the nonreal energy is above this floor while no zero column occurs. Hence the remaining analytic AP input is not an energy statement, but

`NonrealHalfClassSimplexRankOneProjectionExclusionAtP2.`

No unconditional row/column theorem is claimed. □

Theorem 7.26 (Half-class compensation variance for the nonreal packet)**.** *Keep the notation of the preceding theorem and fix $s \in \{+1, -1\}$. Let*

$$H_s = \{a \in \mathbb{F}_P^* : \chi_P(a) = s\}, \qquad h = |H_s| = \frac{P-1}{2},$$

and define the half-class mean

$$\mu_s = \frac{T_0(P) + sT_\chi(P)}{P-1}.$$

For $a \in H_s$ the nonreal residual is exactly

$$R_a = (P-1)\big(\theta(P^2; P, a) - \mu_s\big),$$

and therefore

$$\sum_{a \in H_s} R_a = 0.$$

If a zero column occurs at $a_0 \in H_s$, then

$$R_{a_0} = -(P-1)\mu_s, \qquad \sum_{\substack{a \in H_s \\ a \neq a_0}} R_a = (P-1)\mu_s.$$

Consequently the sharp one-hole variance floor is

$$\sum_{a \in H_s} R_a^2 \geq ((P-1)\mu_s)^2 \frac{h}{h-1}.$$

Equality is attained, at the level of the centered half-class geometry, by the flat compensator in which the non-hole residues all have

$$\theta(P^2; P, a) = \mu_s \frac{h}{h-1}.$$

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-nonreal-halfclass-compensation-variance-router.md.`

Since $T_0 + sT_\chi = 2\sum_{a \in H_s} \theta(P^2; P, a)$, the mean over H_s is μ_s. Substituting this identity into the previous residual formula gives $R_a = (P-1)(\theta_a - \mu_s)$, and summing over H_s gives the centering identity.

If $\theta_{a_0} = 0$, then the displayed value of R_{a_0} follows immediately, and the compensation identity follows from centering. Cauchy's inequality applied to the remaining $h-1$ residues gives

$$\sum_{\substack{a\in H_s\\ a\neq a_0}} R_a^2 \geq \frac{((P-1)\mu_s)^2}{h-1},$$

so adding $R_{a_0}^2$ gives the stated floor. Equality requires all remaining residuals to be equal, i.e. the flat compensator above.

Thus the previous energy floor is not only necessary but convexly sharp. It cannot by itself exclude a zero column, because the flat compensator is a nonnegative centered half-class vector. The remaining self-contained obligation is therefore the actual prime-phase input

`HalfClassCompensationMassNonconcentrationOrColumnCRTPDEC,`

which must either rule out such flat or near-flat compensation for the true prime-induced vector at $x = P^2$, or route persistent compensation phases to a registered ColumnCRT/PDEC or moving-family return. No unconditional row/column theorem is claimed here. □

Theorem 7.27 (Ratio-Fourier lock for half-class compensation)**.** *Let $Q \subset \mathbb{F}_P^*$ be the subgroup of quadratic residues and fix a candidate puncture $a_0 \in H_s$. Put*

$$f_{a_0}(u) = \theta(P^2; P, a_0 u), \qquad u \in Q,$$

so that $u = 1$ is the punctured coordinate. Let

$$z = f_{a_0}(1), \qquad c_{a_0} = \frac{\sum_{u\in Q,\ u\neq 1} f_{a_0}(u)}{h-1}, \qquad \mu_s = \frac{z + (h-1)c_{a_0}}{h}.$$

Then the excess over the one-point variance floor is exactly

$$\sum_{a\in H_s} R_a^2 - (P-1)^2(z-\mu_s)^2\frac{h}{h-1} = (P-1)^2 \sum_{\substack{u\in Q\\ u\neq 1}} \bigl(f_{a_0}(u) - c_{a_0}\bigr)^2.$$

If $\widehat{f}(\psi) = \sum_{u\in Q} f_{a_0}(u)\overline{\psi(u)}$, where ψ ranges over the characters of Q, then the flat compensator

$$f_{a_0}(u) = c_{a_0} \qquad (u \neq 1)$$

is equivalent to the simultaneous lock

$$\widehat{f}(\psi) = z - c_{a_0} \qquad (\psi \neq 1).$$

Moreover,

$$\sum_{\psi\neq 1} \left|\widehat{f}(\psi) - (z - c_{a_0})\right|^2 = h \sum_{\substack{u\in Q\\ u\neq 1}} \bigl(f_{a_0}(u) - c_{a_0}\bigr)^2.$$

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-halfclass-ratio-fourier-lock-router.md.`

The normalization by a_0^{-1} maps the half-class H_s bijectively to Q, because a_0 has the same quadratic character as every element of H_s. Write

$$f_{a_0}(u) = \begin{cases} z, & u = 1, \\ c_{a_0} + d_u, & u \neq 1, \end{cases} \qquad \sum_{u\neq 1} d_u = 0.$$

Substituting this decomposition into $\sum_{a\in H_s} R_a^2 = (P-1)^2 \sum_{u\in Q}(f_{a_0}(u) - \mu_s)^2$ gives the displayed one-point excess identity; the cross term vanishes because the d_u sum to zero.

For the Fourier statement, if $d_u = 0$ for all $u \neq 1$, then for every nontrivial ψ

$$\widehat{f}(\psi) = z + c_{a_0} \sum_{u\neq 1} \overline{\psi(u)} = z - c_{a_0}.$$

Conversely, if all nontrivial coefficients equal $z - c_{a_0}$, Fourier inversion on the finite abelian group Q gives $d_u = 0$ for every $u \neq 1$. Applying Parseval to the function $d_1 = 0$, $d_u = f_{a_0}(u) - c_{a_0}$ for $u \neq 1$, gives the final identity.

Thus a flat zero-column compensator is not merely a low-variance event: after normalization by the missing residue it is an all-frequency lock on the quadratic-residue subgroup. The remaining input is the exclusion of persistent locks of this kind, or their registration as ColumnCRT/PDEC or moving-family returns. No unconditional row/column theorem is claimed here. ☐

Theorem 7.28 (Quadratic-twist pair character orbit lock)**.** *Let $\psi \neq 1$ be a character of the quadratic-residue subgroup Q, and let χ be either extension of ψ to $\mathbb{F}_P^*$. The other extension is $\chi\chi_P$. For $a_0 \in H_s$, define*

$$T_\chi(P) = \sum_{a\in\mathbb{F}_P^*} \theta(P^2; P, a)\overline{\chi(a)}.$$

Then the Q-Fourier coefficient in the preceding theorem satisfies

$$\widehat{f}(\psi) = \chi(a_0)\frac{T_\chi(P) + sT_{\chi\chi_P}(P)}{2}.$$

Consequently the flat compensator condition is equivalent to the simultaneous orbit lock

$$\chi(a_0)\frac{T_\chi(P) + sT_{\chi\chi_P}(P)}{2} = \theta(P^2; P, a_0) - c_{a_0} \qquad (\psi \neq 1).$$

Equivalently,

$$\sum_{\psi\neq 1} \left| \chi(a_0)\frac{T_\chi(P) + sT_{\chi\chi_P}(P)}{2} - (\theta(P^2; P, a_0) - c_{a_0}) \right|^2 = h \sum_{\substack{u\in Q \\ u\neq 1}} \left(f_{a_0}(u) - c_{a_0}\right)^2.$$

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-halfclass-twist-pair-character-lock-router.md.`

Since χ restricts to ψ on Q, for $a = a_0 u$ with $u \in Q$ we have $\overline{\psi(u)} = \chi(a_0)\overline{\chi(a)}$. Also the indicator of H_s is $(1 + s\chi_P(a))/2$. Therefore

$$\widehat{f}(\psi) = \chi(a_0) \sum_{a\in H_s} \theta(P^2; P, a)\overline{\chi(a)} = \chi(a_0)\frac{T_\chi(P) + sT_{\chi\chi_P}(P)}{2}.$$

Substituting this identity into the ratio-Fourier lock theorem gives the orbit lock and Parseval identity.

This reformulation is important for the global route: the obstruction is not a single exceptional character and not a scalar energy excess. A persistent zero-column compensator must synchronize every nontrivial character of Q, or equivalently every quadratic-twist pair of Dirichlet characters modulo P, onto the same a_0-dependent phase orbit. The remaining input is

`QuadraticTwistPairCharacterOrbitLockExclusionOrColumnCRTPDEC.`

No unconditional row/column theorem is claimed here. □

Theorem 7.29 (Log-prime degeneration of exact half-class orbit locks)**.** *For $a \in \mathbb{F}_P^*$, put*

$$S_a(P) = \{\ell \le P^2 : \ell \text{ prime},\ \ell \ne P,\ \ell \equiv a \bmod P\}$$

and $\theta_a = \sum_{\ell \in S_a(P)} \log \ell$. If $a \ne b$ and $\theta_a = \theta_b$, then $S_a(P) = S_b(P) = \varnothing$.

Consequently, for $P \ge 7$, an exact punctured half-class flatness condition

$$\theta(P^2; P, a_0 u) = c \qquad (u \in Q,\ u \ne 1)$$

can hold only in the degenerate case $c = 0$ and

$$S_{a_0 u}(P) = \varnothing \qquad (u \in Q,\ u \ne 1).$$

Thus the exact quadratic-twist pair orbit lock from the preceding theorem is routed to

`PuncturedHalfClassZeroSupportDegeneracyExclusionOrColumnCRTPDEC.`

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-halfclass-log-independence-degeneracy-router.md.`

If $\theta_a = \theta_b$, then

$$\sum_{\ell \in S_a(P)} \log \ell - \sum_{\ell \in S_b(P)} \log \ell = 0.$$

Equivalently,

$$\prod_{\ell \in S_a(P)} \ell = \prod_{\ell \in S_b(P)} \ell.$$

Unique factorization forces the two prime multisets to be identical. Distinct nonzero residues modulo P are disjoint prime supports, so both supports must be empty.

For $P \ge 7$, the punctured set $\{u \in Q : u \ne 1\}$ contains at least two elements. Exact flatness makes all corresponding θ-values equal. Applying the first paragraph to any pair and then to a fixed representative against every other punctured residue gives empty support throughout the punctured half-class and $c = 0$.

Therefore a positive exact flat compensator cannot be the persistent obstruction. The only remaining exact-lock exit is the much sharper zero-support degeneracy: after removing a_0, the same half-class has no prime arrivals up to P^2. This theorem does not exclude that degeneracy and does not prove the row/column theorem unconditionally. □

Theorem 7.30 (Half-class mass versus single-residue capacity)**.** *Let*

$$\Theta_s(P) = \sum_{\chi_P(a)=s} \theta(P^2; P, a) = \frac{T_0(P) + sT_\chi(P)}{2}, \qquad s \in \{\pm 1\}.$$

For every $a \in \mathbb{F}_P^*$,

$$\theta(P^2; P, a) \le P \log(P^2) = 2P \log P.$$

Consequently a punctured half-class zero-support degeneracy can occur only if

$$\min_{s=\pm 1} \Theta_s(P) \le 2P \log P.$$

Equivalently, the sufficient exclusion gate is

$$T_0(P) \left(1 - \frac{|T_\chi(P)|}{T_0(P)}\right) > 4P \log P.$$

Thus the latest exact-lock exit is routed to

QuadraticHalfClassMassBeatsSingleResidueCapacityAtP2,

or else to a named ColumnCRT/PDEC capacity-failure registration.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-halfclass-single-residue-capacity-router.md.`

In a fixed nonzero residue $a \bmod P$, the integers not exceeding P^2 have the form $a + kP$, $0 \le k \le P - 1$. Hence at most P prime terms can occur in this residue before the square barrier, and each carries weight at most $\log(P^2)$. This gives the displayed single-residue capacity bound.

If a punctured half-class zero-support degeneracy occurs in H_s, then all prime weight in that half-class is concentrated in the single unpunctured residue a_0. Therefore

$$\Theta_s(P) = \theta(P^2; P, a_0) \le 2P \log P.$$

Taking the contrapositive gives the half-class capacity exclusion. Finally, since

$$\min_s \Theta_s(P) = \frac{T_0(P) - |T_\chi(P)|}{2},$$

the condition is exactly

$$T_0(P) \left(1 - \frac{|T_\chi(P)|}{T_0(P)}\right) > 4P \log P.$$

This is a strict narrowing of the previous support-degeneracy exit, but it is not yet a proof of the required global half-class mass lower bound. No unconditional row/column theorem is claimed here. □

Theorem 7.31 (Capacity margin as a logarithmic-over-linear projection gap)**.** *The half-class single-residue capacity gate*

$$\min_{s=\pm 1} \Theta_s(P) > 2P \log P$$

is equivalent to

$$1 - \frac{|T_\chi(P)|}{T_0(P)} > \frac{4P \log P}{T_0(P)}.$$

In particular, if a principal Chebyshev lower bound

$$T_0(P) \geq c_0(P)P^2$$

is available, then it is enough to prove

$$1 - \frac{|T_\chi(P)|}{T_0(P)} > \frac{4\log P}{c_0(P)P}.$$

Thus any persistent failure of the capacity gate, in the presence of a P^2-scale principal mass lower bound, forces an ultra-near-one quadratic projection defect

$$\frac{|T_\chi(P)|}{T_0(P)} \geq 1 - O\left(\frac{\log P}{P}\right).$$

Proof. The certificate is recorded in

```
docs/monograph/prime-matrix-halfclass-capacity-margin-factorization-router.md.
```

The identity

$$\min_s \Theta_s(P) = \frac{T_0(P) - |T_\chi(P)|}{2}$$

was proved in the preceding theorem. Therefore

$$\min_s \Theta_s(P) > 2P\log P$$

is equivalent to

$$\frac{T_0(P) - |T_\chi(P)|}{2} > 2P\log P,$$

and division by $T_0(P) > 0$ gives the stated projection-gap threshold. Substituting $T_0(P) \geq c_0(P)P^2$ gives the logarithmic-over-linear sufficient threshold.

This reformulation exposes the scale of the remaining obstruction. The capacity gate does not need a fixed-size half-class margin: a gap of order $\log P/P$, after the principal Chebyshev lower bound is in place, is already enough. Conversely, a counterexample surviving this gate must make the quadratic projection almost equal to the principal mass. The remaining input is

```
QuadraticProjectionGapBeatsLogOverPThresholdAtSquareScale,
```

or a named registration of the ultra-near-one projection defect as a ColumnCRT/PDEC/moving-family exit. No unconditional row/column theorem is claimed here. □

Theorem 7.32 (Low CRT insufficiency and the large splitting residual)**.** *Let*

$$\Theta_s(P) = \sum_{\substack{\ell \leq P^2,\ \ell \neq P \\ \chi_P(\ell) = s}} \log \ell, \qquad s \in \{+1, -1\},$$

where $\chi_P(\ell) = (\ell/P)$. Split

$$L_s(P) = \sum_{\substack{\ell < P \\ \chi_P(\ell) = s}} \log \ell, \qquad G_s(P) = \sum_{\substack{P < \ell \leq P^2 \\ \chi_P(\ell) = s}} \log \ell.$$

Then $\Theta_s(P) = L_s(P) + G_s(P)$, and the projection-gap capacity gate is equivalent to

$$\min_s \Theta_s(P) > 2P \log P.$$

Moreover the complete low CRT layer satisfies the elementary bound

$$L_+(P) + L_-(P) = \vartheta(P-1) \le P \log P < 2P \log P.$$

Consequently the low CRT layer alone cannot close the capacity gate. If the capacity gate fails, then for some sign s

$$G_s(P) \le 2P \log P.$$

Conversely, the large-splitting lower bound

$$\min_s G_s(P) > 2P \log P$$

is sufficient to close the capacity gate.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-quadratic-projection-large-splitting-router.md.`

The decomposition $\Theta_s = L_s + G_s$ is only the partition of the prime sum at the modulus P. The equivalence between the projection-gap gate and $\min_s \Theta_s(P) > 2P \log P$ is the preceding theorem.

For the low layer, every prime counted in $L_+ + L_-$ is $< P$, hence there are fewer than P summands and each summand is at most $\log P$. Thus $L_+ + L_- \le P \log P$. This is strictly below the minority-mass threshold $2P \log P$, so the low CRT period can be entirely absorbed by the failure allowance and cannot by itself force the required contradiction.

If the capacity gate fails, choose s with $\Theta_s(P) = \min_t \Theta_t(P) \le 2P \log P$. Since $L_s(P) \ge 0$, we have $G_s(P) \le 2P \log P$. Conversely, if both large layers satisfy $G_s(P) > 2P \log P$, then $\Theta_s(P) \ge G_s(P) > 2P \log P$ for both signs, which closes the capacity gate. Thus the remaining self-contained attack is

`LargePrimeQuadraticSplittingMinorityMassBeatsTwoPLogPAtSquareScale,`

or the registration of a persistent large-splitting desert as a Siegel/ColumnCRT/PDEC exit. No unconditional row/column theorem is claimed here. ☐

Theorem 7.33 (Large splitting as a beta-gap and zero-packet budget)**.** *Let*

$$G_s(P) = \sum_{\substack{P < \ell \le P^2 \\ \chi_P(\ell) = s}} \log \ell, \qquad H_0(P) = G_+(P) + G_-(P), \qquad H_\chi(P) = G_+(P) - G_-(P).$$

Then the large-splitting gate

$$\min_s G_s(P) > 2P \log P$$

is equivalent to

$$1 - \frac{|H_\chi(P)|}{H_0(P)} > \frac{4P \log P}{H_0(P)}.$$

Consequently, any explicit-formula estimate of the form

$$\frac{|H_\chi(P)|}{H_0(P)} \le R_\beta(P) + E_{\text{zero}}(P)$$

closes the large-splitting gate whenever

$$R_\beta(P) + E_{\mathrm{zero}}(P) < 1 - \frac{4P\log P}{H_0(P)}.$$

For the real-zero shadow model

$$R_\beta(P) = \frac{P^{2\beta} - P^\beta}{\beta H_0(P)}, \qquad \beta = 1 - \delta,$$

the critical no-residual value of δ is asymptotic to $2/P$ when $H_0(P) \sim P^2$.

Proof. The certificate is recorded in

```
docs/monograph/prime-matrix-large-splitting-beta-gap-router.md.
```

The first equivalence is the identity

$$\min_s G_s(P) = \frac{H_0(P) - |H_\chi(P)|}{2}.$$

Thus $\min_s G_s(P) > 2P\log P$ is exactly

$$H_0(P) - |H_\chi(P)| > 4P\log P,$$

and division by $H_0(P) > 0$ gives the stated projection-gap form.

The explicit-formula budget statement is purely algebraic: if the normalized character projection is bounded by $R_\beta + E_{\mathrm{zero}}$, then it is enough for that upper bound to be below the allowed ratio $1 - 4P\log P/H_0(P)$.

For the scale calculation, put $x = P^2$, $\beta = 1 - \delta$, and $H_0(P) \sim P^2$. Near $\delta = 0$,

$$\frac{P^{2\beta} - P^\beta}{\beta H_0(P)} = 1 - \delta(2\log P - 1) + o(\delta \log P) + o(1/P).$$

The required missing ratio is

$$\frac{4P\log P}{H_0(P)} \sim \frac{4\log P}{P}.$$

Equating these first-order quantities gives

$$\delta \sim \frac{4\log P/P}{2\log P - 1} \sim \frac{2}{P}.$$

Hence a persistent failure must come either from a real zero within the $1/P$ scale, or from a non-real zero packet, endpoint, or prime-power residual consuming the same slack. The remaining self-contained target is

```
SquareScaleQuadraticBetaGapAndZeroPacketResidualBudgetAtP2,
```

or the named exit

```
UltraCloseRealZeroOrCoherentZeroPacketLargeSplittingPDEC.
```

No unconditional row/column theorem is claimed here. $\square$

Theorem 7.34 (Page sparsity for ultra-close beta-gap carriers)**.** *Assume the Landau–Page uniqueness input in the following form: for primitive real characters of conductor $q \le Q$, at most one character has a real zero*

$$\beta > 1 - \frac{c_{\mathrm{Page}}}{\log Q}.$$

Let $E(Q)$ be the set of prime moduli $P \le Q$ for which the large-splitting beta-gap failure produces a real-zero carrier

$$\beta_P > 1 - \frac{C(P)}{P}, \qquad C(P) \le C_*.$$

Then

$$\left|E(Q) \cap ((C_*/c_{\mathrm{Page}}) \log Q, Q]\right| \le 1.$$

In particular, after this external Page input and its constant matching are accepted, a positive-density or fixed-period CRT family of such ultra-close real-zero carriers cannot be the persistent obstruction.

Proof. The certificate is recorded in

```
docs/monograph/prime-matrix-beta-gap-page-sparsity-router.md.
```

If $P > (C_*/c_{\mathrm{Page}}) \log Q$, then

$$\frac{C(P)}{P} \le \frac{C_*}{P} < \frac{c_{\mathrm{Page}}}{\log Q}.$$

Hence every zero counted by $E(Q)$ in this range satisfies

$$\beta_P > 1 - \frac{C(P)}{P} > 1 - \frac{c_{\mathrm{Page}}}{\log Q},$$

and so lies in the Page uniqueness region for conductors $q \le Q$. The Page input allows at most one such real-zero carrier. Therefore the high part of $E(Q)$ has cardinality at most one.

This eliminates only the multi-carrier CRT reproduction shape. It does not internalize the Page constant, exclude a moving singleton exceptional carrier, or control non-real zero packets, endpoint terms, or prime-power residuals. The remaining target is

```
PageExceptionalSingletonCarrierOrNonrealZeroPacketResidualBudget,
```

with fallback

```
MovingPageSingletonCarrierPDECOrCoherentZeroPacket.
```

No unconditional row/column theorem is claimed here. □

Theorem 7.35 (Terminal-square downstream frontier synchronization)**.** *The terminal-row square-phase bridge does not leave the row/column program at the older square-phase long-block interface. In the current archive, that interface has already been synchronized with the half-grid and phase-band routers, with the localized P-CRT/AP-zero-packet route, and with the global CRT signed-payload route. Thus*

`SquarePhaseSpecialPhaseLongBlockPDECExclusion`

is an upstream alias rather than the deepest active terminal-row frontier.

Status and reference. The synchronization certificate is recorded in

```
docs/monograph/prime-matrix-terminal-square-phase-downstream-frontier-sync-router.md.
```

It reads the existing terminal CRT atom certificate, the terminal square-phase bridge, the half-grid boundary-word router, the no-slot phase-band router, the localized P-CRT/Linnik–2 barrier, the rank-one/AP-zero-packet routers, the Page-sparsity beta-gap router, and the global CRT signed-payload router.

The result is a bookkeeping theorem: the terminal-square interface is now aligned with the deeper consolidated basis consisting of AP zero-packet residuals, signed-payload construction, same-set PDEC input, and the recorded global incidence/rate/D-structure inputs. No new unconditional contradiction is asserted; the row/column theorem remains open until those downstream inputs are closed or independently accepted. □

Theorem 7.36 (Common source declaration packet for payload and ExactUV)**.** *In the strict internal source lane, the signed-payload obstruction and the ActualEmitterExactUV incidence obstruction have the same missing pre-Cauchy source root. More precisely, the current archive routes*

$$\texttt{NoncircularAtomicBasisWordSignedCoefficientOriginIdentityBeforePushforward}$$

and

$$\texttt{ActualEmitterSourceDomainEntropyLedger}$$
$$\wedge\ \texttt{ExactUVMapFixedPairPolylogFiberBoundLedger}$$

to the single packet

$$\texttt{PreCauchyActualNoncanonicalEmitterSourceDeclarationPacket}.$$

Status and reference. The certificate is recorded in

```
docs/monograph/
prime-matrix-strict-source-declaration-payload-exactuv-
unification-router.md.
```

The packet must be declared before Cauchy, Φ, payment, or any zero-row recovery. Its fields are: an actual noncanonical source declaration line, primitive summand rows, basis-word/signed-coefficient origin identities, prepushforward α/δ equality, source-domain entropy, fixed exact (u, v) fiber bounds, and a named return partition for absent declarations, fiber collapse, entropy deficit, sign conflicts, local-factor zeros, over-budget rows, and canonical leakage.

This is a routing and consolidation statement. The packet has not been proved in the present corpus. The AP zero-packet branch and same-set PDEC branch remain open, as do the model/rate ledgers and D-structure/Rankin acceptance. Therefore no unconditional row/column theorem is claimed here. □

Theorem 7.37 (Internal line-by-line audit)**.** *The author-side line-by-line audit has found no new item classified as a mathematical block. The remaining blocks are referee blocks: Prime Matrix final promotion requires independent acceptance of the D-structure/Tail-log4/finite-verification interface, and RH final promotion requires independent acceptance of all controlled exits in the RH contradiction-field manuscript.*

Proof. The internal audit matrix assigns one of four labels to each link. No link is a math-block.

The terminal entries PM-16 and RH-16 remain referee-blocks. An author-side check cannot replace independent referee verification. Therefore the internal audit is complete, but final unconditional theorem promotion is not. □

Remark 7.38 (No overclaim principle)**.** The monograph may state the shared contradiction-field method and the completed audit closures. It must not state unconditional proofs of RH or of the prime-matrix row/column theorem until the remaining structured inputs have independent acceptance.

Chapter 8

Current Status and Honest Boundary

The purpose of this monograph draft is to make the shared structure visible and auditable. It is not yet a final unconditional theorem manuscript. The precise statuses are maintained in `docs/monograph/claim-status-table.md`. Any future promotion to final theorem status requires independent verification of every reduced structured input and every controlled exit.